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Abstract

Motivated from investment-based asset pricing, we propose a new factor model that consists of the market factor, a size factor, an investment factor, and a return-on-equity factor. The new model [i] outperforms the Carhart (1997) four-factor model in pricing portfolios formed on earnings surprise, idiosyncratic volatility, financial distress, equity issues, as well as on investment and return-on-equity; [ii] performs similarly as the Carhart model in pricing portfolios on momentum as well as on size and book-to-market; but [iii] underperforms in pricing the total accrual deciles. Our model's performance, combined with its clear economic intuition, suggests that it can serve as a new workhorse model for academic research and investment management practice.

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1 Introduction

In an extremely influential article, Fama and French (1996) show that their three-factor model, which includes the market excess return, a factor based on market equity (*SMB*), and a factor based on book-to-market (*HML*), summarizes the state-of-the-art understanding of the cross-section of returns as of the mid-1990s. Over the past 15 years, however, it has become clear that even the Fama-French model fails to explain a wide range of cross-sectional anomalies.¹

We propose a new multifactor model motivated from investment-based asset pricing, which is in turn based on the *q*-theory of investment. In the new model (which we call the *q*-factor model), the expected return of a testing asset in excess of the riskless rate, denoted $E[r^i] - r^f$, is described by the sensitivity of its return to four factors: (i) the market excess return (*MKT*), (ii) the difference between the return on a portfolio of small-market equity stocks and the return on a portfolio of big-market equity stocks (r_{ME}), (iii) the difference between the return on a portfolio of low-investment stocks and the return on a portfolio of high-investment stocks ($r_{\Delta A/A}$), and (iv) the difference between the return on a portfolio of high return-on-equity (*ROE*) stocks and the return on a portfolio of low return-on-equity stocks (r_{ROE}). Formally,

$$E[r^i] - r^f = \beta_{MKT}^i E[MKT] + \beta_{ME}^i E[r_{ME}] + \beta_{\Delta A/A}^i E[r_{\Delta A/A}] + \beta_{ROE}^i E[r_{ROE}], \quad (1)$$

in which $E[MKT]$, $E[r_{ME}]$, $E[r_{\Delta A/A}]$, and $E[r_{ROE}]$ are expected factor premiums, and the factor loadings, β_{MKT}^i , β_{ME}^i , $\beta_{\Delta A/A}^i$, and β_{ROE}^i , are time series slopes from regressing the excess returns of testing asset *i* on *MKT*, r_{ME} , $r_{\Delta A/A}$, and r_{ROE} , respectively. Over the 1972–2011 period, $E[r_{ME}]$ is 0.31% per month ($t = 2.09$), $E[r_{\Delta A/A}]$ 0.44% ($t = 4.73$), and $E[r_{ROE}]$ 0.60% ($t = 4.85$).

Through extensive factor regressions, the *q*-factor model goes a long way toward explaining

¹See, for example, Ritter (1991), Jegadeesh and Titman (1993), Ikenberry, Lakonishok, and Vermaelen (1995), Loughran and Ritter (1995), Spiess and Affleck-Graves (1995), Chan, Jegadeesh, and Lakonishok (1996), Dichev (1998), Griffin and Lemmon (2002), Ang, Hodrick, Xing, and Zhang (2006), Daniel and Titman (2006), and Campbell, Hilscher, and Szilagyi (2008). The evidence is often interpreted as mispricing that arises from investors' over- or underreaction to news. For instance, Campbell et al. suggest that their evidence "is a challenge to standard models of rational asset pricing in which the structure of the economy is stable and well understood by investors (p. 2934)."

many anomalies that the Fama-French model cannot. First, the q -factor model outperforms the Fama-French model and the Carhart (1997) four-factor model in explaining anomalies related to earnings surprise, idiosyncratic volatility, financial distress, net stock issues, composite issuance, as well as investment and *ROE*. For example, the average magnitude of the Fama-French alphas across deciles formed on earnings surprise is 0.17% per month, and the high-minus-low decile has a Fama-French alpha of 0.54% ($t = 4.26$). The Carhart model reduces the average magnitude of alphas to 0.11% and the high-minus-low alpha to 0.32% ($t = 2.43$). The q -factor model reduces the average magnitude of the alphas further to 0.06% and the high-minus-low alpha to 0.14% ($t = 0.92$). Across the idiosyncratic volatility deciles, the average magnitude of the alphas is 0.19% in the Fama-French model, 0.15% in the Carhart model, and 0.10% in the q -factor model. The high-minus-low alphas for the three models are -0.91% ($t = -4.48$), -0.58% ($t = -2.59$), and -0.04% ($t = -0.19$), respectively. Finally, across the Campbell, Hilscher, and Szilagyi (2008) financial distress deciles, the average magnitude of the alphas is 0.25% in the Fama-French model, 0.12% in the Carhart model, and 0.15% in the q -factor model. The high-minus-low alphas for the three models are -1.43% ($t = -5.21$), -0.55% ($t = -2.51$), and 0.02% ($t = 0.07$), respectively.

Second, the q -factor model outperforms the Fama-French model, and performs similarly as the Carhart model in pricing the momentum deciles. The average magnitude of alphas is 0.19% per month, and the winner-minus-loser alpha is 1.13% ($t = 4.42$) in the Fama-French model. The Carhart model reduces the average magnitude of alphas to 0.10% and the winner-minus-loser alpha to 0.05% ($t = 0.44$). The average magnitude of alphas is 0.09% in the q -factor model, and the winner-minus-loser alpha is 0.23% ($t = 0.65$). None of the individual deciles have significant alphas in the q -factor model, in contrast to four in the Fama-French model and three in the Carhart model.

The q -factor model performs similarly as the Fama-French model and the Carhart model in pricing the 25 size and book-to-market portfolios. The average magnitude of alphas is 0.12% in the q -factor model, which is in line with 0.10% in the Fama-French model and 0.11% in the Carhart model. Four individual portfolios have significant alphas in the q -factor model, compared with four

in the Fama-French model and five in the Carhart model. Only one value-minus-growth quintile has a significant alpha in the q -factor model, compared with three in the Fama-French model and two in the Carhart model. The average magnitude of the value-minus-growth alphas is 0.24% in the q -factor model, which is lower than 0.32% in the Fama-French model and 0.29% in the Carhart model.

However, the q -factor model has trouble in explaining the Sloan (1996) total accrual effect. Across the accrual deciles, the average magnitude of the alphas is 0.13% per month in the Fama-French model, 0.11% in the Carhart model, and is 0.14% in the q -factor model. The high-minus-low alphas are -0.29% ($t = -1.96$) and -0.29% ($t = -1.69$) in the Fama-French model and the Carhart model, respectively, but is -0.39% ($t = -2.48$) in the q -factor model. Augmenting the market factor with the investment factor alone reduces the high-minus-low alpha to an insignificant -0.05% . While the investment factor loading goes in the right direction, the ROE factor loading goes in the wrong direction in explaining the accrual effect. Intuitively, high accrual firms invest more, but are also more profitable (and load more on the ROE factor) than low accrual firms.

As noted, we motivate the q -factor model from investment-based asset pricing. Intuitively, investment predicts returns because given expected cash flows, high costs of capital mean low net present values of new capital and low investment, whereas low costs of capital mean high net present values of new capital and high investment. ROE predicts returns because high expected ROE relative to low investment means high discount rates. The high discount rates are necessary to counteract the high expected ROE to induce low net present values of new capital and subsequently low investment. If the discount rates are not high enough to offset the high expected ROE , firms would instead observe high net present values of new capital and invest more. Similarly, low expected ROE relative to high investment (such as small-growth firms in the late 1990s) means low discount rates. If the discount rates are not low enough to counteract the low expected ROE , the firms would instead observe low net present values of new capital and invest less. Finally, we include the size factor primarily to reduce the average magnitude of the alphas across size-related portfolios. As such, the size factor plays only a secondary role in the q -factor model, whereas the

investment and the *ROE* factors are more prominent.

Our central contribution is to provide a new workhorse factor model for estimating expected returns. In particular, we create a new incarnation of Fama and French (1996) by showing that the *q*-factor model summarizes what we know about the cross-section of returns as of the early 2010s. In so doing, we also elaborate a unified conceptual framework in which almost all of the anomalies can be interpreted in an internally consistent and economically meaningful way. The *q*-factor model's performance, combined with its clear economic intuition, suggests that the model can be used in practical applications such as evaluating mutual fund performance, measuring abnormal returns in event studies, estimating costs of capital for capital budgeting and stock valuation, and obtaining expected return estimates for optimal portfolio choice.

The traditional approach in empirical finance and capital markets research in accounting is to look for common factors from the consumption side of the economy (e.g., Breeden, Gibbons, and Litzenberger (1989)). We instead exploit a direct link between stock returns and firm characteristics from the production side. Cochrane (1991) first uses *q*-theory to study asset prices. Berk, Green, and Naik (1999) construct real option models to explain anomalies. Liu, Whited, and Zhang (2009) estimate the characteristics-expected return relations derived from *q*-theory via structural estimation. We use the portfolio approach to build a workhorse factor model. A factor model is more flexible in practice because of its simplicity and the availability of high frequency returns data. Finally, it should be noted that the investment effect and the earnings effect are not new to our work.² However, recognizing their fundamental importance in investment-based asset pricing, we build a new workhorse factor model on these two effects for the cross-section of expected stock returns.

The rest of the paper is organized as follows. Section 2 constructs the new factors. Section 3 tests the *q*-factor model via factor regressions. Section 4 reports specification tests by dropping the

²See, for example, Fairfield, Whisenant, and Yohn (2003), Titman, Wei, and Xie (2004), Anderson and Garcia-Feijóo (2006), Cooper, Gulen, and Schill (2008), Xing (2008), and Polk and Sapienza (2009) for the investment effect, and Ball and Brown (1968), Bernard and Thomas (1989, 1990), Ball, Kothari, and Watts (1993), Chan, Jagadeesh, and Lakonishok (1996), Haugen and Baker (1996), Abarbanell and Bushee (1998), Frankel and Lee (1998), Dechow, Hutton, and Sloan (1999), Piotroski (2000), Fama and French (2006), and Novy-Marx (2012) for the earnings effect.

size factor from the q -factor model. Section 5 interprets the results. Finally, Section 6 concludes. A separate Internet Appendix contains supplementary results.

2 The Explanatory Factors

Monthly returns, dividends, and prices are from the Center for Research in Security Prices (CRSP) and accounting information from the Compustat Annual and Quarterly Fundamental Files. The sample is from January 1972 to December 2011. The starting date is restricted by the availability of quarterly earnings announcement dates as well as quarterly earnings and assets data. We also exclude financial firms and firms with negative book equity.

We measure investment-to-assets ($\Delta A/A$) as the annual change in total assets (Compustat annual item AT) divided by lagged total assets. The change in total assets is in the most comprehensive measure of investment. We measure ROE as income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock.³

We construct the size factor, r_{ME} , the investment factor, $r_{\Delta A/A}$, and the ROE factor, r_{ROE} , from a triple two-by-three-by-three sort on size, $\Delta A/A$, and ROE . Bernard and Thomas (1990) show that the earnings effect in returns seems stronger in small firms than in big firms. Fama and French (2008) show that the investment effect is strong in microcaps and small stocks, but is largely absent in big stocks. As such, we control for size when constructing the investment and the ROE factors. Controlling for size in this way seems a standard practice in constructing the Fama

³Our measure of the book equity is the quarterly version of the annual book equity measure in Davis, Fama, and French (2000). Fama and French (2006) measure shareholders' equity as total assets minus total liabilities. We follow Davis et al. because Compustat quarterly items SEQQ (stockholders' equity) and CEQQ (common equity) have a broader coverage than items ATQ (total assets) and LTQ (total liabilities) before 1980.

and French (1993) value factor, HML , and the Carhart (1997) momentum factor, WML . Finally, sorting on $\Delta A/A$ and ROE independently helps orthogonalize the two new factors.

In June of each year t , we use the median NYSE market equity (stock price times shares outstanding from CRSP) at the end of June to split NYSE, Amex, and NASDAQ stocks into two groups, small and big. Independently, in June of year t , we also break NYSE, Amex, and NASDAQ stocks into three $\Delta A/A$ groups using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked values of $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Also independently, at the beginning of each month, we sort all stocks into three groups based on the NYSE breakpoints for the low 30%, middle 40%, and the high 30% of the ranked values of ROE . Earnings and other accounting variables in Compustat quarterly files are used in the monthly sorts in the months immediately after the most recent public earnings announcement dates (Compustat quarterly item RDQ). For example, if the earnings for the fourth fiscal quarter of year $t - 1$ are publicly announced on March 5 (or March 25) of year t , we use the announced earnings (divided by the book equity from the third quarter of year $t - 1$) to form portfolios at the beginning of April of year t .

Taking the intersections of the two size, the three $\Delta A/A$, and the three ROE groups, we form 18 portfolios. Monthly value-weighted portfolio returns are calculated for the current month, and the portfolios are rebalanced monthly. (The ROE portfolios are reconstructed monthly at the beginning of each month, but the size and the $\Delta A/A$ portfolios are resorted annually in each June.) The size factor, r_{ME} , is the difference (small-minus-big), each month, between the simple average of the returns on the nine small portfolios and the simple average of the returns on the nine big portfolios. Designed to mimic the common variation in returns related to $\Delta A/A$, the investment factor, $r_{\Delta A/A}$, is the difference (low-minus-high), each month, between the simple average of the returns on the six low- $\Delta A/A$ portfolios and the simple average of the returns on the six high- $\Delta A/A$ portfolios. Finally, designed to mimic the common variation in returns related to ROE , the ROE factor is the difference (high-minus-low), each month, between the simple average of the returns on

the six high-*ROE* portfolios and the simple average of the returns on the six low-*ROE* portfolios.⁴

From Panel A of Table 1, the size factor earns an average return of 0.31% per month from January 1972 to December 2011 ($t = 2.09$). The Capital Asset Pricing Model (CAPM) explains this average return, leaving an alpha of 0.24% ($t = 1.64$). Regressing our size factor on the Fama-French three factors produces an *SMB* loading of 0.99. Also, Panel B shows that our size factor and *SMB* have an almost perfect correlation of 0.95. As such, the two size factors are effectively the same.

The investment factor, $r_{\Delta A/A}$, earns an average return of 0.44% per month ($t = 4.73$). Regressing $r_{\Delta A/A}$ on the market factor produces an alpha of 0.51% and a beta of -0.15 , both of which are significant (Panel A). Regressing the investment factor on the Fama-French factors shows a large and significantly positive *HML* loading of 0.40. From Panel B, $r_{\Delta A/A}$ and *HML* have a high correlation of 0.69, suggesting that $r_{\Delta A/A}$ would play a similar role as *HML* in factor regressions.

The *ROE* factor, r_{ROE} , earns an average return of 0.60% per month, which is more than 4.5 standard errors from zero (Panel A). Regressing r_{ROE} on the Fama-French factors produces a low R^2 of only 19%, meaning that r_{ROE} represents an important source of common variation missing from the Fama-French model. Interestingly, even though it is constructed from a triple sort with size controlled, r_{ROE} still has a (relatively) large correlation of -0.30 with r_{ME} (Panel B). A finer sort on size can help reduce the magnitude of the correlation, but it would likely only reinforce the explanatory power of the new factors. Also, Panel B shows that r_{ROE} has a high correlation of 0.50 with the momentum factor, *WML*, meaning that r_{ROE} would play a similar role as *WML* in factor regressions. Finally, the investment and the *ROE* factors have a low and insignificant correlation of 0.05. As such, the triple sort seems successful in orthogonalizing the two new factors.⁵

⁴Let p_{ijk} , with $i = 1, 2$ and $j, k = 1, 2, 3$, be the value-weighted portfolio that contains all the firms in the i_{th} size group, in the j_{th} $\Delta A/A$ group, and in the k_{th} *ROE* group. For example, p_{132} is the portfolio containing all the firms that reside simultaneously in the small-size portfolio, the high- $\Delta A/A$ portfolio, and the median-*ROE* portfolio. The size, the investment, and the *ROE* factors are defined as $r_{ME} \equiv \left(\sum_{j=1}^3 \sum_{k=1}^3 p_{1jk} - \sum_{j=1}^3 \sum_{k=1}^3 p_{2jk} \right) / 9$, $r_{\Delta A/A} \equiv \left(\sum_{i=1}^2 \sum_{k=1}^3 p_{i1k} - \sum_{i=1}^2 \sum_{k=1}^3 p_{i3k} \right) / 6$, and $r_{\Delta ROE} \equiv \left(\sum_{i=1}^2 \sum_{j=1}^3 p_{ij3} - \sum_{i=1}^2 \sum_{j=1}^3 p_{ij1} \right) / 6$, respectively.

⁵When constructing the factors, we construct the $\Delta A/A$ portfolios with annual sorts but the *ROE* portfolios with monthly sorts. There are two reasons for this practice. First, the *ROE* factor is most relevant for explaining momentum, post-earnings-announcement drift, the idiosyncratic volatility effect, and the distress effect (Section 3).

3 Factor Regressions

We use the standard time-series factor regressions to test the q -factor model:

$$r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon^i. \quad (2)$$

If the model's performance is adequate, α_q^i should be economically small and statistically insignificant from zero. As a convention, we use the NYSE breakpoints in constructing testing portfolios. Doing so is consistent with our construction of the q -factors, which is in turn comparable with the construction of *SMB*, *HML*, and *WML*. A good economic reason for using the NYSE breakpoints is to alleviate the impact of microcaps and small stocks. Due to transaction costs and lack of liquidity, the portion of anomalies in microcaps and small stocks might not be exploitable in practice. For completeness, however, we also report in Internet Appendix detailed results from using the NYSE-Amex-NASDAQ breakpoints in constructing a given set of anomaly portfolios if the original paper that documents the anomaly uses such breakpoints (as well as alternative return weighting schemes).

3.1 The Earnings Momentum Deciles

The q -factor model largely explains the post-earnings announcement drift. Following Foster, Olsen, and Shevlin (1984), we measure earnings surprise as Standardized Unexpected Earnings (*SUE*). We calculate *SUE* as the change in the most recently announced quarterly earnings per share from its value announced four quarters ago divided by the standard deviation of this change in quarterly earnings over the prior eight quarters. (We require a minimum of six quarters in calculating *SUE*.)

We use the NYSE breakpoints to rank all NYSE, Amex, and NASDAQ stocks at the beginning of each month based on their most recent past *SUE*. Monthly value-weighted portfolio returns are

Because these testing portfolios are all constructed monthly, we use a similar approach to construct the *ROE* factor. Internet Appendix reports detailed results that the anomalies above do not exist in annually sorted portfolios. None of the high-minus-low portfolios produce mean excess returns or CAPM alphas that are significantly different from zero. Because the targeted anomalies only exist in monthly sorts, it seems natural to also construct the explanatory *ROE* factor with monthly sorts. Second, investment-based asset pricing predicts that *ROE* forecasts future returns to the extent that it forecasts future *ROE* (Section 5.1). Because the most recent *ROE* contains the most up-to-date information about future *ROE*, constructing the *ROE* factor with monthly sorts is consistent with economic theory.

calculated for the current month, and the portfolios are rebalanced at the beginning of next month.

Table 2 shows that consistent with Chan, Jegadeesh, and Lakonishok (1996), high *SUE* stocks earn higher average returns than low *SUE* stocks. The high-minus-low decile earns an average return of 0.43% per month ($t = 3.39$), a CAPM alpha of 0.49% ($t = 4.03$), a Fama-French alpha of 0.54% ($t = 4.26$), and a Carhart alpha of 0.32% ($t = 2.43$). The mean absolute error (m.a.e., calculated as the average absolute value of the alphas) across the deciles are 0.16%, 0.17%, and 0.11% for the CAPM, the Fama-French model, and the Carhart model, respectively. Among the ten deciles, six have significant alphas in the CAPM, five in the Fama-French model, and three in the Carhart model. All three models are strongly rejected by the Gibbons, Ross, and Shanken (1989, GRS) test on the null that the alphas across the ten deciles are jointly zero.

The q -factor model reduces the high-minus-low alpha to insignificance: 0.14% per month, which is within one standard error of zero. The q -factor model derives its explanatory power mostly from the *ROE* factor. The high-minus-low decile has an *ROE* factor loading of 0.49, which is more than 5.5 standard errors from zero, whereas all the other factor loadings are economically small and mostly insignificant. Intuitively, firms that have recently experienced positive earnings surprises are more profitable than firms that have recently experienced negative earnings surprises. The q -factor model also produces a small m.a.e. of 0.06%, which is smaller than 0.11% from the Carhart model. Across the ten deciles, only one out of ten has a significant alpha in the q -factor model, relative to three in the Carhart model. While the traditional models are all rejected by the GRS test, the q -factor model cannot be rejected at the 5% significance level ($p\text{-value} = 0.41$).

3.2 The Price Momentum Deciles

At the beginning of each month t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into deciles based on their prior six-month returns from month $t - 7$ to $t - 2$. Skipping month $t - 1$, we calculate monthly value-weighted returns for the portfolios from month t to $t + 5$. The six-month holding period means that for a given portfolio there exist six sub-portfolios for

each month. As such, we take the simple average of value-weighted returns on the six sub-portfolios as the monthly return of the given portfolio.

From Table 3, the winner-minus-loser decile earns an average return of 0.87% per month ($t = 3.15$). The CAPM alpha and the Fama-French alpha for the winner-minus-loser decile are 0.93% and 1.13%, respectively, which are both more than 3.5 standard errors from zero. The m.a.e. in the CAPM is 0.17%, and that in the Fama-French model is 0.19%. Both models are strongly rejected by the GRS test. The Carhart model performs extremely well for the one-way momentum deciles. The winner-minus-loser alpha is only 0.05%, which is within 0.5 standard errors of zero. The m.a.e. is only 0.10%, but the model is still rejected by the GRS test.

The q -factor model seems largely comparable with the Carhart model. The winner-minus-loser alpha is 0.23% per month, which is within one standard error of zero. The m.a.e. is 0.09%, but the model is rejected by the GRS test. None of the individual deciles have significant alphas in the q -factor model, in contrast to four in the Fama-French model and three in the Carhart model. The average R^2 in the q -factor model is 89%, which is slightly lower than 94% in the Carhart model.

Table 3 shows that the explanatory power of the q -factor model derives almost exclusively from the ROE factor. The loadings of the winner-minus-loser decile on the market and investment factors are economically small and statistically insignificant from zero. The size factor loading is 0.27, but is within 1.5 standard errors from zero. In contrast, the loser decile has a large and significantly negative loading, and the winner decile has a large and significantly positive loading on the ROE factor. The winner-minus-loser decile has a large ROE factor loading of 1.03, which is more than five standard errors from zero. Given the average ROE factor premium of 0.60% per month, the ROE factor loading captures momentum profits about 0.62%.

3.3 The Idiosyncratic Volatility Deciles

Following Ang, Hodrick, Xing, and Zhang (2006), we measure a stock's idiosyncratic volatility ($IVOL$) as the standard deviation of the residuals from regressing the stock's returns on the Fama-

French three factors. Each month we form value-weighted deciles by using the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks on the *IVOL* estimated using daily returns over the previous month. (We require a minimum of 15 daily stock returns.) We hold the value-weighted deciles for the current month, and rebalance the portfolios at the beginning of next month.

Consistent with Ang, Hodrick, Xing, and Zhang (2006), high *IVOL* stocks earn lower average returns than low *IVOL* stocks. From Table 4, the high-minus-low decile earns an average return of -0.54% per month, which is, however, only about 1.5 standard errors from zero. More important, traditional factor loadings often go in the wrong direction in explaining the *IVOL* effect. In the CAPM, for example, the market beta of the high-minus-low decile is 0.91, giving rise to a large CAPM alpha of -0.95% , which is about 3.5 standard errors from zero. In the Fama-French model and the Carhart model, the market and the *SMB* betas are large and positive, going in the wrong direction, but the *HML* and *WML* loadings are large and negative, going in the right direction. The two alphas for the high-minus-low decile are -0.91% and -0.58% , both of which are significant. The m.a.e.'s are 0.20%, 0.19%, and 0.15% for the CAPM, the Fama-French model, and the Carhart model, respectively. All three models are strongly rejected by the GRS test.

The *q*-factor model reduces the high-minus-low alpha to a tiny -0.04% per month ($t = -0.19$). The m.a.e. drops to 0.10% from 0.15% in the Carhart model. Notably, none of the ten *IVOL* deciles have significant alphas in the *q*-factor model. In contrast, four out of ten deciles have significant alphas in all three traditional factor models. Although the market and the size factor loadings go in the wrong direction, the investment and the *ROE* factor loadings both go in the right direction in explaining the *IVOL* effect. The high-minus-low decile has negative loadings of -0.98 and -0.96 on the investment factor and the *ROE* factor, respectively, both of which are more than five standard errors from zero. As such, high *IVOL* stocks, which are relatively small, tend to invest more, but are less profitable than low *IVOL* stocks. Controlling for investment and *ROE* is sufficient to explain the *IVOL* effect. However, the *q*-factor model is still rejected by the GRS test (p-value = 0.02).

3.4 The Financial Distress Deciles

At the beginning of each month, we use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks into deciles on Campbell, Hilscher, and Szilagyi's (2008) failure probability (see Internet Appendix for the detailed definition). Earnings and other accounting data for a fiscal quarter are used in portfolio sorts in the months immediately after the quarter's public earnings announcement dates (Compustat quarterly item RDQ). The starting point of the sample for the failure probability deciles is January 1976, which is restricted by the availability of the quarterly data items required in calculating failure probability. (Campbell et al. start their sample in 1981.) Monthly value-weighted portfolio returns are calculated for the current month, and the portfolios are rebalanced monthly.

From Table 5, more financially distressed firms earn lower average returns than less financially distressed firms. The high-minus-low distress decile earns an average return of -0.57% per month, which is, however, within 1.5 standard errors from zero. More important, controlling for traditional risk measures exacerbates the distress anomaly because more distressed firms appear riskier. In particular, the high-minus-low decile has a CAPM beta of 0.84, giving rise to a CAPM alpha of -1.04% , which is more than three standard errors from zero. In the Fama-French model, all three factor loadings go in the wrong direction in explaining the distress effect. The high-minus-low decile has a market beta of 0.76, an *SMB* loading of 0.97, and an *HML* loading of 0.46. These large and positive risk measures produce a huge Fama-French alpha of -1.43% , which is more than five standard errors from zero. The Carhart model shrinks the high-minus-low alpha to -0.55% because of a negative *WML* loading of -1.02 . Intuitively, most distressed firms tend to be losers, and least distressed firms tend to be winners. The m.a.e. across the deciles is 0.17% in the CAPM, 0.25% in the Fama-French model, and 0.12% in the Carhart model. The CAPM and the Fama-French model are strongly rejected by the GRS test, but the test of the Carhart model is only marginally significant.

The *q*-factor model helps explain the distress effect. The high-minus-low decile has a tiny alpha of 0.02% per month ($t = 0.07$). Going in the right direction in explaining the average returns, more

distressed firms have lower *ROE* factor loadings than less distressed firms. The loading spread across the extreme deciles is -1.79 , which is more than 7.5 standard errors from zero. Intuitively, more distressed firms are less profitable (and load less on the *ROE* factor) than less distressed firms. In particular, profitability enters the failure probability measure with a large and negative coefficient (see Internet Appendix). The *ROE* factor loading of the high-minus-low decile alone is enough to overcome the large and positive loadings on the market and the size factors that go in the wrong direction in explaining the distress effect. The investment factor loading of the high-minus-low decile is only 0.11. However, the *q*-factor model produces an m.a.e. of 0.15%, which is slightly higher than 0.12% in the Carhart model. And the model is rejected by the GRS test ($p\text{-value} = 0.01$).

3.5 The Net Stock Issues Deciles

Fama and French (2008) and Pontiff and Woodgate (2008) show that firms with high net stock issues underperform firms with low net stock issues. Following Fama and French, we measure net stock issues as the natural log of the ratio of the split-adjusted shares outstanding at the fiscal yearend in $t-1$ to the split-adjusted shares outstanding at the fiscal yearend in $t-2$. The split-adjusted shares outstanding is shares outstanding (Compustat annual item CSHO) times the adjustment factor (item AJEX). In June of each year t , we use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks into deciles based on net stock issues for the fiscal year ending in calendar year $t-1$. Because a disproportionately large number of firms have zero net stock issues, we group all the firms with negative net issues into deciles one and two (equal-numbered), and all the firms with zero net issues into decile three. We then sort the firms with positive net issues into the remaining seven (equal-numbered) deciles. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the deciles are rebalanced in June of $t+1$.

From Table 6, firms with high net issues earn lower average returns than firms with low net issues, 0.20% versus 0.88% per month. The high-minus-low decile earns an average return of -0.68% , a CAPM alpha of -0.77% , a Fama-French alpha of -0.62% , and a Carhart alpha of -0.57% , all of

which are more than 3.5 standard errors from zero. Across the ten deciles, six CAPM alphas, six Fama-French alphas, and five Carhart alphas are significantly different from zero. The m.a.e.'s are 0.21%, 0.20%, and 0.17% in the CAPM, the Fama-French model and the Carhart model, respectively. All three factor models are again strongly rejected by the GRS test.

The q -factor model reduces the high-minus-low alpha to -0.32% per month, which is still significant ($t = -2.10$). This alpha represents 44% reduction in magnitude from the Carhart alpha of -0.57% . The m.a.e. also drops from 0.17% in the Carhart model to 0.12% in the q -factor model. However, the q -factor model is still rejected by the GRS test. Both the investment and the *ROE* factors contribute to the q -factor model's explanatory power. The high-minus-low decile has an investment factor loading of -0.66 ($t = -6.33$), going in the right direction in explaining the net issues effect. Intuitively, high net issues firms invest more than low net issues firms. The *ROE* factor loading also moves in the right direction. The high-minus-low decile has an *ROE* factor loading of -0.23 ($t = -3.43$). The evidence suggests that high net issues firms are somewhat less profitable than low net issues firms at the portfolio formation.⁶

3.6 The Composite Issuance Deciles

Following Daniel and Titman (2006), we measure composite issuance as the growth rate in the market equity not attributable to the stock return, $\log(ME_t/ME_{t-5}) - r(t-5, t)$. For June of year t , $r(t-5, t)$ is the cumulative log return on the stock from the last trading day of June in year $t-5$ to the last trading day of June in year t , and ME_t is the total market equity on the last trading day of June in year t from CRSP. Equity issuance such as seasoned equity issues, employee stock option plans, and share-based acquisitions increase the composite issuance, whereas repurchase activities such as share repurchases and cash dividends reduce the composite issuance. In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on composite issuance. Monthly value-weighted portfolio returns are calculated from July of year t to June of

⁶Loughran and Ritter (1995) show that new equity issuers are more profitable than nonissuers. Because net stock issues are new issues net of share repurchases, our evidence is consistent with Lie (2005), who shows that share repurchasing firms exhibit superior operating performance relative to industry peers.

year $t + 1$, and the portfolios are rebalanced in June.

Table 7 shows that high composite issuance firms earn lower average returns than low composite issuance firms. The average return spread is -0.58% per month, which is almost three standard errors from zero. The CAPM beta of the high-minus-low decile is 0.46, which goes in the wrong direction in explaining the average return. As a result, the CAPM alpha is -0.79% , which is more than 4.5 standard errors from zero. In the Fama-French model and the Carhart model, the *HML* betas are -0.67 and -0.70 , which help reduce the high-minus-low alphas in magnitude to -0.50% and -0.40% , ($t = -3.61$ and -2.91), respectively.

The q -factor model reduces the high-minus-low alpha further to -0.21% per month, which is within 1.5 standard errors of zero. The m.a.e. is 0.12% , which is lower than 0.15% for the Carhart model. However, similar to the traditional factor models, the q -factor model is still rejected by the GRS test. The main source of the explanatory power is the investment factor. The high-minus-low decile has an investment factor loading of -1.11 , which is more than 14 standard errors from zero. Although also going in the right direction, the *ROE* factor loading for the high-minus-low decile, -0.11 , is economically small and statistically insignificant.

3.7 The Abnormal Corporate Investment Deciles

Titman, Wei, and Xie (2004) show that firms which increase capital investments earn negative subsequent benchmark-adjusted returns. Following Titman et al., we measure abnormal corporate investment that applies for the portfolio formation year t , as $ACI_{t-1} \equiv CE_{t-1}/[(CE_{t-2} + CE_{t-3} + CE_{t-4})/3] - 1$, in which CE_{t-1} is capital expenditure (Compustat annual item CAPX) scaled by its sales (item SALE) in year $t-1$. The last three-year average capital expenditure aims to project the benchmark investment at the portfolio formation year. Using sales as the deflator implicitly assumes that the benchmark investment grows proportionately with sales. As in Titman et al., we exclude firms with sales less than ten million dollars.

In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks

into deciles based on *ACI* for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are computed from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. Table 8 shows that the *ACI* effect is weak. High *ACI* stocks underperform low *ACI* stocks only by 0.26% per month ($t = 1.57$). The high-minus-low alphas are insignificant in all the factor models. The high-minus-low alpha is -0.11% in the q -factor model, and is somewhat lower in magnitude than -0.16% in the Carhart model. However, the m.a.e. is higher in the q -factor model than in the Carhart model, 0.15% versus 0.12%. Both models are still rejected by the GRS test.

3.8 The Total Accruals Deciles

Table 9 documents a weakness of the q -factor model. Sloan (1996) shows that firms with high total accruals earn lower average returns than firms with low total accruals. Following Sloan, we measure total accruals as changes in noncash working capital minus depreciation expense scaled by total assets averaged over the prior two years. The noncash working capital is the change in noncash current assets minus the change in current liabilities less short-term debt and taxes payable.⁷ In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on total accruals scaled by average total assets (Compustat annual item AT) as of the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June.

From Table 9, high accrual stocks underperform low accrual stocks by 0.30% per month ($t = 1.94$). The CAPM and the Fama-French model both fail to explain this average return, with significant alphas of -0.35% and -0.29% , respectively. The Carhart alpha is -0.29% , but is within 1.7 standard errors of zero. The m.a.e. is 0.13% in the CAPM and in the Fama-French model, and is 0.11% in the Carhart model. All three models are rejected by the GRS test.

The q -factor model underperforms the Fama-French model and the Carhart model. The high-

⁷Specifically, total accruals $\equiv (\Delta CA - \Delta CASH) - (\Delta CL - \Delta STD - \Delta TTP) - DP$, in which ΔCA is the change in current assets (Compustat annual item ACT), $\Delta CASH$ is the change in cash or cash equivalents (item CHE), ΔCL is the change in current liabilities (item LCT), ΔSTD is the change in debt included in current liabilities (item DLC), ΔTTP is the change in income taxes payable (item TXP), and DP is depreciation and amortization expense (item DP).

minus-low alpha is -0.39% per month, which is about 2.5 standard errors from zero. In contrast, the Carhart alpha is only -0.29% ($t = -1.69$). The m.a.e. is 0.14% in the q -factor model, and is higher than 0.11% in the Carhart model. The investment factor loading goes in the right direction as the average returns. The high-minus-low decile has an investment factor loading of -0.56 , which is more than five standard errors from zero.⁸ The trouble of the q -factor model is caused by the ROE factor loading, which goes in the wrong direction in explaining the accrual effect. The ROE factor loading for the high-minus-low decile is 0.34 , which is more than four standard errors from zero. Intuitively, high accrual stocks are more profitable (and load more on the ROE factor) than low accrual stocks. The high-minus-low decile has a size factor loading of 0.42 , also going in the wrong direction. As noted, the ROE factor is critical for the q -factor model to explain cross-sectional predictability related to earnings surprise, idiosyncratic volatility, and financial distress. As such, the difficulty of the q -factor model with the accrual effect appears unavoidable in our effort to build a new workhorse model for the broad cross-section of average returns.

3.9 Industries

Lewellen, Nagel, and Shanken (2010) argue that asset pricing tests are often misleading because apparently strong explanatory power (such as high R^2) provides only weak support for a model. Our tests are (relatively) immune to this critique because we focus on high-minus-low alphas and mean absolute errors from factor regressions as the yardsticks for evaluating factor models. Following Lewellen et al.'s prescription, we also confront the q -factor model with a wide array of testing portfolios. We explore the q -factor model further with ten industry portfolios.

In June of each year t , we assign each NYSE, Amex, and NASDAQ stock to an industry portfolio based on its four-digit SIC code at that time. (We use Compustat SIC codes for the fiscal year ending in calendar year $t - 1$. If Compustat SIC codes are unavailable, we use CRSP SIC codes for June of year t .) The ten-industry classification is from Kenneth French's Web site. We

⁸Consistent with Wu, Zhang, and Zhang (2010), augmenting the CAPM with the investment factor reduces the high-minus-low alpha to -0.05% ($t = -0.32$) and the m.a.e. to 0.09% . Augmenting the Fama-French model with the investment factor reduces the high-minus-low alpha to -0.14% ($t = -0.93$) and the m.a.e. to 0.10% .

exclude financial firms from the last industry portfolio (“Other”). Monthly value-weighted returns are computed from July of year t to June of year $t + 1$.

From Table 10, the CAPM provides an m.a.e. of 0.19% per month across the ten industry portfolios. Two out of ten industries have significant alphas in the CAPM. The Fama-French model produces a similar m.a.e. of 0.21%, and the Carhart model reduces it slightly to 0.18%. The m.a.e. from the q -factor model is somewhat higher, 0.22%. Three out of ten industries have significant alphas in all three multifactor models, but all four factor models are rejected by the GRS test.

3.10 25 Size and Book-to-Market Portfolios

In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on the market equity at the end of June of t . Independently, in June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on book-to-market equity. Book-to-market equity for June of year t is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the market equity at the end of December of $t - 1$.⁹ Taking intersections, we form 25 size and book-to-market portfolios. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced at the end of June.

Table 11 reports factor regressions of 25 size and book-to-market portfolios. Value stocks earn higher average returns than growth stocks. The average value-minus-growth return is 1.01% per month ($t = 4.48$) in the smallest size quintile and 0.19% ($t = 0.89$) in the biggest size quintile. The small-stock value-minus-growth quintile has a CAPM alpha of 1.17% ($t = 5.40$), a Fama-French alpha of 0.69% ($t = 5.44$), and a Carhart alpha of 0.69 ($t = 5.50$). In particular, the small-growth portfolio has a Fama-French alpha of -0.54% , which is more than 4.5 standard errors from zero, as well as a Carhart alpha of -0.48% , which is almost four standard errors from zero.¹⁰ Also, 14

⁹Following Davis, Fama, and French (2000), we measure book equity as stockholders’ book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders’ equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders’ equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

¹⁰The small-growth anomaly is notoriously difficult to explain. Campbell and Vuolteenaho (2004) show that the

out of 25 individual portfolios and four out of five value-minus-growth quintiles have significant alphas in the CAPM. Four out of 25 portfolios and three out of five value-minus-growth quintiles have significant alphas in the Fama-French model. And five out of 25 portfolios and two out of five value-minus-growth quintiles have significant alphas in the Carhart model. The m.a.e. is lowest in the Fama-French model (0.10%), slightly higher in the Carhart model (0.11%), and highest in the CAPM (0.29%). However, all three models are still strongly rejected by the GRS test.

Table 12 shows that the q -factor model's performance seems comparable with that of the Carhart model. The value-minus-growth alpha in the smallest size quintile is 0.58% per month ($t = 2.89$), which has a somewhat smaller magnitude than the Fama-French alpha and the Carhart alpha. The q -factor model does a good job in explaining the small-growth effect. In contrast to the high Fama-French alpha of -0.57% , the q -factor alpha is only -0.25% ($t = -1.46$). However, the small-value portfolio has an alpha of 0.33% ($t = 2.84$) in the q -factor model, in contrast to the Fama-French alpha of only 0.15% ($t = 1.74$). The m.a.e. in the q -factor model is 0.12% , which is comparable with those in the Fama-French model (0.10%) and the Carhart model (0.11%). Four out of 25 individual portfolios have significant alphas in the q -factor model, similar to the performance of the Fama-French model. However, only one out of five value-minus-growth quintiles has a significant alpha in the q -factor model, in contrast to three in the Fama-French model and two in the Carhart model.

The rest of Table 12 shows that the q -factor model's explanatory power derives mostly from the investment factor. Value stocks have significantly higher investment factor loadings than growth stocks. The loading spreads, ranging from 1.18 to 1.57, are all more than eight standard errors from zero. Intuitively, growth firms with high valuation ratios have more growth opportunities and invest more than value firms with low valuation ratios (e.g., Liu, Whited, and Zhang (2009)). As such, growth firms have large and negative loadings, and value firms have large and positive loadings on the (low-minus-high) investment factor. In contrast, the value-minus-growth loadings

small-growth portfolio is particularly risky in their two-beta model, carrying both higher cash flow betas and higher discount rate betas than the small-value portfolio. Their two-beta model fails to explain the small-growth anomaly.

on the market, the size, and the *ROE* factors are mostly small and insignificant.¹¹

3.11 25 $\Delta A/A$ and *ROE* Portfolios

We have shown that the *q*-factor model performs roughly as well as the Carhart model in pricing the momentum deciles as well as the size and book-to-market portfolios. Both sets of portfolios are constructed directly on characteristics underlying the Carhart model. We also examine how the Carhart model performs, in comparison with the *q*-factor model, in explaining the 25 $\Delta A/A$ and *ROE* portfolios, which are constructed directly on characteristics underlying the *q*-factor model.

In June of each year t , we split NYSE, Amex, and NASDAQ stocks into quintiles using the NYSE breakpoints on $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Independently, we sort all stocks, each month, into five *ROE* quintiles based on the NYSE breakpoints of the ranked values of *ROE*. Earnings and other accounting variables in Compustat quarterly files are used in the sorts in the months immediately after the most recent public earnings announcement dates (Compustat quarterly item RDQ). Taking intersections of the $\Delta A/A$ quintiles and the *ROE* quintiles, we obtain the 25 portfolios. We calculate value-weighted portfolio returns for the current month, and rebalance the portfolios monthly.

Table 13 shows that the double sort on $\Delta A/A$ and *ROE* produces large average return spreads. In particular, the high-minus-low *ROE* quintile in the highest $\Delta A/A$ quintile earns an average return of 0.94% per month, which is more than four standard errors from zero. The low-minus-high $\Delta A/A$ quintile in the lowest *ROE* quintile earns an average return of 0.74%, which is more than 3.5 standard errors from zero. Taking the largest spread across the 25 portfolios (between the high-*ROE*-low- $\Delta A/A$ portfolio and the low-*ROE*-high- $\Delta A/A$ portfolio) yields 1.22% ($t = 4.69$).

The traditional factor models largely fail to explain these average returns. In the CAPM, 11 out of 25 portfolios, three high-minus-low *ROE* quintiles, and three low-minus-high $\Delta A/A$ quintiles

¹¹The evidence suggests that the *q*-factor model performs about as well as the Carhart model in explaining the 25 size and book-to-market portfolios. In Internet Appendix, we extend this basic finding to additional portfolios formed on valuation ratios, including market leverage, long-term prior returns, and earnings-to-price.

have significant alphas. In the Fama-French model, 13 individual portfolios, five high-minus-low *ROE* quintiles, and three low-minus-high $\Delta A/A$ quintiles have significant alphas. In the Carhart model, seven individual portfolios, three high-minus-low *ROE* quintiles, and three low-minus-high $\Delta A/A$ quintiles have significant alphas. The m.a.e.'s are 0.25%, 0.26%, and 0.18% per month, respectively, across the three models, which are all strongly rejected by the GRS test.

Table 14 shows that the *q*-factor model does a good job in explaining the 25 $\Delta A/A$ and *ROE* portfolios. Only one out of 25 individual portfolios, one high-minus-low *ROE* quintile, and one low-minus-high $\Delta A/A$ quintile have significant alphas. The m.a.e. is only 0.09% per month, and the model is not rejected by the GRS test (p-value = 0.26). In particular, the high-*ROE*-low- $\Delta A/A$ minus low-*ROE*-high- $\Delta A/A$ portfolio has a tiny alpha of -0.07% , which is within 0.5 standard errors from zero. In contrast, this alpha is 1.37% ($t = 5.54$) in the CAPM, 1.31% ($t = 5.66$) in the Fama-French model, and 0.93% ($t = 4.09$) in the Carhart model.

The rest of the table shows that, naturally, the *ROE* factor loadings explain the average returns for the high-minus-low *ROE* quintiles. The loadings vary from 0.96 to 1.27, which are all more than ten standard errors from zero. Also, the investment factor loadings explain the average returns for the low-minus-high $\Delta A/A$ quintiles. The loadings vary from 1.10 to 1.45, which are all more than seven standard errors from zero. And both factor loadings contribute to the large average return for the high-*ROE*-low- $\Delta A/A$ minus low-*ROE*-high- $\Delta A/A$ portfolio.

4 Specification Tests

As noted, the size factor plays only a secondary role in the *q*-factor model. In this section, we conduct specification tests to quantify the impact of the size factor by estimating:

$$r_t^i - r_t^f = a_q^i + b_{MKT}^i MKT_t + b_{\Delta A/A}^i r_{\Delta A/A,t} + b_{ROE}^i r_{ROE,t} + e_t^i. \quad (3)$$

Table 15 shows why we opt to include the size factor in the *q*-factor model. This table reports

the results for the size deciles. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on the market equity at the end of June. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t+1$, and the portfolios are rebalanced in June. The table shows no average return spread between the small and the big deciles in the 1972–2011 sample. None of the small-minus-big alphas from the traditional factor models are significant. The m.a.e.'s in the CAPM, the Fama-French model, the Carhart model, and the q -factor model are largely comparable. The GRS test fails to reject the CAPM, but do reject the three multifactor models. Because of the high goodness-of-fit in the multifactor models, even small deviations between predict returns and realized returns are significant.

The last eight rows in Table 15 show that without the size factor, the q -factor model produces a small-minus-big alpha of 0.68% per month ($t = 2.17$). This alpha is higher in magnitude than -0.19% in the Fama-French model and -0.17% in the Carhart model. The m.a.e. is 0.34% without the size factor, and is higher than 0.05% in both the Fama-French model and the Carhart model. Including the size factor as in the benchmark q -factor model fixes these shortcomings. The small-minus-big alpha drops to 0.04% and the m.a.e. to 0.08%.

In general, the size factor helps reduce the m.a.e. for the q -factor model, especially when size is involved in forming testing portfolios. Panel A of Table 16 uses the 25 size and book-to-market portfolios as an example. Without the size factor, the m.a.e. increases from 0.12% per month to 0.31%, and the average R^2 decreases from 87% to 78%. The value-minus-growth alphas are not affected, however. The small-stock value-minus-growth alpha even goes down slightly from 0.58% to 0.52% once we drop the size factor, and is still the only significant value-minus-growth alpha.

Panel B of Table 16 reports the results for the 25 $\Delta A/A$ and ROE portfolios. Because size is not used in forming these portfolios, the q -factor model's performance is largely unaffected by dropping the size factor. The m.a.e. goes up slightly from 0.09% per month to 0.12%, and the average R^2 drops slightly from 80% to 79%. Among zero-cost strategies, only the low-minus-high $\Delta A/A$ alpha

in the high-*ROE* quintile is significant, -0.40% . However, its magnitude falls relative to -0.50% in the benchmark *q*-factor model. The high-minus-low *ROE* alpha in the high- $\Delta A/A$ quintile is 0.43% ($t = 2.65$) in the benchmark model, but becomes 0.29% ($t = 1.62$) once we drop the size factor.

Table 17 reports similar results for the one-way deciles. Because size is not used explicitly in forming these portfolios, the *q*-factor model's performance is again largely unchanged by dropping the size factor. Without going through the details, we can report that the m.a.e.'s and the average R^2 s are mostly unaffected by dropping the size factor. The high-minus-low alphas across different sets of deciles are also largely unchanged. In a few cases, dropping the size factor in effect helps the *q*-factor model. For instance, Panel H shows that without the size factor, the *q*-factor model produces a high-minus-low alpha of -0.19% per month and an m.a.e. of 0.11% across the accrual deciles. Both are comparable with those from the Carhart model. In contrast, the high-minus-low alpha is -0.39% and the m.a.e. is 0.14% in the benchmark *q*-factor model.

5 Interpreting the Results

We interpret the *q*-factor model as a parsimonious summary of the cross-section of average returns, a new workhorse that can be used to estimate expected returns in practice. We first show that our empirical results are consistent with investment-based asset pricing (Section 5.1), but we also entertain alternative interpretations based on common risk factors and mispricing (Section 5.2).

5.1 Interpretation from Investment-based Asset Pricing

Our results are consistent with insights from investment-based asset pricing (e.g., Cochrane (1991), Berk, Green, and Naik (1999), Zhang (2005), and Liu, Whitely, and Zhang (2009)).

An Economic Model

We outline a simple two-period model to illustrate the basic intuition. While more complex models provide richer and more subtle predictions, the basic insights from the simple model hold in virtually all of the investment-based theoretical models that we are aware of.

There are two periods, 0 and 1, and heterogeneous firms, indexed by i . Firm i 's operating profits are given by $\Pi_{i0}A_{i0}$ in date 0 and $\Pi_{i1}A_{i1}$ in date 1, in which A_{i0} and A_{i1} are the firm's scale of productive assets, and Π_{i0} and Π_{i1} are the firm's *ROE* in dates 0 and 1, respectively. Firm i starts with assets A_{i0} , invests in date 0, produces in both dates, and exits at the end of date 1 with a liquidation value of $(1 - \delta)A_{i1}$, in which δ is the rate of depreciation. Assets evolve according to $A_{i1} = I_{i0} + (1 - \delta)A_{i0}$, in which I_{i0} is investment. Investment entails quadratic adjustment costs given by $(a/2)(I_{i0}/A_{i0})^2 A_{i0}$, in which $a > 0$ is a constant parameter.

Firm i has a gross discount rate of r_i , which varies across firms. The firm chooses A_{i1} to maximize the market value at the beginning of date 0:

$$\max_{\{A_{i1}\}} \Pi_{i0}A_{i0} - [A_{i1} - (1 - \delta)A_{i0}] - \frac{a}{2} \left[\frac{A_{i1}}{A_{i0}} - (1 - \delta) \right]^2 A_{i0} + \frac{1}{r_i} [\Pi_{i1}A_{i1} + (1 - \delta)A_{i1}]. \quad (4)$$

The market value is date 0's free cash flow, $\Pi_{i0}A_{i0} - I_{i0} - (a/2)(I_{i0}/A_{i0})^2 A_{i0}$, plus the discounted value of date 1's free cash flow, $[\Pi_{i1}A_{i1} + (1 - \delta)A_{i1}] / r_i$. With only two dates the firm does not invest in date 1, so date 1's free cash flow equals the sum of operating profits and the liquidation value.

The tradeoff of firm i is between forgoing date 0's free cash flow and obtaining higher free cash flow in date 1. Setting the first-order derivative of equation (2) with respect to A_{i1} to zero yields:

$$r_i = \frac{\Pi_{i1} + 1 - \delta}{1 + a(I_{i0}/A_{i0})}. \quad (5)$$

This optimality condition is intuitive. The numerator in the right-hand side is the marginal benefit of investment, including the marginal product of capital (*ROE*), Π_{i1} , and the marginal liquidation value of capital, $1 - \delta$. The denominator is the marginal cost of investment, including the marginal purchasing cost of investment (unity) and the marginal adjustment cost, $a(I_{i0}/A_{i0})$. Because the marginal benefit of investment is in date 1's dollar terms and the marginal cost of investment is in date 0's dollar terms, the first-order condition says that the marginal benefit of investment discounted to date 0 should equal the marginal cost of investment. Equivalently, the investment

return, defined as the ratio of the marginal benefit of investment in date 1 divided by the marginal cost of investment in date 0, should equal the discount rate, as in Cochrane (1991).

The Investment Factor

We use the first-order condition (5) to interpret the role of the investment factor and the *ROE* factor in the cross-section of returns. On the investment factor, the equation says that given the expected *ROE*, the expected return decreases with investment-to-assets. We argue that this investment mechanism is consistent with a wide range of cross-sectional predictability patterns including the negative relations of average returns with net stock issues, composite issuance, accruals, valuation ratios, and long-term prior returns (reversal). Figure 1 illustrates the investment mechanism.

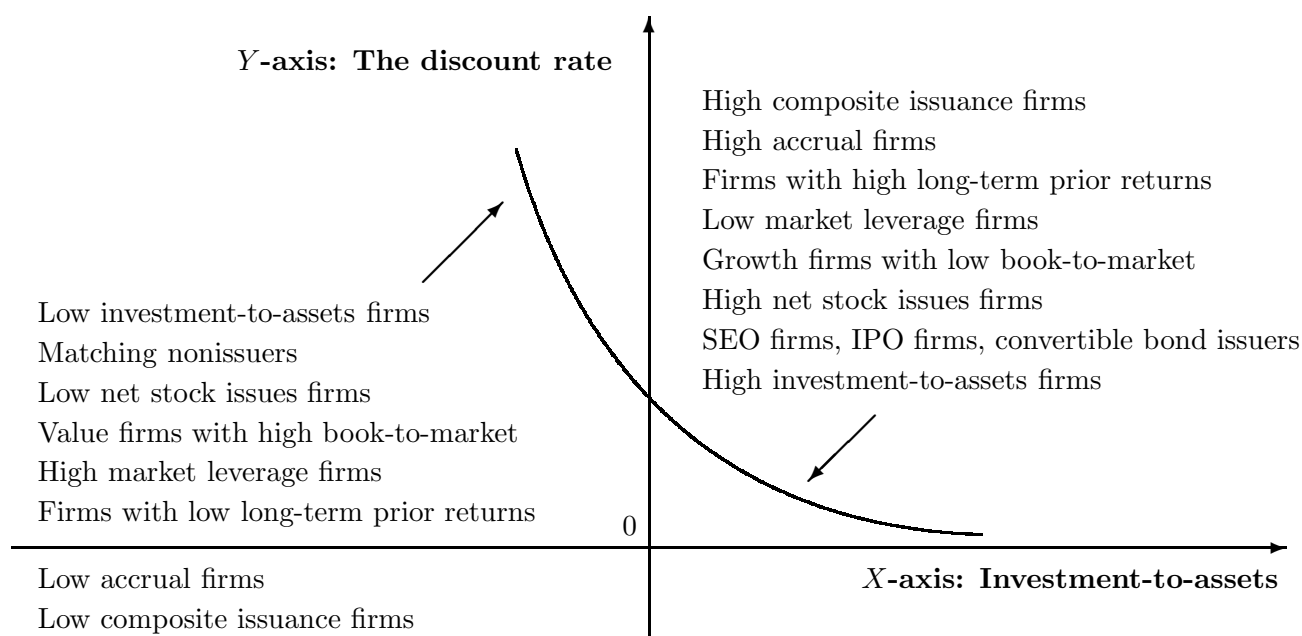
The negative relation between expected returns and investment is intuitive. Firms invest more when their marginal q (the net present value of future cash flows generated from one additional unit of capital) is high. Given expected *ROE* or cash flows, low discount rates give rise to high marginal q and high investment, and high discount rates give rise to low marginal q and low investment. This discount rate intuition is probably most transparent in the capital budgeting language of Brealey, Myers, and Allen (2006). In our setting capital is homogeneous, meaning that there is no difference between project-level costs of capital and firm-level costs of capital. Given expected cash flows, high costs of capital imply low net present values of new projects and in turn low investment, and low costs of capital imply high net present values of new projects and in turn high investment.¹²

The negative investment-expected return relation is conditional on expected *ROE*. Investment is not disconnected with *ROE* because more profitable firms tend to invest more than less profitable firms. This conditional relation provides a natural portfolio interpretation of the investment

¹²The negative investment-discount rate relation has a long tradition in economics. In a world without uncertainty, Fisher (1930) and Fama and Miller (1972, Figure 2.4) show that the interest rate and investment are negatively correlated. Intuitively, the investment demand curve is downward sloping. Extending this insight into a world with uncertainty, Cochrane (1991) and Liu, Whited, and Zhang (2009) demonstrate the negative investment-expected return relation in a dynamic setting with constant returns to scale. Carlson, Fisher, and Giammarino (2004) also predict the negative investment-expected return relation. In their real options model expansion options are riskier than assets in place. Investment converts riskier expansion options into less risky assets in place. As such, high-investment firms are less risky and earn lower expected returns than low-investment firms.

mechanism. Sorting on net stock issues, composite issuance, book-to-market, and other valuation ratios is closer to sorting on investment than sorting on expected *ROE*. Equivalently, these sorts produce wider spreads in investment than in expected *ROE*. As such, we can interpret the average return spreads generated from these diverse sorts using their common implied sort on investment.

Figure 1. The Investment Mechanism



In particular, the negative relation of average returns with equity issues is consistent with the negative investment-expected return relation. The balance-sheet constraint of firms implies that a firm's uses of funds must equal the firm's sources of funds, meaning that, all else equal, issuers must invest more (and earn lower average returns) than nonissuers.¹³ Cooper, Gulen, and Schill (2008) document that asset growth predicts future returns with a negative slope and interpret the evidence as investor underreaction to overinvestment. However, asset growth is the most comprehensive measure of investment-to-assets, in which investment is defined as the change in total assets. As such, the asset growth effect seems to be the premiere manifestation of the investment mechanism.

¹³Lyandres, Sun, and Zhang (2008) show that adding an investment factor to the CAPM and the Fama-French model reduces the magnitude of the underperformance following initial public offerings, seasoned equity offerings, and convertible debt offerings. Lyandres et al. also report the part of Figure 1 that is related to the new issues puzzle.

The value premium is also consistent with the negative investment-expected return relation. Investment-to-assets is an increasing function of marginal q (the denominator of equation (5)), and the marginal q equals the average q under constant returns to scale. The average q and market-to-book equity are highly correlated, and are identical without debt financing. As such, value firms with high book-to-market should invest less, and earn higher average returns than growth firms with low book-to-market. In general, firms with high valuation ratios have more growth opportunities, invest more, and earn lower expected returns than firms with low valuation ratios.

We also include market leverage in this category. Fama and French (1992) measure market leverage as the ratio of total assets to the market equity. Empirically, the q -factor model does a good job in explaining the market leverage-expected return relation (Internet Appendix). Intuitively, because the market equity is in the denominator, high leverage signals fewer growth opportunities, low investment, and high expected returns, whereas low leverage signals more growth opportunities, high investment, and low expected returns. This investment mechanism differs from the standard leverage effect in corporate finance texts. According to the textbook argument, high leverage means a high proportion of asset risk shared by equity holders, inducing high expected equity returns. This argument implicitly assumes that investment policy is fixed and that asset risk does not vary with investment. In contrast, the investment mechanism allows investment and leverage to be jointly determined. As such, market leverage and investment are negatively correlated, giving rise to a positive relation between market leverage and expected returns.

High valuation ratios often result from a stream of positive shocks on fundamentals, and low valuation ratios from a stream of negative shocks on fundamentals. High valuation ratios of growth firms can manifest as high long-term prior returns, and low valuation ratios of value firms as low long-term prior returns. As such, firms with high long-term prior returns should invest more and earn lower average returns than firms with low long-term prior returns. In all, the investment mechanism also helps explain DeBondt and Thaler's (1985) reversal effect (Internet Appendix).

The *ROE* Factor

Equation (5) implies that working in parallel with the investment mechanism is the *ROE* mechanism. Given investment-to-assets, firms with high expected *ROE* should earn higher expected returns than firms with low expected *ROE*. Because expected *ROE* is not observable, we use the current *ROE* as the proxy for expected *ROE*. The *ROE*-expected return relation is consistent with momentum, post-earnings-announcement drift, and the financial distress effect.

Why should high expected *ROE* firms earn higher expected returns than low expected *ROE* firms? We explain the intuition in two ways, discounting and capital budgeting. First, the marginal cost of investment in the denominator of the right-hand side of equation (5) equals marginal q , which in turn equals average q or market-to-book. As such, equation (5) says that the expected return is the expected *ROE* divided by market-to-book. Equivalently, the expected return equals the expected cash flow divided by the market equity. This relation is analogous to the Gordon Growth Model. In a two-period world price equals the expected cash flow divided by the discount rate. High expected cash flows relative to low market equity (or high expected *ROEs* relative to low market-to-book) mean high discount rates. And low expected cash flows relative to high market equity (or low expected *ROEs* relative to high market-to-book) mean low discount rates.¹⁴

From the capital budgeting perspective, equation (5) says that the expected return equals the expected *ROE* divided by an increasing function of investment-to-assets. High expected *ROE* relative to low investment must mean high discount rates. The high discount rates are necessary to offset the high expected *ROE* to induce low net present values of new capital and thus low investment. If the discount rates were not high enough to counteract the high expected *ROE*, firms would instead observe high net present values of new capital and thus invest more. Similarly, low expected *ROE* relative to high investments (such as small-growth firms in the 1990s) must mean

¹⁴This discounting intuition from valuation theory is also noted by Fama and French (2006). Using the residual income model, Fama and French argue that expected stock returns are related to three variables (book-to-market equity, expected profitability, and expected investment). Controlling for book-to-market and expected investment, more profitable firms earn higher expected returns. However, Fama and French do not motivate the *ROE* effect from economic theory or connect the *ROE*-expected return relation to the momentum, earnings, and distress anomalies.

low discount rates. If the discount rates were not low enough to counteract the low expected *ROE*, these firms would instead observe low net present values of new capital and thus invest less.

The *ROE*-expected return relation has important implications. For any sorts that produce wider spreads in expected *ROE* than in investment, the average return patterns across the sorted portfolios can be interpreted using the common implied sort on expected *ROE*. Examples include sorts on momentum, financial distress, and earning surprises. First, momentum sorts should generate an expected *ROE* spread. Intuitively, shocks to earnings are positively correlated with contemporaneous stock returns. Firms with positive earnings surprises are likely to experience immediate stock price increases, whereas firms with negative earnings surprises are likely to experience immediate stock price decreases. As such, winners with high short-term prior returns should have higher expected *ROE* and earn higher average returns than losers with low short-term prior returns.

Second, less financially distressed firms are more profitable (with higher expected *ROE*) and, all else equal, should earn higher average returns than more financially distressed firms. As such, the distress effect is consistent with the positive *ROE*-expected return relation. Finally, sorting on earnings surprise should generate an expected *ROE* spread between extreme portfolios. Intuitively, firms that have experienced large positive earnings surprises should be more profitable than firms that have experienced large negative earnings surprises.

A Few Loose Ends

In Section 2, we sort stocks jointly on $\Delta A/A$ and *ROE* to construct the investment and the *ROE* factors. The practice is consistent with equation (5), which shows that the investment and the *ROE* effects are both conditional in nature. Firms will invest a lot when either the *ROE* of their investment is high, or the cost of capital is low, or both. As such, the negative relation between investment and the cost of capital is conditional on a given level of *ROE*. Similarly, the positive relation between *ROE* and the cost of capital is conditional on a given level of investment. Sorting jointly on $\Delta A/A$ and *ROE* controls for this conditional relation.

Also, the size factor is primarily used to reduce the m.a.e. of the q -factor model across size-sorted portfolios. When size is not involved with constructing the testing portfolios, the performance of the q -factor model is largely unaffected by dropping the size factor (see Section 4). As such, the size factor plays only a secondary role, whereas the investment and the ROE factors are more important.

Finally, equation (5), which is derived from the investment first-order condition, is a nonlinear characteristics-based model. Strictly speaking, the equation does not give rise to a factor model. Nevertheless, we opt to use a linear factor approximation to the nonlinear characteristics model. Stock returns data are available at high frequencies, and are less subject to measurement errors than accounting variables. As such, factor mimicking portfolios often deliver better empirical performance than the underlying economic model itself (e.g., Breeden, Gibbons, and Litzenberger (1989)). Also, estimating the economic model directly involves specification errors in the production and the capital adjustment technologies, and requires a high level of aggregation over the underlying characteristics (e.g., Liu, Whited, and Zhang (2009)). For all these reasons, the q -factor model is more flexible in practice, and can be used to estimate the cost of capital at the firm level.

5.2 Alternative Interpretations

We discuss two alternative interpretations to our empirical results, the common risk factors interpretation as in Fama and French (1993, 1996) and the mispricing interpretation as in, for example, Daniel and Titman (1997). We argue that both alternative interpretations are reasonable, despite representing two polar extremes on the risk-mispricing spectrum. The investment-based interpretation in Section 5.1 seems to provide a healthy balance between the two extremes.

Common Risk Factors

Although we consider the investment, the ROE , and the size factors as common factors, we stop short of claiming common *risk* factors. Fama and French (1993, 1996) argue for the risk-based interpretation of their SMB and HML . Fama and French (1993, p. 4-5) write: “[I]f assets are priced rationally, variables that are related to average returns, such as size and book-to-market equity,

must proxy for sensitivity to common (shared and thus undiversifiable) risk factors in returns. The time-series regressions give direct evidence on this issue. In particular, the slopes and R^2 values show whether mimicking portfolios for risk factors related to size and [book-to-market] capture shared variation in stock and bond returns not explained by other factors.” Fama and French (1996, p. 57) further claim: “[T]he empirical successes of [the three-factor model] suggest that it is an equilibrium pricing model, a three-factor version of Merton’s (1973) intertemporal CAPM (ICAPM) or Ross’s (1976) arbitrage pricing theory (APT). In this view, *SMB* and *HML* mimic combinations of two underlying risk factors or state variables of special hedging concern to investors.”

Consistent with Fama and French (1993, 1996), our time-series regressions provide direct evidence that the investment and the *ROE* factor loadings capture shared variation in average returns across a wide array of anomaly portfolios. To the extent that the q -factors, constructed on all stocks, represent common sources of variation in stock returns, their loadings proxy for covariances between an individual asset’s returns with the factor returns. And covariances are standard measures of risk.

However, we do not pursue the risk factors interpretation for the following reasons. First, Brock (1982) derives the ICAPM and the APT within a general equilibrium production economy. Brock shows that for a common factor to be a risk factor, it must be a source of aggregate shock that affects all firms in the economy, e.g., shocks to total factor productivity, government policy (expenditure and taxation) shocks, and aggregate demand shocks driven by changes of preferences. As such, characteristics-based factors are not the ICAPM or APT risk factors. Firm-specific characteristics, on which the q -factors are based, have no immediate linkages with aggregate shocks that affect the fundamentals of all firms simultaneously. It is well known that ICAPM and APT are silent about the identities of the underlying shocks (state variables). In fact, what shocks drive economic fluctuations is one of the hardest questions in economics (e.g., Cochrane (1994)).

Second, although the investment-based model (see Section 5.1) from which we motivate the investment and the *ROE* factors is consistent with a risk model, the investment-based model is

again silent about what the aggregate shocks are. Lin and Zhang (2012) make this argument in depth. Briefly, the discount rate equation (5), $r_i = (\Pi_{i1} + 1 - \delta)/(1 + aI_{i0}/A_{i0})$, is derived from the investment first-order condition. An equivalent form, $1 + aI_{i0}/A_{i0} = (\Pi_{i1} + 1 - \delta)/r_i$, says that the marginal cost of investment today equals the marginal benefit of investment tomorrow discounted with the discount rate. As such, the net present value of the marginal investment is zero, formalizing the weighted average cost of capital approach to capital budgeting. Different from the consumption (risk) approach that connects the expected return to consumption betas, the investment approach connects the expected return to firm characteristics, such as investment and *ROE*.

In a general equilibrium production economy, the investment first-order condition, $r_i = (\Pi_{i1} + 1 - \delta)/(1 + aI_{i0}/A_{i0})$, and the consumption first-order condition, $E_0[M_1 r_i] = 1$, in which M_1 is a stochastic discount factor, both hold. As such, the (characteristics-based) investment approach and the (covariances-based) consumption approach are not mutually exclusive. Representing two sides of the same coin, the two approaches are internally consistent. However, the crux is that the q -factor model, which is a linear factor representation of $r_i = (\Pi_{i1} + 1 - \delta)/(1 + aI_{i0}/A_{i0})$, is not a risk model of M_1 on economic ground. Neither is the Fama-French model, contrary to their popular (but contentious) interpretation in the existing literature.

Finally, the evidence is mechanical that the investment and the *ROE* factor loadings vary in the same direction as the average returns across the anomaly portfolios. The *only* economic substance is that investment and *ROE* characteristics forecast returns, as predicted by the investment first-order condition. The common variations in average returns that these factors capture are not automatically sources of risk. Despite small differences in statistical properties, on economic ground, time series and cross-sectional regressions are largely equivalent ways of summarizing correlations in the data. If a characteristic shows up significant in cross-sectional regressions, its factor mimicking portfolio is likely to show “explanatory” power in time series regressions. If a factor loading shows up significant in time series regressions, its underlying characteristic is likely significant in cross-sectional regressions. Factor loadings are no more primitive than characteristics, and

characteristics are no more primitive than loadings in “explaining” expected returns.

Mispricing

Anomalies are often interpreted as mispricing. Jegadeesh and Titman (1993, p. 90) write: “The market underreacts to information about the short-term prospects of firms but overreacts to information about their long-term prospects,” and that “investor expectations are systematically biased.” Bernard and Thomas (1990, p. 305) interpret the post-earnings-announcement drift as “consistent with a failure of stock prices to reflect fully the implications of current earnings for future earnings.” Ritter (1991, p. 3) interprets the long-run performance of initial public offerings as “consistent with an IPO market in which (1) investors are periodically overoptimistic about the earnings potential of young growth companies, and (2) firms take advantage of these ‘window of opportunity’.” Cooper, Gulen, and Schill (2008) argue that bias in the capitalization of new investments can lead to investment policy distortions, and interpret the investment effect as saying “such potential distortions are present and economically meaningful (p. 1648).” Our empirical results do not rule out any of these mispricing stories. And we accept the possibility of mispricing in the data.

However, we view our investment-based work as weakening the mispricing interpretation to the anomalies. Our results suggest that firms’ investment decisions are aligned correctly with the discount rate. Firms invest more when their discount rates are low, and invest less when their discount rates are high, all else equal. More profitable firms must have higher discount rates if these firms do not invest more than less profitable firms. To the extent that this alignment between the discount rate and investment policies manifests itself as many empirical relations between firm characteristics and average returns, the relations per se say *nothing* about investor rationality or irrationality. A low discount rate could result from the sentiment of irrationally optimistic investors or the low market prices of risk demanded by rational investors. The investment first-order condition then connects correctly the low discount rate with high investment and low profitability.

It is often argued that the anomalies are “anomalies” precisely because they cannot be explained

by standard risk models. However, it should be noted that the anomalies are “anomalies” in a more fundamental sense that firm characteristics such as investment and earnings are not even present in the standard consumption-based model. The investment approach fills this gaping hole with a single equation (the investment first-order condition). Also, the failure of standard risk models can be due to specification errors in the stochastic discount factor and measurement errors in the risk proxies (see, e.g., the simulation results in Lin and Zhang (2012)). Mispricing is not the only possibility.

6 Conclusion

In his presidential address, Cochrane (2011, p. 1060–1061, original emphasis) writes: “We are going to have to repeat Fama and French’s anomaly digestion, but with many more dimensions. We have a lot of questions to answer: First, which characteristics really provide *independent* information about average returns? Which are subsumed by others? Second, does each new anomaly variable also correspond to a new factor formed on those same anomalies?” “Third, how many of these new factors are really important? Can we again account for N independent dimensions of expected returns with $K < N$ factor exposures?” “[T]he world would be much simpler if betas on only a few factors, important in the covariance matrix of returns, accounted for a larger number of mean characteristics.”

We agree that the Fama and French (1996) anomaly digestion is obsolete, and we offer an update that is long overdue. Our empirical results provide answers to the important questions raised by Cochrane (2011). First, investment and *ROE* provide independent information about average returns. Investment largely subsumes book-to-market, net stock issues, accruals, market leverage, long-term prior returns, earnings-to-price, and composite issuance in forecasting returns. *ROE* largely subsumes short-term prior returns, earnings surprise, and financial distress. And investment and *ROE* both contribute to the idiosyncratic volatility effect. While the q -factor model is by no means perfect, especially when pricing the total accrual deciles, the new model seems to do a good job in summarizing the current understanding of the cross-section of returns.

Second, our extensive factor regressions show that each anomaly variable also corresponds to

a new factor formed on the same variable. In particular, the high-minus-low portfolios often earn significant average returns and alphas from the traditional factor models. However, the evidence is weaker when we form the testing portfolios with the NYSE breakpoints and value-weighted returns than with the NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. Third, we have considered an exhaustive list of about 15 anomaly variables. Among their corresponding factors, we show that the investment and the *ROE* factors are important. The size factor is also useful, especially when pricing size-related portfolios. As such, the 15 anomalies are not all independent. With four factors, the *q*-factor model does a good job in summarizing their average return variations.

Our work has important implications for academic research in finance and accounting. The *q*-factor model can be used as a new workhorse model of expected returns. Any new anomaly variable should be benchmarked against the *q*-factor model to see if the variable provides any incremental information above and beyond investment and *ROE*. More important, the vast anomalies literature in empirical finance and capital markets research in accounting should be reevaluated with the new expected-return benchmark provided by the *q*-factor model. Much work remains to be done.

For example, Agrawal, Jaffe, and Mandelker (1992) document negative abnormal returns for acquiring firms for up to five years following merger announcements. To the extent that acquisition represents an alternative form of capital investment, the investment factor is likely to play a nontrivial role in explaining Agrawal et al.'s evidence. As another example, Michaely, Thaler, and Womack (1995) document that firms that initiate dividends have positive abnormal stock returns for three years after the event, and firms that omit dividends have negative abnormal returns. To the extent that dividend initiation signals strong expected *ROE* and dividend omission weak expected *ROE*, the *ROE* factor is likely to play an important role in explaining Michaely et al.'s evidence.

We emphasize that the *q*-factor model differs from the Fama-French model (and its extension the Carhart model) in a fundamental way. Our reading of the empirical literature suggests that the Fama-French model, a time series model first proposed in Fama and French (1993), is largely moti-

vated by the empirical success of size and book-to-market in cross-sectional regressions in Fama and French (1992). The Carhart model, which augments the Fama-French model with the momentum factor in Carhart (1997), is a response to the failure of the Fama-French model in explaining momentum, as shown in Fama and French (1996). While Carhart does not offer any economic interpretation for *WML*, Fama and French (1993, 1996) suggest that *HML* is a relative distress factor. However, this story has largely been discredited by the literature on the distress effect, which says that more distressed firms earn *lower* average returns than less distress firms, not higher average returns as speculated by Fama and French (e.g., Dichev (1998) and Campbell, Hilscher, and Szilagyi (2008)).

The *q*-factor model is less subject to the data mining critique than the Fama-French model. Building off a rich theoretical literature on investment-based asset pricing (e.g., Cochrane (1991), Berk, Green, and Naik (1999), and Zhang (2005)), we had arrived at the theoretical proposition that investment and *ROE* are two key predictors of cross-sectional returns (e.g., Liu, Whited, and Zhang (2009)) before we conducted any factor regressions. The theory offers clear economic intuition why the investment and the *ROE* factors should work (see Section 5.1). All in all, while the Fama-French model is largely an ad hoc, data mined model, the *q*-factor model is a product of close interaction between theoretical and empirical research in asset pricing. The clear economic intuition increases the likelihood that the good performance of the *q*-factor model can persist in the future.

The *q*-factor model can potentially change the practice of investment management industry. The model can be used to provide expected return estimates for asset allocation, to calculate discount rates for capital budgeting and stock valuation, and to offer empirical benchmarks for evaluating mutual fund performance. Investment companies can also adjust the list of financial products offered to their clients, going beyond traditional styles such as size and book-to-market. The stakes are high.

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Table 1 : Descriptive Statistics of the Size Factor, the Investment Factor, and the ROE Factor (1/1972–12/2011, 480 Months)

Size (market equity, ME) is stock price per share times shares outstanding from CRSP. Investment-to-assets ($\Delta A/A$) is annual change in total assets (Compustat annual item AT) divided by lagged total assets. ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock. In June of each year t , we use the median NYSE size at the end of June to split NYSE, Amex, and NASDAQ stocks into two groups, small and big. Independently, in June of each year t , we also break NYSE, Amex, and NASDAQ stocks into three $\Delta A/A$ groups using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Also independently, each month, we sort all stocks into three groups based on the NYSE breakpoints for the low 30%, middle 40%, and the high 30% of the ranked ROE . Earnings and other accounting variables in Compustat quarterly files are used in the sorts in the months immediately after the most recent public earnings announcement dates (item RDQ). Taking intersections of the two size groups, the three $\Delta A/A$ groups, and the three ROE groups, we form 18 size- $\Delta A/A$ - ROE portfolios. Monthly value-weighted returns on the 18 portfolios are calculated for the current month, and the portfolios are rebalanced monthly. (The ROE portfolios are rebalanced monthly at the beginning of each month, but the size and the $\Delta A/A$ portfolios are rebalanced annually in each June.) The size factor, r_{ME} , is the difference (small-minus-big), each month, between the average returns on the nine small- ME portfolios and the average returns on the nine big- ME portfolios. The investment factor, $r_{\Delta A/A}$, is the difference (low-minus-high), each month, between the average returns on the six low- $\Delta A/A$ portfolios and the average returns on the six high- $\Delta A/A$ portfolios. And the ROE factor, r_{ROE} , is the difference (high-minus-low), each month, between the average returns on the six high- ROE portfolios and the average returns on the six low- ROE portfolios. The data for the Carhart factors are from Kenneth French's Web site. In Panel A, the t -statistics (in parentheses) are adjusted for heteroscedasticity and autocorrelations. In Panel B, the p-values (in parentheses) test that a given correlation is zero.

Panel A: Descriptive statistics								Panel B: Correlation matrix (p-values)						
	Mean	α	β_{MKT}	β_{SMB}	β_{HML}	β_{WML}	R^2		$r_{\Delta A/A}$	r_{ROE}	MKT	SMB	HML	WML
r_{ME}	0.31	0.24	0.17				0.06	r_{ME}	-0.11	-0.30	0.25	0.95	-0.07	0.01
	(2.09)	(1.64)	(4.24)						(0.02)	(0.00)	(0.00)	(0.00)	(0.14)	(0.87)
		0.03	0.02	0.99	0.18		0.93	$r_{\Delta A/A}$		0.05	-0.36	-0.22	0.69	0.04
		(0.92)	(1.61)	(61.46)	(7.43)					(0.23)	(0.00)	(0.00)	(0.00)	(0.38)
$r_{\Delta A/A}$		0.00	0.02	0.99	0.19	0.03	0.94	r_{ROE}			-0.19	-0.37	-0.09	0.50
		(0.06)	(2.34)	(63.30)	(7.67)	(1.93)					(0.00)	(0.00)	(0.05)	(0.00)
	0.44	0.51	-0.15				0.13	MKT				0.28	-0.32	-0.14
	(4.73)	(5.62)	(-5.46)									(0.00)	(0.00)	(0.00)
r_{ROE}		0.32	-0.06	-0.02	0.40		0.50	SMB					-0.23	0.00
		(4.56)	(-3.43)	(-0.78)	(11.80)								(0.00)	(0.92)
		0.26	-0.05	-0.02	0.41	0.05	0.52	HML						-0.16
		(3.64)	(-3.01)	(-0.83)	(11.84)	(2.07)								(0.00)
	0.60	0.65	-0.11				0.03							
	(4.85)	(5.65)	(-2.34)											
		0.78	-0.09	-0.32	-0.20		0.19							
		(6.98)	(-2.08)	(-5.76)	(-2.41)									
		0.52	-0.03	-0.32	-0.10	0.28	0.40							
		(4.80)	(-1.03)	(-4.42)	(-1.50)	(6.47)								

Table 2 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Foster, Olsen, and Shevlin's (1984) Standardized Unexpected Earnings (*SUE*) (1/1972–12/2011, 480 Months)

SUE is the change in the most recently announced quarterly earnings per share from its value announced four quarters ago divided by the standard deviation of the change in quarterly earnings over the prior eight quarters (at least six quarters). We rank all NYSE, Amex, and NASDAQ stocks into deciles at the beginning of each month by their most recent past *SUE* with the NYSE breakpoints. Monthly value-weighted returns on the *SUE* portfolios are calculated for the current month, and the portfolios are rebalanced monthly. We report mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the *q*-factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The *t*-statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r_t^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.35	0.30	0.34	0.27	0.41	0.42	0.61	0.64	0.61	0.78	0.43		
t_{Mean}	1.39	1.23	1.30	1.13	1.81	1.80	2.80	3.00	2.79	3.72	3.39		
α	−0.12	−0.17	−0.14	−0.18	−0.03	−0.02	0.18	0.20	0.19	0.37	0.49	0.16	0.87
β	1.03	1.04	1.06	1.00	0.97	0.96	0.95	0.97	0.95	0.90	−0.12	(0.00)	
t_α	−1.26	−2.09	−1.76	−2.13	−0.41	−0.21	2.33	2.80	2.50	4.97	4.03		
α_{FF}	−0.11	−0.16	−0.14	−0.16	−0.03	−0.01	0.19	0.26	0.20	0.43	0.54	0.17	0.87
b	1.05	1.02	1.03	0.96	0.96	0.97	0.97	0.97	0.95	0.91	−0.13	(0.00)	
s	−0.10	0.06	0.10	0.12	0.04	−0.04	−0.09	−0.10	−0.06	−0.14	−0.04		
h	0.01	−0.04	−0.02	−0.06	−0.02	−0.01	0.01	−0.10	−0.02	−0.08	−0.09		
$t_{\alpha_{FF}}$	−1.15	−1.84	−1.71	−2.02	−0.33	−0.08	2.47	3.65	2.73	5.65	4.26		
α_{CARH}	0.02	−0.07	−0.06	−0.08	−0.01	0.04	0.21	0.16	0.12	0.34	0.32	0.11	0.88
b	1.02	1.00	1.02	0.95	0.96	0.96	0.96	0.99	0.97	0.93	−0.09	(0.02)	
s	−0.10	0.06	0.10	0.12	0.04	−0.04	−0.09	−0.10	−0.06	−0.14	−0.04		
h	−0.04	−0.07	−0.05	−0.09	−0.02	−0.03	0.00	−0.06	0.01	−0.05	−0.01		
w	−0.13	−0.09	−0.09	−0.09	−0.02	−0.04	−0.03	0.11	0.08	0.09	0.22		
$t_{\alpha_{CARH}}$	0.15	−0.82	−0.63	−0.96	−0.12	0.44	2.58	2.20	1.60	4.37	2.43		
α_q	0.06	−0.01	0.06	0.08	0.00	−0.03	0.11	0.02	0.04	0.20	0.14	0.06	0.88
β_{MKT}	1.03	1.00	1.02	0.94	0.97	0.97	0.97	1.01	0.97	0.94	−0.09	(0.41)	
β_{ME}	−0.16	0.03	0.01	0.02	0.01	−0.05	−0.05	−0.04	0.00	−0.05	0.11		
$\beta_{\Delta A/A}$	−0.01	−0.17	−0.12	−0.27	0.03	0.03	0.04	0.09	0.04	0.02	0.02		
β_{ROE}	−0.22	−0.12	−0.21	−0.19	−0.07	0.00	0.10	0.23	0.19	0.27	0.49		
t_{α_q}	0.55	−0.14	0.57	0.81	0.01	−0.29	1.38	0.21	0.49	2.43	0.92		
$t_{\beta_{MKT}}$	32.82	38.00	41.13	34.20	43.91	43.76	45.82	53.24	43.09	36.48	−1.92		
$t_{\beta_{ME}}$	−3.58	0.74	0.31	0.48	0.24	−0.95	−1.39	−1.45	0.00	−1.32	2.04		
$t_{\beta_{\Delta A/A}}$	−0.06	−2.47	−2.01	−3.57	0.49	0.53	0.68	1.53	0.72	0.37	0.23		
$t_{\beta_{ROE}}$	−3.21	−2.53	−4.07	−3.46	−1.71	0.13	1.91	5.92	4.02	7.27	5.78		

Table 3 : Factor Regressions for Monthly Percent Excess Returns of the Price Momentum Deciles (1/1972–12/2011, 480 Months)

At the beginning of each month t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into deciles based on their prior six-month returns from month $t - 2$ to $t - 7$. Skipping month $t - 1$, we calculate monthly value-weighted returns for the portfolios from month t to $t + 5$. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Loser	2	3	4	5	6	7	8	9	Winner	W-L	m.a.e.	$\bar{R}^2$
Mean	0.00	0.27	0.43	0.50	0.44	0.45	0.49	0.54	0.64	0.87	0.87		
t_{Mean}	0.00	0.97	1.76	2.25	2.12	2.25	2.39	2.50	2.71	2.74	3.15		
α	-0.61	-0.23	-0.03	0.07	0.03	0.04	0.08	0.12	0.18	0.32	0.93	0.17	0.86
β	1.35	1.09	1.00	0.94	0.92	0.90	0.90	0.93	1.02	1.21	-0.14	(0.00)	
t_α	-3.50	-1.93	-0.30	0.99	0.45	0.85	1.69	1.93	2.34	2.17	3.59		
α_{FF}	-0.68	-0.28	-0.08	0.03	-0.01	0.01	0.06	0.12	0.21	0.44	1.13	0.19	0.88
b	1.28	1.09	1.03	0.98	0.95	0.94	0.93	0.94	1.00	1.07	-0.22	(0.00)	
s	0.41	0.08	-0.02	-0.06	-0.06	-0.08	-0.08	-0.03	0.04	0.37	-0.04		
h	0.06	0.09	0.11	0.10	0.10	0.08	0.06	0.01	-0.07	-0.34	-0.39		
$t_{\alpha_{FF}}$	-4.28	-2.34	-0.90	0.42	-0.22	0.23	1.03	1.79	2.71	3.33	4.42		
α_{CARH}	-0.02	0.17	0.25	0.24	0.11	0.04	-0.01	-0.04	-0.04	0.03	0.05	0.10	0.94
b	1.14	1.00	0.96	0.93	0.93	0.93	0.94	0.98	1.05	1.15	0.01	(0.00)	
s	0.41	0.08	-0.02	-0.06	-0.06	-0.08	-0.08	-0.03	0.04	0.37	-0.04		
h	-0.18	-0.07	-0.01	0.02	0.06	0.07	0.09	0.07	0.02	-0.19	-0.02		
w	-0.70	-0.47	-0.34	-0.22	-0.13	-0.02	0.07	0.17	0.26	0.43	1.13		
$t_{\alpha_{CARH}}$	-0.21	2.41	4.17	4.21	1.73	0.60	-0.17	-0.69	-0.65	0.33	0.44		
α_q	0.01	0.07	0.11	0.11	0.00	-0.07	-0.10	-0.10	-0.04	0.24	0.23	0.09	0.89
β_{MKT}	1.19	1.04	1.00	0.96	0.94	0.94	0.94	0.97	1.03	1.10	-0.09	(0.00)	
β_{ME}	0.16	-0.04	-0.08	-0.08	-0.06	-0.05	-0.02	0.04	0.12	0.43	0.27		
$\beta_{\Delta A/A}$	-0.34	-0.09	0.01	0.07	0.09	0.14	0.16	0.12	0.02	-0.41	-0.06		
β_{ROE}	-0.75	-0.36	-0.20	-0.09	0.00	0.09	0.15	0.23	0.28	0.28	1.03		
t_{α_q}	0.06	0.39	0.94	1.22	-0.06	-1.20	-1.63	-1.54	-0.44	1.31	0.65		
$t_{\beta_{MKT}}$	24.06	25.57	35.66	40.66	42.32	48.78	54.78	54.32	38.53	27.01	-1.10		
$t_{\beta_{ME}}$	1.49	-0.50	-1.22	-1.73	-1.15	-1.27	-0.58	1.62	3.19	4.69	1.44		
$t_{\beta_{\Delta A/A}}$	-2.41	-0.75	0.15	0.93	1.55	3.03	3.95	2.78	0.24	-3.16	-0.26		
$t_{\beta_{ROE}}$	-6.02	-4.11	-2.96	-1.49	0.03	2.48	4.44	6.69	5.11	3.27	5.40		

Table 4 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Ang, Hodrick, Xing, and Zhang's (2006) Idiosyncratic Volatility (*IVOL*) (1/1972–12/2011, 480 Months)

IVOL is the standard deviation of the residuals from the Fama-French three-factor regression. We form value-weighted deciles each month by using the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks on their *IVOL* computed with daily returns over the previous month (with a minimum of 15 daily observations). We hold the *IVOL* deciles for one month, and rebalance the portfolios monthly. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.44	0.63	0.62	0.66	0.55	0.54	0.64	0.55	0.56	–0.10	–0.54		
t_{Mean}	2.52	3.23	3.00	2.88	2.19	2.02	2.23	1.73	1.50	–0.23	–1.51		
α	0.14	0.26	0.21	0.21	0.07	0.03	0.09	–0.04	–0.09	–0.82	–0.95	0.20	0.83
β	0.67	0.81	0.90	0.99	1.06	1.13	1.22	1.31	1.44	1.58	0.91	(0.00)	
t_α	1.44	3.36	2.72	3.00	0.68	0.30	0.98	–0.36	–0.59	–3.90	–3.44		
α_{FF}	0.08	0.25	0.21	0.20	0.04	0.00	0.11	–0.05	–0.07	–0.83	–0.91	0.19	0.87
b	0.75	0.87	0.93	1.03	1.06	1.11	1.17	1.22	1.28	1.34	0.59	(0.00)	
s	–0.24	–0.23	–0.13	–0.16	0.02	0.11	0.16	0.38	0.62	1.00	1.24		
h	0.17	0.07	0.04	0.06	0.05	0.03	–0.09	–0.06	–0.19	–0.21	–0.38		
$t_{\alpha_{FF}}$	1.04	3.48	2.72	2.54	0.45	0.01	1.24	–0.53	–0.57	–5.30	–4.48		
α_{CARH}	0.03	0.20	0.15	0.16	0.07	0.02	0.21	0.03	0.07	–0.55	–0.58	0.15	0.87
b	0.76	0.88	0.95	1.04	1.06	1.11	1.15	1.20	1.25	1.28	0.52	(0.00)	
s	–0.24	–0.23	–0.13	–0.16	0.02	0.11	0.16	0.38	0.62	1.00	1.24		
h	0.19	0.09	0.06	0.08	0.04	0.02	–0.12	–0.09	–0.24	–0.31	–0.49		
w	0.06	0.05	0.06	0.04	–0.03	–0.02	–0.10	–0.09	–0.14	–0.29	–0.35		
$t_{\alpha_{CARH}}$	0.33	2.60	1.75	2.13	0.75	0.21	2.25	0.32	0.52	–3.11	–2.59		
α_q	–0.15	0.05	0.02	0.05	–0.03	–0.02	0.23	0.08	0.21	–0.19	–0.04	0.10	0.88
β_{MKT}	0.77	0.89	0.95	1.04	1.07	1.11	1.15	1.21	1.26	1.29	0.52	(0.02)	
β_{ME}	–0.15	–0.15	–0.06	–0.10	0.02	0.10	0.13	0.32	0.51	0.73	0.88		
$\beta_{\Delta A/A}$	0.38	0.23	0.16	0.15	0.08	–0.01	–0.26	–0.19	–0.45	–0.60	–0.98		
β_{ROE}	0.20	0.20	0.20	0.17	0.07	0.04	–0.05	–0.15	–0.29	–0.76	–0.96		
t_{α_q}	–1.60	0.58	0.18	0.59	–0.30	–0.18	1.92	0.75	1.50	–1.17	–0.19		
$t\beta_{MKT}$	38.54	47.02	48.02	45.12	45.90	43.02	39.37	46.01	39.20	27.83	9.01		
$t\beta_{ME}$	–4.10	–5.29	–1.95	–1.95	0.40	1.65	2.36	5.48	9.58	10.93	9.94		
$t\beta_{\Delta A/A}$	4.20	3.06	2.05	1.92	1.10	–0.11	–2.62	–2.35	–5.25	–4.90	–5.46		
$t\beta_{ROE}$	3.21	4.14	3.49	3.73	1.54	0.88	–0.83	–2.70	–3.79	–6.49	–6.13		

Table 5 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Campbell, Hilscher, and Szilagyi's (2008) Failure Probability (1/1976–12/2011, 432 Months)

We use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks at the beginning of each month into deciles on the most recent failure probability (see Internet Appendix for detailed variable definition). Earnings and other accounting variables for a fiscal quarter are used in portfolio sorts in the months immediately after the quarter's public earnings announcement dates (Compustat quarterly item RDQ). Monthly value-weighted returns on the portfolios are calculated for the current month, and the portfolios are rebalanced monthly. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME,t}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.69	0.58	0.58	0.62	0.56	0.69	0.62	0.65	0.53	0.11	–0.57		
t_{Mean}	3.00	2.71	2.65	2.76	2.45	2.95	2.22	2.10	1.43	0.22	–1.42		
α	0.20	0.09	0.07	0.09	0.02	0.12	0.01	–0.03	–0.25	–0.84	–1.04	0.17	0.82
β	0.89	0.88	0.91	0.96	0.97	1.02	1.09	1.23	1.41	1.72	0.84	(0.02)	
t_α	1.79	1.13	0.99	1.07	0.24	1.46	0.09	–0.23	–1.58	–3.03	–3.13		
α_{FF}	0.29	0.17	0.10	0.13	0.03	0.05	–0.05	–0.18	–0.40	–1.14	–1.43	0.25	0.84
b	0.85	0.88	0.93	0.97	0.98	1.04	1.10	1.22	1.35	1.60	0.76	(0.00)	
s	–0.01	–0.15	–0.14	–0.12	–0.06	0.04	0.08	0.28	0.50	0.96	0.97		
h	–0.21	–0.10	0.00	–0.03	–0.01	0.13	0.10	0.20	0.11	0.25	0.46		
$t_{\alpha_{FF}}$	2.84	2.17	1.33	1.50	0.45	0.61	–0.47	–1.26	–2.92	–5.18	–5.21		
α_{CARH}	0.08	0.03	0.01	0.08	0.08	0.14	0.16	0.14	–0.03	–0.47	–0.55	0.12	0.88
b	0.89	0.91	0.95	0.98	0.98	1.03	1.06	1.16	1.28	1.48	0.59	(0.05)	
s	–0.04	–0.17	–0.15	–0.12	–0.06	0.05	0.11	0.33	0.55	1.05	1.09		
h	–0.13	–0.05	0.03	–0.02	–0.02	0.10	0.03	0.09	–0.02	0.02	0.15		
w	0.25	0.16	0.10	0.05	–0.06	–0.10	–0.24	–0.36	–0.42	–0.77	–1.02		
$t_{\alpha_{CARH}}$	0.78	0.37	0.10	1.01	1.09	1.60	1.58	1.06	–0.24	–2.53	–2.51		
α_q	–0.01	–0.13	–0.14	–0.04	0.11	0.16	0.24	0.33	0.28	0.01	0.02	0.15	0.87
β_{MKT}	0.89	0.93	0.97	1.00	0.98	1.03	1.06	1.14	1.26	1.46	0.57	(0.01)	
β_{ME}	0.11	–0.05	–0.07	–0.08	–0.09	0.00	–0.02	0.11	0.26	0.54	0.43		
$\beta_{\Delta A/A}$	–0.22	0.09	0.14	0.14	–0.01	0.14	0.04	–0.04	–0.14	–0.12	0.11		
β_{ROE}	0.41	0.29	0.23	0.13	–0.09	–0.16	–0.36	–0.54	–0.78	–1.38	–1.79		
t_{α_q}	–0.09	–1.49	–1.78	–0.49	1.45	1.68	2.16	1.91	2.14	0.04	0.07		
$t_{\beta_{MKT}}$	29.47	47.24	59.12	52.22	46.77	42.97	34.77	28.11	33.32	21.45	6.55		
$t_{\beta_{ME}}$	1.72	–1.32	–2.05	–2.51	–2.88	0.01	–0.44	1.85	3.33	3.09	1.86		
$t_{\beta_{\Delta A/A}}$	–1.62	1.11	2.17	2.47	–0.17	2.12	0.51	–0.31	–1.13	–0.46	0.29		
$t_{\beta_{ROE}}$	5.19	5.64	6.62	2.78	–2.34	–3.28	–5.60	–6.59	–10.21	–7.52	–7.73		

Table 6 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Net Stock Issues (1/1972–12/2011, 480 Months)

We measure net stock issues as the natural log of the ratio of the split-adjusted shares outstanding at the fiscal yearend in $t-1$ divided by the split-adjusted shares outstanding at the fiscal yearend in $t-2$. The split-adjusted shares outstanding is the Compustat shares outstanding (Compustat annual item CSHO) times the Compustat adjustment factor (item ADJEX_C). In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on the net stock issues for the fiscal year ending in calendar year $t-1$. Because a disproportionately large number of firms have zero net stock issues, we group all the firms with negative net issues into deciles one and two (equal-numbered), and the firms with zero net issues into decile three. We then sort the firms with positive net stock issues into the remaining seven deciles (equal-numbered). Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.88	0.58	0.66	0.55	0.47	0.51	0.70	0.58	0.20	0.20	-0.68		
t_{Mean}	4.07	2.85	3.14	2.50	2.15	2.15	2.88	2.11	0.73	0.75	-4.11		
α	0.47	0.18	0.30	0.14	0.05	0.05	0.24	0.06	-0.31	-0.29	-0.77	0.21	0.83
β	0.91	0.88	0.80	0.89	0.94	1.01	1.02	1.15	1.12	1.09	0.19	(0.00)	
t_α	4.41	2.43	2.56	1.56	0.57	0.61	2.71	0.64	-2.66	-3.09	-4.74		
α_{FF}	0.34	0.18	0.17	0.14	0.04	0.11	0.29	0.20	-0.22	-0.29	-0.62	0.20	0.85
b	0.95	0.92	0.85	0.94	0.95	1.00	0.98	1.07	1.06	1.02	0.08	(0.00)	
s	0.06	-0.16	0.01	-0.19	-0.02	-0.05	0.09	0.05	0.09	0.28	0.22		
h	0.26	0.05	0.26	0.06	0.02	-0.11	-0.12	-0.30	-0.19	-0.08	-0.34		
$t_{\alpha_{FF}}$	3.45	2.48	1.34	1.55	0.48	1.39	3.36	2.33	-2.20	-3.03	-4.06		
α_{CARH}	0.30	0.19	0.16	0.10	0.03	0.08	0.25	0.20	-0.13	-0.27	-0.57	0.17	0.85
b	0.96	0.91	0.85	0.95	0.95	1.00	0.99	1.08	1.04	1.02	0.06	(0.00)	
s	0.06	-0.16	0.01	-0.19	-0.02	-0.05	0.09	0.05	0.09	0.28	0.22		
h	0.28	0.04	0.27	0.07	0.02	-0.11	-0.10	-0.30	-0.23	-0.08	-0.36		
w	0.04	-0.01	0.01	0.03	0.00	0.03	0.04	0.00	-0.10	-0.02	-0.05		
$t_{\alpha_{CARH}}$	3.05	2.59	1.16	1.19	0.40	1.02	2.87	2.12	-1.25	-2.73	-3.68		
α_q	0.21	-0.01	0.05	-0.02	-0.07	0.09	0.26	0.33	0.09	-0.11	-0.32	0.12	0.85
β_{MKT}	0.95	0.95	0.86	0.95	0.96	1.00	0.99	1.08	1.02	0.99	0.04	(0.00)	
β_{ME}	0.10	-0.11	0.04	-0.13	0.01	-0.04	0.10	0.00	0.00	0.24	0.14		
$\beta_{\Delta A/A}$	0.35	0.25	0.40	0.16	0.08	-0.17	-0.11	-0.45	-0.47	-0.31	-0.66		
β_{ROE}	0.10	0.14	0.06	0.18	0.10	0.08	0.02	-0.05	-0.24	-0.13	-0.23		
t_{α_q}	2.04	-0.12	0.36	-0.27	-0.75	1.03	2.56	3.13	0.87	-1.14	-2.10		
$t_{\beta_{MKT}}$	34.19	46.63	26.74	41.88	33.68	47.62	46.40	36.06	36.29	38.99	1.14		
$t_{\beta_{ME}}$	1.80	-3.22	0.56	-3.34	0.28	-1.38	2.56	-0.12	-0.10	6.17	1.79		
$t_{\beta_{\Delta A/A}}$	4.54	5.85	3.36	2.60	1.27	-2.69	-1.50	-6.34	-6.63	-6.09	-6.33		
$t_{\beta_{ROE}}$	1.59	3.65	1.01	4.96	2.43	2.06	0.47	-0.84	-4.69	-4.33	-3.43		

Table 7 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Composite Issuance (1/1972–12/2011, 480 Months)

Composite issuance is the growth rate in the market equity not attributable to the stock return, $\log(ME_t/ME_{t-5}) - r(t-5, t)$. For June of year t , $r(t-5, t)$ is the cumulative log return on the stock from the last trading day of June in year $t-5$ to the last trading day of June in year t , and ME_t is the market equity on the last trading day of June in year t from CRSP. In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on composite issuance. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.69	0.73	0.68	0.39	0.54	0.40	0.51	0.58	0.40	0.11	-0.58		
t_{Mean}	3.64	3.77	3.69	1.87	2.46	1.62	2.05	2.10	1.41	0.38	-2.92		
α	0.36	0.35	0.32	0.00	0.13	-0.06	0.02	0.06	-0.13	-0.43	-0.79	0.19	0.82
β	0.73	0.83	0.79	0.88	0.89	1.03	1.09	1.14	1.18	1.19	0.46	(0.00)	
t_α	3.18	3.85	3.43	-0.05	1.69	-0.67	0.17	0.69	-1.36	-4.32	-4.68		
α_{FF}	0.19	0.28	0.26	-0.06	0.13	0.03	0.15	0.12	-0.04	-0.31	-0.50	0.16	0.85
b	0.83	0.90	0.86	0.95	0.93	1.00	1.02	1.10	1.12	1.10	0.27	(0.00)	
s	-0.12	-0.16	-0.21	-0.21	-0.14	-0.02	0.06	0.07	0.11	0.19	0.31		
h	0.38	0.18	0.17	0.16	0.03	-0.20	-0.29	-0.14	-0.21	-0.28	-0.67		
$t_{\alpha_{FF}}$	1.81	3.15	3.06	-0.74	1.80	0.39	1.60	1.34	-0.46	-3.49	-3.61		
α_{CARH}	0.13	0.27	0.26	-0.09	0.15	0.07	0.14	0.13	0.01	-0.27	-0.40	0.15	0.85
b	0.84	0.90	0.86	0.96	0.92	0.99	1.02	1.10	1.11	1.09	0.24	(0.00)	
s	-0.12	-0.16	-0.21	-0.21	-0.14	-0.02	0.06	0.07	0.11	0.19	0.31		
h	0.40	0.18	0.17	0.17	0.03	-0.21	-0.29	-0.14	-0.23	-0.30	-0.70		
w	0.06	0.02	0.01	0.03	-0.01	-0.03	0.01	0.00	-0.05	-0.04	-0.10		
$t_{\alpha_{CARH}}$	1.23	2.91	2.71	-1.01	1.90	0.75	1.44	1.35	0.08	-2.95	-2.91		
α_q	0.02	0.17	0.02	-0.27	0.03	0.19	0.19	0.14	0.02	-0.19	-0.21	0.12	0.86
β_{MKT}	0.84	0.91	0.89	0.96	0.94	0.98	1.03	1.09	1.11	1.09	0.25	(0.01)	
β_{ME}	-0.08	-0.15	-0.13	-0.11	-0.10	-0.06	0.04	0.09	0.10	0.16	0.24		
$\beta_{\Delta A/A}$	0.63	0.36	0.43	0.26	0.11	-0.38	-0.44	-0.28	-0.36	-0.49	-1.11		
β_{ROE}	0.07	0.04	0.18	0.24	0.11	-0.07	0.07	0.07	0.02	-0.05	-0.11		
t_{α_q}	0.17	1.77	0.21	-3.10	0.34	2.01	1.55	1.46	0.17	-1.87	-1.35		
$t_{\beta_{MKT}}$	30.02	35.89	37.47	35.31	42.91	39.44	35.88	42.55	46.51	42.78	6.76		
$t_{\beta_{ME}}$	-1.80	-3.36	-2.75	-2.87	-2.73	-1.46	0.79	2.18	2.58	4.19	3.52		
$t_{\beta_{\Delta A/A}}$	9.92	5.89	5.60	3.56	1.72	-5.78	-5.58	-5.07	-7.65	-9.28	-14.10		
$t_{\beta_{ROE}}$	1.10	0.87	3.32	4.25	2.07	-1.44	1.47	1.66	0.59	-0.91	-1.38		

Table 8 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Abnormal Corporate Investment (1/1972–12/2011, 480 Months)

We measure abnormal corporate investment (ACI) that applies for the portfolio formation year t , as $ACI_{t-1} \equiv 3CE_{t-1}/(CE_{t-2} + CE_{t-3} + CE_{t-4}) - 1$, in which CE_{t-1} is capital expenditure (Compustat annual item CAPX) scaled by its sales (item SALE) in year $t-1$. In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on ACI for the fiscal year ending in calendar year $t-1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.63	0.69	0.61	0.48	0.60	0.37	0.37	0.42	0.60	0.38	-0.26		
t_{Mean}	2.23	2.77	2.62	2.28	2.95	1.75	1.71	1.84	2.56	1.33	-1.57		
α	0.12	0.22	0.14	0.07	0.20	-0.03	-0.05	-0.02	0.14	-0.13	-0.24	0.11	0.85
β	1.14	1.03	1.03	0.91	0.88	0.89	0.92	0.96	1.03	1.12	-0.03	(0.05)	
t_α	0.94	2.38	1.72	0.95	2.46	-0.43	-0.58	-0.21	1.63	-1.17	-1.43		
α_{FF}	0.15	0.30	0.22	0.13	0.20	-0.05	0.00	-0.02	0.18	-0.13	-0.28	0.14	0.87
b	1.03	0.98	1.00	0.91	0.92	0.93	0.95	0.99	1.00	1.08	0.05	(0.00)	
s	0.42	0.09	-0.02	-0.11	-0.16	-0.11	-0.22	-0.12	0.01	0.17	-0.25		
h	-0.17	-0.18	-0.16	-0.09	0.04	0.07	-0.04	0.03	-0.10	-0.04	0.13		
$t_{\alpha_{FF}}$	1.44	3.24	2.81	1.68	2.65	-0.64	-0.02	-0.21	2.20	-1.15	-1.64		
α_{CARH}	0.10	0.31	0.24	0.05	0.21	-0.04	0.04	0.01	0.13	-0.06	-0.16	0.12	0.87
b	1.04	0.98	1.00	0.93	0.92	0.93	0.95	0.99	1.02	1.06	0.02	(0.00)	
s	0.41	0.09	-0.02	-0.11	-0.16	-0.11	-0.22	-0.12	0.01	0.17	-0.25		
h	-0.15	-0.18	-0.17	-0.06	0.04	0.06	-0.05	0.02	-0.08	-0.06	0.09		
w	0.06	-0.01	-0.02	0.08	-0.01	-0.01	-0.04	-0.03	0.06	-0.07	-0.13		
$t_{\alpha_{CARH}}$	0.91	3.11	2.81	0.69	2.64	-0.53	0.43	0.13	1.57	-0.51	-0.91		
α_q	0.13	0.37	0.32	0.02	0.15	-0.16	-0.07	-0.13	0.17	0.02	-0.11	0.15	0.87
β_{MKT}	1.05	0.98	1.00	0.93	0.92	0.94	0.98	1.00	1.00	1.06	0.02	(0.00)	
β_{ME}	0.39	0.06	-0.06	-0.08	-0.16	-0.06	-0.23	-0.07	0.04	0.10	-0.29		
$\beta_{\Delta A/A}$	-0.23	-0.29	-0.28	-0.03	0.10	0.12	0.12	0.09	-0.18	-0.11	0.12		
β_{ROE}	0.02	-0.02	-0.04	0.13	0.05	0.13	0.02	0.12	0.08	-0.18	-0.20		
t_{α_q}	0.90	2.79	3.27	0.23	1.96	-1.83	-0.71	-1.49	1.84	0.22	-0.56		
$t_{\beta_{MKT}}$	31.22	35.74	43.65	43.60	37.81	35.12	41.97	41.61	48.16	35.06	0.35		
$t_{\beta_{ME}}$	9.96	1.49	-1.57	-2.15	-4.47	-1.27	-5.77	-2.02	0.92	2.23	-4.52		
$t_{\beta_{\Delta A/A}}$	-2.10	-2.77	-3.31	-0.50	2.16	1.91	1.95	1.06	-2.56	-1.61	0.80		
$t_{\beta_{ROE}}$	0.36	-0.34	-0.81	4.02	1.28	2.56	0.46	2.21	1.63	-2.94	-2.16		

Table 9 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Total Accruals (1/1972–12/2011, 480 Months)

Following Sloan (1996), we measure total accruals (TAC) as changes in noncash working capital minus depreciation expense scaled by average total assets (Compustat annual item AT) in the prior two years. The noncash working capital is the change in noncash current assets minus the change in current liabilities less short-term debt and taxes payable. Specifically, $TAC \equiv (\Delta CA - \Delta CASH) - (\Delta CL - \Delta STD - \Delta TP) - DP$, in which ΔCA is the change in current assets (item ACT), $\Delta CASH$ is the change in cash or cash equivalents (item CHE), ΔCL is the change in current liabilities (item LCT), ΔSTD is the change in debt included in current liabilities (item DLC), ΔTP is the change in income taxes payable (item TXP), and DP is depreciation and amortization expense (item DP). In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on total accruals scaled by average total assets for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.53	0.53	0.59	0.57	0.54	0.51	0.54	0.39	0.33	0.23	-0.30		
t_{Mean}	1.83	2.19	2.82	2.75	2.44	2.53	2.48	1.71	1.30	0.75	-1.94		
α	0.00	0.07	0.17	0.15	0.13	0.12	0.12	-0.04	-0.17	-0.34	-0.35	0.13	0.85
β	1.16	1.00	0.92	0.92	0.90	0.86	0.93	0.95	1.09	1.27	0.11	(0.01)	
t_α	0.04	0.67	2.35	1.96	1.72	1.61	1.74	-0.44	-1.93	-2.93	-2.27		
α_{FF}	0.09	0.17	0.19	0.14	0.11	0.10	0.19	-0.01	-0.08	-0.20	-0.29	0.13	0.87
b	1.10	0.97	0.93	0.93	0.93	0.89	0.92	0.94	1.05	1.12	0.02	(0.01)	
s	0.06	-0.04	-0.09	-0.03	-0.11	-0.10	-0.10	0.02	0.03	0.37	0.31		
h	-0.19	-0.19	-0.01	0.04	0.08	0.07	-0.11	-0.06	-0.18	-0.37	-0.19		
$t_{\alpha_{FF}}$	0.78	1.76	2.64	1.66	1.38	1.33	2.58	-0.13	-0.98	-2.08	-1.96		
α_{CARH}	0.15	0.15	0.12	0.15	0.14	0.08	0.17	0.00	-0.03	-0.14	-0.29	0.11	0.87
b	1.09	0.98	0.95	0.93	0.93	0.90	0.93	0.94	1.04	1.11	0.02	(0.01)	
s	0.06	-0.04	-0.09	-0.03	-0.11	-0.10	-0.10	0.02	0.03	0.37	0.31		
h	-0.21	-0.18	0.01	0.03	0.07	0.08	-0.10	-0.06	-0.20	-0.39	-0.19		
w	-0.06	0.02	0.07	-0.02	-0.03	0.02	0.02	-0.01	-0.05	-0.06	0.00		
$t_{\alpha_{CARH}}$	1.16	1.50	1.66	1.67	1.61	1.06	2.34	-0.01	-0.35	-1.44	-1.69		
α_q	0.27	0.23	0.22	0.07	0.11	-0.04	0.12	-0.10	-0.10	-0.11	-0.39	0.14	0.87
β_{MKT}	1.11	0.99	0.94	0.95	0.93	0.91	0.93	0.95	1.05	1.11	0.00	(0.01)	
β_{ME}	-0.05	-0.11	-0.11	-0.04	-0.10	-0.05	-0.05	0.05	0.05	0.38	0.42		
$\beta_{\Delta A/A}$	-0.22	-0.13	0.02	0.14	0.10	0.17	-0.14	-0.05	-0.28	-0.77	-0.56		
β_{ROE}	-0.23	-0.10	-0.05	0.03	0.00	0.13	0.13	0.12	0.09	0.12	0.34		
t_{α_q}	1.90	1.83	2.80	0.71	1.25	-0.50	1.60	-1.17	-1.03	-1.11	-2.48		
$t\beta_{MKT}$	28.38	37.55	45.92	34.75	41.62	49.39	41.47	40.04	36.27	41.60	-0.06		
$t\beta_{ME}$	-0.80	-2.20	-3.13	-1.20	-2.58	-1.76	-1.61	1.55	1.21	11.39	6.94		
$t\beta_{\Delta A/A}$	-2.12	-1.05	0.42	1.99	1.75	2.33	-3.23	-0.88	-3.77	-14.70	-5.31		
$t\beta_{ROE}$	-2.39	-1.29	-1.54	0.53	0.04	2.69	2.99	2.82	1.65	2.22	4.16		

Table 10 : Factor Regressions for Monthly Percent Excess Returns of Ten Industry Portfolios (1/1972–12/2011, 480 Months)

The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. We assign each NYSE, Amex, and NASDAQ stock to an industry portfolio at the end of June of year t based on its four-digit SIC code at that time. We use Compustat SIC codes for the fiscal year ending in calendar year $t - 1$. If Compustat SIC codes are unavailable, we use CRSP SIC codes for June of year t . Monthly value-weighted returns are computed from July of year t to June of year $t + 1$. The ten-industry classification is from Kenneth French's Web site. We exclude financial firms from the last industry portfolio ("Other"). We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations.

	NoDur	Durbl	Manuf	Enrgy	HiTec	Telcm	Shops	Hlth	Utils	Other	m.a.e.	$\bar{R}^2$
Mean	0.66	0.35	0.54	0.73	0.48	0.50	0.51	0.54	0.49	0.28		
t_{Mean}	3.05	1.11	2.20	2.94	1.47	2.16	1.92	2.37	2.54	1.04		
α	0.31	-0.16	0.07	0.37	-0.09	0.15	0.07	0.17	0.26	-0.22	0.19	0.64
β	0.78	1.13	1.04	0.80	1.27	0.78	0.99	0.80	0.51	1.12	(0.00)	
t_α	2.29	-0.93	0.74	1.94	-0.58	0.94	0.46	1.15	1.63	-2.49		
α_{FF}	0.24	-0.46	-0.02	0.28	0.18	0.14	0.02	0.37	0.06	-0.31	0.21	0.69
b	0.83	1.22	1.08	0.89	1.10	0.83	0.98	0.80	0.64	1.11	(0.00)	
s	-0.10	0.17	-0.02	-0.24	0.22	-0.22	0.12	-0.32	-0.19	0.21		
h	0.17	0.57	0.18	0.24	-0.61	0.06	0.07	-0.32	0.45	0.13		
$t_{\alpha_{FF}}$	1.82	-3.14	-0.25	1.49	1.34	0.92	0.11	2.66	0.40	-3.76		
α_{CARH}	0.24	-0.23	0.00	0.13	0.29	0.22	0.09	0.31	-0.01	-0.29	0.18	0.69
b	0.83	1.17	1.08	0.92	1.08	0.81	0.97	0.81	0.66	1.10	(0.00)	
s	-0.10	0.17	-0.02	-0.24	0.22	-0.22	0.12	-0.32	-0.19	0.21		
h	0.17	0.49	0.18	0.29	-0.65	0.03	0.04	-0.30	0.47	0.13		
w	0.00	-0.24	-0.02	0.16	-0.12	-0.08	-0.08	0.06	0.07	-0.02		
$t_{\alpha_{CARH}}$	1.82	-1.61	-0.06	0.72	2.18	1.29	0.65	2.09	-0.05	-3.35		
α_q	-0.08	-0.32	-0.15	0.09	0.57	0.42	-0.15	-0.04	-0.06	-0.37	0.22	0.69
β_{MKT}	0.86	1.16	1.09	0.91	1.09	0.80	0.99	0.87	0.62	1.11	(0.00)	
β_{ME}	0.02	0.14	0.02	-0.20	0.08	-0.32	0.20	-0.17	-0.12	0.22		
$\beta_{\Delta A/A}$	0.35	0.55	0.24	0.44	-0.94	0.07	0.03	-0.05	0.58	0.13		
β_{ROE}	0.31	-0.24	0.13	0.15	-0.31	-0.37	0.24	0.42	0.07	0.05		
t_{α_q}	-0.60	-1.66	-1.57	0.46	3.53	2.42	-0.99	-0.22	-0.32	-4.19		
$t_{\beta_{MKT}}$	24.30	21.79	48.05	17.40	25.30	20.44	21.51	17.68	17.56	46.73		
$t_{\beta_{ME}}$	0.34	1.35	0.56	-2.49	1.18	-5.70	2.20	-2.47	-1.91	3.81		
$t_{\beta_{\Delta A/A}}$	3.81	3.60	3.16	3.38	-7.90	0.73	0.35	-0.38	4.90	1.92		
$t_{\beta_{ROE}}$	3.93	-2.02	2.12	1.49	-3.44	-4.05	2.85	4.66	0.78	1.29		

Table 11 : Descriptive Statistics for Monthly Percent Excess Returns of 25 Size and Book-to-Market Portfolios (1/1972–12/2011, 480 Months)

The data for the one-month Treasury bill rate and the Carhart factors are from Kenneth French's Web site. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. The stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on market equity at the end of June of t . Independently, in June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on book-to-market equity. Book-to-market for June of year t is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the market equity at the end of December of $t - 1$. Taking intersections, we form 25 size and book-to-market portfolios. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced at the end of June. We report mean percent excess returns, the CAPM alphas (α), the intercepts (α_{FF}) from the Fama-French three-factor regressions, and the intercepts (α_{CARH}) from the Carhart four-factor regressions, as well as t -statistics adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. $\bar{R}^2$ is the average goodness-of-fit across the 25 portfolios. p_{GRS} is p-value for the GRS test on the null that the alphas of all the testing portfolios are jointly zero.

	Low	2	3	4	High	H–L		Low	2	3	4	High	H–L
	Mean							t_{Mean}					
Small	0.07	0.69	0.81	0.96	1.08	1.01		0.17	1.91	2.41	3.03	3.16	4.48
2	0.28	0.70	0.83	0.86	0.96	0.68		0.78	2.19	2.89	3.21	3.16	2.94
3	0.36	0.69	0.75	0.73	1.02	0.66		1.06	2.37	2.89	2.84	3.81	2.76
4	0.49	0.56	0.74	0.73	0.84	0.35		1.60	2.18	2.86	2.97	3.16	1.52
Big	0.38	0.51	0.56	0.60	0.56	0.19		1.60	2.33	2.74	2.91	2.44	0.89
	α (m.a.e. = 0.29)							t_{α} ($\bar{R}^2 = 0.73$, $p_{GRS} = 0$)					
Small	−0.57	0.13	0.29	0.48	0.59	1.17		−2.42	0.62	1.56	2.63	2.90	5.40
2	−0.35	0.15	0.33	0.40	0.48	0.83		−1.98	0.98	2.29	2.83	2.55	3.70
3	−0.24	0.17	0.28	0.30	0.57	0.81		−1.67	1.38	2.20	2.31	3.32	3.57
4	−0.08	0.07	0.27	0.30	0.41	0.49		−0.59	0.65	2.26	2.35	2.59	2.14
Big	−0.07	0.09	0.17	0.25	0.21	0.28		−0.76	1.06	1.71	2.08	1.30	1.31
	α_{FF} (m.a.e. = 0.10)							$t_{\alpha_{FF}}$ ($\bar{R}^2 = 0.89$, $p_{GRS} = 0$)					
Small	−0.54	0.00	0.10	0.19	0.15	0.69		−4.74	0.04	1.21	2.74	1.74	5.44
2	−0.23	0.03	0.08	0.07	0.03	0.25		−2.82	0.39	1.14	0.93	0.30	2.30
3	−0.10	0.03	0.03	−0.02	0.14	0.23		−1.28	0.44	0.32	−0.20	1.25	1.76
4	0.13	−0.03	0.05	0.01	0.03	−0.10		1.59	−0.35	0.49	0.14	0.27	−0.71
Big	0.16	0.09	0.12	0.01	−0.15	−0.31		2.67	1.04	1.27	0.06	−1.25	−2.29
	α_{CARH} (m.a.e. = 0.11)							$t_{\alpha_{CARH}}$ ($\bar{R}^2 = 0.89$, $p_{GRS} = 0$)					
Small	−0.48	0.02	0.09	0.19	0.21	0.69		−3.93	0.24	1.09	2.67	2.30	5.50
2	−0.19	0.07	0.09	0.11	0.03	0.21		−2.32	0.93	1.32	1.52	0.29	1.86
3	−0.05	0.04	0.08	0.00	0.16	0.21		−0.66	0.51	0.79	−0.02	1.38	1.55
4	0.14	−0.02	0.09	0.04	0.11	−0.03		1.66	−0.26	0.92	0.34	0.90	−0.19
Big	0.17	0.06	0.10	−0.02	−0.12	−0.30		2.86	0.76	0.99	−0.16	−0.94	−2.04

Table 12 : Q -factor Regressions for Monthly Percent Excess Returns of 25 Size and Book-to-Market Portfolios (1/1972–12/2011, 480 Months)

The data for the one-month Treasury bill rate and the Carhart factors are from Kenneth French's Web site. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. The stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on market equity at the end of June of t . Independently, in June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into quintiles on book-to-market equity. Book-to-market for June of year t is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the market equity at the end of December of $t - 1$. Taking intersections, we form 25 size and book-to-market portfolios. Monthly value-weighted portfolio returns are calculated, and the portfolios are rebalanced at the end of June. We report the q -factor regressions: $r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. $\bar{R}^2$ is the average goodness-of-fit across the 25 portfolios. p_{GRS} is p-value for the GRS test on the null that the alphas of all the testing portfolios are jointly zero.

	Low	2	3	4	High	H–L		Low	2	3	4	High	H–L
	α_q (m.a.e. = 0.12)							t_{α_q} ($\bar{R}^2 = 0.87$, $p_{GRS} = 0$)					
Small	–0.25	0.28	0.29	0.32	0.33	0.58		–1.46	2.25	2.79	4.08	2.84	2.89
2	–0.16	0.08	0.03	0.09	0.08	0.24		–1.31	0.95	0.43	0.86	0.69	1.21
3	–0.03	–0.03	–0.05	–0.02	0.15	0.18		–0.31	–0.31	–0.48	–0.15	1.18	1.09
4	0.15	–0.15	0.01	0.04	0.08	–0.08		1.25	–1.61	0.13	0.33	0.59	–0.38
Big	0.09	–0.05	0.09	0.02	–0.01	–0.10		1.25	–0.56	0.98	0.12	–0.07	–0.54
	β_{MKT}							$t_{\beta_{MKT}}$					
Small	1.10	0.96	0.92	0.88	0.96	–0.15		25.10	27.75	35.43	39.25	27.03	–2.56
2	1.14	1.02	1.01	0.94	1.00	–0.14		32.92	51.42	45.04	43.45	33.88	–2.44
3	1.10	1.05	1.01	0.94	1.01	–0.09		38.45	50.60	33.09	37.85	27.14	–1.63
4	1.10	1.07	1.05	0.98	1.02	–0.08		33.93	42.89	31.28	30.96	26.71	–1.35
Big	0.98	0.98	0.93	0.86	0.89	–0.09		52.66	40.39	30.43	31.70	25.17	–2.05
	β_{ME}							$t_{\beta_{ME}}$					
Small	1.13	1.19	1.09	1.01	0.99	–0.13		16.77	17.02	21.82	36.12	16.12	–1.22
2	0.92	0.93	0.81	0.73	0.86	–0.06		16.54	32.94	15.21	13.68	13.68	–0.63
3	0.72	0.66	0.51	0.43	0.49	–0.23		14.34	16.41	6.75	7.24	5.19	–1.76
4	0.40	0.32	0.25	0.20	0.19	–0.21		6.27	7.26	3.30	3.76	2.32	–1.52
Big	–0.23	–0.09	–0.20	–0.10	–0.12	0.11		–7.93	–2.65	–5.28	–1.65	–2.01	1.43
	$\beta_{\Delta A/A}$							$t_{\beta_{\Delta A/A}}$					
Small	–0.65	–0.34	–0.11	0.17	0.53	1.18		–5.27	–3.81	–1.38	3.09	5.91	8.51
2	–0.76	–0.16	0.24	0.42	0.62	1.37		–9.77	–3.11	3.79	6.18	9.03	10.62
3	–0.77	0.03	0.38	0.54	0.80	1.57		–11.21	0.43	3.45	6.13	8.90	11.73
4	–0.71	0.17	0.39	0.60	0.78	1.49		–8.33	2.38	3.88	5.41	7.30	9.43
Big	–0.40	0.12	0.30	0.61	0.82	1.22		–9.47	2.03	4.93	4.69	6.75	8.42
	β_{ROE}							$t_{\beta_{ROE}}$					
Small	–0.40	–0.40	–0.31	–0.26	–0.37	0.03		–3.29	–4.92	–5.21	–7.61	–7.24	0.26
2	–0.04	–0.09	–0.03	–0.10	–0.18	–0.14		–0.49	–2.21	–0.76	–2.06	–3.06	–1.22
3	0.02	0.04	0.03	–0.09	–0.16	–0.18		0.33	0.83	0.50	–1.33	–2.11	–1.72
4	0.06	0.08	–0.01	–0.14	–0.17	–0.23		0.92	1.55	–0.08	–1.84	–2.14	–1.86
Big	0.15	0.15	–0.04	–0.07	–0.26	–0.41		4.58	3.36	–0.75	–0.86	–3.23	–4.34

Table 13 : Descriptive Statistics for Monthly Percent Excess Returns of 25 $\Delta A/A$ and ROE Portfolios (1/1972–12/2011, 480 Months)

The 25 $\Delta A/A$ and ROE portfolios are the intersections of quintiles formed on $\Delta A/A$ and quintiles formed on ROE . $\Delta A/A$ is annual change in total assets (Compustat annual item AT) divided by lagged total assets. In each June we break NYSE, Amex, and NASDAQ stocks into five $\Delta A/A$ quintiles using the NYSE breakpoints. ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock. Each month, we sort all stocks into five ROE quintiles based on the NYSE breakpoints of the ranked ROE . Earnings and other accounting variables in Compustat quarterly files are used in the sorts in the months immediately after the most recent public earnings announcement dates (item RDQ). Taking intersections of the $\Delta A/A$ quintiles and the ROE quintiles, we obtain the 25 testing portfolios. The portfolio returns are value-weighted. We report mean percent excess returns, the CAPM alphas (α), the intercepts (α_{FF}) from the Fama-French three-factor regressions, and the intercepts (α_{CARH}) from the Carhart four-factor regressions, as well as t -statistics adjusted for heteroscedasticity and autocorrelations. For each factor model, we report the mean absolute error (m.a.e., the average magnitude of the alphas) across the testing portfolios, the average goodness-of-fit across the 25 portfolios ($\bar{R}^2$), and the p -value (p_{GRS}) for the GRS test on the null that the alphas of all the testing portfolios are jointly zero. The table entries at the intersection of the L–H rows and the H–L columns are for the high- ROE and low- $\Delta A/A$ portfolio minus the low- ROE and high- $\Delta A/A$ portfolio, a zero-investment portfolio with the largest average return spread across the double sorts.

	Low	2	3	4	High	H–L	Low	2	3	4	High	H–L
	Mean						t_{Mean}					
High $\Delta A/A$	–0.29	0.02	0.27	0.45	0.64	0.94	–0.80	0.08	0.88	1.64	2.18	4.36
4	0.16	0.49	0.53	0.49	0.57	0.41	0.53	1.93	2.22	2.03	2.62	1.94
3	0.30	0.25	0.67	0.67	0.59	0.30	1.00	1.07	3.10	3.16	2.69	1.34
2	0.50	0.70	0.52	0.67	0.76	0.26	1.68	3.30	2.52	3.10	3.44	1.13
Low $\Delta A/A$	0.44	0.60	0.88	0.71	0.93	0.48	1.27	2.18	3.58	2.85	3.51	2.09
L–H	0.74	0.58	0.61	0.26	0.28	1.22	3.75	3.28	3.01	1.30	1.38	4.69
	α (m.a.e. = 0.25)						t_{α} ($\bar{R}^2 = 0.73$, $p_{GRS} = 0$)					
High $\Delta A/A$	–0.90	–0.51	–0.26	–0.08	0.10	1.01	–5.09	–4.04	–1.85	–0.63	0.78	4.81
4	–0.34	0.05	0.08	0.03	0.17	0.51	–2.31	0.36	0.67	0.33	1.61	2.52
3	–0.19	–0.16	0.28	0.27	0.20	0.39	–1.19	–1.25	2.41	2.82	1.70	1.88
2	0.02	0.33	0.16	0.30	0.36	0.34	0.13	2.67	1.34	2.36	2.91	1.56
Low $\Delta A/A$	–0.12	0.14	0.45	0.29	0.47	0.58	–0.65	0.94	3.54	2.04	3.14	2.62
L–H	0.78	0.64	0.71	0.36	0.36	1.37	3.92	3.53	3.50	1.87	1.74	5.54
	α_{FF} (m.a.e. = 0.26)						$t_{\alpha_{FF}}$ ($\bar{R}^2 = 0.76$, $p_{GRS} = 0$)					
High $\Delta A/A$	–0.90	–0.49	–0.20	0.06	0.39	1.29	–5.85	–3.79	–1.59	0.53	3.65	6.86
4	–0.46	–0.06	0.01	0.11	0.31	0.77	–3.14	–0.41	0.11	1.13	3.27	4.10
3	–0.42	–0.30	0.14	0.26	0.25	0.67	–2.75	–2.41	1.33	2.74	2.28	3.49
2	–0.18	0.11	0.03	0.22	0.41	0.59	–1.04	0.98	0.30	1.68	3.35	2.73
Low $\Delta A/A$	–0.32	–0.05	0.27	0.14	0.41	0.73	–2.12	–0.38	2.13	0.99	2.81	3.41
L–H	0.58	0.44	0.47	0.08	0.02	1.31	3.31	2.72	2.52	0.41	0.09	5.66
	α_{CARH} (m.a.e. = 0.18)						$t_{\alpha_{CARH}}$ ($\bar{R}^2 = 0.77$, $p_{GRS} = 0$)					
High $\Delta A/A$	–0.66	–0.32	–0.08	0.06	0.30	0.96	–4.52	–2.67	–0.67	0.51	2.58	5.08
4	–0.22	0.07	0.03	0.10	0.18	0.39	–1.41	0.54	0.23	1.06	1.97	2.15
3	–0.23	–0.17	0.15	0.21	0.15	0.37	–1.49	–1.24	1.29	2.19	1.33	1.98
2	0.00	0.16	–0.01	0.18	0.31	0.32	–0.02	1.43	–0.08	1.49	2.58	1.32
Low $\Delta A/A$	–0.09	0.11	0.28	0.10	0.27	0.36	–0.58	0.86	2.13	0.66	1.84	1.68
L–H	0.57	0.43	0.37	0.04	–0.03	0.93	3.07	2.47	1.98	0.21	–0.17	4.09

Table 14 : Q -factor Regressions for Monthly Percent Excess Returns of 25 $\Delta A/A$ and ROE Portfolios (1/1972–12/2011, 480 Months)

$\Delta A/A$ is annual change in total assets (Compustat annual item AT) divided by lagged total assets. ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock. In June of each year t , we use the NYSE breakpoints to break NYSE, Amex, and NASDAQ stocks into quintiles on $\Delta A/A$ for the fiscal year ending in calendar year $t-1$. Independently, each month, we sort all stocks into ROE quintiles based on the NYSE breakpoints. Earnings and other accounting variables in Compustat quarterly files are used in the sorts in the months immediately after the most recent public earnings announcement dates (item RDQ). Taking the intersections of the $\Delta A/A$ quintiles and the ROE quintiles, we form 25 $\Delta A/A$ and ROE portfolios. We calculate monthly value-weighted portfolio returns. We report the q -factor regressions: $r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas. $\bar{R}^2$ is the average goodness-of-fit across the 25 portfolios. p_{GRS} is the p-value for the GRS test on the null that the alphas of all the 25 testing portfolios are jointly zero.

	Low	2	3	4	High	H–L	Low	2	3	4	High	H–L
	α_q (m.a.e. = 0.09)						t_{α_q} ($\bar{R}^2 = 0.80$, $p_{GRS} = 0.26$)					
High $\Delta A/A$	–0.04	–0.01	0.04	0.07	0.39	0.43	–0.34	–0.08	0.35	0.57	3.00	2.65
4	0.07	0.19	0.10	0.01	0.01	–0.06	0.48	1.30	0.79	0.10	0.08	–0.35
3	–0.03	–0.01	0.03	0.06	–0.11	–0.08	–0.19	–0.07	0.25	0.63	–0.94	–0.45
2	0.12	0.20	–0.17	–0.04	–0.06	–0.18	0.68	1.86	–1.30	–0.28	–0.51	–0.90
Low $\Delta A/A$	0.13	0.08	0.09	–0.11	–0.11	–0.25	1.06	0.65	0.74	–0.75	–0.82	–1.32
L–H	0.18	0.09	0.05	–0.18	–0.50	–0.07	1.06	0.55	0.29	–1.01	–2.84	–0.40
	β_{MKT}						$t_{\beta_{MKT}}$					
High $\Delta A/A$	1.09	1.04	1.05	1.08	1.08	–0.01	22.56	23.21	29.57	41.92	33.73	–0.15
4	1.02	0.92	1.01	1.03	0.94	–0.08	31.02	21.49	30.75	38.09	40.25	–1.90
3	1.05	0.92	0.95	0.93	0.93	–0.11	29.58	24.72	33.54	37.77	32.50	–2.59
2	1.09	0.84	0.88	0.90	1.00	–0.09	33.17	27.53	33.13	24.30	34.79	–2.44
Low $\Delta A/A$	1.13	1.08	1.03	1.01	1.10	–0.03	39.80	38.51	25.74	22.88	36.91	–0.87
L–H	0.04	0.03	–0.02	–0.07	0.02	0.01	0.77	0.62	–0.27	–1.33	0.41	0.18
	β_{ME}						$t_{\beta_{ME}}$					
High $\Delta A/A$	0.31	0.08	0.20	0.14	0.01	–0.30	4.58	0.99	4.16	3.37	0.23	–4.19
4	0.12	0.17	–0.03	–0.13	–0.11	–0.23	1.53	3.57	–0.58	–3.75	–2.69	–2.22
3	0.12	–0.10	–0.03	–0.01	–0.06	–0.17	1.67	–1.61	–0.63	–0.24	–1.30	–2.13
2	–0.01	0.12	–0.03	0.08	–0.12	–0.11	–0.17	2.24	–0.74	0.96	–2.36	–1.96
Low $\Delta A/A$	0.44	0.02	0.11	0.10	0.23	–0.21	8.88	0.35	1.89	1.41	3.77	–3.07
L–H	0.12	–0.06	–0.09	–0.03	0.21	–0.09	1.49	–0.88	–1.25	–0.35	2.56	–0.87
	$\beta_{\Delta A/A}$						$t_{\beta_{\Delta A/A}}$					
High $\Delta A/A$	–0.81	–0.57	–0.58	–0.54	–0.92	–0.11	–8.14	–7.13	–6.73	–7.39	–10.18	–0.91
4	0.00	–0.06	0.06	–0.14	–0.15	–0.15	–0.05	–0.67	0.80	–1.90	–2.26	–1.28
3	0.34	0.26	0.52	0.15	0.13	–0.21	3.13	2.91	6.57	2.62	1.63	–1.95
2	0.61	0.58	0.55	0.46	0.24	–0.37	5.25	7.20	6.48	4.71	3.07	–3.32
Low $\Delta A/A$	0.41	0.60	0.73	0.55	0.53	0.12	6.80	6.32	8.43	4.72	7.21	1.31
L–H	1.22	1.16	1.32	1.10	1.45	1.34	11.43	11.40	11.56	7.21	12.24	10.32
	β_{ROE}						$t_{\beta_{ROE}}$					
High $\Delta A/A$	–0.80	–0.35	–0.09	0.14	0.28	1.08	–9.25	–5.82	–1.45	2.96	5.01	10.46
4	–0.68	–0.24	–0.08	0.19	0.40	1.08	–9.73	–3.86	–1.13	3.73	8.96	14.65
3	–0.56	–0.40	–0.01	0.21	0.39	0.96	–8.68	–4.08	–0.12	4.37	6.06	10.62
2	–0.63	–0.30	0.09	0.13	0.50	1.14	–9.18	–4.92	1.49	2.55	9.46	13.41
Low $\Delta A/A$	–0.87	–0.39	–0.06	0.14	0.40	1.27	–18.04	–5.75	–0.84	1.53	6.31	19.07
L–H	–0.07	–0.04	0.03	0.00	0.12	1.20	–0.68	–0.45	0.33	–0.02	1.48	10.87

Table 15 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Market Equity (1/1972–12/2011, 480 Months)

The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. Market equity is stock price per share times shares outstanding from CRSP. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on market equity at the end of June of t . Monthly value-weighted portfolio returns are calculated, and the portfolios are rebalanced at the end of June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$), and the q -factor regressions without the size factor ($r_t^i - r_t^f = a_q^i + b_{MKT}^i MKT_t + b_{\Delta A/A}^i r_{\Delta A/A,t} + b_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Small	2	3	4	5	6	7	8	9	Big	S-B	m.a.e.	$\bar{R}^2$
Mean	0.65	0.66	0.62	0.67	0.67	0.59	0.63	0.58	0.55	0.41	0.24		
t_{Mean}	1.77	1.92	1.96	2.21	2.28	2.19	2.35	2.28	2.32	2.01	0.88		
α	0.12	0.09	0.06	0.13	0.13	0.08	0.12	0.08	0.09	0.00	0.12	0.09	0.83
β	1.16	1.26	1.22	1.21	1.19	1.13	1.12	1.10	1.02	0.91	0.25	(0.71)	
t_α	0.58	0.53	0.45	0.97	1.19	0.88	1.53	1.03	1.61	-0.02	0.49		
α_{FF}	-0.08	-0.07	-0.08	0.01	0.00	-0.01	0.06	0.04	0.07	0.11	-0.19	0.05	0.95
b	0.94	1.05	1.05	1.05	1.06	1.04	1.05	1.04	0.99	0.93	0.01	(0.01)	
s	1.25	1.16	0.96	0.90	0.74	0.54	0.40	0.32	0.15	-0.27	1.53		
h	0.12	0.06	0.07	0.02	0.08	0.06	0.02	0.02	0.01	-0.16	0.28		
$t_{\alpha_{FF}}$	-0.82	-1.18	-1.48	0.20	0.09	-0.11	1.11	0.53	1.22	2.93	-1.72		
α_{CARH}	-0.07	-0.06	-0.09	0.02	0.00	0.02	0.07	0.05	0.06	0.10	-0.17	0.05	0.95
b	0.94	1.05	1.05	1.04	1.06	1.03	1.05	1.04	0.99	0.93	0.01	(0.03)	
s	1.25	1.16	0.96	0.90	0.74	0.54	0.40	0.32	0.15	-0.27	1.53		
h	0.12	0.06	0.07	0.02	0.08	0.05	0.02	0.01	0.01	-0.15	0.27		
w	-0.01	-0.01	0.00	-0.01	0.00	-0.03	0.00	-0.01	0.01	0.01	-0.02		
$t_{\alpha_{CARH}}$	-0.68	-0.96	-1.60	0.38	0.04	0.37	1.12	0.68	1.15	2.45	-1.47		
α_q	0.14	0.09	-0.04	0.07	0.04	0.01	0.08	0.10	0.13	0.10	0.04	0.08	0.95
β_{MKT}	0.93	1.03	1.04	1.04	1.05	1.03	1.05	1.04	0.98	0.94	-0.02	(0.00)	
β_{ME}	1.10	1.05	0.92	0.85	0.71	0.51	0.38	0.28	0.13	-0.28	1.37		
$\beta_{\Delta A/A}$	-0.06	-0.14	-0.10	-0.16	-0.09	-0.04	-0.05	-0.05	-0.04	-0.12	0.06		
β_{ROE}	-0.39	-0.27	-0.09	-0.10	-0.05	-0.04	-0.03	-0.10	-0.07	0.03	-0.42		
t_{α_q}	1.16	1.02	-0.75	1.32	0.70	0.13	1.19	1.33	2.02	2.53	0.35		
$t_{\beta_{MKT}}$	30.24	46.03	67.77	64.42	79.83	60.95	69.50	57.64	56.66	81.33	-0.60		
$t_{\beta_{ME}}$	29.02	35.19	31.56	41.70	30.28	16.11	19.00	8.47	3.53	-17.35	40.51		
$t_{\beta_{\Delta A/A}}$	-0.62	-2.24	-2.87	-4.95	-1.81	-0.95	-1.10	-0.86	-0.75	-4.16	0.88		
$t_{\beta_{ROE}}$	-5.05	-4.74	-2.82	-2.90	-1.55	-1.29	-1.08	-2.97	-2.16	1.62	-6.22		
a_q	0.65	0.57	0.38	0.47	0.37	0.25	0.26	0.23	0.18	-0.02	0.68	0.34	0.85
b_{MKT}	1.07	1.17	1.16	1.15	1.14	1.10	1.10	1.07	1.00	0.91	0.16	(0.00)	
$b_{\Delta A/A}$	-0.10	-0.18	-0.14	-0.19	-0.12	-0.06	-0.07	-0.06	-0.05	-0.11	0.01		
b_{ROE}	-0.74	-0.60	-0.38	-0.37	-0.28	-0.20	-0.15	-0.19	-0.11	0.12	-0.86		
t_{a_q}	2.43	2.50	2.26	2.77	2.67	2.22	2.45	2.11	2.30	-0.37	2.17		
$t_{b_{MKT}}$	17.95	22.37	26.27	26.81	29.52	36.61	47.68	50.03	59.06	52.17	2.26		
$t_{b_{\Delta A/A}}$	-0.62	-1.21	-1.12	-1.45	-0.94	-0.76	-0.78	-0.69	-0.67	-1.97	0.03		
$t_{b_{ROE}}$	-5.31	-5.37	-4.04	-4.31	-3.46	-4.09	-3.30	-3.99	-2.60	3.38	-5.34		

Table 16 : Alphas from Q -factor Regressions without the Size Factor, Monthly Percent Excess Returns of 25 Size and Book-to-Market Portfolios as well as 25 $\Delta A/A$ and ROE Portfolios (1/1972–12/2011, 480 Months)

See the captions of Tables 11 and 13 for the constructions of the 25 size and book-to-market portfolios as well as the 25 $\Delta A/A$ and ROE portfolios, respectively. We report the alphas from the three-factor version of the q -factor regressions: $r_t^i - r_t^f = a_q^i + b_{MKT}^i MKT_t + b_{\Delta A/A}^i r_{\Delta A/A,t} + b_{ROE}^i r_{ROE,t} + e_t^i$. See the caption of Table 1 for the description of $r_{\Delta A/A}$ and r_{ROE} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. $\bar{R}^2$ is the average goodness-of-fit across a given set of 25 testing portfolios. p_{GRS} is the p-value from the GRS test on the null that the alphas of a given set of 25 portfolios are jointly zero. The data for the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site.

Panel A: 25 size and book-to-market portfolios												
	Low	2	3	4	High	H–L	Low	2	3	4	High	H–L
	a_q (m.a.e. = 0.31)						t_{a_q} ($\bar{R}^2 = 0.78$, $p_{GRS} = 0$)					
Small	0.27	0.83	0.79	0.79	0.79	0.52	0.87	2.73	2.89	3.61	3.91	2.31
2	0.27	0.51	0.41	0.42	0.48	0.20	1.21	2.62	2.80	3.08	2.72	1.06
3	0.30	0.28	0.18	0.18	0.38	0.08	1.68	1.83	1.44	1.52	2.52	0.39
4	0.34	0.00	0.13	0.13	0.17	–0.17	1.95	0.03	1.08	1.09	1.21	–0.73
Big	–0.01	–0.09	0.00	–0.03	–0.06	–0.05	–0.17	–0.99	0.00	–0.25	–0.40	–0.27
Panel B: 25 $\Delta A/A$ and ROE portfolios												
	Low	2	3	4	High	H–L	Low	2	3	4	High	H–L
	a_q (m.a.e. = 0.12)						t_{a_q} ($\bar{R}^2 = 0.79$, $p_{GRS} = 0.06$)					
High $\Delta A/A$	0.10	0.03	0.14	0.14	0.39	0.29	0.71	0.21	0.99	1.02	3.13	1.62
4	0.12	0.27	0.09	–0.05	–0.04	–0.17	0.85	1.83	0.73	–0.50	–0.41	–0.90
3	0.03	–0.06	0.01	0.05	–0.13	–0.16	0.19	–0.36	0.10	0.58	–1.21	–0.93
2	0.12	0.26	–0.18	0.00	–0.12	–0.24	0.65	2.30	–1.39	–0.02	–0.91	–1.23
Low $\Delta A/A$	0.34	0.09	0.14	–0.06	–0.01	–0.35	2.28	0.72	1.13	–0.43	–0.06	–1.88
L–H	0.24	0.06	0.01	–0.20	–0.40	–0.11	1.35	0.38	0.03	–1.04	–2.03	–0.61

Table 17 : Alphas from Q -factor Regressions without the Size Factor, Monthly Percent Excess Returns of One-way Deciles

See Tables 2 to 10 for the constructions of the testing portfolios. We report the alphas from the three-factor version of the q -factor regressions: $r_t^i - r_t^f = a_q^i + b_{MKT}^i MKT_t + b_{\Delta A/A}^i r_{\Delta A/A,t} + b_{ROE}^i r_{ROE,t} + e_t^i$. See the caption of Table 1 for the description of $r_{\Delta A/A}$ and r_{ROE} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. $\bar{R}^2$ is the average goodness-of-fit across a given set of deciles. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. The data for the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. The sample for the failure probability deciles is from January 1976 to December 2011 (432 months). All the other portfolios are from January 1972 to December 2011 (480 months).

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Panel A: <i>SUE</i>													
a_q	-0.01	0.00	0.06	0.08	0.01	-0.05	0.09	0.00	0.04	0.18	0.19	0.05	0.88
t_{a_q}	-0.07	0.02	0.64	0.90	0.06	-0.50	1.14	-0.04	0.49	2.16	1.24	(0.61)	
Panel B: Price momentum													
a_q	0.09	0.05	0.08	0.08	-0.03	-0.09	-0.10	-0.08	0.02	0.44	0.36	0.11	0.88
t_{a_q}	0.43	0.27	0.60	0.76	-0.37	-1.38	-1.81	-1.26	0.20	1.88	0.91	(0.00)	
Panel C: <i>IVOL</i>													
a_q	-0.22	-0.02	-0.01	0.00	-0.02	0.03	0.29	0.23	0.45	0.15	0.37	0.14	0.86
t_{a_q}	-2.28	-0.19	-0.12	0.02	-0.18	0.28	2.48	1.99	2.71	0.70	1.35	(0.01)	
Panel D: Failure probability													
a_q	0.04	-0.16	-0.17	-0.08	0.07	0.16	0.24	0.38	0.40	0.26	0.22	0.20	0.87
t_{a_q}	0.25	-1.82	-2.06	-0.91	0.92	1.69	2.11	2.29	3.00	0.96	0.60	(0.00)	
Panel E: Net stock issues													
a_q	0.26	-0.06	0.07	-0.08	-0.06	0.07	0.30	0.32	0.09	0.00	-0.25	0.13	0.85
t_{a_q}	2.51	-0.80	0.50	-0.93	-0.68	0.81	2.87	3.17	0.84	0.05	-1.62	(0.00)	
Panel F: Composite issuance													
a_q	-0.02	0.10	-0.04	-0.32	-0.02	0.16	0.21	0.18	0.06	-0.11	-0.09	0.12	0.85
t_{a_q}	-0.17	0.96	-0.40	-3.65	-0.22	1.68	1.75	1.78	0.63	-1.17	-0.62	(0.01)	
Panel G: Abnormal corporate investment													
a_q	0.31	0.40	0.29	-0.02	0.08	-0.19	-0.17	-0.16	0.19	0.07	-0.24	0.19	0.86
t_{a_q}	1.84	3.13	3.08	-0.22	0.85	-1.99	-1.63	-1.89	2.03	0.66	-1.22	(0.00)	
Panel H: Total accruals (scaled by average total assets)													
a_q	0.25	0.18	0.17	0.06	0.06	-0.07	0.10	-0.08	-0.07	0.06	-0.19	0.11	0.86
t_{a_q}	1.75	1.49	2.12	0.56	0.67	-0.82	1.29	-0.91	-0.79	0.54	-1.04	(0.10)	
Panel I: Industries													
	NoDur	Durbl	Manuf	Enrgy	HiTec	Telcm	Shops	Hlth	Tils	Other			
a_q	-0.07	-0.25	-0.13	0.00	0.61	0.27	-0.06	-0.12	-0.11	-0.27	0.19		
t_{a_q}	-0.54	-1.27	-1.46	0.00	3.82	1.44	-0.38	-0.70	-0.65	-2.79	(0.00)		

Internet Appendix: Not for Publication

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Abstract

This Internet Appendix reports supplementary results for our manuscript “Digesting anomalies: An investment approach.”

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IA.1 The Definition of Failure Probability

We construct the failure probability (distress) measure following Campbell, Hilscher, and Szilagyi (2008, the third column in Table IV):

$$\begin{aligned} \text{Distress}(t) \equiv & -9.164 - 20.264 NIMTAAVG_t + 1.416 TLMTA_t - 7.129 EXRETAVG_t \\ & + 1.411 SIGMA_t - 0.045 RSIZE_t - 2.132 CASHMTA_t + 0.075 MB_t - 0.058 PRICE_t \end{aligned} \quad (1)$$

$$NIMTAAVG_{t-1,t-12} \equiv \frac{1-\phi^3}{1-\phi^{12}} (NIMTA_{t-1,t-3} + \dots + \phi^9 NIMTA_{t-10,t-12}) \quad (2)$$

$$EXRETAVG_{t-1,t-12} \equiv \frac{1-\phi}{1-\phi^{12}} (EXRET_{t-1} + \dots + \phi^{11} EXRET_{t-12}), \quad (3)$$

in which $\phi = 2^{-1/3}$. *NIMTA* is net income (Compustat quarterly item NIQ) divided by the sum of market equity and total liabilities (item LTQ). The moving average *NIMTAAVG* is designed to capture the idea that a long history of losses is a better predictor of bankruptcy than one large quarterly loss in a single month. $EXRET \equiv \log(1 + R_{it}) - \log(1 + R_{S\&P500,t})$ is the monthly log excess return on each firm's equity relative to the S&P 500 index. The moving average *EXRETAVG* is designed to capture the idea that a sustained decline in stock market value is a better predictor of bankruptcy than a sudden stock price decline in a single month.

TLMTA is the ratio of total liabilities (Compustat quarterly item LTQ) divided by the sum of market equity and total liabilities. *SIGMA* is the annualized three-month rolling sample standard deviation: $\sqrt{\frac{252}{N-1} \sum_{k \in \{t-1, t-2, t-3\}} r_k^2}$, in which k is the index of trading days in months $t-1, t-2$, and $t-3$, r_k^2 is the firm-level daily return, and N is the total number of trading days in the three-month period. *SIGMA* is treated as missing if there are less than five nonzero observations over the three months in the rolling window. *RSIZE* is the relative size of each firm measured as the log ratio of its market equity to that of the S&P 500 index. *CASHMTA*, used to capture the liquidity position of the firm, is the ratio of cash and short-term investments (item CHEQ) divided by the sum of market equity and total liabilities. *MB* is the market-to-book equity, in which book equity is measured in the same way as the denominator of *ROE*. Following Campbell et al., we add 10% of the

difference between market and book equity to the book equity to alleviate measurement issues for extremely small book equity values. For firm-month observations that still have negative book equity after this adjustment, we replace these negative values with \$1 to ensure that the market-to-book ratios for these firms are in the right tail of the distribution. *PRICE* is each firm's log price per share, truncated above at \$15. We further eliminate stocks with prices less than \$1 at the portfolio formation date. Following Campbell et al., we winsorize the variables in the right-hand side of equation (1) at the 5th and 95th percentiles of their pooled distribution across all firm-month observations.

IA.2 Alternative Testing Portfolios

In the manuscript, we construct testing portfolios using the NYSE breakpoints and value-weighted returns. As noted, this empirical design alleviates the impact of microcaps and small stocks. However, some prominent papers that first document the anomalies use the NYSE-Amex-NASDAQ breakpoints and equal-weighted returns. In Internet Appendix, we evaluate the performance of difference factor models, using alternative constructions of the testing portfolios. We find that the alternative constructions tend to increase the magnitude of the anomalies. However, the basic result that the *q*-factor model is competitive with the traditional factor models remains unchanged.

IA.2.1 The Earnings Momentum Deciles, Equal-weighted Six-month Returns

Chan, Jegadeesh, and Lakonishok (1996) use equal-weighted six-month holding-period returns after portfolio formation when constructing the *SUE* deciles. (Chan et al. also use the NYSE breakpoints.) We redo the tests in Section 3.1 while matching exactly their research design.

Specifically, at the beginning of each month t , we rank all NYSE, Amex, and NASDAQ stocks into deciles by their most recent past *SUE* with the NYSE breakpoints. Monthly equal-weighted six-month holding-period returns on the *SUE* deciles are calculated from month t to $t + 5$, and the portfolios are rebalanced at the beginning of month $t + 1$. Table IA.1 shows that equal-weighting returns makes the *SUE* effect stronger. The high-minus-low *SUE* decile earns an average return

of 0.74% per month, which is more than six standard errors from zero. The CAPM beta and the Fama-French factor loadings are all negative, going in the wrong direction in explaining the average returns. The high-minus-low alpha rises to 0.88% in the Fama-French model, which is more than eight standard errors from zero. The Carhart model helps with a positive *WML* loading. The high-minus-low alpha drops to 0.60%, which is still more than five standard errors from zero. The *q*-factor model outperforms the Carhart model in reducing the high-minus-low alpha to 0.36% per month, albeit still significant ($t = 2.80$). However, the m.a.e. is 0.41%, which is larger than 0.34% in the Carhart model. Finally, all the factor models are strongly rejected by the GRS test.

IA.2.2 The Idiosyncratic Volatility Deciles, NYSE-Amex-NASDAQ Breakpoints

Ang, Hodrick, Xing, and Zhang (2006) use all NYSE-Amex-NASDAQ breakpoints to form deciles on idiosyncratic volatility. We redo the tests in Section 3.3 using this alternative construction of the *IVOL* deciles. Table IA.2 shows that using the NYSE-Amex-NASDAQ breakpoints makes the *IVOL* effect stronger. The high-minus-low decile earns an average return of -1.34% per month, which is more than three standard errors from zero. In contrast, this average return is only -0.54% ($t = -1.51$) with the NYSE breakpoints (see Table 4).

The traditional factor models fail to explain the *IVOL* effect. In the CAPM, four out of ten deciles have significant alphas. The m.a.e. is 0.41% per month. And the high-minus-low decile has an alpha of -1.73% , which is more than 4.5 standard errors from zero. In the Fama-French model, five out of ten deciles have significant alphas. The m.a.e. is also 0.41%. The high-minus-low decile has a Fama-French alpha of -1.77% , which is more than six standard errors from zero. In the Carhart model, four out of ten deciles have significant alphas. The m.a.e. drops to 0.29%. The high-minus-low decile has a Carhart alpha of -1.37% , which is more than four standard errors from zero.

Although outperforming the traditional models, the *q*-factor model also fails to explain the *IVOL* effect. Two out of ten deciles have significant alphas. The m.a.e. drops further to 0.18%.

All four factor models are strongly rejected by the GRS test. The high-minus-low alpha in the q -factor model is -0.68% , which is slightly more than two standard errors from zero. Although still large, this alpha represents a reduction of 50% in magnitude from the Carhart alpha of -1.37% .

IA.2.3 The Financial Distress Deciles, NYSE-Amex-NASDAQ Breakpoints

Campbell, Hilscher, and Szilagyi (2008) use all NYSE-Amex-NASDAQ breakpoints to form deciles on failure probability. We redo the tests in Section 3.4 using this alternative construction of the distress deciles. Table IA.3 shows that using the NYSE-Amex-NASDAQ breakpoints makes the distress effect stronger. The high-minus-low decile earns an average return of -0.96% per month, which is slightly more than two standard errors from zero. In contrast, this average return is only -0.57% ($t = -1.42$) with the NYSE breakpoints (see Table 5).

The traditional factor models fail to explain the distress effect. In the CAPM, four out of ten deciles have significant alphas. The m.a.e. is 0.33% per month, and the high-minus-low decile has an alpha of -1.45% , which is more than 3.5 standard errors from zero. In the Fama-French model, six out of ten deciles have significant alphas. The m.a.e. is 0.43% , and the high-minus-low alpha inflates to -1.92% , which is about six standard errors from zero. In the Carhart model, three out of ten deciles have significant alphas. The m.a.e. drops to 0.21% , and the high-minus-low alpha drops to -0.93% , which is more than three standard errors from zero.

The q -factor model again outperforms the traditional factor models. Two out of ten deciles have significant alphas. The m.a.e. drops further to 0.18% . The high-minus-low alpha falls to -0.28% , which is within one standard error of zero. Although still sizable, this alpha represents a reduction of 70% in magnitude from the Carhart alpha of -0.93% .

IA.2.4 The Composite Issuance Deciles, NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns

Daniel and Titman (2006) document the negative relation between composite issuance and average stock returns using the Fama-MacBeth (1973) cross-sectional regressions. This practice is similar

to a portfolio approach with all NYSE-Amex-NASDAQ breakpoints and equal-weighted portfolio returns. Table IA.4 reports factor regressions of the composite issuance deciles constructed from this alternative design. Doing so again increases the magnitude of the anomaly. The average return of the high-minus-low decile is -0.74% per month, which is more than three standard errors from zero.

The CAPM produces a large high-minus-low alpha of -0.96% per month ($t = -4.88$) and a large m.a.e. of 0.46% . In the Fama-French model, the *HML* loading of the high-minus-low decile -0.40 , which goes in the right direction in explaining the average return. However, using equal-weighted returns produces a large *SMB* loading of 0.70 , which goes in the wrong direction. As such, the high-minus-low decile has a Fama-French alpha of -0.84% , which is 5.5 standard errors from zero. The m.a.e. is reduced to 0.26% . Relative to the Fama-French model, the Carhart model reduces the high-minus-low alpha in magnitude to -0.64% ($t = -3.79$), but increases the m.a.e. to 0.36% . Both the Fama-French model and the Carhart model are rejected by the GRS test.

The *q*-factor model reduces the high-minus-low alpha in magnitude further to -0.28% per month, which is within 1.5 standard errors from zero. The m.a.e. is 0.36% , which is the same as that in the Carhart model. The investment factor continues to help with a loading of -0.75 for the high-minus-low decile, which is more than five standard errors from zero. Using equal-weighted returns also brings the *ROE* factor to work. The high-minus-low decile has an *ROE* factor loading of -0.63 , which is more than 4.5 standard errors from zero.

IA.2.5 The Abnormal Corporate Investment Deciles, NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns

We show in Section 3.7 that Titman, Wei, and Xie's (2004) abnormal corporate investment does not produce a significant average return spread with the NYSE breakpoints and value-weighted returns. Table IA.5 shows that with the NYSE-Amex-NASDAQ breakpoints and equal-weighting, the high-minus-low decile earns an average return of -0.39% per month, which is more than four standard errors from zero. The high-minus-low alphas are significant in both the CAPM and the

Fama-French model, which are in turn rejected by the GRS test.

The q -factor model performs about as well as the Carhart model. The high-minus-low alpha is -0.30% in the q -factor model, which is slightly higher than -0.28% in the Carhart model. The m.a.e. is 0.34% in the q -factor model, which is comparable with 0.33% in the Carhart model. However, both models are still strongly rejected by the GRS test. Also, the m.a.e. is 0.19% in the Fama-French model, which is lower than those from the Carhart model and the q -factor model.

IA.2.6 The Total Accruals Deciles, NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns

Table IA.6 reports the results for the accrual deciles constructed with the NYSE-Amex-NASDAQ breakpoints and equal-weighted returns, as in Sloan (1996). The average return of the high-minus-low decile is -0.78% per month, which is more than five standard errors from zero. This average return is more than doubled in magnitude from -0.30% with the NYSE breakpoints and value-weighted returns. The traditional factor loadings of the high-minus-low decile are all close to zero, giving rise to alphas that are close in magnitude to the average return.

As in the benchmark case with the NYSE breakpoints and value-weighted returns, the q -factor model struggles to explain the accrual effect. The high-minus-low alpha is -0.92% per month, which is almost six standard errors from zero. This alpha is larger in magnitude than the Fama-French alpha of -0.76% and the Carhart alpha of -0.75% . In addition, the m.a.e. is 0.47% in the q -factor model, which is higher than 0.25% in the Fama-French model and 0.38% in the Carhart model. The culprit is again the *ROE* factor loading. The high-minus-low decile has an *ROE* factor loading of 0.39 , which is almost 4.5 standard errors from zero, going in the wrong direction in explaining the average return. Although going in the right direction, the investment factor loading for the high-minus-low decile is only -0.24 .

IA.2.7 The Price Momentum Deciles, NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns

Jegadeesh and Titman (1993) use all NYSE-Amex-NASDAQ breakpoints to sort stocks into price momentum portfolios, and equal-weight the portfolio returns. We redo the tests in Section 3.2 using this alternative design. However, when equal-weighting returns, we also follow Jegadeesh and Titman in excluding stocks with price under \$5. Including these stocks produces a large and negative January effect, which reduces greatly the magnitude of the equal-weighted momentum profits. (Value-weighting returns as in Section 3.2 is immune to this issue.)

Table IA.7 shows that the alternative design increases the average winner-minus-loser return to 1.32% per month, which is more than 5.5 standard errors from zero. The CAPM and the Fama-French model produce alphas of 1.36% and 1.50%, respectively, both of which are more than six standard errors from zero. The Carhart model reduces the winner-minus-loser alpha to 0.56% ($t = 4.15$). The m.a.e. is 0.13%, dropping from 0.29% in the Fama-French model.

The q -factor model again performs similarly as the Carhart model. The winner-minus-loser alpha is 0.61% per month. Although large, this alpha is within 1.8 standard errors from zero. The m.a.e. is 0.11%, which is comparable to 0.13% in the Carhart model. The average R^2 is 93%, which is again comparable to 96% in the Carhart model. Finally, three out of ten deciles have significant alphas in the q -factor model, in contrast to six in the Carhart model.

IA.2.8 25 Size and Momentum Portfolios

At the beginning of each month t , we use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks into quintiles on their prior six-month returns from month $t-7$ to $t-2$, skipping month $t-1$. Independently, in June of each year t , we also use NYSE breakpoints to sort stocks into quintiles on the market equity at the end of June. We form 25 portfolios each month from taking the intersections of the size and the momentum quintiles, and compute value-weighted portfolio returns for the subsequent six months from month t to $t+5$.

Table IA.8 reports large momentum profits. From Panel A, the average winner-minus-loser return varies from 0.51% ($t = 2.31$) to 1.02% per month ($t = 5.46$). The CAPM alphas of the winner-minus-loser quintiles are all significantly positive across the size quintiles. In particular, the small-stock winner-minus-loser quintile earns an alpha of 1.04%, which is more than six standard errors from zero. Consistent with Fama and French (1996), their three-factor model exacerbates momentum. The small-stock winner-minus-loser quintile earns a Fama-French alpha of 1.22% ($t = 7.20$). The m.a.e. across the 25 testing portfolios is 0.26% in the CAPM and 0.22% in the Fama-French model, and the average magnitude of the winner-minus-loser alphas is 0.74% in the CAPM and 0.90% in the Fama-French model. Both models are strongly rejected by the GRS test.

Including *WML* into the Fama-French model as in Carhart (1997) improves the performance substantially. The m.a.e. across the 25 portfolios drops from 0.22% per month in the Fama-French model to only 0.11% in the Carhart model, and the average magnitude of the winner-minus-loser alphas drops from 0.90% to only 0.25%. However, three out of five winner-minus-loser quintiles still have significant alphas in the Carhart model. In particular, the small-stock winner-minus-loser has a Carhart alpha of 0.55%, which is more than 4.5 standard errors from zero. In addition, six out of 25 portfolios have significant Carhart alphas. And the model is again rejected by the GRS test.

Table IA.9 reports the *q*-factor regressions. The m.a.e. across the 25 size and momentum portfolios is 0.11% per month, which is identical to that in the Carhart model. However, the average magnitude of the winner-minus-loser alphas is 0.19% in the *q*-factor model, which is lower than 0.25% in the Carhart model. Only one out of five winner-minus-loser alphas is significant, compared with three in the Carhart model. And two out of 25 individual portfolios have significant alphas in the *q*-factor model, relative to six in the Carhart model. Overall, the performance of the *q*-factor model seems largely comparable with that of the Carhart model.

The rest of Table IA.9 shows that the explanatory power of the *q*-factor model derives exclusively from the *ROE* factor. The loadings of the winner-minus-loser quintiles on the market, size,

and investment factors are all economically small and statistically insignificant from zero. In contrast, losers have large and significantly negative loadings, and winners have large and significantly positive loadings on r_{ROE} . Across the winner-minus-loser quintiles, the ROE factor loadings vary from 0.72 to 0.92, which are all at least 4.5 standard errors from zero. Given the average ROE factor return of 0.60%, these loadings capture momentum profits that range from 0.43% to 0.55%.

Table IA.10 shows that using the NYSE-Amex-NASDAQ breakpoints and equal-weighted returns increases momentum profits by about ten basis points on average. For example, the average winner-minus-loser return across the five size quintiles is 0.80% per month, compared to 0.70% with the NYSE breakpoints and value-weighted returns (see Table IA.8). The changes in the CAPM alphas and the Fama-French alphas have similar magnitudes.

Focusing on the performance of the q -factor model relative to the Carhart model, we count nine out of 25 individual portfolios and two out of five winner-minus-loser portfolios that have significant alphas in the Carhart model. The m.a.e. is 0.11% per month, and the Carhart model is rejected by the GRS test. In comparison, Table IA.11 shows that two out of 25 individual portfolios and one out of five winner-minus-loser portfolios have significant alphas in the q -factor model. The m.a.e. is 0.13% per month, and the model is again rejected by the GRS test. Similar to Table IA.9, the ROE factor loadings again provide the sole source of explanatory power for the q -factor model.

IA.2.9 The Book-to-Market Deciles

In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on book-to-market equity. Book-to-market for June of year t is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the market equity at the end of December of $t - 1$. Monthly value-weighted portfolio returns are computed from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. The stockholders' equity is the value reported by Compustat (item

SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

Table IA.12 shows that the value-minus-growth decile earns an average return of 0.68% per month, which is more than 2.5 standard errors from zero. The value-minus-growth alpha in the CAPM is 0.72% ($t = 2.88$). The m.a.e. is 0.23%, and the model is rejected by the GRS test. Both the Fama-French model and the Carhart model succeed in explaining the value premium. The value-minus-growth alpha is -0.03% ($t = -0.19$) and the m.a.e. is 0.07% in both models, which are in turn not rejected by the GRS test. The q -factor model delivers a value-minus-growth alpha of 0.19% per month, which is within one standard error of zero. The m.a.e. is 0.09%, and the model is not rejected by the GRS test (p-value = 0.24). As in the Fama-French model and in the Carhart model, none of the individual deciles has a significant alpha in the q -factor model.

IA.2.10 The Investment-to-Assets Deciles

We measure $\Delta A/A$ as annual change in total assets (Compustat annual item AT) divided by lagged total assets. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are computed from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. Table IA.13 shows that the high-minus-low decile earns an average return of -0.41% per month ($t = -2.32$). The CAPM alpha is -0.49% , which is more than 2.5 standard errors from zero. The m.a.e. for the CAPM is 0.17%, and the CAPM is rejected by the GRS test.

The Fama-French model explains the investment effect with a high-minus-low alpha of -0.14% per month, which is within one standard error from zero. The Carhart model also does well with a small high-minus-low alpha of -0.07% . The m.a.e. is 0.12% in the Fama-French model and 0.10% in the Carhart model, but both models are still rejected by the GRS test. The q -factor model's

performance is largely comparable. The high-minus-low alpha is 0.14% ($t = 1.03$), and the m.a.e. is 0.09%. However, the model is also rejected by the GRS test.

When documenting the investment effect, Cooper, Gulen, and Schill (2008) use the NYSE-Amex-NASDAQ breakpoints and both equal-weighted and value-weighted returns. Table IA.14 uses such breakpoints and equal-weighted returns. The high-minus-low decile earns an average return of -1.49% per month ($t = -6.65$). All the factor models fail to explain such a large spread. The Carhart model produces a high-minus-low alpha of -1.30% , which is almost six standard errors from zero, and an m.a.e. of 0.43% . The q -factor model produces a high-minus-low alpha of -1.47% ($t = -5.89$) and an m.a.e. of 0.49% . The reason why all the factor models fail to explain the equal-weighted returns across the investment deciles is that microcaps dominate the equal-weighted investment effect (e.g., Fama and French (2008)). In contrast, all the common factors are based on value-weighted returns, on which the impact of microcaps is negligible.

Using value-weighted returns limits the influence of microcaps. Table IA.15 shows that with value-weighted returns, the average return of the high-minus-low decile is only -0.56% per month ($t = -2.80$), dropping from -1.49% with equal-weighted returns. The CAPM continues to fail, but the Fama-French model and the Carhart model explain the value-weighted investment effect. The high-minus-low alphas are -0.29% and -0.23% , respectively, both of which are insignificant. The m.a.e. is 0.13% in the Fama-French model and 0.12% in the Carhart model. The q -factor model reduces the high-minus-low alpha somewhat to -0.16% , which is within one standard error of zero, and the m.a.e. is 0.11% , which is comparable to 0.12% in the Carhart model.

IA.2.11 The *ROE* Deciles

At the beginning of each month, we sort all stocks into deciles based on the NYSE breakpoints of the ranked values of *ROE*. Earnings and other accounting variables in Compustat quarterly files are used in the monthly sorts in the months immediately after the most recent public earnings announcement dates (Compustat quarterly item RDQ). We calculate monthly value-weighted

portfolio returns for the current month, and the portfolios are rebalanced monthly. The measure of *ROE* is the same as in Section 2.

Table IA.16 shows that high *ROE* stocks earn higher returns on average than low *ROE* stocks. The high-minus-low decile earns an average return of 0.81% per month, which is more than three standard errors from zero. The CAPM, the Fama-French model, and the Carhart model all fail to explain the *ROE* effect, with the high-minus-low alphas of 0.97%, 1.19%, and 0.86%, respectively, which are close to or more than four standard errors from zero. The m.a.e.'s are 0.18%, 0.25%, and 0.16%, respectively, and the models are all rejected by the GRS test.

The *q*-factor model seems to outperform the traditional factor models. The high-minus-low alpha is only 0.03% per month ($t = 0.25$). The m.a.e. is 0.10%, dropping from 0.16% in the Carhart model. However, the model is still rejected by the GRS test.

IA.3 Additional Value Deciles

In Section 3.10, we have shown that the *q*-factor model performs roughly as well as the Fama-French model and the Carhart model in explaining the average returns across the 25 size and book-to-market portfolios. We extend this basic finding to additional testing portfolios, which Fama and French (1996) show that their three-factor model does a good job in explaining.

Table IA.17 reports the results for the market leverage (A/ME) deciles. We measure A/ME as the ratio of total book assets (Compustat annual item AT) to the market equity (price per share times number of shares outstanding from CRSP). In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on A/ME , in which A is total assets for the fiscal year ending in calendar year $t - 1$ and ME is the market equity at the end of December of $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June. The high-minus-low decile earns an average return of 0.42% per month ($t = 1.78$). The high-minus-low alpha is 0.40% ($t = 1.66$) in the CAPM,

which also produces an m.a.e. of 0.21%. The Fama-French model reduces the m.a.e. to 0.11%, but the high-minus-low alpha becomes significantly negative, -0.32% ($t = -2.40$). The Carhart model reduces the m.a.e. further to 0.07%, and the high-minus-low alpha is -0.23% , which is insignificant. In the q -factor model, the m.a.e. is 0.06%, and the high-minus-low alpha is -0.10% . Also, both the CAPM and the Fama-French model are rejected by the GRS test, whereas the Carhart model and the q -factor model are not, with p-values 0.05 and 0.32, respectively.

Table IA.18 shows that the q -factor model is again comparable with the Carhart model in explaining the reversal (prior 13–60 month returns) deciles. At the beginning of each month t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on the prior returns from month $t - 13$ to $t - 60$. Monthly value-weighted portfolio returns are computed, and the portfolios are rebalanced monthly. The average return and the alphas for the high-minus-low deciles are all insignificant. The m.a.e. is somewhat high, 0.21% per month in the CAPM, but is largely comparable among the other three models, around 0.11%. Finally, while the CAPM and the Fama-French model are rejected by the GRS test, the Carhart model and the q -factor model are not.

Table IA.19 reports the results for the earnings-to-price (E/P) deciles. In June of each year t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into deciles based on E/P , calculated as total earnings before extraordinary items (Compustat annual item IB) for the last fiscal year end in $t - 1$ divided by the market equity at the end of December of year $t - 1$. Stocks with negative earnings for the last fiscal year end in $t - 1$ are excluded. We calculate monthly value-weighted portfolio returns from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. Table IA.19 shows that high E/P stocks earn higher average returns than low E/P stocks. The average return spread of 0.58% is more than 2.5 standard errors from zero. The CAPM fails to explain the E/P effect. The model is rejected by the GRS test, and the high-minus-low alpha is 0.68% ($t = 2.98$). Both the Fama-French model and the Carhart model do a good job. The high-minus-low alphas are both close to zero, and the GRS test fails to reject the models. The q -factor model also performs well. The high-minus-low alpha is somewhat higher, 0.17%, but is

within one standard error of zero. The model is not rejected by the GRS test either. All three multifactor models deliver comparable m.a.e.'s, around 0.10%.

IA.4 Annually Rebalanced Testing Portfolios on Earnings Surprise, Idiosyncratic Volatility, Financial Distress, and Momentum

We show that the original earnings surprise, momentum, idiosyncratic volatility, and financial distress anomalies vanish, once we change the rebalancing frequency of the testing portfolios from monthly to annual. We use the breakpoints and weighting schemes of returns following the original papers that document these anomalies.

Specifically, in June of each year t , we use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks into ten deciles based on the ranked values of SUE measured at the fiscal yearend of $t - 1$. Monthly equal-weighted returns are calculated from July of year t to June of $t + 1$ and the portfolios are rebalanced in June. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles based on the prior returns from July of year $t - 1$ to December of year $t - 1$. Monthly equal-weighted returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles based on the ranked values of $IVOL$ calculated with the information up to December of year $t - 1$. Finally, in June of year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles based on the Campbell, Hilscher, and Szilagyi (2008) failure probability calculated at the fiscal yearend of $t - 1$. Monthly value-weighted returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. Due to data limitations, we start the sample in July 1972 for the SUE deciles and in July 1976 for the distress deciles. The samples for the $IVOL$ deciles and for the momentum deciles both start in January 1972, as in most testing portfolios studied in the paper.

Table IA.20 shows that none of the annually rebalanced portfolios earn significant average re-

turns or CAPM alphas. All of the high-minus-low average returns and CAPM alphas are within 1.5 standard errors of zero. In particular, Panel B shows that winners with high prior returns earn insignificantly lower returns on average than losers with low prior returns. This evidence suggests that momentum profits are short-lived, consistent with Chan, Jegadeesh, and Lakonishok (1996).

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Table IA.1 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Foster, Olsen, and Shevlin's (1984) Standardized Unexpected Earnings (SUE), Equal-weighted Six-month Holding-period Returns (1/1972–12/2011, 480 Months)

SUE is the change in the most recently announced quarterly earnings per share from its value announced four quarters ago divided by the standard deviation of the change in quarterly earnings over the prior eight quarters (at least six quarters). At the beginning of each month t , we rank all NYSE, Amex, and NASDAQ stocks into deciles by their most recent past SUE with the NYSE breakpoints. Monthly equal-weighted six-month holding-period returns on the SUE portfolios are calculated from month t to $t + 5$, and the portfolios are rebalanced monthly. We report mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.51	0.53	0.69	0.76	0.91	1.04	1.07	1.13	1.13	1.24	0.74		
t_{Mean}	1.47	1.66	2.12	2.35	2.83	3.30	3.49	3.76	3.74	4.20	6.24		
α	-0.03	0.02	0.16	0.24	0.40	0.52	0.56	0.62	0.62	0.73	0.76	0.39	0.75
β	1.18	1.14	1.15	1.15	1.15	1.14	1.13	1.13	1.13	1.13	-0.05	(0.00)	
t_α	-0.16	0.11	1.03	1.53	2.55	3.54	3.97	4.56	4.42	5.46	7.06		
α_{FF}	-0.29	-0.23	-0.09	-0.01	0.14	0.28	0.34	0.45	0.45	0.59	0.88	0.29	0.92
b	1.06	1.03	1.04	1.03	1.04	1.04	1.03	1.01	1.02	1.03	-0.04	(0.00)	
s	0.95	0.90	0.91	0.92	0.89	0.85	0.79	0.77	0.76	0.68	-0.28		
h	0.31	0.30	0.30	0.30	0.31	0.29	0.27	0.17	0.16	0.13	-0.18		
$t_{\alpha_{FF}}$	-2.65	-2.65	-1.08	-0.11	1.76	3.67	4.69	6.76	6.58	7.85	8.29		
α_{CARH}	0.06	0.03	0.14	0.22	0.32	0.42	0.47	0.55	0.52	0.66	0.60	0.34	0.94
b	0.99	0.97	0.99	0.99	1.00	1.01	1.01	0.99	1.01	1.01	0.02	(0.00)	
s	0.95	0.90	0.91	0.92	0.89	0.85	0.79	0.77	0.76	0.68	-0.28		
h	0.19	0.21	0.22	0.22	0.25	0.24	0.22	0.14	0.14	0.11	-0.08		
w	-0.37	-0.28	-0.24	-0.24	-0.19	-0.15	-0.13	-0.10	-0.07	-0.07	0.30		
$t_{\alpha_{CARH}}$	0.46	0.31	1.47	2.17	3.81	5.13	5.95	7.27	6.63	7.83	5.27		
α_q	0.26	0.19	0.26	0.34	0.40	0.49	0.47	0.55	0.51	0.62	0.36	0.41	0.93
β_{MKT}	0.98	0.96	0.99	0.98	1.00	1.00	1.01	1.00	1.01	1.02	0.03	(0.00)	
β_{ME}	0.73	0.73	0.75	0.74	0.75	0.74	0.71	0.70	0.71	0.65	-0.08		
$\beta_{\Delta A/A}$	-0.03	-0.01	0.07	0.09	0.16	0.15	0.17	0.07	0.06	0.03	0.06		
β_{ROE}	-0.69	-0.53	-0.49	-0.50	-0.40	-0.33	-0.25	-0.20	-0.14	-0.08	0.61		
t_{α_q}	1.44	1.26	1.90	2.40	3.57	4.67	4.76	5.59	5.33	5.95	2.80		
$t_{\beta_{MKT}}$	27.30	30.24	34.24	32.92	36.22	38.13	37.65	40.12	43.43	38.27	1.35		
$t_{\beta_{ME}}$	10.28	10.62	10.86	12.07	11.41	12.22	10.94	13.25	15.62	10.14	-2.10		
$t_{\beta_{\Delta A/A}}$	-0.20	-0.04	0.66	0.76	1.80	1.88	2.03	0.95	0.81	0.36	0.77		
$t_{\beta_{ROE}}$	-6.75	-6.27	-6.54	-6.43	-7.09	-6.13	-4.78	-3.57	-2.47	-1.33	8.01		

Table IA.2 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Ang, Hodrick, Xing, and Zhang's (2006) Idiosyncratic Volatility (*IVOL*), All NYSE-Amex-NASDAQ Breakpoints (1/1972–12/2011, 480 Months)

IVOL is the standard deviation of the residuals from the Fama-French three-factor regression. We form value-weighted deciles each month by sorting all stocks on their *IVOL* computed using daily returns over the previous month (with a minimum of 15 daily observations). We use the NYSE-Amex-NASDAQ breakpoints. We hold the *IVOL* deciles for one month, and rebalance the portfolios monthly. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.51	0.51	0.55	0.55	0.63	0.39	0.39	0.03	–0.22	–0.83	–1.34		
t_{Mean}	2.87	2.52	2.33	2.02	2.05	1.09	1.00	0.08	–0.48	–1.67	–3.13		
α	0.18	0.11	0.08	0.02	0.05	–0.24	–0.28	–0.68	–0.96	–1.54	–1.73	0.41	0.79
β	0.72	0.88	1.05	1.17	1.29	1.39	1.49	1.58	1.62	1.57	0.85	(0.00)	
t_α	2.26	1.76	1.24	0.21	0.43	–1.55	–1.48	–3.39	–3.77	–4.85	–4.72		
α_{FF}	0.17	0.11	0.10	0.06	0.13	–0.16	–0.18	–0.60	–0.95	–1.61	–1.77	0.41	0.86
b	0.78	0.91	1.04	1.11	1.18	1.21	1.27	1.33	1.34	1.29	0.51	(0.00)	
s	–0.23	–0.14	–0.04	0.20	0.34	0.59	0.76	0.89	1.19	1.29	1.51		
h	0.08	0.02	–0.04	–0.13	–0.24	–0.29	–0.39	–0.39	–0.30	–0.17	–0.26		
$t_{\alpha_{FF}}$	2.43	2.00	1.51	0.72	1.37	–1.41	–1.27	–3.87	–5.00	–6.21	–6.20		
α_{CARH}	0.12	0.11	0.14	0.11	0.16	–0.06	–0.01	–0.40	–0.58	–1.25	–1.37	0.29	0.87
b	0.79	0.91	1.04	1.10	1.17	1.19	1.23	1.29	1.26	1.21	0.42	(0.00)	
s	–0.23	–0.14	–0.04	0.20	0.34	0.59	0.76	0.90	1.19	1.29	1.51		
h	0.10	0.03	–0.05	–0.15	–0.25	–0.32	–0.44	–0.45	–0.43	–0.30	–0.40		
w	0.05	0.01	–0.04	–0.05	–0.03	–0.11	–0.18	–0.21	–0.38	–0.37	–0.43		
$t_{\alpha_{CARH}}$	1.65	1.92	1.98	1.34	1.72	–0.50	–0.07	–2.76	–2.72	–4.25	–4.22		
α_q	0.02	0.03	0.13	0.13	0.28	0.14	0.23	–0.07	–0.12	–0.66	–0.68	0.18	0.87
β_{MKT}	0.79	0.92	1.04	1.11	1.17	1.20	1.24	1.29	1.26	1.21	0.42	(0.01)	
β_{ME}	–0.17	–0.11	–0.05	0.16	0.28	0.44	0.58	0.67	0.84	0.86	1.03		
$\beta_{\Delta A/A}$	0.23	0.07	–0.09	–0.25	–0.45	–0.54	–0.73	–0.77	–0.81	–0.59	–0.81		
β_{ROE}	0.13	0.10	0.01	–0.03	–0.11	–0.32	–0.43	–0.58	–0.97	–1.22	–1.35		
t_{α_q}	0.29	0.55	1.59	1.35	2.77	1.09	1.46	–0.47	–0.57	–2.30	–2.08		
$t\beta_{MKT}$	46.50	51.75	41.73	43.37	55.92	40.39	30.25	30.03	21.19	17.90	5.63		
$t\beta_{ME}$	–5.48	–3.70	–0.87	3.51	7.50	8.21	8.10	9.25	9.33	7.65	8.52		
$t\beta_{\Delta A/A}$	3.03	1.21	–1.26	–3.93	–7.00	–6.20	–6.16	–5.84	–5.29	–2.77	–3.10		
$t\beta_{ROE}$	2.54	2.64	0.37	–0.71	–2.08	–4.16	–5.03	–6.17	–6.79	–6.56	–6.09		

Table IA.3 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Campbell, Hilscher, and Szilagyi's (2008) Failure Probability, All NYSE-Amex-NASDAQ Breakpoints (1/1976–12/2011, 432 Months)

We use the NYSE-Amex-NASDAQ breakpoints to sort all stocks at the beginning of each month into deciles on the most recent failure probability. (Appendix IA.1 contains detailed the variable definition.) Earnings and other accounting variables for a fiscal quarter are used in portfolio sorts in the months immediately after the quarter's public earnings announcement dates (Compustat quarterly item RDQ). Monthly value-weighted returns on the portfolios are calculated for the current month, and the portfolios are rebalanced monthly. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.72	0.55	0.61	0.55	0.61	0.55	0.55	0.41	−0.04	−0.24	−0.96		
t_{Mean}	3.11	2.65	2.81	2.48	2.50	1.89	1.72	1.05	−0.08	−0.42	−2.07		
α	0.22	0.06	0.09	0.01	0.04	−0.11	−0.15	−0.41	−0.97	−1.23	−1.45	0.33	0.79
β	0.89	0.88	0.93	0.97	1.04	1.17	1.27	1.48	1.68	1.77	0.88	(0.00)	
t_α	2.12	0.86	1.42	0.14	0.45	−0.80	−1.00	−1.99	−3.60	−3.53	−3.77		
α_{FF}	0.31	0.14	0.15	0.03	0.04	−0.13	−0.20	−0.50	−1.14	−1.61	−1.92	0.43	0.82
b	0.85	0.88	0.94	0.97	1.03	1.15	1.21	1.36	1.55	1.60	0.75	(0.00)	
s	0.00	−0.13	−0.15	−0.03	0.02	0.15	0.29	0.63	0.77	1.33	1.33		
h	−0.20	−0.12	−0.06	−0.02	−0.02	0.00	−0.02	−0.06	0.06	0.29	0.49		
$t_{\alpha_{FF}}$	3.07	2.09	2.28	0.37	0.50	−0.99	−1.36	−2.79	−4.97	−5.80	−5.96		
α_{CARH}	0.09	0.01	0.10	0.04	0.20	0.12	0.10	−0.09	−0.54	−0.84	−0.93	0.21	0.87
b	0.89	0.90	0.95	0.97	1.00	1.10	1.15	1.28	1.44	1.45	0.56	(0.00)	
s	−0.03	−0.14	−0.16	−0.03	0.04	0.19	0.32	0.69	0.85	1.43	1.46		
h	−0.12	−0.07	−0.05	−0.03	−0.08	−0.09	−0.12	−0.20	−0.15	0.02	0.14		
w	0.25	0.15	0.06	−0.02	−0.18	−0.29	−0.34	−0.47	−0.69	−0.88	−1.14		
$t_{\alpha_{CARH}}$	0.98	0.19	1.56	0.58	2.14	1.05	0.68	−0.59	−3.10	−3.35	−3.29		
α_q	0.02	−0.13	0.00	0.03	0.27	0.28	0.38	0.36	−0.09	−0.26	−0.28	0.18	0.86
β_{MKT}	0.89	0.93	0.97	0.98	1.00	1.10	1.14	1.25	1.42	1.43	0.54	(0.00)	
β_{ME}	0.12	−0.05	−0.11	−0.04	−0.05	0.00	0.09	0.33	0.40	0.82	0.70		
$\beta_{\Delta A/A}$	−0.20	0.05	0.05	0.02	−0.16	−0.14	−0.34	−0.51	−0.36	−0.12	0.08		
β_{ROE}	0.39	0.26	0.15	−0.03	−0.21	−0.46	−0.58	−0.90	−1.19	−1.64	−2.03		
t_{α_q}	0.11	−1.63	0.05	0.38	1.95	1.99	2.48	1.89	−0.33	−0.70	−0.59		
$t_{\beta_{MKT}}$	28.87	46.78	55.44	53.34	38.52	32.68	32.00	26.42	21.43	17.55	5.49		
$t_{\beta_{ME}}$	1.59	−1.48	−3.49	−1.49	−1.04	−0.03	1.23	2.91	2.16	4.34	2.80		
$t_{\beta_{\Delta A/A}}$	−1.44	0.76	0.99	0.46	−1.55	−1.29	−2.13	−2.46	−1.41	−0.39	0.19		
$t_{\beta_{ROE}}$	5.32	6.00	3.98	−0.75	−2.94	−6.57	−7.51	−8.62	−8.04	−6.83	−6.98		

Table IA.4 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Composite Issuance, All NYSE-Amex-NASDAQ Breakpoints and Equal-weighted Returns (1/1972–12/2011, 480 Months)

We measure composite issuance as the growth rate in the market equity not attributable to the stock return, $\log(ME_t/ME_{t-5}) - r(t-5, t)$. For June of year t , $r(t-5, t)$ is the cumulative log return on the stock from the last trading day of June in year $t-5$ to the last trading day of June in year t , and ME_t is the market equity on the last trading day of June in year t from CRSP. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles on composite issuance for the fiscal year ending in calendar year $t-1$. Monthly equal-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	1.06	0.99	0.92	1.01	0.98	1.02	1.06	0.93	0.75	0.32	-0.74		
t_{Mean}	4.42	4.34	3.63	3.71	3.40	3.11	2.96	2.63	2.03	0.78	-3.14		
α	0.67	0.60	0.50	0.56	0.51	0.50	0.49	0.35	0.16	-0.29	-0.96	0.46	0.73
β	0.86	0.86	0.93	0.99	1.04	1.14	1.25	1.27	1.30	1.34	0.48	(0.00)	
t_α	5.45	5.32	4.01	4.30	3.67	3.06	2.69	2.04	0.86	-1.28	-4.88		
α_{FF}	0.35	0.31	0.20	0.29	0.28	0.23	0.25	0.18	0.00	-0.49	-0.84	0.26	0.90
b	0.88	0.88	0.93	0.95	0.96	1.04	1.10	1.11	1.12	1.15	0.27	(0.00)	
s	0.47	0.43	0.54	0.64	0.73	0.88	1.02	0.98	1.04	1.17	0.70		
h	0.53	0.48	0.48	0.40	0.28	0.35	0.25	0.12	0.07	0.13	-0.40		
$t_{\alpha_{FF}}$	4.44	4.74	2.74	4.23	3.78	2.60	2.58	1.77	0.04	-3.16	-5.50		
α_{CARH}	0.44	0.38	0.30	0.39	0.44	0.42	0.43	0.35	0.21	-0.20	-0.64	0.36	0.91
b	0.86	0.87	0.90	0.93	0.92	1.00	1.07	1.08	1.08	1.09	0.22	(0.00)	
s	0.47	0.43	0.54	0.64	0.73	0.88	1.02	0.98	1.04	1.17	0.70		
h	0.50	0.46	0.44	0.36	0.22	0.28	0.19	0.07	0.00	0.03	-0.47		
w	-0.10	-0.07	-0.11	-0.10	-0.17	-0.20	-0.19	-0.18	-0.22	-0.30	-0.21		
$t_{\alpha_{CARH}}$	5.10	5.59	4.15	5.14	5.16	4.38	3.94	3.08	1.96	-1.19	-3.79		
α_q	0.36	0.25	0.20	0.35	0.46	0.46	0.58	0.47	0.37	0.08	-0.28	0.36	0.90
β_{MKT}	0.85	0.87	0.90	0.91	0.93	1.00	1.05	1.08	1.08	1.08	0.23	(0.00)	
β_{ME}	0.44	0.43	0.53	0.61	0.62	0.75	0.87	0.83	0.86	0.91	0.47		
$\beta_{\Delta A/A}$	0.55	0.53	0.44	0.29	0.16	0.20	-0.04	-0.08	-0.16	-0.20	-0.75		
β_{ROE}	-0.13	-0.03	-0.07	-0.13	-0.29	-0.37	-0.42	-0.42	-0.51	-0.76	-0.63		
t_{α_q}	3.39	2.80	1.69	3.25	3.45	3.75	4.26	3.44	2.70	0.42	-1.40		
$t_{\beta_{MKT}}$	26.88	32.90	28.54	37.46	31.38	29.58	29.41	33.98	32.10	23.15	4.56		
$t_{\beta_{ME}}$	5.88	6.57	6.54	11.31	9.32	9.18	11.13	13.35	14.02	11.05	7.34		
$t_{\beta_{\Delta A/A}}$	6.51	7.80	4.63	3.40	1.45	2.09	-0.41	-0.79	-1.63	-1.38	-5.31		
$t_{\beta_{ROE}}$	-1.97	-0.65	-1.14	-2.18	-3.98	-5.51	-5.66	-5.52	-6.38	-6.55	-4.94		

Table IA.5 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Abnormal Corporate Investment, All NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns (1/1972–12/2011, 480 Months)

We measure abnormal corporate investment (ACI) that applies for the portfolio formation year t , as $ACI_{t-1} \equiv 3CE_{t-1}/(CE_{t-2} + CE_{t-3} + CE_{t-4}) - 1$, in which CE_{t-1} is capital expenditure (Compustat annual item CAPX) scaled by its sales (item SALE) in year $t-1$. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles on ACI for the fiscal year ending in calendar year $t-1$. Monthly equal-weighted portfolio returns are calculated from July of year t to June of year $t+1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of $r_{\Delta A/A}$, r_{ROE} , and r_{ME} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	1.08	1.02	0.98	0.97	0.95	0.87	0.89	0.89	0.87	0.69	-0.39		
t_{Mean}	3.01	3.12	3.25	3.38	3.34	3.18	3.23	3.06	2.91	2.12	-4.01		
α	0.55	0.51	0.47	0.48	0.47	0.40	0.40	0.39	0.37	0.18	-0.38	0.42	0.76
β	1.17	1.12	1.11	1.07	1.06	1.04	1.07	1.11	1.12	1.14	-0.03	(0.00)	
t_α	2.78	2.93	3.32	3.54	3.64	3.45	3.48	3.09	2.65	1.08	-4.02		
α_{FF}	0.25	0.22	0.23	0.24	0.22	0.16	0.17	0.16	0.13	-0.06	-0.31	0.19	0.91
b	1.05	1.02	1.03	1.00	1.02	0.99	1.03	1.05	1.03	1.04	-0.01	(0.00)	
s	1.04	0.91	0.79	0.71	0.62	0.62	0.57	0.63	0.76	0.85	-0.20		
h	0.38	0.36	0.30	0.32	0.35	0.33	0.33	0.32	0.31	0.29	-0.09		
$t_{\alpha_{FF}}$	2.06	2.59	3.13	3.35	3.11	2.78	2.39	2.22	1.70	-0.68	-3.30		
α_{CARH}	0.43	0.33	0.42	0.38	0.37	0.28	0.33	0.32	0.29	0.16	-0.28	0.33	0.93
b	1.01	1.00	0.99	0.97	0.99	0.97	1.00	1.02	1.00	0.99	-0.02	(0.00)	
s	1.04	0.91	0.79	0.71	0.62	0.62	0.57	0.63	0.76	0.85	-0.20		
h	0.32	0.32	0.23	0.28	0.30	0.29	0.28	0.26	0.25	0.21	-0.10		
w	-0.20	-0.11	-0.19	-0.14	-0.15	-0.12	-0.16	-0.16	-0.17	-0.23	-0.03		
$t_{\alpha_{CARH}}$	3.10	3.35	5.50	4.66	4.63	5.23	4.59	4.38	3.67	1.52	-2.68		
α_q	0.52	0.36	0.43	0.36	0.34	0.25	0.29	0.27	0.32	0.22	-0.30	0.34	0.92
β_{MKT}	1.01	0.99	0.99	0.97	0.98	0.96	1.00	1.02	0.99	0.99	-0.01	(0.00)	
β_{ME}	0.88	0.81	0.68	0.63	0.56	0.57	0.52	0.57	0.68	0.71	-0.17		
$\beta_{\Delta A/A}$	0.23	0.24	0.14	0.24	0.24	0.23	0.23	0.21	0.12	0.10	-0.13		
β_{ROE}	-0.44	-0.26	-0.29	-0.22	-0.19	-0.16	-0.19	-0.19	-0.27	-0.40	0.03		
t_{α_q}	2.57	2.75	3.30	2.98	3.10	3.08	2.80	2.66	3.06	1.51	-2.49		
$t_{\beta_{MKT}}$	22.55	29.83	31.20	37.49	34.09	38.79	40.51	36.05	34.37	28.56	-0.48		
$t_{\beta_{ME}}$	9.33	9.17	8.19	10.69	7.31	8.68	7.53	7.71	8.85	8.52	-3.70		
$t_{\beta_{\Delta A/A}}$	1.36	2.06	1.28	2.53	2.61	3.26	2.66	2.57	1.34	0.88	-1.46		
$t_{\beta_{ROE}}$	-3.96	-3.70	-4.01	-3.42	-3.28	-3.50	-3.58	-3.21	-4.74	-4.98	0.44		

Table IA.6 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Total Accruals, All NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns (1/1972–12/2011, 480 Months)

Following Sloan (1996), we measure total accruals (TAC) as changes in noncash working capital minus depreciation expense scaled by average total assets (Compustat annual item AT) in the prior two years. The noncash working capital is the change in noncash current assets minus the change in current liabilities less short-term debt and taxes payable. Specifically, $TAC \equiv (\Delta CA - \Delta CASH) - (\Delta CL - \Delta STD - \Delta TP) - DP$, in which ΔCA is the change in current assets (item ACT), $\Delta CASH$ is the change in cash or cash equivalents (item CHE), ΔCL is the change in current liabilities (item LCT), ΔSTD is the change in debt included in current liabilities (item DLC), ΔTP is the change in income taxes payable (item TXP), and DP is depreciation and amortization expense (item DP). In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles on total accruals scaled by average total assets for the fiscal year ending in calendar year $t - 1$. Monthly equal-weighted portfolio returns are calculated from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$), and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	1.21	0.99	1.12	1.06	0.95	0.90	0.83	0.84	0.80	0.42	-0.78		
t_{Mean}	2.84	2.92	3.56	3.55	3.30	3.13	2.75	2.63	2.39	1.08	-5.31		
α	0.62	0.47	0.62	0.56	0.49	0.43	0.34	0.33	0.26	-0.16	-0.78	0.43	0.71
β	1.30	1.16	1.11	1.09	1.03	1.05	1.09	1.13	1.20	1.29	-0.01	(0.00)	
t_α	2.39	2.59	3.96	3.95	3.55	3.23	2.32	2.17	1.56	-0.77	-5.42		
α_{FF}	0.41	0.23	0.38	0.33	0.25	0.18	0.11	0.13	0.09	-0.35	-0.76	0.25	0.90
b	1.09	1.04	1.02	1.00	0.96	0.98	0.98	1.00	1.04	1.10	0.01	(0.00)	
s	1.24	0.92	0.81	0.77	0.73	0.75	0.85	0.88	0.96	1.12	-0.12		
h	0.12	0.27	0.28	0.29	0.32	0.31	0.26	0.18	0.10	0.11	0.00		
$t_{\alpha_{FF}}$	2.37	2.06	4.30	4.33	3.07	2.78	1.62	1.75	1.12	-2.74	-5.00		
α_{CARH}	0.67	0.44	0.55	0.49	0.40	0.33	0.26	0.28	0.27	-0.08	-0.75	0.38	0.91
b	1.04	0.99	0.98	0.97	0.92	0.95	0.95	0.97	1.00	1.04	0.01	(0.00)	
s	1.24	0.92	0.81	0.77	0.73	0.75	0.85	0.88	0.96	1.12	-0.12		
h	0.03	0.20	0.22	0.24	0.26	0.26	0.21	0.13	0.04	0.02	-0.01		
w	-0.27	-0.22	-0.18	-0.17	-0.16	-0.15	-0.15	-0.16	-0.19	-0.28	-0.02		
$t_{\alpha_{CARH}}$	3.27	3.43	5.34	5.86	4.64	4.62	3.42	3.28	3.14	-0.53	-4.70		
α_q	0.99	0.56	0.63	0.54	0.45	0.39	0.32	0.36	0.37	0.07	-0.92	0.47	0.91
β_{MKT}	1.04	1.00	0.98	0.97	0.92	0.94	0.94	0.97	1.00	1.04	-0.01	(0.00)	
β_{ME}	0.92	0.73	0.66	0.65	0.63	0.64	0.75	0.77	0.83	0.95	0.02		
$\beta_{\Delta A/A}$	-0.09	0.15	0.19	0.21	0.19	0.17	0.05	-0.04	-0.17	-0.33	-0.24		
β_{ROE}	-0.84	-0.52	-0.42	-0.36	-0.32	-0.32	-0.28	-0.30	-0.34	-0.44	0.39		
t_{α_q}	4.14	3.59	5.19	5.14	4.51	4.34	2.97	3.26	2.85	0.31	-5.57		
$t_{\beta_{MKT}}$	20.02	28.48	33.02	38.04	38.48	35.79	39.11	35.34	31.46	22.07	-0.27		
$t_{\beta_{ME}}$	10.29	9.90	11.90	11.49	11.16	12.04	15.30	12.92	12.53	8.83	0.31		
$t_{\beta_{\Delta A/A}}$	-0.53	1.22	2.16	2.68	2.74	2.52	0.56	-0.47	-1.60	-1.86	-2.27		
$t_{\beta_{ROE}}$	-5.85	-6.00	-5.81	-6.43	-6.69	-6.42	-4.86	-4.93	-5.09	-3.74	4.45		

Table IA.7 : Factor Regressions for Monthly Percent Excess Returns of One-way Momentum Deciles, All NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns (1/1972–12/2011, 480 Months)

At the beginning of each month t , we use the NYSE-Amex-NASDAQ breakpoints to split all NYSE, Amex, and NASDAQ stocks into deciles based on their prior six-month returns from month $t-2$ to $t-7$. Skipping month $t-1$, we calculate monthly equal-weighted returns for the portfolios from month t to $t+5$. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Loser	2	3	4	5	6	7	8	9	Winner	W-L	m.a.e.	$\bar{R}^2$
Mean	-0.17	0.31	0.50	0.62	0.66	0.72	0.73	0.85	0.93	1.15	1.32		
t_{Mean}	-0.44	0.97	1.74	2.30	2.57	2.86	2.88	3.18	3.13	3.16	5.89		
α	-0.81	-0.23	0.01	0.15	0.21	0.27	0.28	0.37	0.41	0.55	1.36	0.33	0.78
β	1.41	1.18	1.08	1.03	1.00	1.00	1.01	1.06	1.14	1.34	-0.08	(0.00)	
t_α	-4.20	-1.61	0.08	1.31	1.86	2.53	2.58	3.25	2.98	2.88	6.37		
α_{FF}	-0.94	-0.41	-0.19	-0.05	0.01	0.09	0.11	0.24	0.33	0.56	1.50	0.29	0.92
b	1.25	1.09	1.02	0.97	0.95	0.93	0.93	0.95	0.99	1.09	-0.16	(0.00)	
s	0.92	0.73	0.65	0.59	0.57	0.59	0.62	0.68	0.81	1.01	0.09		
h	0.05	0.20	0.26	0.28	0.27	0.24	0.19	0.11	-0.02	-0.25	-0.31		
$t_{\alpha_{FF}}$	-6.65	-4.20	-2.32	-0.77	0.11	1.49	2.27	4.72	4.76	4.59	6.89		
α_{CARH}	-0.32	-0.02	0.08	0.12	0.12	0.12	0.07	0.12	0.11	0.23	0.56	0.13	0.96
b	1.12	1.00	0.96	0.94	0.92	0.93	0.94	0.97	1.03	1.16	0.04	(0.00)	
s	0.93	0.73	0.65	0.60	0.57	0.59	0.62	0.68	0.81	1.01	0.09		
h	-0.16	0.06	0.16	0.22	0.23	0.22	0.20	0.15	0.06	-0.14	0.02		
w	-0.65	-0.42	-0.29	-0.19	-0.12	-0.04	0.04	0.12	0.23	0.34	0.99		
$t_{\alpha_{CARH}}$	-2.57	-0.24	1.38	2.07	2.03	2.04	1.35	2.45	1.95	2.42	4.15		
α_q	-0.17	-0.01	0.03	0.05	0.03	0.06	0.03	0.13	0.20	0.44	0.61	0.11	0.93
β_{MKT}	1.15	1.02	0.97	0.94	0.93	0.92	0.93	0.96	1.00	1.12	-0.03	(0.00)	
β_{ME}	0.62	0.57	0.56	0.55	0.55	0.59	0.62	0.69	0.82	1.01	0.39		
$\beta_{\Delta A/A}$	-0.46	-0.11	0.05	0.14	0.17	0.15	0.13	0.06	-0.09	-0.39	0.08		
β_{ROE}	-0.85	-0.46	-0.27	-0.15	-0.06	-0.01	0.05	0.08	0.10	0.11	0.96		
t_{α_q}	-0.64	-0.07	0.19	0.46	0.33	0.81	0.55	2.21	2.33	2.83	1.76		
$t_{\beta_{MKT}}$	23.40	28.87	31.70	35.19	35.75	39.02	44.38	45.34	43.01	33.63	-0.44		
$t_{\beta_{ME}}$	4.70	5.41	5.95	6.79	7.08	9.34	13.40	21.20	22.91	14.14	2.15		
$t_{\beta_{\Delta A/A}}$	-2.01	-0.78	0.49	1.59	2.19	2.44	2.91	1.36	-1.66	-4.20	0.28		
$t_{\beta_{ROE}}$	-5.48	-5.42	-4.15	-3.10	-1.48	-0.22	1.62	2.27	1.78	1.21	5.01		

Table IA.8 : Descriptive Statistics for Monthly Percent Excess Returns of 25 Size and Momentum Portfolios (1/1972–12/2011, 480 Months)

At the beginning of month t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into quintiles based on their prior six-month returns from month $t - 2$ to $t - 7$ (skipping month $t - 1$). Independently, in June of each year t , we use the NYSE breakpoints to split on NYSE, Amex, and NASDAQ stocks into quintiles based on market equity observed at the end of June. Taking intersections, we form 25 size and momentum portfolios, and we calculate value-weighted returns for the portfolios from month t to $t + 5$. We report mean percent excess returns, the CAPM alphas (α), the intercepts (α_{FF}) from the Fama-French three-factor regressions, and the intercepts (α_{CARH}) from the Carhart four-factor regressions, as well as t -statistics adjusted for heteroscedasticity and autocorrelations. For each factor model, we report the mean absolute error (m.a.e., the average magnitude of the alphas) across the testing portfolios, the average goodness-of-fit ($\bar{R}^2$) across the testing portfolios, and the p-value (p_{GRS}) from the Gibbons, Ross, and Shanken (1989) test on the null that the alphas of all the portfolios are jointly zero. The data for the one-month Treasury bill rate (r^f) and the Carhart factor are from Kenneth French's Web site.

	Loser	2	3	4	Winner	W-L	Loser	2	3	4	Winner	W-L
	Mean						t_{Mean}					
Small	0.10	0.68	0.79	0.89	1.12	1.02	0.24	2.09	2.60	2.87	2.99	5.46
2	0.16	0.67	0.76	0.78	0.98	0.81	0.44	2.23	2.76	2.84	2.94	4.01
3	0.24	0.61	0.70	0.71	0.90	0.66	0.69	2.21	2.75	2.84	2.91	2.95
4	0.33	0.57	0.63	0.63	0.84	0.51	1.00	2.16	2.69	2.68	2.83	2.31
Big	0.16	0.44	0.39	0.45	0.68	0.52	0.54	2.01	1.97	2.23	2.62	2.35
	α (m.a.e. = 0.26)						t_{α} ($\bar{R}^2 = 0.78, p_{GRS} = 0$)					
Small	-0.50	0.19	0.33	0.41	0.54	1.04	-2.22	1.07	1.94	2.39	2.54	6.10
2	-0.46	0.16	0.27	0.29	0.40	0.86	-2.59	1.10	2.19	2.32	2.51	4.65
3	-0.36	0.12	0.24	0.25	0.35	0.71	-2.13	1.07	2.28	2.46	2.53	3.40
4	-0.25	0.08	0.19	0.17	0.31	0.56	-1.65	0.84	2.02	2.20	2.26	2.67
Big	-0.35	0.02	-0.01	0.05	0.20	0.55	-2.37	0.26	-0.17	0.88	1.83	2.51
	α_{FF} (m.a.e. = 0.22)						$t_{\alpha_{FF}}$ ($\bar{R}^2 = 0.89, p_{GRS} = 0$)					
Small	-0.76	-0.09	0.07	0.19	0.45	1.22	-6.25	-1.28	1.10	2.82	3.86	7.20
2	-0.66	-0.08	0.05	0.10	0.37	1.03	-5.74	-0.97	0.84	1.64	3.74	5.79
3	-0.53	-0.09	0.03	0.09	0.36	0.88	-3.86	-1.09	0.34	1.31	3.34	4.32
4	-0.33	-0.07	0.03	0.06	0.36	0.70	-2.35	-0.77	0.31	0.83	2.91	3.25
Big	-0.34	0.03	0.00	0.09	0.34	0.68	-2.26	0.34	-0.07	1.44	3.26	3.05
	α_{CARH} (m.a.e. = 0.11)						$t_{\alpha_{CARH}}$ ($\bar{R}^2 = 0.93, p_{GRS} = 0$)					
Small	-0.37	0.03	0.10	0.12	0.18	0.55	-2.82	0.45	1.42	1.70	1.84	4.69
2	-0.19	0.09	0.08	0.03	0.08	0.27	-2.13	1.15	1.25	0.49	0.96	2.25
3	0.03	0.13	0.09	0.00	0.04	0.01	0.31	1.79	1.08	-0.02	0.41	0.08
4	0.22	0.17	0.10	-0.03	0.00	-0.22	2.38	2.00	1.19	-0.35	0.01	-1.97
Big	0.20	0.29	0.07	-0.03	0.02	-0.18	2.08	5.03	1.16	-0.50	0.22	-1.40

Table IA.9 : Q -factor Regressions for Monthly Percent Excess Returns of 25 Size and Momentum Portfolios (1/1972–12/2011, 480 Months)

At the beginning of month t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into quintiles based on their prior six-month returns from month $t - 2$ to $t - 7$ (skipping month $t - 1$). Independently, in June of each year t , we use the NYSE breakpoints to split on NYSE, Amex, and NASDAQ stocks into quintiles based on market equity observed at the end of June. Taking intersections, we form 25 size and momentum portfolios, and we calculate value-weighted returns for the portfolios from month t to $t+5$. We report the q -factor regressions: $r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. $\bar{R}^2$ is the average goodness-of-fit across the testing portfolios. p_{GRS} is p-value for the Gibbons, Ross, and Shanken (1989) test on the null that the alphas of all the testing portfolios are jointly zero. The data for the one-month Treasury bill rate (r^f) and the Carhart factor are from Kenneth French's Web site.

	Loser	2	3	4	Winner	W-L	Loser	2	3	4	Winner	W-L
	α_q (m.a.e. = 0.11)						t_{α_q} ($\bar{R}^2 = 0.90$, $p_{GRS} = 0$)					
Small	-0.13	0.12	0.14	0.19	0.41	0.54	-0.72	1.33	1.95	2.34	2.66	2.17
2	-0.11	0.02	0.02	-0.02	0.16	0.28	-0.66	0.23	0.28	-0.27	1.37	1.06
3	0.04	0.03	-0.01	-0.07	0.11	0.07	0.23	0.24	-0.14	-0.98	0.75	0.23
4	0.20	0.05	-0.04	-0.11	0.15	-0.05	1.13	0.44	-0.46	-1.47	0.86	-0.18
Big	0.13	0.16	-0.03	-0.09	0.15	0.03	0.67	1.60	-0.45	-1.37	1.13	0.09
	β_{MKT}						$t_{\beta_{MKT}}$					
Small	1.04	0.92	0.88	0.90	1.04	0.00	24.85	36.72	39.26	40.11	30.79	-0.04
2	1.16	0.99	0.96	0.97	1.10	-0.06	27.00	38.01	52.62	49.32	37.18	-0.89
3	1.15	1.01	0.98	0.98	1.09	-0.06	23.12	34.53	36.04	43.59	30.72	-0.81
4	1.16	1.04	1.00	1.01	1.08	-0.08	22.11	35.82	33.30	48.67	24.62	-1.01
Big	1.07	0.96	0.92	0.94	1.04	-0.03	22.68	37.03	47.68	55.96	31.42	-0.42
	β_{ME}						$t_{\beta_{ME}}$					
Small	0.98	0.93	0.93	0.96	1.19	0.21	15.09	19.83	26.31	37.15	14.74	1.57
2	0.76	0.77	0.75	0.78	0.96	0.21	8.27	11.34	18.21	20.01	16.42	1.49
3	0.50	0.47	0.46	0.53	0.74	0.24	4.51	5.89	6.19	10.58	10.50	1.41
4	0.17	0.19	0.20	0.27	0.53	0.35	1.70	2.48	2.48	7.32	5.03	1.77
Big	-0.26	-0.25	-0.19	-0.13	0.01	0.26	-3.14	-5.59	-5.38	-6.04	0.12	1.96
	$\beta_{\Delta A/A}$						$t_{\beta_{\Delta A/A}}$					
Small	-0.13	0.11	0.15	0.09	-0.27	-0.13	-0.92	1.57	2.70	1.84	-2.59	-0.71
2	-0.18	0.12	0.16	0.13	-0.27	-0.10	-1.31	1.74	3.02	3.09	-3.41	-0.49
3	-0.20	0.16	0.24	0.18	-0.22	-0.02	-1.47	1.95	3.15	3.30	-2.05	-0.09
4	-0.17	0.19	0.29	0.22	-0.25	-0.08	-1.34	2.12	3.52	3.75	-1.95	-0.35
Big	-0.23	0.01	0.07	0.10	-0.24	-0.01	-1.79	0.12	1.67	2.47	-2.40	-0.05
	β_{ROE}						$t_{\beta_{ROE}}$					
Small	-0.82	-0.33	-0.17	-0.09	-0.03	0.79	-7.17	-6.80	-4.54	-2.04	-0.34	5.54
2	-0.67	-0.16	-0.01	0.10	0.23	0.90	-5.98	-2.96	-0.15	2.94	3.28	5.55
3	-0.64	-0.15	0.03	0.16	0.28	0.92	-5.15	-2.25	0.50	4.15	3.75	4.99
4	-0.63	-0.16	0.05	0.16	0.26	0.89	-5.89	-2.40	0.88	3.78	3.23	5.19
Big	-0.45	-0.13	0.04	0.19	0.27	0.72	-4.80	-2.21	1.02	5.78	3.75	4.76

Table IA.10 : Descriptive Statistics for Monthly Percent Excess Returns of 25 Size and Momentum Portfolios, All NYSE-Amex-NASDAQ Breakpoints for Momentum, Equal-weighted Returns (1/1972–12/2011, 480 Months)

At the beginning of month t , we use the NYSE-Amex-NASDAQ breakpoints to split all NYSE, Amex, and NASDAQ stocks into quintiles based on their prior six-month returns from month $t - 2$ to $t - 7$ (skipping month $t - 1$). Independently, in June of each year t , we use the NYSE breakpoints to split on NYSE, Amex, and NASDAQ stocks into quintiles based on market equity observed at the end of June. Taking intersections, we form 25 size and momentum portfolios, and we calculate equal-weighted returns for the portfolios from month t to $t+5$. We report mean percent excess returns, the CAPM alphas (α), the intercepts (α_{FF}) from the Fama-French three-factor regressions, and the intercepts (α_{CARH}) from the Carhart four-factor regressions, as well as t -statistics adjusted for heteroscedasticity and autocorrelations. For each factor model, we report the mean absolute error (m.a.e., the average magnitude of the alphas) across the testing portfolios, the average goodness-of-fit ($\bar{R}^2$) across the testing portfolios, and the p-value (p_{GRS}) from the Gibbons, Ross, and Shanken (1989) test on the null that the alphas of all the portfolios are jointly zero. The data for the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site.

	Loser	2	3	4	Winner	W-L	Loser	2	3	4	Winner	W-L
	Mean						t_{Mean}					
Small	0.06	0.60	0.79	0.94	1.18	1.13	0.16	2.06	2.86	3.26	3.39	7.08
2	0.10	0.62	0.73	0.81	0.99	0.89	0.28	2.09	2.64	2.92	2.97	4.94
3	0.17	0.59	0.70	0.74	0.95	0.79	0.46	2.07	2.67	2.86	3.01	3.59
4	0.31	0.52	0.64	0.67	0.84	0.54	0.88	1.94	2.69	2.76	2.70	2.25
Big	0.08	0.42	0.45	0.52	0.75	0.68	0.23	1.72	2.13	2.43	2.60	2.71
	α (m.a.e. = 0.26)						t_{α} ($\bar{R}^2 = 0.78, p_{GRS} = 0$)					
Small	-0.48	0.16	0.37	0.49	0.63	1.12	-2.65	0.98	2.35	3.05	3.20	7.46
2	-0.52	0.11	0.24	0.31	0.40	0.92	-2.93	0.75	1.85	2.43	2.45	5.36
3	-0.46	0.09	0.21	0.26	0.38	0.84	-2.49	0.69	2.00	2.48	2.64	4.02
4	-0.31	0.03	0.18	0.19	0.29	0.59	-1.81	0.23	2.00	2.33	1.92	2.61
Big	-0.49	-0.04	0.02	0.09	0.24	0.73	-2.84	-0.50	0.38	1.65	1.77	2.92
	α_{FF} (m.a.e. = 0.22)						$t_{\alpha_{FF}}$ ($\bar{R}^2 = 0.88, p_{GRS} = 0$)					
Small	-0.73	-0.10	0.12	0.28	0.54	1.27	-7.18	-1.28	1.69	4.39	5.48	8.51
2	-0.68	-0.12	0.01	0.13	0.37	1.05	-5.49	-1.32	0.18	2.21	3.76	6.02
3	-0.59	-0.12	0.02	0.12	0.38	0.97	-3.67	-1.25	0.21	1.89	3.78	4.61
4	-0.37	-0.12	0.03	0.10	0.35	0.72	-2.27	-1.13	0.42	1.51	2.66	3.04
Big	-0.48	-0.09	-0.02	0.09	0.36	0.84	-2.65	-0.96	-0.24	1.48	2.78	3.28
	α_{CARH} (m.a.e. = 0.11)						$t_{\alpha_{CARH}}$ ($\bar{R}^2 = 0.93, p_{GRS} = 0$)					
Small	-0.37	0.04	0.17	0.21	0.31	0.68	-3.96	0.58	2.28	3.21	3.50	7.33
2	-0.20	0.10	0.09	0.08	0.11	0.31	-2.13	1.28	1.27	1.19	1.31	2.95
3	0.05	0.17	0.12	0.03	0.11	0.06	0.41	2.29	1.66	0.52	1.19	0.45
4	0.29	0.18	0.13	0.03	0.02	-0.27	2.43	2.14	1.57	0.34	0.18	-1.95
Big	0.16	0.22	0.09	-0.01	0.03	-0.13	1.30	3.47	1.34	-0.15	0.26	-0.90

Table IA.11 : Q -factor Regressions for Monthly Percent Excess Returns of 25 Size and Momentum Portfolios, All NYSE-Amex-NASDAQ Breakpoints for Momentum, Equal-weighted Returns (1/1972–12/2011, 480 Months)

At the beginning of month t , we use the NYSE-Amex-NASDAQ breakpoints to split all NYSE, Amex, and NASDAQ stocks into quintiles based on their prior six-month returns from month $t - 2$ to $t - 7$ (skipping month $t - 1$). Independently, in June of each year t , we use the NYSE breakpoints to split on NYSE, Amex, and NASDAQ stocks into quintiles based on market equity observed at the end of June. Taking intersections, we form 25 size and momentum portfolios, and we calculate equal-weighted returns for the portfolios from month t to $t+5$. We report the q -factor regressions: $r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$. See Table 1 for the description of $r_{\Delta A/A}$, r_{ROE} , and r_{ME} . The t -statistics are adjusted for heteroscedasticity and autocorrelations. m.a.e. is the average magnitude of the alphas across the testing portfolios. F_{GRS} and p_{GRS} are the test statistic and its p-value from the Gibbons, Ross, and Shanken (1989) test on the null that the alphas of all the testing portfolios are jointly zero. The data for the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site.

	Loser	2	3	4	Winner	W-L	Loser	2	3	4	Winner	W-L
	α_q (m.a.e. = 0.13)						t_{α_q} ($\bar{R}^2 = 0.89, p_{GRS} = 0$)					
Small	-0.30	0.03	0.14	0.24	0.46	0.76	-1.65	0.27	1.39	2.89	3.60	3.49
2	-0.15	0.01	0.00	0.03	0.19	0.34	-0.72	0.11	0.02	0.40	1.66	1.31
3	0.12	0.06	0.02	-0.02	0.19	0.07	0.47	0.46	0.18	-0.22	1.46	0.21
4	0.31	0.07	-0.01	-0.05	0.18	-0.13	1.28	0.53	-0.11	-0.65	1.05	-0.38
Big	0.19	0.08	-0.03	-0.03	0.26	0.08	0.82	0.63	-0.38	-0.59	1.60	0.22
	β_{MKT}						$t_{\beta_{MKT}}$					
Small	0.99	0.85	0.82	0.85	1.01	0.02	24.52	26.94	27.56	32.47	33.77	0.42
2	1.14	1.00	0.98	0.99	1.13	-0.01	26.20	34.94	40.11	43.48	41.52	-0.20
3	1.16	1.03	1.01	1.00	1.12	-0.04	22.42	29.18	33.52	43.44	35.02	-0.57
4	1.19	1.06	1.01	1.03	1.11	-0.09	19.93	28.99	34.94	43.39	26.44	-1.00
Big	1.16	1.03	0.98	0.98	1.07	-0.09	21.14	35.41	49.10	56.67	29.38	-1.15
	β_{ME}						$t_{\beta_{ME}}$					
Small	0.82	0.80	0.82	0.89	1.08	0.26	8.55	10.19	11.55	20.74	21.88	2.19
2	0.74	0.74	0.74	0.80	0.97	0.23	6.11	8.69	10.58	16.47	21.12	1.64
3	0.51	0.48	0.49	0.59	0.77	0.26	3.80	4.53	5.67	12.30	15.04	1.54
4	0.18	0.17	0.24	0.31	0.56	0.39	1.30	1.67	3.11	6.97	6.20	1.76
Big	-0.18	-0.14	-0.07	0.02	0.25	0.43	-1.73	-1.89	-1.38	0.84	2.83	2.35
	$\beta_{\Delta A/A}$						$t_{\beta_{\Delta A/A}}$					
Small	-0.08	0.13	0.18	0.10	-0.20	-0.12	-0.49	1.20	1.96	1.49	-2.21	-0.71
2	-0.31	0.10	0.16	0.10	-0.29	0.02	-1.72	0.99	2.12	2.07	-4.20	0.08
3	-0.46	0.09	0.17	0.09	-0.29	0.16	-2.16	0.84	2.06	1.87	-3.45	0.62
4	-0.34	0.10	0.23	0.12	-0.32	0.02	-1.73	0.94	2.86	2.35	-2.83	0.07
Big	-0.43	0.05	0.11	0.07	-0.39	0.05	-2.60	0.45	1.76	1.83	-3.40	0.18
	β_{ROE}						$t_{\beta_{ROE}}$					
Small	-0.52	-0.21	-0.09	-0.02	0.02	0.55	-5.30	-3.44	-1.74	-0.44	0.26	4.39
2	-0.60	-0.20	-0.02	0.07	0.19	0.79	-5.23	-3.33	-0.49	2.17	2.73	5.49
3	-0.73	-0.21	-0.01	0.14	0.24	0.97	-4.99	-2.73	-0.22	3.63	3.38	5.19
4	-0.76	-0.21	0.03	0.16	0.21	0.97	-5.48	-3.08	0.51	3.77	2.65	5.15
Big	-0.64	-0.17	0.02	0.13	0.17	0.82	-5.76	-2.76	0.51	3.76	2.11	4.94

Table IA.12 : Factor Regressions for Monthly Percent Excess Returns of One-way Book-to-Market Deciles (1/1972–12/2011, 480 Months)

Book equity is stockholders' book equity, plus balance sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. The stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on book-to-market equity. Book-to-market for June of year t is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the market equity at the end of December of $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced at the end of June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.30	0.49	0.54	0.51	0.59	0.66	0.65	0.70	0.73	0.98	0.68		
t_{Mean}	1.15	2.12	2.39	2.09	2.76	3.00	3.02	3.15	3.12	3.54	2.73		
α	-0.18	0.04	0.10	0.05	0.18	0.24	0.26	0.32	0.34	0.54	0.72	0.23	0.80
β	1.06	1.00	0.98	1.00	0.91	0.92	0.85	0.84	0.87	0.96	-0.10	(0.03)	
t_α	-1.75	0.54	1.22	0.59	1.81	2.75	2.41	2.71	2.51	3.03	2.88		
α_{FF}	0.10	0.13	0.12	-0.05	0.09	0.13	0.02	0.03	-0.01	0.07	-0.03	0.07	0.87
b	0.99	0.97	0.97	1.03	0.95	0.95	0.93	0.93	0.96	1.06	0.07	(0.15)	
s	-0.17	-0.02	0.02	0.08	-0.02	0.07	0.11	0.15	0.21	0.41	0.57		
h	-0.53	-0.17	-0.04	0.20	0.20	0.22	0.46	0.56	0.65	0.86	1.39		
$t_{\alpha_{FF}}$	1.54	1.81	1.40	-0.59	0.90	1.44	0.27	0.32	-0.11	0.59	-0.19		
α_{CARH}	0.11	0.13	0.10	-0.07	0.06	0.10	0.04	-0.01	0.03	0.08	-0.03	0.07	0.87
b	0.98	0.97	0.97	1.03	0.96	0.96	0.92	0.93	0.96	1.06	0.07	(0.22)	
s	-0.17	-0.02	0.02	0.08	-0.02	0.07	0.11	0.15	0.21	0.41	0.57		
h	-0.53	-0.17	-0.04	0.20	0.20	0.22	0.45	0.57	0.64	0.86	1.39		
w	-0.01	-0.01	0.01	0.02	0.03	0.03	-0.02	0.04	-0.04	-0.01	0.00		
$t_{\alpha_{CARH}}$	1.63	1.84	1.22	-0.81	0.64	1.16	0.45	-0.09	0.28	0.63	-0.19		
α_q	0.03	-0.03	-0.01	-0.17	-0.02	0.15	0.10	0.03	0.12	0.22	0.19	0.09	0.84
β_{MKT}	1.02	1.00	0.99	1.02	0.96	0.94	0.89	0.89	0.91	1.00	-0.02	(0.24)	
β_{ME}	-0.13	0.02	0.06	0.13	0.00	0.05	0.09	0.17	0.17	0.32	0.44		
$\beta_{\Delta A/A}$	-0.59	-0.09	0.02	0.25	0.36	0.27	0.46	0.60	0.65	0.90	1.49		
β_{ROE}	0.18	0.16	0.13	0.10	0.03	-0.10	-0.15	-0.09	-0.24	-0.33	-0.51		
t_{α_q}	0.38	-0.33	-0.07	-1.84	-0.24	1.67	0.96	0.30	1.07	1.47	0.98		
$t_{\beta_{MKT}}$	48.71	53.76	34.28	36.38	32.13	38.33	33.92	33.17	34.03	27.06	-0.36		
$t_{\beta_{ME}}$	-4.01	0.69	1.78	3.29	-0.05	1.50	1.74	3.25	3.34	4.73	5.11		
$t_{\beta_{\Delta A/A}}$	-12.27	-1.71	0.28	3.33	4.98	4.60	4.62	5.98	7.25	9.52	12.39		
$t_{\beta_{ROE}}$	4.70	3.67	2.69	1.63	0.47	-2.16	-2.05	-1.36	-3.25	-5.12	-5.90		

Table IA.13 : Factor Regressions for Monthly Percent Excess Returns of One-way Investment-to-Assets ($\Delta A/A$) Deciles (1/1972–12/2011, 480 Months)

$\Delta A/A$ is annual change in total assets (Compustat annual item AT) divided by lagged total assets. In June of each year t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the deciles. The numbers (in parentheses) are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.69	0.72	0.65	0.54	0.52	0.56	0.52	0.51	0.49	0.28	–0.41		
t_{Mean}	2.41	3.13	3.23	2.69	2.59	2.64	2.43	2.24	1.84	0.91	–2.32		
α	0.19	0.29	0.26	0.16	0.14	0.15	0.10	0.06	–0.01	–0.30	–0.49	0.17	0.84
β	1.09	0.95	0.86	0.83	0.85	0.90	0.92	1.00	1.12	1.27	0.18	(0.00)	
t_α	1.53	3.06	2.81	1.91	1.81	1.84	1.37	0.81	–0.12	–2.65	–2.75		
α_{FF}	0.02	0.15	0.14	0.11	0.07	0.10	0.13	0.17	0.19	–0.12	–0.14	0.12	0.87
b	1.08	0.99	0.93	0.86	0.89	0.94	0.96	0.98	1.02	1.15	0.07	(0.00)	
s	0.33	0.09	–0.06	–0.04	–0.05	–0.11	–0.20	–0.09	0.05	0.19	–0.14		
h	0.26	0.26	0.27	0.12	0.14	0.14	–0.01	–0.21	–0.42	–0.40	–0.67		
$t_{\alpha_{FF}}$	0.22	1.74	1.50	1.28	0.99	1.34	1.73	2.46	2.21	–1.39	–0.98		
α_{CARH}	0.00	0.23	0.09	0.11	0.06	0.06	0.08	0.14	0.17	–0.08	–0.07	0.10	0.87
b	1.09	0.97	0.94	0.86	0.89	0.95	0.97	0.98	1.03	1.14	0.05	(0.02)	
s	0.33	0.09	–0.07	–0.04	–0.05	–0.11	–0.20	–0.09	0.05	0.19	–0.14		
h	0.27	0.23	0.29	0.12	0.14	0.15	0.01	–0.20	–0.42	–0.42	–0.69		
w	0.03	–0.08	0.05	0.00	0.01	0.03	0.05	0.04	0.02	–0.05	–0.07		
$t_{\alpha_{CARH}}$	–0.02	2.56	0.92	1.32	0.80	0.83	1.05	1.88	1.97	–0.87	–0.46		
α_q	–0.12	0.04	–0.14	0.03	0.01	–0.06	0.03	0.11	0.36	0.02	0.14	0.09	0.89
β_{MKT}	1.12	1.03	0.97	0.87	0.89	0.96	0.96	0.99	1.01	1.14	0.02	(0.01)	
β_{ME}	0.29	0.04	–0.03	–0.03	–0.04	–0.05	–0.15	–0.05	0.03	0.16	–0.13		
$\beta_{\Delta A/A}$	0.62	0.64	0.66	0.29	0.27	0.26	0.05	–0.31	–0.81	–0.73	–1.35		
β_{ROE}	–0.11	–0.14	0.12	–0.02	0.00	0.14	0.13	0.18	0.05	0.03	0.14		
t_{α_q}	–1.06	0.50	–1.59	0.33	0.08	–0.74	0.33	1.57	3.72	0.24	1.03		
$t_{\beta_{MKT}}$	41.75	45.59	38.03	37.10	41.02	50.35	48.76	44.01	52.01	36.81	0.57		
$t_{\beta_{ME}}$	5.48	1.04	–0.60	–0.47	–1.14	–1.54	–5.36	–1.44	0.64	3.14	–1.93		
$t_{\beta_{\Delta A/A}}$	9.93	12.24	10.25	4.82	5.35	3.86	0.72	–6.44	–10.79	–11.61	–15.01		
$t_{\beta_{ROE}}$	–2.14	–2.96	2.47	–0.40	0.07	2.53	3.05	4.93	1.51	0.59	2.06		

Table IA.14 : Factor Regressions for Monthly Percent Excess Returns of One-way Investment-to-Assets ($\Delta A/A$) Deciles, All NYSE-Amex-NASDAQ Breakpoints, Equal-weighted Returns (1/1972–12/2011, 480 Months)

$\Delta A/A$ is annual change in total assets (Compustat annual item AT) divided by lagged total assets. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to split stocks into deciles on $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Monthly equal-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the deciles. The numbers (in parentheses) are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	1.53	1.30	1.19	1.05	0.99	0.93	0.82	0.79	0.50	0.04	–1.49		
t_{Mean}	3.18	3.46	3.75	3.72	3.58	3.37	2.90	2.59	1.48	0.11	–6.65		
α	0.94	0.79	0.71	0.59	0.54	0.46	0.33	0.27	–0.06	–0.59	–1.53	0.53	0.71
β	1.31	1.13	1.06	1.00	1.00	1.03	1.07	1.14	1.25	1.41	0.10	(0.00)	
t_α	2.92	3.46	4.20	4.30	4.06	3.63	2.66	2.01	–0.40	–2.96	–6.92		
α_{FF}	0.67	0.50	0.39	0.32	0.27	0.25	0.14	0.10	–0.19	–0.69	–1.35	0.35	0.89
b	1.07	0.98	0.99	0.94	0.95	0.95	0.98	1.03	1.10	1.21	0.14	(0.00)	
s	1.50	1.16	0.83	0.71	0.69	0.69	0.72	0.77	0.85	1.01	–0.49		
h	0.20	0.32	0.45	0.38	0.37	0.26	0.22	0.16	0.06	–0.05	–0.25		
$t_{\alpha_{FF}}$	2.91	3.50	4.17	4.36	4.23	3.74	2.33	1.48	–2.10	–5.03	–6.12		
α_{CARH}	0.95	0.70	0.58	0.46	0.41	0.35	0.25	0.24	0.01	–0.35	–1.30	0.43	0.90
b	1.01	0.94	0.95	0.91	0.92	0.93	0.95	1.00	1.05	1.13	0.13	(0.00)	
s	1.50	1.16	0.83	0.71	0.69	0.69	0.72	0.77	0.85	1.01	–0.49		
h	0.10	0.25	0.38	0.34	0.32	0.23	0.18	0.11	–0.01	–0.17	–0.27		
w	–0.30	–0.21	–0.20	–0.14	–0.14	–0.10	–0.12	–0.15	–0.22	–0.35	–0.06		
$t_{\alpha_{CARH}}$	3.64	4.25	5.07	5.67	5.73	4.86	3.92	3.45	0.15	–2.56	–5.95		
α_q	1.34	0.89	0.62	0.46	0.40	0.36	0.29	0.29	0.13	–0.13	–1.47	0.49	0.90
β_{MKT}	1.02	0.94	0.96	0.92	0.91	0.92	0.94	0.99	1.05	1.13	0.11	(0.00)	
β_{ME}	1.12	0.92	0.68	0.61	0.62	0.62	0.65	0.70	0.73	0.79	–0.33		
$\beta_{\Delta A/A}$	0.05	0.24	0.43	0.38	0.28	0.13	0.00	–0.11	–0.30	–0.58	–0.63		
β_{ROE}	–1.07	–0.68	–0.46	–0.31	–0.23	–0.18	–0.17	–0.20	–0.32	–0.55	0.51		
t_{α_q}	4.61	4.75	4.29	4.53	3.83	3.77	3.42	3.25	0.93	–0.60	–5.89		
$t_{\beta_{MKT}}$	17.64	26.46	28.71	34.82	36.92	42.03	40.41	38.57	32.47	24.73	2.60		
$t_{\beta_{ME}}$	12.57	15.46	8.19	9.74	11.19	13.85	11.68	11.33	8.76	7.37	–2.91		
$t_{\beta_{\Delta A/A}}$	0.27	1.69	3.68	4.43	3.53	1.82	0.01	–1.58	–2.67	–3.36	–3.85		
$t_{\beta_{ROE}}$	–6.88	–6.68	–5.59	–5.64	–4.03	–3.56	–3.78	–4.21	–4.65	–4.75	4.82		

Table IA.15 : Factor Regressions for Monthly Percent Excess Returns of One-way Investment-to-Assets ($\Delta A/A$) Deciles, All NYSE-Amex-NASDAQ Breakpoints, Value-weighted Returns (1/1972–12/2011, 480 Months)

$\Delta A/A$ is annual change in total assets (Compustat annual item AT) divided by lagged total assets. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to split stocks into deciles on $\Delta A/A$ for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the deciles. The numbers (in parentheses) are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.77	0.64	0.75	0.60	0.55	0.52	0.55	0.50	0.46	0.22	–0.56		
t_{Mean}	2.38	2.45	3.43	3.12	2.71	2.48	2.56	1.98	1.60	0.67	–2.80		
α	0.24	0.19	0.33	0.22	0.16	0.11	0.12	0.01	–0.08	–0.38	–0.62	0.19	0.84
β	1.19	1.00	0.93	0.83	0.86	0.91	0.95	1.07	1.19	1.33	0.14	(0.00)	
t_α	1.45	1.54	3.52	2.75	2.36	1.34	1.87	0.13	–0.68	–2.93	–3.09		
α_{FF}	0.09	0.06	0.18	0.15	0.09	0.09	0.18	0.14	0.15	–0.20	–0.29	0.13	0.87
b	1.10	1.02	0.98	0.89	0.90	0.95	0.96	1.00	1.08	1.19	0.10	(0.00)	
s	0.63	0.16	0.06	–0.10	–0.06	–0.15	–0.14	0.05	0.06	0.24	–0.40		
h	0.15	0.23	0.30	0.17	0.17	0.07	–0.09	–0.27	–0.47	–0.42	–0.57		
$t_{\alpha_{FF}}$	0.62	0.48	2.01	1.99	1.42	1.23	2.78	1.75	1.66	–1.86	–1.63		
α_{CARH}	0.07	0.07	0.19	0.17	0.08	0.04	0.17	0.11	0.15	–0.16	–0.23	0.12	0.87
b	1.10	1.02	0.98	0.88	0.91	0.96	0.96	1.01	1.08	1.18	0.08	(0.00)	
s	0.63	0.16	0.06	–0.10	–0.06	–0.15	–0.14	0.05	0.06	0.24	–0.40		
h	0.16	0.23	0.29	0.16	0.17	0.09	–0.08	–0.26	–0.47	–0.43	–0.59		
w	0.02	–0.01	–0.01	–0.02	0.01	0.05	0.01	0.03	0.00	–0.05	–0.07		
$t_{\alpha_{CARH}}$	0.39	0.59	2.05	2.02	1.26	0.55	2.32	1.41	1.61	–1.46	–1.07		
α_q	0.06	–0.10	0.01	–0.04	0.07	–0.08	0.12	0.17	0.38	–0.09	–0.16	0.11	0.88
β_{MKT}	1.13	1.06	1.02	0.92	0.90	0.96	0.96	1.00	1.06	1.19	0.05	(0.00)	
β_{ME}	0.53	0.14	0.04	–0.09	–0.06	–0.09	–0.10	0.06	0.02	0.22	–0.30		
$\beta_{\Delta A/A}$	0.46	0.64	0.68	0.51	0.28	0.18	–0.12	–0.47	–0.91	–0.73	–1.19		
β_{ROE}	–0.28	–0.10	–0.05	0.04	–0.06	0.17	0.14	0.10	0.00	0.05	0.33		
t_{α_q}	0.39	–0.94	0.06	–0.47	1.09	–0.93	1.74	1.82	3.66	–0.81	–0.85		
$t_{\beta_{MKT}}$	23.46	42.77	44.87	45.60	49.90	48.02	54.40	44.04	37.43	39.19	1.05		
$t_{\beta_{ME}}$	6.92	2.95	1.03	–2.73	–2.22	–2.59	–4.09	1.45	0.45	4.64	–3.31		
$t_{\beta_{\Delta A/A}}$	3.95	10.00	12.02	9.72	6.36	2.34	–2.71	–6.97	–12.89	–11.17	–8.80		
$t_{\beta_{ROE}}$	–2.36	–2.15	–0.94	0.76	–1.71	2.78	4.26	1.95	0.11	0.93	2.56		

Table IA.16 : Factor Regressions for Monthly Percent Excess Returns of One-way *ROE* Deciles (1/1972–12/2011, 480 Months)

ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by one-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock. At the beginning of each month, we sort all stocks into deciles based on the NYSE breakpoints of the ranked values of *ROE*. Earnings and other accounting variables in Compustat quarterly files are used in the sorts in the months immediately after the most recent public earnings announcement dates (item RDQ). We calculate monthly value-weighted portfolio returns for the current month, and the portfolios are rebalanced monthly. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the *q*-factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the deciles. The numbers (in parentheses) are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The *t*-statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	−0.12	0.31	0.32	0.36	0.55	0.42	0.57	0.50	0.54	0.69	0.81		
t_{Mean}	−0.32	1.07	1.23	1.67	2.65	1.80	2.60	2.26	2.41	2.88	3.08		
α	−0.73	−0.20	−0.14	−0.06	0.14	−0.02	0.13	0.06	0.10	0.24	0.97	0.18	0.84
β	1.34	1.12	1.02	0.92	0.91	0.98	0.97	0.97	0.97	0.99	−0.35	(0.01)	
t_α	−3.93	−1.70	−1.41	−0.63	1.79	−0.26	1.80	0.84	1.37	2.40	3.96		
α_{FF}	−0.78	−0.30	−0.25	−0.15	0.08	−0.06	0.13	0.13	0.22	0.41	1.19	0.25	0.86
b	1.18	1.11	1.03	0.91	0.93	0.98	0.97	0.97	0.95	0.96	−0.22	(0.00)	
s	0.73	0.21	0.15	0.19	0.00	0.07	−0.01	−0.12	−0.13	−0.18	−0.91		
h	−0.07	0.15	0.19	0.14	0.12	0.05	0.00	−0.11	−0.22	−0.30	−0.23		
$t_{\alpha_{FF}}$	−4.73	−2.63	−2.39	−1.61	0.95	−0.69	1.90	1.74	3.42	4.36	5.36		
α_{CARH}	−0.55	−0.08	−0.07	−0.06	0.12	−0.04	0.15	0.08	0.12	0.31	0.86	0.16	0.87
b	1.14	1.07	1.00	0.90	0.93	0.97	0.97	0.98	0.97	0.98	−0.15	(0.00)	
s	0.73	0.21	0.15	0.19	0.00	0.07	−0.01	−0.12	−0.13	−0.18	−0.91		
h	−0.15	0.08	0.12	0.11	0.10	0.05	−0.01	−0.09	−0.19	−0.27	−0.12		
w	−0.24	−0.23	−0.20	−0.10	−0.04	−0.01	−0.02	0.05	0.10	0.11	0.35		
$t_{\alpha_{CARH}}$	−3.46	−0.75	−0.66	−0.64	1.42	−0.51	2.20	1.06	1.75	3.45	3.99		
α_q	0.00	0.21	0.27	0.04	0.14	−0.02	0.09	−0.08	0.07	0.04	0.03	0.10	0.90
β_{MKT}	1.12	1.06	0.97	0.89	0.93	0.98	0.98	1.00	0.97	1.02	−0.10	(0.02)	
β_{ME}	0.37	−0.01	−0.07	0.11	−0.04	0.03	0.00	−0.05	−0.05	−0.04	−0.41		
$\beta_{\Delta A/A}$	−0.32	0.06	0.02	0.10	0.19	0.07	0.01	−0.02	−0.26	−0.21	0.11		
β_{ROE}	−1.01	−0.67	−0.63	−0.27	−0.14	−0.07	0.05	0.24	0.27	0.49	1.51		
t_{α_q}	0.02	2.27	3.14	0.47	1.62	−0.22	1.17	−0.97	0.83	0.42	0.25		
$t_{\beta_{MKT}}$	33.22	42.75	31.04	35.44	39.77	46.98	47.09	54.39	49.86	57.02	−2.75		
$t_{\beta_{ME}}$	6.12	−0.20	−1.22	3.29	−1.28	1.13	−0.14	−1.73	−2.05	−1.24	−6.43		
$t_{\beta_{\Delta A/A}}$	−3.99	0.86	0.34	1.82	3.21	1.12	0.24	−0.29	−4.69	−3.76	1.01		
$t_{\beta_{ROE}}$	−16.84	−17.03	−10.90	−5.76	−2.99	−1.61	1.32	7.44	8.44	12.46	20.78		

Table IA.17 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Market Leverage (A/ME) (1/1972–12/2011, 480 Months)

We measure A/ME as the ratio of total book assets (item AT) to the market equity. In June of each year t , we use the NYSE breakpoints to sort NYSE, Amex, and NASDAQ stocks into deciles on A/ME for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted portfolio returns are calculated from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on the one-month Treasury bill rate (r^f), the Fama-French factors, and the momentum factor are from Kenneth French's Web site. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.32	0.39	0.54	0.60	0.73	0.54	0.70	0.68	0.73	0.74	0.42		
t_{Mean}	1.24	1.74	2.42	2.79	3.44	2.43	3.38	3.04	2.89	2.43	1.78		
α	-0.16	-0.05	0.10	0.18	0.31	0.12	0.32	0.30	0.32	0.24	0.40	0.21	0.80
β	1.07	0.96	0.97	0.92	0.93	0.91	0.83	0.85	0.91	1.12	0.05	(0.00)	
t_α	-1.54	-0.63	1.18	2.09	3.44	1.18	3.29	2.49	2.22	1.45	1.66		
α_{FF}	0.12	0.05	0.09	0.12	0.21	-0.06	0.13	0.00	-0.06	-0.20	-0.32	0.11	0.87
b	0.98	0.94	0.98	0.95	0.98	0.98	0.90	0.96	1.03	1.21	0.23	(0.00)	
s	-0.10	-0.08	-0.02	0.01	-0.05	0.04	0.03	0.08	0.12	0.40	0.50		
h	-0.55	-0.17	0.01	0.13	0.21	0.37	0.38	0.58	0.73	0.80	1.35		
$t_{\alpha_{FF}}$	1.80	0.65	1.10	1.46	2.63	-0.72	1.69	0.04	-0.62	-1.79	-2.40		
α_{CARH}	0.13	0.07	0.07	0.06	0.17	-0.05	0.05	0.02	0.00	-0.09	-0.23	0.07	0.87
b	0.97	0.94	0.99	0.96	0.99	0.98	0.92	0.95	1.02	1.19	0.21	(0.05)	
s	-0.10	-0.08	-0.02	0.01	-0.05	0.04	0.03	0.08	0.12	0.40	0.50		
h	-0.56	-0.18	0.02	0.15	0.23	0.37	0.40	0.58	0.71	0.76	1.32		
w	-0.01	-0.02	0.02	0.07	0.04	-0.01	0.08	-0.01	-0.07	-0.11	-0.10		
$t_{\alpha_{CARH}}$	1.94	0.89	0.83	0.66	2.06	-0.57	0.71	0.17	0.05	-0.82	-1.68		
α_q	0.10	-0.10	-0.07	0.03	0.06	-0.12	0.09	0.02	0.02	0.01	-0.10	0.06	0.85
β_{MKT}	1.01	0.97	1.00	0.95	0.99	0.97	0.89	0.91	0.98	1.15	0.14	(0.32)	
β_{ME}	-0.09	-0.02	0.03	0.03	0.00	0.07	0.05	0.09	0.10	0.29	0.38		
$\beta_{\Delta A/A}$	-0.65	-0.12	0.11	0.23	0.37	0.49	0.47	0.60	0.73	0.81	1.46		
β_{ROE}	0.13	0.20	0.15	0.04	0.09	-0.03	-0.03	-0.08	-0.16	-0.39	-0.52		
t_{α_q}	1.12	-1.30	-0.77	0.40	0.80	-1.11	1.03	0.20	0.15	0.06	-0.48		
$t_{\beta_{MKT}}$	44.42	40.00	39.07	36.50	35.50	41.32	37.56	25.77	27.67	31.58	2.86		
$t_{\beta_{ME}}$	-2.20	-0.71	0.91	0.89	0.05	1.47	1.20	1.30	1.44	3.28	3.19		
$t_{\beta_{\Delta A/A}}$	-12.46	-2.26	2.01	3.13	5.16	4.49	5.80	6.32	7.26	7.02	10.26		
$t_{\beta_{ROE}}$	3.11	4.77	3.46	0.79	1.85	-0.47	-0.48	-1.00	-2.15	-5.01	-5.48		

Table IA.18 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Prior 13–60 Month Returns (Reversal) (1/1972–12/2011, 480 Months)

The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. At the beginning of each month t , we use the NYSE breakpoints to split the NYSE, Amex, and NASDAQ stocks into deciles on the prior returns from month $t-13$ to $t-60$. Monthly value-weighted portfolio returns are computed, and the portfolios are rebalanced monthly. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across the testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p -values for the GRS test on the null that the alphas across the deciles for a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H–L	m.a.e.	$\bar{R}^2$
Mean	0.86	0.77	0.83	0.57	0.67	0.63	0.63	0.51	0.43	0.49	–0.37		
t_{Mean}	2.50	3.02	3.54	2.65	3.26	3.02	3.12	2.45	1.74	1.58	–1.46		
α	0.31	0.30	0.40	0.17	0.29	0.24	0.24	0.11	–0.02	–0.07	–0.38	0.21	0.79
β	1.21	1.03	0.94	0.89	0.85	0.86	0.87	0.87	0.99	1.23	0.02	(0.00)	
t_α	1.63	2.17	3.26	1.86	3.08	2.59	2.66	1.20	–0.22	–0.54	–1.50		
α_{FF}	0.01	0.10	0.24	0.01	0.15	0.16	0.21	0.14	0.05	0.18	0.17	0.12	0.83
b	1.14	1.03	0.96	0.92	0.93	0.91	0.92	0.90	1.00	1.14	0.00	(0.03)	
s	0.85	0.37	0.19	0.14	–0.06	–0.09	–0.16	–0.16	–0.16	–0.06	–0.91		
h	0.41	0.32	0.29	0.28	0.30	0.18	0.10	–0.01	–0.10	–0.49	–0.90		
$t_{\alpha_{FF}}$	0.05	0.80	2.19	0.14	1.69	1.82	2.22	1.48	0.50	1.69	0.93		
α_{CARH}	0.15	0.14	0.22	0.03	0.10	0.09	0.14	0.07	–0.01	0.18	0.03	0.11	0.83
b	1.10	1.02	0.97	0.91	0.94	0.92	0.94	0.92	1.01	1.14	0.03	(0.14)	
s	0.85	0.37	0.19	0.14	–0.06	–0.09	–0.16	–0.16	–0.16	–0.06	–0.91		
h	0.36	0.30	0.30	0.28	0.31	0.20	0.12	0.01	–0.08	–0.49	–0.85		
w	–0.15	–0.04	0.02	–0.02	0.05	0.07	0.07	0.07	0.06	0.00	0.15		
$t_{\alpha_{CARH}}$	0.99	1.08	1.88	0.33	1.16	1.02	1.38	0.73	–0.10	1.59	0.15		
α_q	0.28	0.14	0.16	–0.04	0.01	0.03	0.00	–0.05	–0.15	0.17	–0.10	0.10	0.84
β_{MKT}	1.10	1.02	0.97	0.92	0.93	0.91	0.94	0.92	1.02	1.15	0.05	(0.18)	
β_{ME}	0.70	0.31	0.17	0.14	–0.01	–0.03	–0.08	–0.09	–0.07	–0.02	–0.71		
$\beta_{\Delta A/A}$	0.41	0.41	0.43	0.40	0.45	0.27	0.24	0.08	–0.05	–0.72	–1.13		
β_{ROE}	–0.53	–0.20	–0.03	–0.05	0.09	0.12	0.22	0.21	0.26	0.20	0.73		
t_{α_q}	1.95	1.15	1.41	–0.35	0.05	0.31	–0.03	–0.51	–1.63	1.40	–0.54		
$t_{\beta_{MKT}}$	30.46	28.03	31.07	34.10	37.42	33.30	32.67	36.25	41.85	38.24	0.95		
$t_{\beta_{ME}}$	11.30	5.19	3.40	2.81	–0.24	–0.81	–1.84	–2.13	–1.83	–0.39	–8.41		
$t_{\beta_{\Delta A/A}}$	4.74	4.93	4.52	4.28	4.91	3.12	3.35	1.16	–0.74	–10.59	–9.55		
$t_{\beta_{ROE}}$	–8.33	–3.31	–0.39	–0.76	1.47	1.77	3.46	3.99	4.35	3.48	7.15		

Table IA.19 : Factor Regressions for Monthly Percent Excess Returns of Deciles Formed on Earnings-to-Price (E/P) (1/1972–12/2011, 480 Months)

The data on the one-month Treasury bill rate (r^f) and the Carhart factors are from Kenneth French's Web site. Earnings-to-price used in the annual sorts in June of year t are total earnings before extraordinary items (Compustat annual item IB) for the last fiscal year end in $t - 1$ divided by the market equity at the end of December of year $t - 1$. In June of each year t , we use the NYSE breakpoints to split all NYSE, Amex, and NASDAQ stocks into deciles based on E/P . Stocks with negative earnings for the last fiscal year end in $t - 1$ are excluded. We calculate monthly value-weighted portfolio returns, and the portfolios are rebalanced annually in June. We report the mean monthly percent excess returns, the CAPM regressions ($r_t^i - r_t^f = \alpha^i + \beta^i MKT_t + \epsilon_t^i$), the Fama-French three-factor regressions ($r_t^i - r_t^f = \alpha_{FF}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + \epsilon_t^i$), the Carhart four-factor regressions ($r_t^i - r_t^f = \alpha_{CARH}^i + b^i MKT_t + s^i SMB_t + h^i HML_t + w^i WML_t + \epsilon_t^i$) and the q -factor regressions ($r_t^i - r_t^f = \alpha_q^i + \beta_{MKT}^i MKT_t + \beta_{ME}^i r_{ME,t} + \beta_{\Delta A/A}^i r_{\Delta A/A,t} + \beta_{ROE}^i r_{ROE,t} + \epsilon_t^i$). m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. See the caption of Table 1 for the description of r_{ME} , $r_{\Delta A/A}$, and r_{ROE} .

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Mean	0.31	0.39	0.52	0.53	0.47	0.62	0.71	0.74	0.72	0.89	0.58		
t_{Mean}	1.02	1.61	2.33	2.60	2.22	2.88	3.49	3.47	3.27	3.40	2.54		
α	-0.24	-0.08	0.10	0.14	0.07	0.23	0.33	0.37	0.34	0.44	0.68	0.23	0.79
β	1.20	1.04	0.92	0.87	0.89	0.86	0.83	0.82	0.84	0.99	-0.22	(0.01)	
t_α	-1.90	-1.05	1.19	1.46	0.66	2.22	3.22	3.26	2.61	3.10	2.98		
α_{FF}	0.06	0.04	0.14	0.15	0.00	0.08	0.18	0.17	0.07	0.09	0.04	0.10	0.83
b	1.08	1.02	0.93	0.89	0.94	0.93	0.91	0.90	0.93	1.08	0.00	(0.22)	
s	0.00	-0.12	-0.10	-0.10	-0.09	-0.02	-0.05	0.00	0.10	0.22	0.22		
h	-0.59	-0.23	-0.06	0.00	0.16	0.30	0.32	0.41	0.52	0.66	1.25		
$t_{\alpha_{FF}}$	0.61	0.63	1.81	1.78	-0.04	0.95	1.94	1.57	0.67	0.76	0.23		
α_{CARH}	0.10	0.03	0.11	0.13	0.00	0.09	0.14	0.15	0.08	0.10	-0.01	0.09	0.84
b	1.07	1.02	0.94	0.89	0.94	0.93	0.92	0.91	0.92	1.08	0.01	(0.42)	
s	0.01	-0.12	-0.10	-0.10	-0.09	-0.02	-0.05	0.00	0.10	0.22	0.22		
h	-0.61	-0.23	-0.05	0.00	0.16	0.30	0.33	0.42	0.52	0.66	1.26		
w	-0.05	0.01	0.04	0.01	-0.01	0.00	0.04	0.02	-0.01	-0.01	0.04		
$t_{\alpha_{CARH}}$	1.10	0.42	1.18	1.58	0.05	0.97	1.47	1.34	0.74	0.82	-0.03		
α_q	0.11	-0.10	-0.14	-0.03	-0.18	-0.04	0.00	0.07	0.15	0.28	0.17	0.11	0.82
β_{MKT}	1.11	1.04	0.96	0.92	0.95	0.93	0.90	0.89	0.88	0.99	-0.12	(0.14)	
β_{ME}	-0.04	-0.04	0.01	-0.04	0.00	0.03	0.04	0.04	0.09	0.21	0.24		
$\beta_{\Delta A/A}$	-0.66	-0.24	0.07	0.15	0.25	0.42	0.43	0.50	0.46	0.37	1.03		
β_{ROE}	-0.01	0.23	0.31	0.16	0.19	0.07	0.16	0.05	-0.11	-0.12	-0.11		
t_{α_q}	0.87	-1.23	-1.57	-0.35	-1.90	-0.38	0.02	0.64	1.21	1.85	0.71		
$t_{\beta_{MKT}}$	35.32	47.99	43.74	36.40	35.78	34.97	34.20	25.93	25.64	21.31	-1.93		
$t_{\beta_{ME}}$	-0.77	-1.24	0.21	-1.14	0.01	0.74	0.70	0.58	1.50	1.97	1.87		
$t_{\beta_{\Delta A/A}}$	-8.45	-4.17	0.89	1.83	2.69	4.96	5.20	5.11	5.01	3.05	5.80		
$t_{\beta_{ROE}}$	-0.11	5.77	5.24	2.82	2.57	1.11	2.39	0.65	-1.31	-1.16	-0.77		

Table IA.20 : Descriptive Statistics for Monthly Percent Excess Returns on Annually Rebalanced Deciles Formed on Standardized Unexpected Earnings, Idiosyncratic Volatility, and Campbell, Hilscher, and Szilagyi's (2008) Failure Probability and on Annually Rebalanced 25 Size and Momentum Portfolios

In June of each year t , we use the NYSE breakpoints to sort all NYSE, Amex, and NASDAQ stocks into ten deciles based on SUE at the fiscal year end of $t - 1$. Monthly equal-weighted returns are computed from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles based on the prior returns from July of year $t - 1$ to December of year $t - 1$. Monthly equal-weighted returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. In June of each year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles on $IVOL$ calculated with the information up to December of year $t - 1$. Separately, in June of year t , we use the NYSE-Amex-NASDAQ breakpoints to sort all stocks into deciles based on the Campbell, Hilscher, and Szilagyi (2008) failure probability calculated at the fiscal yearend of $t - 1$. Monthly value-weighted returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June. We report average returns and t -statistics in monthly percent and the CAPM regression. m.a.e. is the average magnitude of the alphas across a given set of testing portfolios. The numbers (in parentheses) beneath the m.a.e.'s are the p-values for the GRS test on the null that the alphas across all the deciles from a given factor model are jointly zero. $\bar{R}^2$ is the average goodness-of-fit across the deciles. The t -statistics are adjusted for heteroscedasticity and autocorrelations. The data on one-month Treasury bill and the market factor are from Kenneth French's Web site.

	Low	2	3	4	5	6	7	8	9	High	H-L	m.a.e.	$\bar{R}^2$
Panel A: The SUE deciles (7/1972-12/2011, 474 Months)													
Mean	0.86	0.82	0.89	0.94	0.91	0.92	0.94	0.84	0.93	0.92	0.06		
t_{Mean}	2.57	2.54	2.82	2.91	2.90	2.97	3.15	2.84	3.11	3.16	0.60		
α	0.34	0.31	0.39	0.43	0.41	0.42	0.45	0.35	0.44	0.42	0.08	0.40	0.75
β	1.16	1.13	1.13	1.14	1.12	1.11	1.10	1.11	1.10	1.12	-0.04	(0.03)	
t_α	2.05	2.05	2.55	2.76	2.69	2.88	3.11	2.62	3.11	3.23	0.84		
Panel B: The price momentum deciles (1/1972-12/2011, 480 Months)													
Mean	0.69	0.69	0.70	0.78	0.76	0.77	0.69	0.70	0.59	0.44	-0.25		
t_{Mean}	1.77	2.11	2.34	2.81	2.79	3.06	2.78	2.65	2.00	1.21	-1.19		
α	0.08	0.16	0.20	0.31	0.30	0.32	0.24	0.22	0.06	-0.18	-0.26	0.21	0.77
β	1.35	1.18	1.09	1.04	1.02	1.00	1.00	1.06	1.17	1.37	0.02	(0.00)	
t_α	0.39	0.94	1.38	2.32	2.47	2.98	2.43	2.09	0.44	-0.98	-1.19		
Panel C: Ten idiosyncratic volatility deciles (1/1972-12/2011, 480 Months)													
Mean	0.45	0.51	0.46	0.59	0.65	0.61	0.34	0.49	0.35	0.32	-0.13		
t_{Mean}	2.49	2.41	1.89	2.15	2.13	1.80	0.92	1.24	0.76	0.70	-0.36		
α	0.10	0.09	-0.02	0.07	0.08	0.00	-0.32	-0.17	-0.36	-0.35	-0.45	0.16	0.80
β	0.77	0.94	1.07	1.16	1.25	1.35	1.47	1.45	1.56	1.47	0.71	(0.08)	
t_α	1.51	1.63	-0.29	0.80	0.68	0.00	-1.73	-0.92	-1.30	-1.33	-1.47		
Panel D: Ten failure probability deciles (7/1976-12/2011, 426 Months)													
Mean	0.58	0.48	0.55	0.51	0.63	0.65	0.52	0.84	0.68	0.80	0.22		
t_{Mean}	2.41	2.22	2.54	2.28	2.51	2.43	1.77	2.38	1.65	1.70	0.61		
α	0.06	0.00	0.08	0.00	0.08	0.08	-0.08	0.17	-0.12	0.01	-0.06	0.07	0.80
β	0.98	0.90	0.89	0.97	1.03	1.09	1.13	1.27	1.52	1.51	0.53	(0.51)	
t_α	0.59	0.02	0.99	0.03	0.99	0.65	-0.55	0.93	-0.53	0.03	-0.17		

Digesting Anomalies: An Investment Approach

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Abstract

An empirical q -factor model consisting of the market factor, a size factor, an investment factor, and a profitability factor largely summarizes the cross section of average stock returns. A comprehensive examination of nearly 80 anomalies reveals that about one-half of the anomalies are insignificant in the broad cross section. More importantly, with a few exceptions, the q -factor model's performance is at least comparable to, and in many cases better than that of the Fama-French (1993) 3-factor model and the Carhart (1997) 4-factor model in capturing the remaining significant anomalies. (*JEL* G12, G14)

In a highly influential article, Fama and French (1996) show that, except for momentum, their 3-factor model, which consists of the market factor, a factor based on market equity (small-minus-big, SMB), and a factor based on book-to-market equity (high-minus-low, HML), summarizes the cross section of average stock returns as of the mid-1990s. Over the past 2 decades, however, it has become clear that the Fama-French model fails to account for a wide array of asset pricing anomalies.¹

Our contribution is to construct a new empirical model that does a good job in summarizing the cross section of average stock returns. In particular, many (but not all) of the anomalies that prove challenging for the Fama-French model can be captured.² Our model is in part inspired by investment-based asset pricing, which is in turn built on the neoclassical q -theory of investment. In our model (dubbed the q -factor model), the expected return of an asset in excess of the risk-free rate, denoted $E[r^i] - r^f$, is described by the sensitivities of its returns to 4 factors: the market excess return (MKT), the difference between the return on a portfolio of small size stocks and the return on a portfolio of big size stocks (r_{ME}), the difference between the return on a portfolio of low investment stocks and the return on a portfolio of high investment stocks ($r_{I/A}$), and the difference between the return on a portfolio of high profitability (return on equity, ROE) stocks and the return

¹See, for example, Ball and Brown (1968); Bernard and Thomas (1990); Ritter (1991); Jegadeesh and Titman (1993); Ikenberry, Lakonishok, and Vermaelen (1995); Loughran and Ritter (1995); Chan, Jegadeesh, and Lakonishok (1996); Sloan (1996); Ang, Hodrick, Xing, and Zhang (2006); Daniel and Titman (2006); Campbell, Hilscher, and Szilagyi (2008); Cooper, Gulen, and Schill (2008); and Hafzalla, Lundholm, and Van Winkle (2011).

²The need for a new factor model is evident in Cochrane (2011, p. 1060–61, original emphasis): “We are going to have to repeat Fama and French’s anomaly digestion, but with many more dimensions. We have a lot of questions to answer: First, which characteristics really provide *independent* information about average returns? Which are subsumed by others? Second, does each new anomaly variable also correspond to a new factor formed on those same anomalies?... Third, how many of these new factors are really important? Can we again account for N independent dimensions of expected returns with $K < N$ factor exposures?... [T]he world would be much simpler if betas on only a few factors, important in the covariance matrix of returns, accounted for a larger number of mean characteristics.”

on a portfolio of low profitability stocks (r_{ROE}). Formally,

$$E[r^i] - r^f = \beta_{\text{MKT}}^i E[\text{MKT}] + \beta_{\text{ME}}^i E[r_{\text{ME}}] + \beta_{\text{I/A}}^i E[r_{\text{I/A}}] + \beta_{\text{ROE}}^i E[r_{\text{ROE}}], \quad (1)$$

in which $E[\text{MKT}]$, $E[r_{\text{ME}}]$, $E[r_{\text{I/A}}]$, and $E[r_{\text{ROE}}]$ are expected factor premiums, and β_{MKT}^i , β_{ME}^i , $\beta_{\text{I/A}}^i$, and β_{ROE}^i are the factor loadings on MKT, r_{ME} , $r_{\text{I/A}}$, and r_{ROE} , respectively.

We construct the q -factors from a triple 2-by-3-by-3 sort on size, investment-to-assets, and ROE. From January 1972 to December 2012, the size factor earns an average return of 0.31% per month ($t = 2.12$); the investment factor 0.45% ($t = 4.95$); and the ROE factor 0.58% ($t = 4.81$). The investment factor has a high correlation of 0.69 with HML, and the ROE factor has a high correlation of 0.50 with the Carhart (1997) momentum factor (up-minus-down, UMD). The alphas of HML and UMD in the q -factor model are small and insignificant, but the alphas of the investment and the ROE factors in the Carhart model (that augments the Fama-French model with UMD) are large and significant. As such, HML and UMD might be noisy versions of the q -factors.

To evaluate the relative performance of the q -factor model, we start with a wide array of nearly 80 variables that cover all major categories of anomalies. Following Fama and French (1996), we construct testing deciles based on the breakpoints from the New York Stock Exchange (NYSE), and calculate value-weighted decile returns. Surprisingly, the high-minus-low deciles formed on about one-half of the anomaly variables, including the vast majority of variables related to trading frictions, have average returns that are insignificant at the 5% level. As such, echoing Schwert (2003) and Harvey, Liu, and Zhu (2013), we suggest that many claims in the anomalies literature seem exaggerated.

More importantly, in the playing field consisting of 35 anomalies that are significant in the broad cross section, the q -factor model performs well, compared to the Fama-French and Carhart models. Across the 35 high-minus-low deciles, the average magnitude of the alphas is 0.20% per month in the q -factor model, in contrast to 0.33% in the Carhart model and 0.55% in the Fama-French model. Five high-minus-low alphas are significant at the 5% level in the q -factor model, in contrast to 19 in the Carhart model and 27 in the Fama-French model. In addition, the q -factor model is rejected by the Gibbons, Ross, and Shanken (1989, GRS) test in 20 sets of deciles. In contrast, the Carhart model is rejected in 24, and the Fama-French model in 28 sets of deciles.

In particular, the q -factor model outperforms the Fama-French and Carhart models in capturing momentum. The high-minus-low earnings momentum decile has a Fama-French alpha of 0.55% per month and a Carhart alpha of 0.34%, both of which are significant. The alpha in the q -factor model (the q -alpha) is 0.16% ($t = 1.12$). The high-minus-low price momentum decile has a Fama-French alpha of 1.12% ($t = 4.47$) and a Carhart alpha of 0.06% ($t = 0.51$). The q -alpha is 0.24% ($t = 0.71$). The q -factor model performs similarly as the other 2 models in fitting the 25 size and book-to-market portfolios. The average magnitude of the alphas across the 25 portfolios is 0.11% in the q -factor model, which is close to 0.10% in the Fama-French model and 0.11% in the Carhart model. However, the q -factor model underperforms the Fama-French and Carhart models in capturing the operating accrual anomaly and the R&D-to-market anomaly.

Intuitively, investment predicts returns because given expected cash flows, high costs of capital imply low net present values of new capital and low investment, and low costs of capital imply high net present values of new capital and high investment. ROE predicts

returns because high expected ROE relative to low investment must imply high discount rates. The high discount rates are necessary to offset the high expected ROE to induce low net present values of new capital and low investment. If the discount rates were not high enough, firms would instead observe high net present values of new capital and invest more. Conversely, low expected ROE relative to high investment must imply low discount rates. If the discount rates were not low enough to counteract the low expected ROE, firms would instead observe low net present values of new capital and invest less.

The traditional approach in asset pricing is to look for common factors from the consumption side of the economy (e.g., Breeden, Gibbons, and Litzenberger 1989). We instead exploit a direct link between stock returns and firm characteristics from the production side, following Cochrane (1991). Berk, Green, and Naik (1999); Carlson, Fisher, and Giammarino (2004); and Zhang (2005) construct fully specified dynamic models for the cross section of expected stock returns. Liu, Whited, and Zhang (2009) estimate the characteristics-expected return relation derived from q -theory via generalized method of moments. Kogan and Papanikolaou (2013) relate the investment and profitability effects to embodied technology shocks. We differ by using the Black, Jensen, and Scholes (1972) portfolio approach to build a new factor model. A factor model is more flexible in practice because of its simplicity and the availability of high frequency returns data. Finally, it should be noted that the investment and profitability effects are not new to our work.³ However, recognizing their fundamental importance in investment-based asset pricing, we build a new workhorse model on these effects for the cross section of expected stock returns.⁴

³See Fairfield, Whisenant, and Yohn (2003); Titman, Wei, and Xie (2004); Cooper, Gulen, and Schill (2008); Xing (2008); and Polk and Sapienza (2009) for the investment effect, and Ball and Brown (1968); Bernard and Thomas (1990); Chan, Jagadeesh, and Lakonishok (1996); Haugen and Baker (1996); Piotroski (2000); Fama and French (2006); and Novy-Marx (2013) for the earnings (profitability) effect.

⁴Fama and French (2013, 2014), circulated after the first draft of our work dated October 2012, have

1 Background

The q -factor model is in part inspired from investment-based asset pricing. In this section, we use a simple economic model to illustrate the key intuitions behind the q -factor model.

1.1 An economic model

Consider a 2-period stochastic general equilibrium model as in Lin and Zhang (2013). There are 2 dates, 0 and 1. The economy is populated by a representative household and heterogeneous firms, indexed by $i = 1, 2, \dots, N$. The representative household maximizes its expected utility, $U(C_0) + \rho E_0[U(C_1)]$, in which ρ is time preference, and C_0 and C_1 are consumption in dates 0 and 1, respectively. Firms produce a single commodity to be consumed or invested. Firm i starts with productive assets, A_{i0} , and produces in both dates. Firms exit at the end of date 1, with a liquidation value of zero, meaning that the depreciation rate of assets is 100%.

Firms differ in date-0 assets, A_{i0} , and date-0 profitability, Π_{i0} , which is known at the beginning of date 0. The operating cash flow of firm i is $\Pi_{it}A_{it}$, for $t = 0, 1$, in which the firm's stochastic date-1 profitability, Π_{i1} , is subject to a vector of aggregate shocks affecting all firms simultaneously, as well as a vector of firm-specific shocks affecting only firm i . Let I_{i0} denote investment for date 0, then $A_{i1} = I_{i0}$, because A_{i0} depreciates fully at the beginning of date 1. Investment entails quadratic adjustment costs, $(a/2)(I_{i0}/A_{i0})^2 A_{i0}$, in which $a > 0$ is a constant parameter.

The household side is standard. Let P_{it} and D_{it} denote the ex-dividend equity and recently incorporated variables that resemble our 2 q -factors into their 3-factor model to form a 5-factor asset pricing model. Their 2013 draft adds only a profitability factor, and the 2014 draft subsequently adds an investment factor.

dividend for firm i , respectively. The first principle for consumption says that $P_{i0} = E_0[M_1(P_{i1} + D_{i1})]$ or $E_0[M_1 r_{i1}^S] = 1$, in which $r_{i1}^S \equiv (P_{i1} + D_{i1})/P_{i0}$ is the stock return, and $M_1 \equiv \rho U'(C_1)/U'(C_0)$ is the stochastic discount factor. On the production side, firm i uses the date-0 operating cash flow to pay investment and adjustment costs. If the free cash flow, $D_{i0} \equiv \Pi_{i0}A_{i0} - I_{i0} - (a/2)(I_{i0}/A_{i0})^2 A_{i0}$, is positive, the firm distributes it back to the household. A negative D_{i0} means external equity. At date 1, the firm uses assets, A_{i1} , to obtain the operating cash flow, $\Pi_{i1}A_{i1}$, which is in turn distributed as dividends, D_{i1} . With only 2 dates, the firm does not invest in date 1, $I_{i1} = 0$, and the ex-dividend equity value, P_{i1} , is zero. Taking the household's stochastic discount factor, M_1 , as given, firm i chooses I_{i0} to maximize the cum-dividend equity value at the beginning of date 0:

$$P_{i0} + D_{i0} \equiv \max_{\{I_{i0}\}} \Pi_{i0}A_{i0} - I_{i0} - \frac{a}{2} \left(\frac{I_{i0}}{A_{i0}} \right)^2 A_{i0} + E_0[M_1 \Pi_{i1}A_{i1}]. \quad (2)$$

The first principle for investment is given by:

$$1 + a \frac{I_{i0}}{A_{i0}} = E_0[M_1 \Pi_{i1}]. \quad (3)$$

Intuitively, to obtain an extra unit of assets at the beginning of date 1, firm i must pay the purchasing price of unity and the marginal adjustment costs. The marginal benefit of this extra unit of assets over period 1 is the marginal product of assets (profitability), Π_{i1} . Discounting this marginal benefit back to date 0 using M_1 yields marginal q , which equals the marginal costs of investment.

Using the definition of D_{i0} , we can derive the ex-dividend equity value from Equation (2) as $P_{i0} = E_0[M_1 \Pi_{i1}A_{i1}]$ at the optimum. As such, we can rewrite the stock return as

$r_{i1}^S = (P_{i1} + D_{i1})/P_{i0} = \Pi_{i1}A_{i1}/E_0[M_1\Pi_{i1}A_{i1}] = \Pi_{i1}/E_0[M_1\Pi_{i1}]$. Equation (3) then implies $r_{i1}^S = \Pi_{i1}/(1 + a(I_{i0}/A_{i0}))$. Intuitively, firm i will keep investing until the marginal costs of investment at date 0, $1 + a(I_{i0}/A_{i0})$, equal the marginal benefit of investment at date 1, Π_{i1} , discounted to date 0 with the stock return, r_{i1}^S , as the discount rate. Equivalently, the ratio of the date-1 marginal benefit of investment over the date-0 marginal costs of investment should equal the discount rate.⁵

1.2 Implications

Taking the expectation on both sides of $r_{i1}^S = \Pi_{i1}/(1 + a(I_{i0}/A_{i0}))$, yields:

$$E_0[r_{i1}^S] = \frac{E_0[\Pi_{i1}]}{1 + a(I_{i0}/A_{i0})}. \quad (4)$$

This equation predicts that, all else equal, high investment stocks should earn lower expected returns than low investment stocks earn, and that, all else equal, high expected profitability stocks should earn higher expected returns than low expected profitability stocks earn. When expected returns vary over time and across firms, stock prices would adjust in a way that connects expected returns to investment and profitability per Equation (4). In particular, stock prices would not adjust to give rise to a cross-sectionally constant discount rate, which would mean that investment and profitability do not predict returns in the cross

⁵We have opted to derive the characteristics-return relation in the simplest setting possible. However, the relation holds under substantially more general conditions. Cochrane (1991) derives this relation in a dynamic setting. Incorporating taxes and debt, Liu, Whited, and Zhang (2009) show that the investment return (the ratio of the next period marginal benefit of investment over the current period marginal costs of investment) equals the weighted average cost of capital (WACC), $w_{it}r_{it+1}^{Ba} + (1 - w_{it})r_{it+1}^S$, in which w_{it} is the market leverage at the beginning of period t , and r_{it+1}^{Ba} is the after-tax corporate bond return over period t . This equivalence in effect provides the microfoundation for the standard WACC approach to capital budgeting in corporate finance. With the notations in our 2-period setup, the present value of the marginal project is $\Pi_{i1}/[w_{i0}r_{i1}^{Ba} + (1 - w_{i0})r_{i1}^S]$, which uses the WACC as the discount rate. The costs of the marginal project remain $1 + a(I_{i0}/A_{i0})$. The first principle requires the marginal costs to be equal to the present value of the marginal project, meaning that its net present value is zero.

section. A cross-sectionally constant discount rate can only arise if all firms are equally risky, and stock prices follow a random walk.

1.2.1 The investment channel. Equation (4) predicts that given the expected profitability, expected returns decrease with investment-to-assets. This investment channel is consistent with many cross-sectional patterns including the negative relations of average returns with net stock issues, composite issuance, accruals, valuation ratios, and long-term prior returns (reversal). Figure 1 illustrates this insight.

The negative investment-expected return relation is intuitive. Firms invest more when their marginal q (the net present value of future cash flows generated from an additional unit of assets) is high. Given expected profitability or cash flows, low discount rates imply high marginal q and high investment, and high discount rates imply low marginal q and low investment. This intuition is probably most transparent in the capital budgeting context. In our setting, assets are homogeneous, meaning no difference between project costs of capital and firm costs of capital. Given expected cash flows, high costs of capital imply low net present values of new projects and in turn low investment, and low costs of capital imply high net present values of new projects and in turn high investment.⁶

The negative investment-return relation is conditional on expected profitability. Invest-

⁶The negative investment-expected return relation (a downward-sloping investment demand curve) has a long tradition in financial economics. Fisher (1930) and Fama and Miller (1972, Figure 2.4) show that the interest rate and investment are negatively correlated. As noted, Cochrane (1991) and Liu, Whited, and Zhang (2009) extend this insight into a dynamic world with uncertainty. In our simple economic model, project-level discount rates equal firm-level expected returns. This equivalence no longer holds in more general settings with project heterogeneity as in real options models such as Berk, Green, and Naik (1999); Carlson, Fisher, and Giammarino (2004); and Kogan and Papanikolaou (2013). However, these real options models also predict the negative investment-expected return relation. Intuitively, expansion options are riskier than assets in place. Investment converts riskier expansion options into less risky assets in place, causing high investment firms to be less risky and earn lower expected returns than low investment firms. A notable exception is Gomes, Kogan, and Zhang (2003), in which projects are assumed to be distributed randomly across firms, implying a flat investment-expected return relation. However, as pointed out in Zhang (2005), this flat relation gives rise to a counterfactual prediction that value stocks have higher cash flow durations than growth stocks, inconsistent with the evidence in Dechow, Sloan, and Soliman (2004).

ment is linked to profitability because firms that are more profitable tend to invest more than less profitable firms. This conditional relation provides a natural portfolio interpretation for the investment channel. Sorting on net stock issues, composite issuance, book-to-market, and other valuation ratios is closer to sorting on investment than in expected profitability. These sorts produce wider cross-sectional expected return spreads associated with investment than those associated with expected profitability. As such, we can interpret these diverse sorts using their common implied sort on investment.

The negative relation between average returns and equity issues is consistent with the negative investment-expected return relation. The balance-sheet constraint of firms implies that a firm's uses of funds must equal its sources of funds, meaning that, all else equal, issuers must invest more and earn lower average returns than nonissuers.⁷ Cooper, Gulen, and Schill (2008) show that asset growth predicts future returns with a negative slope. However, asset growth is the most comprehensive measure of investment-to-assets, in which investment is measured as the change in total assets. As such, the asset growth effect seems to be the premier manifestation of the investment channel.

The value premium is also consistent with the negative investment-return relation. Investment increases in marginal q (the denominator of Equation (4)), and the marginal q equals the average q under constant returns to scale. The average q and market-to-book are highly correlated and are identical without debt financing. Consequently, value firms with high book-to-market (low market-to-book) should invest less and earn higher expected returns than growth firms with low book-to-market (high market-to-book). In general,

⁷Lyandres, Sun, and Zhang (2008) show that adding an investment factor to the Capital Asset Pricing Model (CAPM) and the Fama-French model reduces the magnitude of the underperformance following initial public offerings, seasoned equity offerings, and convertible debt offerings. Lyandres et al. also report the part of Figure 1 that is related to the new issues puzzle.

compared to firms with low valuation ratios, firms with high valuation ratios should have more growth opportunities, invest more, and earn lower expected returns.

High valuation ratios often result from a stream of positive shocks on fundamentals, and low valuation ratios from a stream of negative shocks on fundamentals. High valuation ratios of growth firms can manifest as high long-term prior returns, and low valuation ratios of value firms as low long-term prior returns. As such, firms with high long-term prior returns should invest more and earn lower expected returns than firms with low long-term prior returns, meaning that the investment channel also helps interpret the De Bondt and Thaler (1985) long-term reversal effect.

1.2.2 The profitability channel. Equation (4) also gives rise to the profitability channel that works in parallel with the investment channel. Given investment-to-assets, firms with high expected profitability should earn higher expected returns than firms with low expected profitability. The profitability-expected return relation is consistent with momentum, post-earnings-announcement drift, and the financial distress effect.

Why should high expected profitability firms earn higher expected returns than low expected profitability firms? Equation (4) says that the expected return equals the expected profitability divided by an increasing function of investment-to-assets. From the capital budgeting perspective, high expected profitability relative to low investment must mean high discount rates, which are necessary to offset the high expected profitability to induce low net present values of new capital and low investment. If the discount rates are not high enough to counteract the high expected profitability, firms would instead observe high net present values of new capital and invest more. Similarly, low expected profitability relative to high investments must mean low discount rates (as in small-growth firms in the 1990s).

If the discount rates are not low enough to counteract the low expected profitability, these firms would instead observe low net present values of new capital and invest less.

The profitability-expected return relation is also consistent with the standard discounting model. The marginal costs of investment in the denominator of the right-hand side of Equation (4) equal marginal q , which in turn equals average q or market-to-book. As such, the expected return equals the expected profitability divided by market-to-book. Multiplying the numerator and the denominator with book equity equates the expected return with the ratio of the expected cash flows over the market equity. This relation is analogous to the Gordon Growth model. In a 2-period world, the equity value equals the expected cash flows divided by the discount rate. High expected cash flows relative to low market equity (high expected profitability relative to low market-to-book) imply high discount rates. Analogously, low expected cash flows relative to high market equity (low expected profitability relative to high market-to-book) imply low discount rates (e.g., Berk 1995).

The profitability-expected return relation has important implications. For any sorts that produce wider cross-sectional expected return spreads associated with expected profitability than with investment, their average returns can be interpreted with the common implied sort on expected profitability. Examples include momentum, distress, and earning surprises. In particular, momentum winners have higher expected profitability and should earn higher expected returns than momentum losers. Intuitively, shocks to profitability are positively correlated with stock returns. Firms with positive profitability shocks tend to experience immediate stock price increases, whereas firms with negative profitability shocks tend to experience immediate stock price decreases.

In addition, less financially distressed firms are more profitable, meaning higher ex-

pected profitability, and, all else equal, should earn higher expected returns than more financially distressed firms. As such, the distress effect is consistent with the positive profitability-expected return relation. Finally, sorting on earnings surprises should produce an expected profitability spread between extreme portfolios. Intuitively, all else equal, firms that have experienced large positive shocks to earnings tend to be more profitable than firms that have experienced large negative shocks to earnings.⁸

1.3 Limitations

We implement the economic model in Equation (4) with factor regressions. We construct factor mimicking portfolios on investment and profitability (as a proxy for expected profitability) in a way that is analogous to the size and book-to-market factors in Fama and French (1993, 1996). We then use the investment and profitability factors as right-hand side variables in factor regressions. The factor approach often delivers better empirical performance than the economic model itself. One reason is that stock returns data are available at high frequencies and are less subject to measurement errors than accounting variables. Perhaps more important, implementing the economic model directly via structural estimation involves specification errors in the production and capital adjustment technologies, specification errors that are absent in the factor model.

However, although in part inspired by Equation (4) derived from q -theory, the q -factor model is largely a reduced form, empirical model. In particular, the factor model requires

⁸We have so far only described anomaly variables that are directly related to investment and profitability. In our empirical tests, we confront the q -factor model with a substantially broader set of anomalies. As argued in Lin and Zhang (2013), the consumption model and the investment model of asset pricing are equivalent in general equilibrium, delivering identical expected returns. While the consumption model says that consumption risks are sufficient for accounting for expected returns, the investment model says that characteristics are sufficient. We take the latter prediction seriously and confront the q -factor model with a wide array of anomaly variables that are not directly related to investment and profitability.

that returns of stocks with similar investment (and returns of stocks with similar profitability) comove together, a prediction that Equation (4) does not make, at least not directly. To ground the q -factor model more rigorously in theories of investment-based asset pricing, one would need to go beyond the first principle in Equation (4). To this end, one would need to specify fully a dynamic investment model to quantify the comovement behind the q -factors, as well as the sources of the cross-sectional heterogeneity in investment, profitability, and their factor loadings. While offering some useful guidance, our empirical work makes little theoretical contribution in this direction.

2 Factors

Monthly returns, dividends, and prices are from the Center for Research in Security Prices (CRSP), and accounting information is from the Compustat Annual and Quarterly Fundamental Files. The sample is from January 1972 to December 2012. The starting date is restricted by the availability of quarterly earnings announcement dates, as well as quarterly earnings and book equity data. Financial firms and firms with negative book equity are excluded from the sample.

2.1 Factor construction

We measure investment-to-assets, I/A , as the annual change in total assets (Compustat annual item AT) divided by 1-year-lagged total assets. We measure profitability as ROE, which is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity.⁹ We construct the q -factors from a triple 2-by-3-by-3 sort

⁹Our measure of the book equity is the quarterly version of the annual book equity measure in Davis, Fama, and French (2000). In particular, book equity is shareholders' equity, plus balance-sheet deferred taxes

on size, I/A, and ROE. Because both the investment and earnings effects in the data are stronger in small firms than in big firms (e.g., Bernard and Thomas 1990; Fama and French 2008), we control for size when constructing the investment and ROE factors. Sorting jointly with size is also standard in constructing the value factor, HML, and the momentum factor, UMD. HML is from a double 2-by-3 sort on size and book-to-market, and UMD is from a double 2-by-3 sort on size and prior 2–12 month returns. Finally, sorting on investment and ROE independently helps orthogonalize the 2 new factors.

Specifically, at the end of June of each year t , we use the median NYSE size (stock price per share times shares outstanding from CRSP) to split NYSE, Amex, and NASDAQ stocks into 2 groups, small and big. Independently, at the end of June of year t , we break NYSE, Amex, and NASDAQ stocks into 3 I/A groups, using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked values of I/A for the fiscal year ending in calendar year $t - 1$.

In addition, independently, at the beginning of each month, we sort all stocks into 3 groups based on the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked values of ROE. Earnings data in Compustat quarterly files are used in the months immediately after the most recent public quarterly earnings announcement dates (Compustat quarterly item RDQ). For example, if the earnings for the fourth fiscal quarter of year $t - 1$ are publicly announced on March 5 (or March 25) of year t , we use the

and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ) in that order as shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock. Fama and French (2006) measure shareholders' equity as total assets minus total liabilities. We follow Davis et al. because Compustat quarterly items SEQQ (stockholders' equity) and CEQQ (common equity) have a broader coverage than items ATQ (total assets) and LTQ (total liabilities) before 1980.

announced earnings (divided by the book equity from the third quarter of year $t - 1$) to form portfolios at the beginning of April of year t . In addition, for a firm to enter the factor construction, we require the end of the fiscal quarter that corresponds to its most recently announced quarterly earnings to be within 6 months prior to the portfolio formation. We impose this restriction to exclude stale earnings information.

Taking the intersections of the 2 size, 3 I/A, and 3 ROE groups, we form 18 portfolios. Monthly value-weighted portfolio returns are calculated for the current month, and the portfolios are rebalanced monthly. (The ROE portfolios are rebalanced monthly at the beginning of each month, and the size and I/A portfolios are rebalanced annually at the end of each June.) The size factor, r_{ME} , is the difference (small-minus-big), each month, between the simple average of the returns on the 9 small size portfolios and the simple average of the returns on the 9 big size portfolios. Designed to mimic the common variation in returns related to I/A, the investment factor, $r_{I/A}$, is the difference (low-minus-high), each month, between the simple average of the returns on the 6 low I/A portfolios and the simple average of the returns on the 6 high I/A portfolios. Finally, designed to mimic the common variation in returns related to ROE, the ROE factor is the difference (high-minus-low), each month, between the simple average of the returns on the 6 high ROE portfolios and the simple average of the returns on the 6 low ROE portfolios.

2.2 Empirical properties

From Panel A of Table 1, the size factor earns an average return of 0.31% per month from January 1972 to December 2012 ($t = 2.12$). For comparison, the average return of SMB is 0.19% ($t = 1.35$). From Panel B, our size factor and SMB have an almost perfect

correlation of 0.95. The data for SMB, HML, and UMD are from Kenneth French's Web site. We construct the market factor, MKT, as the value-weighted market return minus the 1-month Treasury bill rate from CRSP.

The investment factor, $r_{I/A}$, earns an average return of 0.45% per month ($t = 4.95$). Its CAPM alpha is 0.52% ($t = 5.93$, Panel A). The Fama-French alpha is 0.33% ($t = 4.85$), and the Carhart alpha 0.28% ($t = 3.85$). Panel B shows that the investment factor and HML have a significant correlation of 0.69, suggesting that the investment factor would play a similar role as HML in factor regressions.

The ROE factor, r_{ROE} , earns an average return of 0.58% per month ($t = 4.81$). The Fama-French alpha of 0.77% ($t = 6.94$) and the R^2 of only 20% suggest that r_{ROE} represents an important source of common return variation missing from the Fama-French model. The Carhart model cannot capture the average ROE factor return either, with an alpha of 0.50% ($t = 4.75$). From Panel B, r_{ROE} has a high correlation of 0.50 with UMD, suggesting that r_{ROE} would play a similar role as UMD in factor regressions. Finally, the investment and the ROE factors have a small, insignificant correlation of 0.06, meaning that the triple sort succeeds in orthogonalizing the 2 q -factors.

Raising an important concern on data mining in empirical finance, Harvey, Liu, and Zhu (2013) argue that the standard significance criterion with a t -statistic (testing that the average return is zero) greater than 2 is inappropriate for newly discovered factors. Instead, new factors must clear a much higher hurdle, with t -statistics greater than 3. Harvey et al. also argue that factors based on first principles should have a lower hurdle of significance than empirically motivated factors. The investment and ROE factors, with at least some connection to investment-based asset pricing and t -statistics of 4.95 and 4.81, respectively,

seem to clear their high hurdle.

The average returns of HML and UMD are 0.40% per month ($t = 2.60$) and 0.72% ($t = 3.41$), respectively, in our sample. More importantly, the alphas of HML and UMD in the q -factor model are only 0.06% and 0.13%, respectively; both of which are within 1 standard error from zero. As such, the q -factor model captures the average returns of HML and UMD, but the Fama-French and Carhart models cannot capture the average returns of the 2 q -factors. The evidence suggests that HML and UMD might be noisy versions of the investment and ROE factors, respectively.

2.3 Methodological issues

A few clarifications on our factor construction are in order. First, we design the q -factors to account for anomalies in the broad cross section of returns, as opposed to anomalies that exist only in extremely small and highly illiquid stocks. This design is reflected in our use of NYSE breakpoints and value-weighted portfolio returns in forming the q -factors. Consequently, the stocks that reside in the q -factors represent the broad cross section. The percentages of the total market capitalization for stocks in the size, investment, and ROE factors are 92.76%, 50.42%, and 58.07% of our entire sample, respectively. The percentage for the size factor is not 100% because we require stocks to have valid investment and ROE data. These percentages are largely in line with those for SMB, HML, and UMD, which are 97.20%, 66.92%, and 54.48%, respectively, in our sample.

Second, sorting on investment and ROE jointly is consistent with the economic model in Equation (4), which shows that the investment and ROE effects are conditional in nature. Firms will invest a lot if either the ROE of their investment is high or the cost of

capital is low or both. As such, the negative relation between investment and the cost of capital is conditional on a given level of ROE. Investment and the cost of capital could be positively correlated unconditionally, if large investment delivers disproportionately high ROE. Analogously, the positive relation between ROE and the cost of capital is conditional on a given level of investment. ROE and the cost of capital could be negatively correlated unconditionally, if high ROE comes with disproportionately large investment. Sorting on investment and ROE jointly controls for these conditional relations.

Third, when constructing the q -factors, we form the investment portfolios annually but the ROE portfolios monthly. This design is also consistent with the economic model in Equation (4). The equation implies, for the most part, contemporaneous correlations between investment and anomaly variables, such as equity issues, accruals, and book-to-market, which the investment factor is hypothesized to capture. Because these anomaly portfolios are all constructed annually, it seems natural to use the same frequency to construct the investment factor. Analogously, Equation (4) implies contemporaneous correlations between ROE and anomaly variables, such as price momentum, earnings surprise, and financial distress, which the ROE factor is hypothesized to capture. Because these anomaly portfolios are all constructed monthly, it seems natural to adopt the same frequency to construct the ROE factor. In addition, Equation (4) says that ROE predicts future stock returns to the extent that it predicts future ROE. Because the most recent ROE contains the most up-to-date information about future ROE, we use the latest ROE in our monthly sorts.

3 Empirical Results

To evaluate the performance of the q -factor model, we use time-series factor regressions:

$$r_t^i - r_t^f = \alpha_q^i + \beta_{\text{MKT}}^i \text{MKT}_t + \beta_{\text{ME}}^i r_{\text{ME},t} + \beta_{\text{I/A}}^i r_{\text{I/A},t} + \beta_{\text{ROE}}^i r_{\text{ROE},t} + \epsilon^i. \quad (5)$$

If the model is well specified, α_q^i should be economically small and statistically insignificant from zero. Because we neutralize the 2 q -factors against size, we include the size factor in the q -factor model. Including the size factor also brings the q -factor model to the same footing, in terms of the number of factors, as the Carhart model. Finally, Equation (4) is primarily a cross-sectional model. As such, we include the market factor to capture the common variation in returns over time, while accounting for the cross-sectional variation in average returns with the q -factors.

3.1 Testing portfolios and overall performance

We report our key empirical results in this subsection.

3.1.1 The playing field. Table 2 lists the 74 primary anomalies that we study. Adding 6 momentum-reversal variables (momentum strategies for holding periods longer than 6 months, see Section 3.2.1), we examine in total 80 anomaly variables. Our goal is to be conceptually comprehensive yet empirically parsimonious. To be comprehensive, we cover all the major anomaly categories, including momentum, value-versus-growth, investment, profitability, intangibles, as well as trading frictions.

To be parsimonious, we select the most important variables in each category to avoid (excessive) redundancy. For example, we use Daniel and Titman's (2006) composite is-

suance and Pontiff and Woodgate's (2008) net stock issues as representative examples of equity financing anomalies, but do not separately study Loughran and Ritter's (1995) underperformance anomaly following seasoned equity offerings or Ikenberry, Lakonishok, and Vermaelen's (1995) overperformance anomaly following open market share repurchases. In addition, we have examined 2 different holding periods (1-month and 6-month) for many anomaly portfolios that are monthly rebalanced, such as Ang et al.'s (2006) total, idiosyncratic, and systematic volatilities; Campbell, Hilscher, and Szilagyi's (2008) failure probability; and Balakrishnan, Bartov, and Faurel's (2010) return on assets. However, we only report for a given variable the holding period that delivers the higher average return spread across the deciles. Doing so raises the hurdle on the q -factor model.

Our scope of nearly 80 anomalies is comparable to that of McLean and Pontiff (2013), who use 82 variables (but do not provide the detailed list). Green, Hand, and Zhang (2013) identify more than 330 return predictive signals but use only 60 in their tests. Harvey, Liu, and Zhu (2013) identify 314 factors, but many are macroeconomic risk factors, such as Treasury bill return and aggregate consumption growth. In contrast, we examine firm-level anomalies. Moreover, our universe covers most (if not all) of the major categories of anomalies studied in Harvey et al.'s work.

We detail the variable definition and the construction of testing portfolios in Appendix A. As the benchmark procedure in constructing these portfolios, we use NYSE breakpoints and value-weighted portfolio returns. Doing so is consistent with the construction of the q -factors and SMB, HML, and UMD. Another important reason for the benchmark practice is to alleviate the impact of microcaps, which are defined as stocks with market capitalization below the 20th NYSE percentile. As shown in Fama and French (2008), despite accounting

for about 60% of the total number of stocks, microcaps are on average only about 3% of the market capitalization of the NYSE-Amex-NASDAQ universe. Because of transaction costs and lack of liquidity, the portion of anomalies in microcaps is unlikely to be exploitable in practice.¹⁰

When constructing annually sorted testing portfolios, such as the book-to-market deciles, we follow the Fama and French (1993) timing. At the end of June of each year t , we sort all stocks into deciles, using NYSE breakpoints for book-to-market measured at the fiscal year end in calendar year $t - 1$, and calculate value-weighted decile returns from July of year t to June of $t + 1$. To construct monthly sorted testing portfolios involving latest earnings data, such as the ROE deciles, we follow our timing in constructing the ROE factor. In particular, earnings data in Compustat quarterly files are used in the months immediately after the quarterly earnings announcement dates. Finally, to construct monthly sorted testing portfolios involving quarterly accounting data other than earnings, such as the failure probability deciles, we follow the accounting literature in imposing a 4-month lag between the sorting variable and holding period returns (e.g., Hirshleifer et al. 2004). Unlike earnings, other quarterly data items might not be available upon earnings announcement dates. As such, we impose the 4-month lag to guard against potential look-ahead bias.

3.1.2 Insignificant anomalies in the broad cross section. Out of our list of anomalies, 38 are insignificant in the broad cross section. Table 3 reports the average high-minus-low decile returns and their t -statistics for the insignificant anomalies.

¹⁰We document that extreme deciles under the benchmark procedure assign only modest portfolio weights to microcaps, but those under an alternative procedure with NYSE-Amex-NASDAQ breakpoints and equal-weighted returns assign excessively large weights (see the Online Appendix). For example, across the 10 momentum variables, the loser deciles allocate on average 10.85% of the portfolio weights to microcaps under the benchmark procedure, but 63.58% under the alternative procedure. For the winner deciles, the comparison is between 5.17% and 52.50%.

Most importantly, 12 out of 13 anomalies in the trading frictions category are insignificant at the 5% level. The only significant anomaly is Ang et al.'s (2006) systematic volatility. However, their idiosyncratic volatility anomaly is not. The high-minus-low decile earns on average -0.54% per month ($t = -1.56$). Even Jegadeesh's (1990) short-term reversal anomaly earns an insignificant -0.31% ($t = -1.39$). Other notable examples are Diether, Malloy, and Scherbina's (2002) dispersion of analysts' earnings forecasts and Amihud's (2002) illiquidity. The evidence suggests that trading frictions play a limited role in the broad cross section.

Outside trading frictions, Titman, Wei, and Xie's (2004) abnormal investment anomaly yields -0.27% per month, Richardson et al.'s (2005) total accrual anomaly yields -0.19% , and Piotroski's (2000) F -score earns 0.37% . All 3 are insignificant. Gompers, Ishii, and Metrick's (2003) corporate governance index earns a tiny 0.03% ($t = 0.09$), and Francis et al.'s (2005) accrual quality measure earns -0.18% ($t = -0.79$).

It is surprising that 38 anomalies (about one-half across all categories), including 12 out of 13 anomalies in the trading frictions category, are insignificant in the broad cross section. Lending support to Harvey, Liu, and Zhu (2013), we suggest that many claims in the anomalies literature are likely exaggerated. While Harvey et al. adopt a sophisticated multiple testing framework to raise the statistical hurdle for significance, we use simple NYSE breakpoints and value-weighted portfolio returns to ensure that the anomalies we study are relevant for the broad cross section.

3.1.3 Significant anomalies: Pricing errors and tests of overall performance.

Table 4 reports the overall performance of various factor models in fitting the 35 significant anomalies in the broad cross section. The q -factor model performs well relative to the

Carhart model and even more so relative to the Fama-French model. In particular, across the 35 high-minus-low deciles, the average magnitude of the q -alphas is 0.20% per month, which is lower than 0.33% in the Carhart model and 0.55% in the Fama-French model. Moreover, only 5 out of 35 high-minus-low deciles have q -alphas that are significant at the 5% level. In contrast, 19 high-minus-low deciles have significant Carhart alphas, and 27 have significant Fama-French alphas.

To measure a model's overall performance, we also use the mean absolute value of alphas across the deciles formed on a given anomaly variable, as well as the p -value associated with the GRS test (on the null that the alphas are jointly zero across a given set of deciles). Table 4 shows that the q -factor model has the lowest average magnitude of alphas across all 35 sets of deciles, 0.11% per month. The average magnitude is 0.12% in the Carhart model and 0.16% in the Fama-French model. In addition, the GRS test rejects the q -factor model at the 5% level in 20 out of 35 sets of deciles, the Carhart model in 24 sets of deciles, and the Fama-French model in 28 sets of deciles.

Looking at different categories of anomalies, we observe that the q -factor model outperforms the Fama-French and Carhart models in all except for the value-versus-growth category, in which the 3 models are largely comparable. Across the 9 anomalies in the momentum category, the average magnitude of the high-minus-low alphas is 0.19% per month in the q -factor model, 0.29% in the Carhart model, and 0.85% in the Fama-French model. Out of 9 high-minus-low deciles, 2 have significant alphas in the q -factor model, in contrast to 5 in the Carhart model and all 9 in the Fama-French model. Across the 9 sets of momentum deciles, the average magnitude of the alphas is 0.10% in both the q -factor model and the Carhart model but 0.19% in the Fama-French model. The GRS test rejects the q -factor

model and the Carhart model in 6 out of 9 sets of deciles, but rejects the Fama-French model in all 9.

Across the 5 value-versus-growth anomalies, the high-minus-low net payout yield decile has significant alphas in all 3 models. However, the high-minus-low q -alpha is 0.36% per month, which is lower than the Carhart alpha, 0.49%, and the Fama-French alpha, 0.52%. All 3 models are rejected by the GRS test across the net payout yield deciles. In addition, none of the high-minus-low alphas for the book-to-market, earnings-to-price, cash flow-to-price, and duration deciles are significant in the q -factor model, the Carhart model, or the Fama-French model. The GRS statistics are mostly insignificant. However, while the q -factor model is rejected across the cash flow-to-price deciles, the Carhart and Fama-French models are not.

In the investment category, only 1 out of 11 high-minus-low q -alpha is significant, compared to 6 Carhart alphas and 10 Fama-French alphas. The average magnitude of the high-minus-low alphas is 0.22% per month in the q -factor model, which is lower than 0.30% in the Carhart model and 0.38% in the Fama-French model. However, all 3 models are rejected by the GRS test in most sets of deciles in the investment category. The q -factor model performs the best in the profitability category. None of the 5 high-minus-low alphas are significant in the q -factor model, but all 5 are significant in the Carhart and Fama-French models. The average magnitude of the high-minus-low alphas is 0.12% in the q -factor model, which is substantially lower than 0.61% in the Carhart model and 0.95% in the Fama-French model. The q -factor model is rejected by the GRS test in only 1 out of 5 sets of deciles, but the Carhart and Fama-French models are rejected in all 5.

However, we emphasize that the q -factor model underperforms the Fama-French and

Carhart models in fitting Sloan's (1996) operating accrual anomaly (see Section 3.2.2) and Chan, Lakonishok, and Sougiannis's (2001) R&D-to-market anomaly. In particular, the high-minus-low R&D-to-market decile earns an average return of 0.63% per month ($t = 2.31$). The Fama-French alpha is 0.22% ($t = 0.93$), the Carhart alpha is 0.31% ($t = 1.40$), but the q -alpha is 0.60% ($t = 2.40$).

Finally, as noted, we include the size factor in the q -factor model to bring it to the same footing as the Carhart model in terms of the number of factors. Including the size factor helps the q -factor model fit the average returns across the size deciles. The small-minus-big decile earns an average return of 0.24% per month ($t = 0.90$). The Fama-French and Carhart alphas are small and insignificant. The q -alpha is tiny, 0.03% ($t = 0.29$), but would be 0.66% ($t = 2.16$) without the size factor.

More importantly, however, the incremental effect of the size factor in capturing the 35 significant anomalies is rather limited, probably because the anomaly variables are not directly related to size. Table 5 replicates Table 4, but for an alternative 3-factor q -model without the size factor. As noted, the magnitude of the high-minus-low alphas averaged across the 35 anomalies is 0.20% per month in the 4-factor q -model. Dropping the size factor raises it to 0.23%, which still beats 0.33% in the Carhart model and 0.55% in the Fama-French model. The mean absolute alpha averaged across the deciles is 0.11% in the 4-factor q -model. Dropping the size factor raises it to only 0.12%, compared to 0.12% in the Carhart model and 0.16% in the Fama-French model.

Overall, except for the operating accrual anomaly and the R&D-to-market anomaly, the q -factor model performs as well as, and often outperforms the Fama-French and Carhart models across major categories of anomalies. The q -factor model beats the Carhart model

and by a bigger margin the Fama-French model in the momentum category. The q -factor model also outperforms in the investment category and dominates the 2 competing models in the profitability category. The 3 models are largely comparable in the value-versus-growth category. Finally, the size factor plays only a limited role in the q -factor model's success.

3.1.4 Q -factor loadings and economic characteristics. To shed light on the sources of the good performance of the q -factor model, we examine the q -factor loadings and their underlying characteristics, including size, investment-to-assets, and ROE for the high-minus-low deciles. We align the timing of characteristics with the timing of portfolio returns as in the construction of the q -factors. In particular, for size and investment-to-assets, we align returns from July of year t to June of $t + 1$ with size at the end of June of year t and investment-to-assets for the fiscal year ending in calendar year $t - 1$. For ROE, we align returns for month t with the ROE calculated with the most recently announced quarterly earnings.

Firm-level characteristics are aggregated to the portfolio level as in Fama and French (1995). Size is the average market capitalization (in billions of dollars) across all firms in a given decile. Investment-to-assets (in percent) is the sum of changes in total assets across all firms in a given decile divided by the sum of 1-year-lagged assets across the same set of firms. Finally, ROE (in percent) is the sum of the most recently announced quarterly earnings across all firms in a given decile divided by the sum of their 1-quarter-lagged book equity.

Table 6 shows that many significant anomalies in Table 4 are basically different combinations of the investment and ROE effects. The first 9 columns in the upper panel suggest

that price and earnings momentum are mainly linked to the ROE factor. Across the 9 high-minus-low deciles, the ROE factor loadings vary from 0.18 to 1.48, all of which are significant. Consistent with the pattern in ROE factor loadings, the spreads in ROE characteristic between winners and losers are economically large, ranging from 1.49% to 6.58% per quarter. In contrast, the investment factor loadings are mostly insignificant for the high-minus-low deciles.

The next 5 columns (from “B/M” to “Dur”) in the upper panel of Table 6 show that the value-versus-growth anomalies are mainly linked to the investment factor. The loadings vary from -0.85 to 1.45 across the high-minus-low deciles, all of which are more than 5 standard errors from zero. Accordingly, the corresponding spreads in investment-to-assets vary from -14.43% to 3.95% per annum. In contrast, the ROE factor loadings for the high-minus-low deciles are mostly insignificant.

The last 4 columns in the upper panel and the first 7 columns in the lower panel show that the investment, equity financing, inventory, and accrual anomalies are primarily linked to the investment factor. The net operating assets (NOA) anomaly is one exception, for which none of the factor loadings are significant for the high-minus-low decile. Leaving aside NOA and operating accruals (OA, which we examine in depth in Section 3.2.2), the investment factor loadings vary from -0.65 to -1.37 across the high-minus-low deciles, all of which are more than 5 standard errors from zero. The corresponding spreads in investment-to-assets are also economically large, varying from 11.12% to 83.89% per annum. In addition, similar to the value-versus-growth deciles, the ROE factor loadings for the high-minus-low deciles in the investment category are mostly insignificant.

The next 5 columns (from “ROE” to “FP”) show that the profitability and distress

anomalies are primarily captured by the ROE factor. The ROE factor loadings of the high-minus-low deciles vary from -1.62 to 1.50 , all of which are more than 7 standard errors from zero. The corresponding ROE spreads are large, ranging from -8.74% per quarter to 16.95% . In contrast, the investment factor loadings of the high-minus-low deciles are all small and often insignificant.

We comment on 3 results from the remaining columns in the lower panel. First, the organizational capital (OC) effect can be captured by the q -factor model. The high-minus-low decile has an investment factor loading of 0.35 ($t = 3.52$) and an ROE factor loading of 0.51 ($t = 7.12$). Consistent with the loadings, despite being more profitable by 1.52% per quarter, high OC firms invest less than low OC firms by 13.77% per annum. As a result, the q -factor model does a good job in fitting the average return spread of 0.56% per month ($t = 4.07$), leaving an alpha of 0.09% ($t = 0.66$, Table 4). In contrast, the Carhart alpha is 0.40% ($t = 2.97$), and the Fama-French alpha is 0.61% ($t = 4.52$).

Second, column “R&D/M” in Table 6 sheds light on why the q -factor model fails to capture the R&D-to-market anomaly. High R&D firms have lower ROE than low R&D firms, with a spread of 2.80% per quarter. Intuitively, because R&D is expensed per standard accounting practice, high R&D expenses give rise to artificially low ROE. As a result, the ROE factor loading of the high-minus-low R&D/M decile is -0.58 ($t = -4.10$), which goes in the wrong direction in fitting the average returns. Adjusting the earnings data for R&D firms by capitalizing instead of expensing R&D might improve the q -factor model’s performance in fitting this anomaly.

Third, the ROE factor largely captures Ang et al.’s (2006) systematic volatility (Svol) effect. As shown in Table 4, the high-minus-low Svol decile earns a significant average return

of -0.60% per month. Whereas the Carhart model fails to capture the Svol effect, leaving a significant alpha of -0.62% , the q -factor model delivers an insignificant alpha of -0.37% ($t = -1.42$). From the last column in Table 6, the high-minus-low decile has an ROE factor loading of -0.43 ($t = -3.54$) and an ROE spread of -0.64% per quarter. In contrast, both the investment factor loading and the investment-to-assets spread are insignificant.¹¹

Overall, many anomalies are basically different manifestations of the investment and ROE effects. The ROE factor is the main source of the q -factor model's empirical power to capture anomalies in the momentum and profitability categories. The investment factor is the main source of the q -factor model's power to capture anomalies in the value-versus-growth and investment categories. Finally, a combination of the 2 factors helps the q -factor model fit the other anomalies.

3.2 Detailed results for selected anomalies

In this subsection, we report more detailed results for earnings and price momentum, which are classic anomalies to the Fama-French model. We also present detailed results for the accrual anomaly, which the q -factor model fails to capture, as well as the 25 size and book-to-market portfolios, which are the key testing portfolios for Fama and French (1993, 1996).¹²

3.2.1 Earnings momentum (SUE-1) and price momentum (R6-6). Table 7 reports

¹¹Ang et al.'s (2006) idiosyncratic volatility (Ivol) anomaly has received more attention in the literature than their Svol effect. As shown in Table 3, the high-minus-low Ivol decile earns an average return of -0.54% per month ($t = -1.56$). In the Online Appendix, we show that the high-minus-low Ivol decile has a Fama-French alpha of -0.92% ($t = -4.66$) and a Carhart alpha of -0.59% ($t = -2.75$). In contrast, the q -alpha is only -0.08% ($t = -0.39$). Differing from the Svol effect, both q -factors help capture the Ivol anomaly. The investment and ROE factor loadings of the high-minus-low Ivol decile are both -0.94 and are more than 5 standard errors from zero.

¹²The q -factor model's performance across the Fama-French (1997) 10 industry portfolios is largely comparable to that of the Fama-French model and that of the Carhart model (see the Online Appendix).

factor regressions for individual SUE-1 and R6-6 deciles. Across the SUE-1 deciles, 5 have significant alphas in the Fama-French model versus 3 in the Carhart model and 1 in the q -factor model. Across the R6-6 deciles, 4 alphas are significant in the Fama-French model versus 3 in the Carhart model and zero in the q -factor model.

The q -factor model captures momentum via the ROE factor. Moving from SUE-1 decile 1 to 10, the ROE factor loadings increase from -0.22 to 0.26 , and moving from R6-6 decile 1 to 10, the loadings increase from -0.74 to 0.28 . Accordingly, moving from SUE-1 decile 1 to 10, ROE increases from -0.80% to 5.00% per quarter, and moving from R6-6 decile 1 to 10, it increases from -0.71% to 3.43% . Intuitively, earnings and price momentum winners are more profitable and load more heavily on the ROE factor than earnings and price momentum losers, respectively.

Chan, Jegadeesh, and Lakonishok (1996) show that momentum profits are short-lived. To examine this issue, we calculate the returns of earnings and price momentum deciles for holding periods longer than 6 months. At the beginning of month t , we sort all stocks on their most recent SUE and calculate value-weighted decile returns for the holding periods from month t to $t + 11$, from $t + 12$ to $t + 35$, and from $t + 36$ to $t + 59$. In addition, at the beginning of month t , we sort all stocks on their prior 6-month returns from $t - 7$ to $t - 2$, skipping month $t - 1$, and calculate value-weighted decile returns for the holding periods from month t to $t + 11$, from $t + 12$ to $t + 35$, and from $t + 36$ to $t + 59$. We call these portfolios designed to capture the reversal of momentum profits momentum-reversal deciles.

Confirming the short-lived nature of momentum, Table 8 shows that none of the earnings momentum-reversal high-minus-low deciles earn significant average returns. Except

for the 12-month holding period, the price momentum-reversal high-minus-low deciles do not earn significant average returns either. More importantly, the table shows that the q -factor model succeeds in capturing the short-lived nature of momentum. In particular, the q -factor model does not produce significant alphas for any of the momentum-reversal high-minus-low deciles.

3.2.2 Operating accruals (OA) and percent operating accruals (POA). Sloan (1996) shows that firms with high operating accruals earn lower average returns than firms with low operating accruals. Accruals are often scaled by lagged assets or average assets over the prior 2 years. Hafzalla, Lundholm, and Van Winkle (2011) show that scaling accruals by the absolute value of earnings (percent accruals) is more effective in selecting firms for which the differences between sophisticated and naive earnings forecasts are the most extreme.

Panel A of Table 9 shows why the q -factor model fails to fit the OA anomaly. The low OA decile, despite having lower investment-to-assets (13.87% per annum) than the high OA decile (24.02%), has a large, negative investment factor loading (-0.56), which is close to that of the high OA decile (-0.58). More important, the ROE factor loading is -0.26 for the low decile and 0.03 for the high decile, and the spread is more than 4 standard errors from zero (Table 6). Intuitively, because earnings equal operating cash flows plus accruals, high OA firms tend to be more profitable and load more heavily on the ROE factor than low OA firms. This pattern in ROE factor loadings goes in the wrong direction in capturing the OA anomaly.

Panel B shows why the q -factor model performs better for the POA anomaly. The ROE factor loadings are largely flat across the deciles. Intuitively, because POA scales accruals

with earnings, the earnings in the denominator work against the positive correlation between accruals and earnings in the numerator. In addition, the investment factor loadings fall from 0.35 for the low POA decile to -0.55 for the high POA decile, going in the right direction in fitting the average returns.

3.2.3 25 size (ME) and book-to-market (B/M) portfolios. Table 10 reports that the high-minus-low B/M quintile earns an average return of 1.02% per month ($t = 4.59$) in small stocks but only 0.16% ($t = 0.79$) in big stocks. The Fama-French model does a good job in fitting the average returns of the 25 portfolios. The average magnitude of the alphas is only 0.10%, but their model is still rejected by the GRS test. Only 5 out of 25 portfolios have significant alphas. In particular, the small-growth portfolio has an alpha of -0.54% ($t = -4.84$).¹³ The performance of the Carhart model is largely comparable.

The q -factor model's performance is also similar. The average magnitude of the alphas is 0.11% per month, which is close to 0.10% in the Fama-French model and 0.11% in the Carhart model. Out of 25, 4 portfolios have significant alphas in the q -factor model, compared with 5 in the Fama-French model and 5 in the Carhart model. More importantly, only 1 out of 5 high-minus-low B/M quantiles has a significant alpha in the q -factor model, compared with 3 in the Fama-French model and 2 in the Carhart model. The small-growth portfolio has an insignificant q -alpha of only -0.25% ($t = -1.48$), but the small-value q -alpha is 0.32% ($t = 2.72$).

The q -factor model's performance is mostly from the investment factor. From the panel denoted " $\beta_{I/A}$," value stocks have significantly higher investment factor loadings than growth stocks. The loadings spreads range from 1.19 to 1.56 across the size quintiles.

¹³The small-growth anomaly is notoriously difficult to capture (e.g., Fama and French 1993, 1996; Davis, Fama, and French 2000; and Campbell and Vuolteenaho 2004).

The panel denoted “I/A” shows that value stocks invest less than growth stocks, and the investment-to-assets spreads vary from 6.14% per annum to 13.94% across the size quintiles. In contrast, the value-minus-growth loadings on the other factors are mostly small and insignificant.

3.3 Sharpe ratios

In the mean-variance framework, a factor model can account for all the anomalies if and only if the mean-variance efficient portfolio from combining all the anomaly portfolios lies in the span of the factors. As such, the mean-variance efficient combination of the factors should have a Sharpe ratio that is greater than or equal to the maximum Sharpe ratio from combining all the anomaly portfolios. Complementing the results from factor regressions, Table 11 reports Sharpe ratios for different (combinations of) factors and anomaly portfolios.

Panel A reports the monthly Sharpe ratio for each individual factor, calculated as the mean return divided by its volatility. The investment and ROE factors have the highest Sharpe ratios, 0.24 and 0.22, compared with the Sharpe ratios of 0.13 and 0.16, respectively, for HML and UMD. Following MacKinlay (1995), we also calculate the maximum Sharpe ratio achievable from a given factor model as $\sqrt{\mu_f' V_f^{-1} \mu_f}$, in which μ_f is the vector of mean factor returns, and V_f is the variance-covariance matrix of the factor returns. Panel B shows that the Fama-French model produces a maximum Sharpe ratio of 0.21, the Carhart model 0.30, and the q -factor model 0.43.

Panel C reports the Sharpe ratios for the high-minus-low deciles and the maximum Sharpe ratios from combining deciles formed on a given anomaly variable. In general, these Sharpe ratios are lower than the maximum Sharpe ratio of 0.43 for the q -factor model.

However, once we pool together 28 out of the 35 high-minus-low deciles that cover the entire sample from January 1972 to December 2012, we obtain a maximum Sharpe ratio of 0.48. We also obtain a maximum Sharpe ratio of 1.60 from pooling together all 28 sets of deciles that have valid data for the entire sample.¹⁴

Overall, consistent with factor regressions, the Sharpe ratios in Table 11 suggest that the q -factor model is more effective than the Carhart and Fama-French models, but that the q -factor model is by no means perfect in capturing all the anomalies.

The Sharpe ratios also help in interpreting the q -factor model, which we view as a parsimonious empirical model for estimating expected stock returns. In particular, we wish to emphasize that the q -factor model is silent about the debate between rational asset pricing and mispricing. This interpretation is somewhat weaker than the risk factors interpretation per Fama and French (1993, 1996).

Consistent with Fama and French (1993, 1996), our factor regressions provide direct evidence that the q -factors capture shared variation in returns across a wide array of anomaly portfolios. To the extent that the q -factors, constructed on economic fundamentals, represent common variation in stock returns, their loadings represent covariances between an individual asset's returns with the factor returns. The Sharpe ratios in Table 11 lend further support to the common variation captured by the q -factors. If stocks with similar investment or stocks with similar profitability do not comove together, it would be possible

¹⁴Calculating the maximum Sharpe ratio achievable from pooling together all the 35 sets of deciles on their common sample starting in February 1986 (the starting date of the sample for the Svol deciles) is infeasible. The reason is that the number of observations is not large enough to estimate the large number of parameters in the variance-covariance matrix. Instead, we calculate the maximum Sharpe ratios from pooling together all the deciles in each category of anomalies. The estimates are 0.67 across the 9 sets of momentum deciles (from "SUE-1" to "I-Mom") on their common sample starting in January 1977; 0.49 across the 5 sets of value-versus-growth deciles on the sample starting in July 1972; 0.65 across the 11 sets of investment deciles on the full sample; 0.49 over the 5 sets of profitability deciles on the sample starting in January 1976; and 0.59 across the remaining 5 sets of deciles on the sample starting in February 1986.

to diversify away the variances of the q -factors, giving rise to extremely high Sharpe ratios. Table 11 shows otherwise. The Sharpe ratios for the q -factors are high, but not excessively high, indicating comovement in returns.

However, our evidence is not inconsistent with mispricing. If waves of investor sentiment affect stocks with similar investment or stocks with similar profitability simultaneously, the q -factor model would work in the data as well. Moreover, under the mispricing hypothesis, trading opportunities with mild Sharpe ratios can persist, even though opportunities with extreme Sharpe ratios are arbitrated away. MacKinlay (1995), in particular, argues that the maximum Sharpe ratio for the Fama-French model is too high to be consistent with rational asset pricing. As shown in Table 11, the maximum Sharpe ratio for the q -factor model is even higher. Finally, perhaps most important, our work is silent about the structural sources behind the common variation in the q -factors and the heterogeneity in their loadings across the diverse anomaly portfolios (Section 1.3).

4 Conclusion

Our examination of nearly 80 anomalies yields 2 major findings. About one-half of the anomalies earn insignificant average returns for the high-minus-low deciles formed with NYSE breakpoints and value-weighted returns. The evidence suggests that many claims in the anomalies literature seem exaggerated, likely by excessively weighting on microcaps. More importantly, an empirical q -factor model consisting of the market factor, a size factor, an investment factor, and a profitability factor outperforms the Fama-French and Carhart models in capturing many (but not all) of the significant anomalies. Many seemingly unrelated anomalies turn out to be different manifestations of the investment and profitability

effects. These empirical results highlight the importance of understanding the driving forces behind the q -factors and their broad empirical power in the cross section.

Accepted Manuscript

Figure Legends

Figure 1

The investment channel

Accepted Manuscript

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Appendix A

A.1 Variable Definition and Portfolio Construction

For each anomaly variable, we describe its detailed definition and portfolio construction.

A.1.1 Momentum

This category includes 10 momentum variables, including SUE-1, SUE-6, Abr-1, Abr-6, RE-1, RE-6, R6-1, R6-6, R11-1, and I-Mom, as well as 6 momentum-reversal variables: SUE-12, SUE-13-36, SUE-37-60, R6-12, R6-13-36, and R6-37-60.

A.1.1.1 SUE-1, SUE-6, SUE-12, SUE-13-36, and SUE-37-60. Following Foster, Olsen, and Shevlin (1984), we measure earnings surprise as standardized unexpected earnings (SUE). We calculate SUE as the change in the most recently announced quarterly earnings per share from its value 4 quarters ago, divided by the standard deviation of this change in quarterly earnings over the prior 8 quarters (6 quarters minimum).

For a firm to enter our portfolio formation, we require the end of the fiscal quarter that corresponds to the most recently announced earnings to be within 6 months prior to the portfolio formation. We do so to exclude stale information on earnings. We impose this restriction when constructing the ROE factor and portfolios that use quarterly earnings data (including SUE, Abr, ROE, ROA, NEI, and RS). To avoid potentially erroneous records, we also require the earnings announcement date to be after the corresponding fiscal quarter end.

At the beginning of each month t , we split all NYSE, Amex, and NASDAQ stocks into deciles based on their most recent past SUE using NYSE breakpoints. Monthly value-weighted portfolio returns are calculated, separately, for the current month t (SUE-1), from month t to $t+5$ (SUE-6), from month t to $t+11$ (SUE-12), from month $t+12$ to $t+35$ (SUE-13-36), and from month $t+36$ to $t+59$ (SUE-37-60). The deciles are rebalanced monthly. The holding period that is longer than 1 month as in, for instance, SUE-6, means that for a given SUE-6 decile in each month there exist 6 subdeciles, each of which is initiated in a different month in the prior 6-month period. We take the simple average of the subdecile returns as the monthly return of the SUE-6 decile.

A.1.1.2 R6-1, R6-6, R6-12, R6-13-36, R6-37-60, and R11-1. At the beginning of each month t , we split all stocks into deciles based on their prior 6-month returns from month $t-7$ to $t-2$ using NYSE breakpoints. Skipping month $t-1$, we calculate monthly value-weighted decile returns, separately, for month t (R6-1), from month t to $t+5$ (R6-6), from month t to $t+11$ (R6-12), from month $t+12$ to $t+35$ (R6-13-36), and from month $t+36$ to $t+59$ (R6-37-60). All the deciles are rebalanced at the beginning of month $t+1$. The holding period that is longer than 1 month as in, for instance, R6-6, means that for a given R6-6 decile in each month there exist 6 subdeciles, each of which is initiated in a different month in the prior 6-month period. We take the simple average of the subdeciles returns as the monthly return of the R6-6 decile.

To construct the R11-1 deciles as in Fama and French (1996), we split all stocks into deciles at the beginning of each month t based on their prior 11-month returns from month $t-12$ to $t-2$ using NYSE breakpoints. Skipping month $t-1$, we calculate monthly value-weighted decile returns for month t , and the deciles are rebalanced at the beginning

of month $t + 1$. Because we exclude financial firms, these decile returns are different from those posted on Kenneth French's Web site.

A.1.1.3 Abr-1 and Abr-6. Following Chan, Jegadeesh, and Lakonishok (1996), we also measure earnings surprise as cumulative abnormal stock return (Abr) around the latest quarterly earnings announcement date:

$$\text{Abr}_i = \sum_{d=-2}^{+1} r_{id} - r_{md}, \quad (\text{A1})$$

in which r_{id} is stock i 's return on day d (with the earnings announced on day 0), and r_{md} is the return on the market index. We cumulate returns until 1 (trading) day after the announcement date to account for the possibility of 1-day-delayed stock price reaction to earnings news. r_{md} is the value-weighted return on the market index.

At the beginning of each month t , we split all stocks into deciles based on their most recent past Abr using NYSE breakpoints. Monthly value-weighted decile returns are calculated for the current month t (Abr-1), and, separately, from month t to $t + 5$ (Abr-6). The deciles are rebalanced monthly. The 6-month holding period for Abr-6 means that for a given decile in each month there exist 6 subdeciles, each of which is initiated in a different month in the prior 6-month period. We take the simple average of the subdecile returns as the monthly return of the Abr-6 decile.

A.1.1.4 RE-1 and RE-6. Following Chan, Jegadeesh, and Lakonishok (1996), we also measure earnings surprise as changes in analysts' forecasts of earnings. The earnings forecast data are from the Institutional Brokers' Estimate System (IBES). Because analysts' forecasts are not necessarily revised each month, we construct a 6-month moving average of past changes in analysts' forecasts:

$$\text{RE}_{it} = \sum_{j=1}^6 \frac{f_{it-j} - f_{it-j-1}}{p_{it-j-1}}, \quad (\text{A2})$$

in which f_{it-j} is the consensus mean forecast (IBES unadjusted file, item MEANEST) issued in month $t - j$ for firm i 's current fiscal year earnings (fiscal period indicator = 1), and p_{it-j-1} is the prior month's share price (unadjusted file, item PRICE). We require a minimum of 4 monthly forecast changes when constructing RE.

At the beginning of each month t , we split all stocks into deciles, based on their RE using NYSE breakpoints. Monthly value-weighted decile returns are calculated for the current month t (RE-1), and, separately, from month t to $t + 5$ (RE-6). The deciles are rebalanced monthly. The 6-month holding period for RE-6 means that for a given decile in each month there exist 6 subdeciles, each of which is initiated in a different month in the prior 6-month period. We take the simple average of the subdecile returns as the monthly return of the RE-6 decile. Because of the availability of analysts' forecasts data, the sample for RE-1 and RE-6 decile returns starts in January 1977.

A.1.1.5 I-Mom. Moskowitz and Grinblatt (1999) document profitable industry momentum strategies that buy stocks in past winning industries and sell stocks in past losing industries. We use the Fama-French 49-industry classifications. Excluding financial firms from the sample leaves 45 industries. At the beginning of each month t , we sort industries based on their prior six-month value-weighted returns from $t - 6$ to $t - 1$. Following Moskowitz and Grinblatt, we do not skip month $t - 1$ when measuring industry momentum. We form 9 portfolios ($9 \times 5 = 45$), each of which contains 5 different industries. We define the return of a given portfolio as the equal-weighted average of the 5 industry returns within the portfolio. We hold the 9 portfolios for 6 months from t to $t + 5$, and rebalance the portfolios at the beginning of $t + 1$. For a given I-Mom portfolio in each month there exist 6 subportfolios, each of which is initiated in a different month in the prior 6-month period. We take the simple average of the 6 subportfolio returns as the monthly return of the I-Mom portfolio.

A.1.2 Value-versus-growth

We start by describing the construction of the 25 size (ME) and book-to-market (B/M) portfolios. In addition, Table 2 lists 12 anomaly variables in this category including B/M, A/ME, Rev, E/P, EF/P, CF/P, D/P, O/P, NO/P, SG, LTG, and Dur.

A.1.2.1 The 25 ME and B/M Portfolios. At the end of June of each year t , we split stocks into quintiles based on the June-end ME (price times shares outstanding from CRSP). Independently, we split stocks into quintiles based on B/M, which is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the ME (from Compustat or CRSP) at the end of December of $t - 1$. We use NYSE breakpoints for both ME and B/M. Taking intersections, we form 25 ME and B/M portfolios and calculate monthly value-weighted portfolio returns from July of year t to June of $t + 1$.

Following Davis, Fama, and French (2000), we measure book equity as stockholders' book equity, plus balance-sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if available. If not, stockholders' equity is the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets (item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

A.1.2.2 B/M. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on B/M, which is the book equity for the fiscal year ending in calendar year $t - 1$ divided by the ME (from Compustat or CRSP) at the end of December of $t - 1$. We calculate monthly value-weighted decile returns from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Following Davis, Fama, and French (2000), we measure book equity as stockholders' book equity, plus balance-sheet deferred taxes and investment tax credit (Compustat annual item TXDITC) if available, minus the book value of preferred stock. Stockholders' equity is the value reported by Compustat (item SEQ), if it is available. If not, we measure stockholders' equity as the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or the book value of assets

(item AT) minus total liabilities (item LT). Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock.

A.1.2.3 A/ME. We follow Fama and French (1992) and measure A/ME as the ratio of total book assets to ME. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on A/ME. Total assets (Compustat annual item AT) are from the fiscal year ending in calendar year $t - 1$ and the ME (price per share times shares outstanding from Compustat or CRSP) is at the end of December of $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.2.4 Rev. To capture the De Bondt and Thaler (1985) long-term reversal (Rev) effect, at the beginning of each month t , we use NYSE breakpoints to split stocks into deciles based on the prior returns from month $t - 60$ to $t - 13$. Monthly value-weighted decile returns are computed for month t , and the deciles are rebalanced at the beginning of $t + 1$. To be included in a portfolio for month t , a stock must have a valid price at the end of $t - 61$ and a valid return for $t - 13$. In addition, any missing returns from month $t - 60$ to $t - 14$ must be -99.0 , which is the CRSP code for a missing price.

A.1.2.5 E/P. To construct the Basu (1983) earnings-to-price (E/P) deciles, we use NYSE breakpoints to split stocks into deciles based on E/P at the end of June of each year t . E/P is calculated as income before extraordinary items (Compustat annual item IB) for the fiscal year ending in calendar year $t - 1$ divided by the ME (from Compustat or CRSP) at the end of December of $t - 1$. Stocks with negative earnings are excluded. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.2.6 EF/P. Following Elgers, Lo, and Pfeiffer (2001), we measure analysts' earnings forecasts-to-price (EF/P) as the consensus median forecasts (IBES unadjusted file, item MEDEST) for the current fiscal year (fiscal period indicator = 1) divided by share price (unadjusted file, item PRICE). At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on EF/P estimated with forecasts in month $t - 1$. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of $t + 1$. Because the earnings forecast data start in January 1976, the EF/P decile returns start in February 1976.

A.1.2.7 CF/P. We measure cash flows (CF) as income before extraordinary items (Compustat annual item IB), plus equity's share of depreciation (item DP), plus deferred taxes (if available, item TXDI). The equity's share is defined as ME divided by total assets (item AT) minus book equity plus ME. ME is share price times shares outstanding from Compustat or CRSP. Book equity is the stockholder's book equity, plus balance-sheet deferred taxes and investment tax credit (if available, item TXDITC), minus the book value of preferred stock. Depending on availability, we use redemption (item PSTKRV), liquidating (item PSTKL), or par value (item PSTK) for the book value of preferred stock. Stockholder's equity is the

value reported by Compustat (item SEQ), or the book value of common equity (item CEQ) plus the par value of preferred stock (item PSTK), or total assets minus total liabilities (item LT).

At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles, based on CF for the fiscal year ending in calendar year $t - 1$ divided by the ME at the end of December of $t - 1$. We exclude firms with negative CFs. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.2.8 D/P. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on their dividend yields, which are the total dividends paid out from July of year $t - 1$ to June of t divided by the ME (from CRSP) at the end of June of t . Following Fama and French (1988), we calculate monthly dividends as the begin-of-month ME times the difference between cum- and ex-dividend returns. Monthly dividends are then accumulated from July of $t - 1$ to June of t . We exclude firms that do not pay dividends. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.2.9 O/P and NO/P. As in Boudoukh et al. (2007), total payouts are dividends on common stock (Compustat annual item DVC) plus repurchases. Repurchases are the total expenditure on the purchase of common and preferred stocks (item PRSTKC) plus any reduction (negative change over the prior year) in the value of the net number of preferred stocks outstanding (item PSTKRV). Net payouts equal total payouts minus equity issuances, which are the sale of common and preferred stock (item SSTK) minus any increase (positive change over the prior year) in the value of the net number of preferred stocks outstanding (item PSTKRV).

At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on total payouts (O/P) (or net payouts, NO/P) for the fiscal year ending in calendar year $t - 1$ divided by the ME at the end of December of $t - 1$. We exclude firms with non-positive total payouts (zero net payouts). Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the data on total expenditure of common and preferred stocks start in 1971, the O/P (NO/P) decile returns start in July 1972.

A.1.2.10 SG. Following Lakonishok, Shleifer, and Vishny (1994), we measure sales growth (SG) in June of year t as the weighted average of the annual SG ranks for the prior 5 years, $\sum_{j=1}^5 (6 - j) \times \text{Rank}(t - j)$. The SG for year $t - j$ is the growth rate in sales (COMPUSTAT annual item SALE) from fiscal year ending in $t - j - 1$ to fiscal year ending in $t - j$. Only firms with data for all 5 prior years are used to determine the annual SG ranks. For each year from $t - 5$ to $t - 1$, we rank stocks into deciles based on their annual SG, and then assign rank i ($i = 1, \dots, 10$) to a firm if its annual SG falls into the i^{th} decile. At the end of June of each year t , we use NYSE breakpoints to assign stocks into deciles based on SG, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.2.11 LTG. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on analysts' consensus median forecast of the long-term earnings growth rate (IBES item MEDEST, fiscal period indicator = 0). Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of $t + 1$. Because the long-term growth forecast data start in December 1981, the LTG decile returns start in January 1982.

A.1.2.12 Dur. Following Dechow, Sloan, and Soliman (2004), we calculate firm-level equity duration as:

$$\text{Dur} = \frac{\sum_{t=1}^T t \times \text{CD}_t / (1+r)^t}{\text{ME}} + \left(T + \frac{1+r}{r} \right) \frac{P - \sum_{t=1}^T \text{CD}_t / (1+r)^t}{\text{ME}}, \quad (\text{A3})$$

in which CD_t is the net cash distribution in year t , ME is market equity, T is the length of forecasting period, and r is the cost of equity. ME is price per share times shares outstanding (Compustat annual item PRCC.F times item CSHO). Net cash distribution, $\text{CD}_t = \text{BE}_{t-1}(\text{ROE}_t - g_t)$, in which BE_{t-1} is the book equity at the end of year $t - 1$, ROE_t is return on equity in year t , and g_t is the book equity growth in t . We use autoregressive processes to forecast ROE and book equity growth in future years. We model ROE as a first-order autoregressive process with an autocorrelation coefficient of 0.57 and a long-run mean of 0.12, and the growth in book equity as a first-order autoregressive process with an autocorrelation coefficient of 0.24 and a long-run mean of 0.06. For the starting year ($t = 0$), we measure ROE as income before extraordinary items (item IB) divided by 1-year-lagged book equity (item CEQ), and the book equity growth rate as the annual change in sales (item SALE). Nissim and Penman (2001) show that past sales growth is a better indicator of future book equity growth than past book equity growth. Finally, we use a forecasting period of $T = 10$ years and a cost of equity of $r = 0.12$.

At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on Dur for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.3 Investment

Table 2 lists 14 anomaly variables in this category, including ACI, I/A, NOA, $\Delta\text{PI}/\text{A}$, IG, NSI, CEI, NXF, IvG, IvC, OA, TA, POA, and PTA.

A.1.3.1 ACI. Following Titman, Wei, and Xie (2004), we measure ACI at the end of June of year t as $\text{CE}_{t-1} / [(\text{CE}_{t-2} + \text{CE}_{t-3} + \text{CE}_{t-4}) / 3] - 1$, in which CE_{t-j} is capital expenditure (Compustat annual item CAPX) scaled by sales (item SALE) for the fiscal year ending in calendar year $t - j$. The last 3-year average capital expenditure is designed to project the benchmark investment at the portfolio formation year. We exclude firms with sales less than 10 million dollars. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on their ACI. Monthly value-weighted decile returns are computed from July of year t to June of $t + 1$.

A.1.3.2 I/A. Following Cooper, Gulen, and Schill (2008), we measure investment-to-assets, I/A , for the portfolio formation year t as total assets (Compustat annual item AT) for the fiscal year ending in calendar year $t - 1$ divided by total assets for the fiscal year ending in $t - 2$ minus 1. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on I/A , and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.3.3 NOA. Following Hirshleifer, Hou, Teoh, and Zhang (2004), we define net operating assets (NOA) as operating assets minus operating liabilities. Operating assets are total assets (COMPUSTAT annual item AT) minus cash and short-term investment (item CHE). Operating liabilities are total assets minus debt included in current liabilities (item DLC, zero if missing), minus long-term debt (item DLTT, zero if missing), minus minority interests (item MIB, zero if missing), minus preferred stocks (item PSTK, zero if missing), and minus common equity (item CEQ). At the end of June of each year t , we use NYSE breakpoints to assign stocks into deciles based on NOA for the fiscal year ending in calendar year $t - 1$ scaled by total assets for the fiscal year ending in $t - 2$, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.3.4 $\Delta PI/A$. Following Lyandres, Sun, and Zhang (2008), we measure $\Delta PI/A$ as changes in gross property, plant, and equipment (Compustat annual item PPEGT) plus changes in inventory (item INVT) scaled by lagged total assets (item AT). At the end of June of each year t , we use NYSE breakpoints to assign stocks into deciles based on $\Delta PI/A$ for the fiscal year ending in calendar year $t - 1$, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.3.5 IG. Following Xing (2008), we measure investment growth, IG, for the portfolio formation year t as the growth rate in capital expenditure (Compustat annual item CAPX) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in $t - 1$. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on IG, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.3.6 NSI. Following Fama and French (2008), at the end of June of year t , we measure net stock issues (NSI) as the natural log of the ratio of the split-adjusted shares outstanding at the fiscal year ending in calendar year $t - 1$ to the split-adjusted shares outstanding at the fiscal year ending in $t - 2$. The split-adjusted shares outstanding is shares outstanding (Compustat annual item CSHO) times the adjustment factor (item AJEX). At the end of June of each year t , we use NYSE breakpoints to sort all stocks into deciles based on NSI. We exclude firms with zero NSI. Monthly value-weighted decile returns are from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.3.7 CEI. Following Daniel and Titman (2006), we measure CEI as the growth rate in the ME not attributable to the stock return, $\log(ME_t/ME_{t-5}) - r(t-5, t)$. For the portfolio formation at the end of June of year t , $r(t-5, t)$ is the cumulative log return on the stock from the last trading day of June in year $t - 5$ to the last trading day of June in year t , and ME_t is the ME on the last trading day of June in year t from CRSP. Equity issuance such as

seasoned equity issues, employee stock option plans, and share-based acquisitions increase the composite issuance, whereas repurchase activities such as share repurchases and cash dividends reduce the composite issuance. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles on CEI, and calculate monthly value-weighted decile returns from July of year t to June of year $t + 1$.

A.1.3.8 NXF. Following Bradshaw, Richardson, and Sloan (2006), we measure net external financing (NXF) as the sum of net equity financing and net debt financing. Net equity financing is the proceeds from the sale of common and preferred stocks (Compustat annual item SSTK) less cash payments for the repurchases of common and preferred stocks (item PRSTKC) less cash payments for dividends (item DV). Net debt financing is the cash proceeds from the issuance of long-term debt (item DLTIS) less cash payments for long-term debt reductions (item DLTR) plus the net changes in current debt (item DLCCH, zero if missing). At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on NXF for the fiscal year ending in calendar year $t - 1$ scaled by the average of total assets for fiscal years ending in $t - 2$ and $t - 1$. We exclude firms with zero NXF. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the coverage of the financing data starts in 1971, the NXF decile returns start in July 1972.

A.1.3.9 IvG. Following Belo and Lin (2011), we define inventory growth (IvG) for the portfolio formation year t as the growth rate in inventory (Compustat annual item INVT) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in $t - 1$. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on IvG. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the decile returns are rebalanced in June of $t + 1$.

A.1.3.10 IvC. Following Thomas and Zhang (2002), we define inventory changes (IvC) for the portfolio formation year t as the change in inventory (Compustat annual item INVT) from the fiscal year ending in calendar year $t - 2$ to the fiscal year ending in $t - 1$, scaled by the average of total assets for fiscal years ending in $t - 2$ and $t - 1$. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on IvC. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.3.11 OA. Prior to 1988, we use the balance-sheet approach of Sloan (1996) to measure operating accruals (OA) as changes in noncash working capital minus depreciation, in which the noncash working capital is changes in noncash current assets minus changes in current liabilities less short-term debt and taxes payable. In particular, OA equals $(\Delta CA - \Delta CASH) - (\Delta CL - \Delta STD - \Delta TP) - DP$, in which ΔCA is the change in current assets (Compustat annual item ACT), $\Delta CASH$ is the change in cash or cash equivalents (item CHE), ΔCL is the change in current liabilities (item LCT), ΔSTD is the change in debt included in current liabilities (item DLC, zero if missing), ΔTP is the change in income taxes payable (item TXP, zero if missing), and DP is depreciation and amortization (item DP, zero if missing). Starting from 1988, we follow Hribar and Collins (2002) to measure

OA using the statement of cash flows as net income (item NI) minus net cash flow from operations (item OANCF). Doing so helps mitigate measurement errors that can arise from nonoperating activities such as acquisitions and divestitures. Data from the statement of cash flows are only available since 1988. To construct the OA deciles, at the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on OA for the fiscal year ending in calendar year $t - 1$ scaled by total assets (item AT) for the fiscal year ending in $t - 2$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.3.12 TA. Prior to 1988, we use the balance-sheet approach in Richardson, Sloan, Soliman, and Tuna (2005) to measure total accruals (TA) as $\Delta WC + \Delta NCO + \Delta FIN$. ΔWC is the change in net non-cash working capital. Net non-cash working capital is current operating asset (COA) minus current operating liabilities (COL), with COA = current assets (Compustat annual item ACT) minus cash and short-term investments (item CHE) and COL = current liabilities (item LCT) minus debt in current liabilities (item DLC, zero if missing). ΔNCO is the change in net non-current operating assets. Net non-current operating assets is non-current operating assets (NCOA) minus non-current operating liabilities (NCOL), with NCOA = total assets (item AT) minus current assets (item ACT) minus investments and advances (item IVAO, zero if missing), and NCOL = total liabilities (item LT) minus current liabilities (item LCT) minus long-term debt (item DLTT, zero if missing). ΔFIN is the change in net financial assets. Net financial assets is financial assets (FINA) minus financial liabilities (FINL), with FINA = short-term investments (item IVST, zero if missing) plus long-term investments (item IVAO, zero if missing), and FINL = long-term debt (item DLTT, zero if missing) plus debt in current liabilities (item DLC, zero if missing) plus preferred stock (item PSTK, zero if missing).

Starting from 1988, we use the cash flow approach to measure TA as net income (Compustat annual item NI) minus total operating, investing, and financing cash flows (items OANCF, IVNCF, and FINCF) plus sales of stocks (item SSTK, zero if missing) minus stock repurchases and dividends (items PRSTKC and DV, zero if missing). Data from the statement of cash flows are only available since 1988. To construct the TA deciles, we use NYSE breakpoints to sort stocks at the end of June of each year t into deciles based on TA for the fiscal year ending in calendar year $t - 1$ scaled by total assets (Compustat annual item AT) for the fiscal year ending in $t - 2$. We calculate monthly value-weighted decile returns from July of year t to June of $t + 1$, and rebalance the deciles in June of $t + 1$.

A.1.3.13 POA. Accruals are traditionally scaled by total assets. Hafzalla, Lundholm, and Van Winkle (2011) show that scaling accruals by the absolute value of earnings (percent accruals) is more effective in selecting firms for which the differences between sophisticated and naive forecasts of earnings are the most extreme. To construct the percent operating accruals (POA) deciles, at the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on OA scaled by the absolute value of net income (Compustat annual item NI) for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.3.14 PTA. We use NYSE breakpoints to sort stocks at the end of June of each year t into deciles based on total accruals (TA) scaled by the absolute value of net income (Compustat annual item NI) for the fiscal year ending in calendar year $t - 1$. We calculate value-weighted decile returns from July of year t to June of $t + 1$, and rebalance the deciles in June of $t + 1$.

A.1.4 Profitability

Table 2 lists 14 anomaly variables in this category, including ROE, ROA, RNA, PM, ATO, CTO, GP/A, F , TES, TI/BI, RS, NEI, FP, and O .

A.1.4.1 ROE. We measure ROE as income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity. Book equity is shareholders' equity, plus balance-sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of preferred stock (item PSTKQ), or total assets (item ATQ) minus total liabilities (item LTQ), in that order, as shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of preferred stock.

At the beginning of each month t , we use NYSE breakpoints to sort all stocks into deciles based on ROE computed with the most recently announced quarterly earnings. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced monthly. Earnings data in Compustat quarterly files are used in the monthly sorts in the months immediately after the most recent public earnings announcement dates (item RDQ). If the earnings for the fourth fiscal quarter of year $t - 1$ are publicly announced on March 5 (or March 25) of year t , we use the announced earnings (divided by the book equity from the third quarter of year $t - 1$) to form portfolios at the beginning of April of year t . For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to the most recently announced earnings to be within 6 months prior to the portfolio formation. This restriction is imposed to exclude stale earnings information.

A.1.4.2 ROA. We measure ROA as income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged total assets (item ATQ). At the beginning of each month t , we use NYSE breakpoints to sort all stocks into deciles based on ROA computed with the most recently announced quarterly earnings. Monthly value-weighted decile returns are calculated for month t , and the deciles are rebalanced at the beginning of $t + 1$. For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to the most recently announced earnings to be within 6 months prior to the portfolio formation to exclude stale earnings information.

A.1.4.3 RNA, PM, and ATO. Following Soliman (2008), we use DuPont analysis to decompose ROE as $\text{RNA} + \text{FLEV} \times \text{SPREAD}$, in which ROE is return on equity, RNA is return on net operating assets, FLEV is financial leverage, and SPREAD is the difference between return on net operating assets and borrowing costs. We further decompose $\text{RNA} = \text{PM} \times \text{ATO}$, in which PM is profit margin (operating income/sales) and ATO is asset turnover (sales/net operating assets).

Following Soliman (2008), we use annual sorts to form RNA, PM, and ATO deciles. At the end of June of year t , we measure RNA as operating income after depreciation (Compustat annual item OIADP) for the fiscal year ending in calendar year $t - 1$ divided by net operating assets (NOA) for the fiscal year ending in $t - 2$. NOA is operating assets minus operating liabilities. Operating assets are total assets (item AT) less cash and short-term investments (item CHE) less other investment and advances (item IVAO, zero if missing). Operating liabilities are total assets (item AT), less the long- and short-term portions of debt (items DLTT and DLC, zero if missing), less minority interest (item MIB, zero if missing), less book value of preferred equity (items PSTK, zero if missing), less book value of common equity (items CEQ). PM is operating income after depreciation (item OIADP) divided by sales (item SALE) for the fiscal year ending in calendar year $t - 1$. ATO is sales (item SALE) for the fiscal year ending in calendar year $t - 1$ divided by net operating assets (NOA) for the fiscal year ending in $t - 2$. At the end of June of each year t , we use NYSE breakpoints to sort stocks into 3 sets of deciles based on RNA, PM, and ATO. We exclude firms with nonpositive NOA for the fiscal year ending in calendar year $t - 2$ when forming the RNA and the ATO deciles. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.4.4 CTO. To construct the Haugen and Baker (1996) capital turnover (CTO) deciles, at the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on sales (Compustat annual item SALE) divided by total assets (item AT) for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.4.5 GP/A. Following Novy-Marx (2013), we measure gross profits-to-assets (GP/A) as total revenue (Compustat annual item REVT) minus cost of goods sold (item COGS) divided by total assets (item AT, the denominator is current, not lagged, total assets). At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on GP/A for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.4.6 F . Piotroski (2000) classifies each firm's fundamental signal as either good or bad depending on the signal's implication for future stock prices and profitability. An indicator variable for a particular signal is 1 if its realization is good and zero if it is bad. The aggregate signal, denoted F , is the sum of the 9 binary signals. F is designed to measure the overall quality, or strength, of the firm's financial position. Nine fundamental signals are chosen to measure 3 areas of a firm's financial condition, profitability, liquidity, and operating efficiency.

Four variables are selected to measure profitability-related factors:

- ROA is income before extraordinary items (Compustat annual item IB) scaled by 1-year-lagged total assets (item AT). If the firm's ROA is positive, the indicator variable F_{ROA} equals 1, and zero otherwise.

- CFO is cash flow from operation scaled by 1-year-lagged total assets (Compustat annual item AT). Cash flow from operation is item OANCF, available after 1988. Prior to that, we use funds from operation (item FOPT) minus the annual change in working capital (item WCAP). If the firm's CFO is positive, the indicator variable F_{CFO} equals 1, and zero otherwise.
- ΔROA is the current year's ROA less the prior year's ROA. If ΔROA is positive, the indicator variable $F_{\Delta ROA}$ is 1, and zero otherwise.
- Sloan (1996) shows that earnings driven by positive accrual adjustments are a bad signal about future earnings. As such, the indicator F_{ACC} equals 1, if CFO is greater than ROA, and zero otherwise.

Three variables are selected to measure changes in capital structure and a firm's ability to meet future debt obligations. Piotroski (2000) assumes that an increase in leverage, a deterioration of liquidity, or the use of external financing is a bad signal about financial risk.

- $\Delta LEVER$ is the change in the ratio of total long-term debt (Compustat annual item DLTT) to average total assets over the prior 2 years. $F_{\Delta LEVER}$ is 1 if the firm's leverage ratio falls (i.e., $\Delta LEVER < 0$) in the year preceding portfolio formation, and zero otherwise.
- $\Delta LIQUID$ measures the change in a firm's current ratio between the current and prior years, in which the current ratio is the ratio of current assets (Compustat annual item ACT) to current liabilities (item LCT). An improvement in liquidity ($\Delta LIQUID > 0$) is a good signal about the firm's ability to service current debt obligations. The indicator $F_{\Delta LIQUID}$ equals 1 if the firm's liquidity improves, and zero otherwise.
- The indicator, EQ , equals 1 if the firm does not issue common equity in the year prior to portfolio formation, and zero otherwise. The issuance of common equity is sales of common and preferred stocks (Compustat annual item SSTK) minus any increase in preferred stock (item PSTK). Issuing equity is interpreted as a bad signal (inability to generate sufficient internal funds to service future obligations).

The remaining 2 signals are designed to measure changes in the efficiency of the firm's operations that reflect 2 key constructs underlying the decomposition of return on assets.

- $\Delta MARGIN$ is the firm's current gross margin ratio, measured as gross margin (Compustat annual item SALE minus item COGS) scaled by sales (item SALE), less the prior year's gross margin ratio. An improvement in margins signifies a potential improvement in factor costs, a reduction in inventory costs, or a rise in the price of the firm's product. The indicator $F_{\Delta MARGIN}$ equals 1 if $\Delta MARGIN > 0$ and zero otherwise.
- $\Delta TURN$ is the firm's current year asset turnover ratio, measured as total sales (Compustat annual item SALE) scaled by 1-year-lagged total assets (item AT), minus the prior year's asset-turnover ratio. An improvement in asset turnover ratio signifies greater productivity from the asset base. The indicator, $F_{\Delta TURN}$, equals 1 if $\Delta TURN$ is positive, and zero otherwise.

Piotroski (2000) forms a composite score, F , as the sum of the individual binary signals:

$$F = F_{\text{ROA}} + F_{\Delta\text{ROA}} + F_{\text{CFO}} + F_{\text{ACC}} + F_{\Delta\text{MARGIN}} + F_{\Delta\text{TURN}} + F_{\Delta\text{LEVER}} + F_{\Delta\text{LIQUID}} + EQ. \quad (\text{A4})$$

At the end of June of each year t , we sort stocks based on F for the fiscal year ending in calendar year $t - 1$ to form 7 portfolios: low ($F = 0, 1, 2$), 3, 4, 5, 6, 7, and high ($F = 8, 9$). Because extreme F scores are rare, we combine scores 0, 1, and 2 into the low portfolio, and scores 8 and 9 into the high portfolio. Monthly value-weighted portfolio returns are calculated from July of year t to June of $t + 1$, and the portfolios are rebalanced in June of $t + 1$. Because the data on equity offering start in 1971, the F portfolio returns start in July 1972.

A.1.4.7 TES. Following Thomas and Zhang (2011), we measure tax expense surprise (TES) as changes in tax expense, which is tax expense per share (Compustat quarterly item TXTQ/(item CSHPRQ times item AJEXQ)) in quarter q minus tax expense per share in quarter $q - 4$, scaled by assets per share (item ATQ/(item CSHPRQ times item AJEXQ)) in quarter $q - 4$. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on their TES calculated with Compustat quarterly data items from at least 4 months ago. We exclude firms with zero TES (most of these firms pay no taxes). We calculate deciles returns for the subsequent 3 months from t to $t + 2$, and the portfolios are rebalanced at the beginning of month $t + 1$. The 3-month holding period means that in each month for any given TES decile there exist 3 subdeciles. We take the simple average of the value-weighted subdecile returns as the monthly return of the TES decile. For sufficient data coverage, we start the TES deciles in January 1976.

A.1.4.8 TI/BI. Following Green, Hand, and Zhang (2013), we measure taxable income-to-book income (TI/BI) as pretax income (Compustat annual item PI) divided by net income (item NI). At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles, based on TI/BI for the fiscal year ending in calendar year $t - 1$. We exclude firms with negative or zero net income. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$.

A.1.4.9 RS. Following Jegadeesh and Livnat (2006), we measure revenue surprise (RS) as the change in revenue per share (Compustat quarterly item SALEQ/(item CSHPRQ times item AJEXQ)) from its value 4 quarters ago divided by the standard deviation of this change in quarterly revenue per share over the prior 8 quarters (6 quarters minimum). Jegadeesh and Livnat argue that quarterly revenue data are available when earnings are announced. As such, we use the same timing convention as for the SUE-1 deciles. At the beginning of each month t , we use NYSE breakpoints to split stocks into deciles, based on RS at the most recent quarterly earnings announcement date (item RDQ). Monthly value-weighted deciles returns are calculated for the current month t . To avoid stale information, we require the end of the fiscal quarter that corresponds to the most recent quarterly earnings announcement to be within 6 months prior to the portfolio formation.

A.1.4.10 NEI. We follow Barth, Elliott, and Finn (1999) and Green, Hand, and Zhang (2013) in measuring NEI as the number of consecutive quarters (up to 8 quarters) with an

increase in earnings (Compustat quarterly item IBQ) over the same quarter in the prior year. At the beginning of each month t , we sort stocks into 9 portfolios (with $NEI = 0, 1, 2, \dots, 7$, and 8, respectively) based on NEI calculated at the most recent earnings announcement date (item RDQ). To avoid stale information, we require the end of the fiscal quarter that corresponds to the most recent earnings announcement to be within 6 months prior to the portfolio formation. We calculate monthly value-weighted portfolio returns for the current month t and rebalance the portfolios at the beginning of $t + 1$.

A.1.4.11 FP. We construct failure probability (FP) following Campbell, Hilscher, and Szilagyi (2008, the third column in Table IV):

$$\begin{aligned} FP_t \equiv & -9.164 - 20.264 NIMTAAVG_t + 1.416 TLMTA_t - 7.129 EXRETAVG_t \\ & + 1.411 SIGMA_t - 0.045 RSIZE_t - 2.132 CASHMTA_t + 0.075 MB_t - 0.058 PRICE_t \end{aligned} \quad (A5)$$

in which

$$NIMTAAVG_{t-1,t-12} \equiv \frac{1 - \phi^3}{1 - \phi^{12}} (NIMTA_{t-1,t-3} + \dots + \phi^9 NIMTA_{t-10,t-12}) \quad (A6)$$

$$EXRETAVG_{t-1,t-12} \equiv \frac{1 - \phi}{1 - \phi^{12}} (EXRET_{t-1} + \dots + \phi^{11} EXRET_{t-12}), \quad (A7)$$

and $\phi = 2^{-1/3}$. $NIMTA$ is net income (Compustat quarterly item NIQ) divided by the sum of market equity (share price times the number of shares outstanding from CRSP) and total liabilities (item LTQ). The moving average $NIMTAAVG$ captures the idea that a long history of losses is a better predictor of bankruptcy than 1 large quarterly loss in a single month. $EXRET \equiv \log(1 + R_{it}) - \log(1 + R_{S\&P500,t})$ is the monthly log excess return on each firm's equity relative to the S&P 500 index. The moving average $EXRETAVG$ captures the idea that a sustained decline in stock market value is a better predictor of bankruptcy than a sudden stock price decline in a single month.

$TLMTA$ is the ratio of total liabilities divided by the sum of market equity and total liabilities. $SIGMA$ is the annualized 3-month rolling sample standard deviation: $\sqrt{\frac{252}{N-1} \sum_{k \in \{t-1, t-2, t-3\}} r_k^2}$, in which k is the index of trading days in months $t-1, t-2$, and $t-3$, r_k^2 is the firm-level daily return, and N is the total number of trading days in the 3-month period. $SIGMA$ is treated as missing if there are less than five nonzero observations over the three months in the rolling window. $RSIZE$ is the relative size of each firm measured as the log ratio of its market equity to that of the S&P 500 index. $CASHMTA$, used to capture the liquidity position of the firm, is the ratio of cash and short-term investments (Compustat quarterly item CHEQ) divided by the sum of market equity and total liabilities (item LTQ).

MB is the market-to-book equity, in which book equity is measured in the same way as the denominator of ROE. Following Campbell et al., we add 10% of the difference between the market equity and the book equity to the book equity to alleviate measurement issues for extremely small book equity values. For firm-month observations that still have negative book equity after this adjustment, we replace these negative values with \$1, to ensure that

the market-to-book ratios for these firms are in the right tail of the distribution. *PRICE* is each firm's log price per share, truncated above at \$15. We further eliminate stocks with prices less than \$1 at the portfolio formation date. We winsorize the variables on the right-hand side of Equation (A5) at the 5th and 95th percentiles of their pooled distribution across all firm-month observations.

To construct the FP deciles, at the beginning of each month t , we use NYSE breakpoints to split stocks into deciles based on FP calculated with accounting data from the fiscal quarter ending at least 4 months ago. Because unlike earnings, other quarterly data items in the definition of FP might not be available upon earnings announcement, we impose a 4-month gap between the fiscal quarter end and portfolio formation to guard against look-ahead bias. We calculate decile returns for the subsequent 6 months after the portfolio formation from month t to $t+5$ and rebalance the deciles at the beginning of $t+1$. (Holding the decile returns for only the current month t yields an insignificant average return of -0.47% per month ($t = -1.25$) for the high-minus-low FP decile.) Because of the 6-month holding period, there exist 6 subdeciles for a given FP decile in each month. We take the simple average of the value-weighted subdecile returns as the monthly return of the FP decile. Because of the limited data coverage, we start the FP deciles in January 1976.

A.1.4.12 O. We follow Ohlson (1980, Model 1 in Table 4) and Dichev (1998) to construct *O*-score:

$$\begin{aligned} & -1.32 - 0.407 \log(TA) + 6.03 TLTA - 1.43 WCTA + 0.076 CLCA \\ & - 1.72 OENEG - 2.37 NITA - 1.83 FUTL + 0.285 INTWO - 0.521 CHIN, \end{aligned}$$

in which TA is total assets (Compustat annual item AT). $TLTA$ is the leverage ratio defined as the book value of debt (item DLC plus item DLTT) divided by total assets. $WCTA$ is working capital divided by total assets (item ACT minus item LCT)/item AT. $CLCA$ is current liability (item LCT) divided by current assets (item ACT). $OENEG$ is 1 if total liabilities (item LT) exceeds total assets (item AT) and is zero otherwise. $NITA$ is net income (item NI) divided by total assets. $FUTL$ is the fund provided by operations (item PI) divided by total liabilities (item LT). $INTWO$ is equal to 1 if net income (item NI) is negative for the last 2 years, and zero otherwise. $CHIN$ is $(NI_t - NI_{t-1})/(|NI_t| + |NI_{t-1}|)$, in which NI_t is net income (item NI).

At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on *O*-score for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.5 Intangibles

Table 2 lists 11 anomaly variables in this category, including OC/A, BC/A, Ad/M, RD/S, RD/M, RC/A, H/N, OL, G , AccQ, and Ind.

A.1.5.1 OC/A. Following Eisfeldt and Papanikolaou (2013), we construct a measure of organization capital using Selling, General, and Administrative (SG&A) expenses (Compustat annual item XSGA). The stock of organization capital, OC, is constructed with the

perpetual inventory method:

$$OC_{it} = (1 - \delta)OC_{it-1} + SG\&A_{it}/CPI_t, \quad (A8)$$

in which CPI_t is the consumer price index during year t and δ is the annual depreciation rate of OC. The initial stock of OC is $OC_{i0} = SG\&A_{i0}/(g + \delta)$, in which $SG\&A_{i0}$ is the first valid SG&A observation (zero or positive) for firm i , and g is the long-term growth rate of SG&A. Following Eisfeldt and Papanikolaou, we assume a depreciation rate of 15% for OC, and a long-term growth rate of 10% for SG&A. Missing SG&A values after the starting date are treated as zero. For portfolio formation at the end of June of year t , we require SG&A to be nonmissing for the fiscal year ending in calendar year $t - 1$ because this SG&A value receives the highest weight in OC. In addition, we exclude firms with zero OC. We form organization capital-to-assets (OC/A) by scaling OC with total assets (item AT) from the same fiscal year.

Following Eisfeldt and Papanikolaou (2013), we industry-standardize OC/A using the Fama-French (1997) 17-industry classification. We demean a firm's OC/A by its industry mean and then divide the demeaned OC/A by the standard deviation of OC/A within its industry. When computing industry mean and standard deviation, we winsorize OC/A at the 1 and 99 percentiles of all firms each year. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on OC/A for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.5.2 BC/A. Following Belo, Lin, and Vitorino (2014), we construct brand capital (BC) by accumulating advertising expenses (Compustat annual item XAD) via the perpetual inventory method:

$$BC_{it} = (1 - \delta)BC_{it-1} + XAD_{it}, \quad (A9)$$

in which δ is the annual depreciation rate of BC. The initial stock of BC is $BC_{i0} = XAD_{i0}/(g + \delta)$, in which XAD_{i0} is first valid XAD (zero or positive) for firm i , and g is the long-term growth rate of XAD. We assume a depreciation rate of 50% for BC, and a long-term growth rate of 10% for XAD. Missing values of XAD after the starting date are treated as zero. For the portfolio formation at the end of June of year t , we exclude firms with zero BC and require XAD to be nonmissing for the fiscal year ending in calendar year $t - 1$. We form brand capital-to-assets (BC/A) by scaling BC with total assets (item AT) from the same fiscal year.

At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on BC/A for the fiscal year ending in calendar year $t - 1$. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because sufficient XAD data start in 1972, the BC/A decile returns start in July 1973.

A.1.5.3 Ad/M. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on advertising expenses-to-market (Ad/M), which is advertising expenses (Compustat annual item XAD) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from Compustat or CRSP) at the end of December of $t - 1$. We keep

only firms with positive advertising expenses. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$. Because sufficient XAD data start in 1972, the BC/A decile returns start in July 1973.

A.1.5.4 RD/S. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on R&D-to-sales (RD/S), which is R&D expenses (Compustat annual item XRD) divided by sales (item SALE) for the fiscal year ending in calendar year $t - 1$. We keep only firms with positive R&D expenses. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$. Because the accounting treatment of R&D expenses was standardized in 1975 (by Financial Accounting Standards Board Statement No. 2), the RD/S decile returns start in July 1976.

A.1.5.5 RD/M. At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on R&D-to-market (RD/M), which is R&D expenses (Compustat annual item XRD) for the fiscal year ending in calendar year $t - 1$ divided by the market equity (from Compustat or CRSP) at the end of December of $t - 1$. We keep only firms with positive R&D expenses. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. Because the accounting treatment of R&D expenses was standardized in 1975, the RD/M decile returns start in July 1976.

A.1.5.6 RC/A. Following Li (2011), we measure R&D capital (RC) as a weighted average of annual R&D expenses (Compustat annual item XRD) over the last 5 years with a depreciation rate of 20%:

$$RC_{it} = XRD_{it} + 0.8 XRD_{it-1} + 0.6 XRD_{it-2} + 0.4 XRD_{it-3} + 0.2 XRD_{it-4}. \quad (A10)$$

We scale RC with total assets (item AT) to form R&D capital-to-assets (RC/A). At the end of June of each year t , we use NYSE breakpoints to split stocks into deciles based on RC/A for the fiscal year ending in calendar year $t - 1$. We keep only firms with positive RC. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$. For portfolio formation at the end of June of year t , we require R&D expenses to be nonmissing for the fiscal year ending in calendar year $t - 1$, because this value of R&D expenses receives the highest weight in RC. Finally, because RC requires past 5 years of R&D expenses data and the accounting treatment of R&D expenses was standardized in 1975, the RC/A decile returns start in July 1980.

A.1.5.7 H/N. Following Belo, Lin, and Bazdresch (2014), at the end of June of year t , we measure the firm-level hiring rate (H/N) as $(N_{t-1} - N_{t-2}) / (0.5N_{t-1} + 0.5N_{t-2})$, in which N_{t-j} is the number of employees (Compustat annual item EMP) from the fiscal year ending in calendar year $t - j$. At the end of June of year t , we use NYSE breakpoints to sort stocks into deciles based on H/N. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$.

A.1.5.8 OL. Following Novy-Marx (2011), we measure operating leverage (OL) as operating costs scaled by total assets (Compustat annual item AT, the denominator is current, not lagged, total assets). Operating costs are the cost of goods sold (item COGS) plus selling, general, and administrative expenses (item XSGA). At the end of June of year t , we use NYSE breakpoints to sort stocks into deciles based on OL for the fiscal year ending in calendar year $t - 1$, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.5.9 G . The data for Gompers, Ishii, and Metrick's (2003) firm-level corporate governance index (G , from September 1990 to December 2006) are from Andrew Metrick's Web site. To form the G portfolios, we use the following breakpoints: $G \leq 5, 6, 7, 8, 9, 10, 11, 12, 13$, and ≥ 14 (see Table VI, Gompers et al.). We rebalance the portfolios in the months immediately following each publication of the G -index, and calculate monthly value-weighted portfolio returns between 2 adjacent publication dates. The first months following the publication dates are September 1990, July 1993, July 1995, February 1998, November 1999, January 2002, January 2004, and January 2006.

A.1.5.10 AccQ. Following Francis et al. (2005), we estimate accrual quality (AccQ) with the following cross-sectional regression (all variables are scaled by lagged total assets):

$$TCA_{it} = \phi_{0,i} + \phi_{1,i}CFO_{it-1} + \phi_{2,i}CFO_{it} + \phi_{3,i}CFO_{it+1} + \phi_{4,i}\Delta REV_{it} + \phi_{5,i}PPE_{it} + v_{it}, \quad (A11)$$

in which TCA_{it} is firm i 's total current accruals in year t , CFO_{it} is cash flow from operations in t , ΔREV_{it} is change in revenues (Compustat annual item SALE) between $t - 1$ and t , and PPE_{it} is gross property, plant, and equipment (item PPEGT) in t . $TCA_{it} = \Delta CA_{it} - \Delta CL_{it} - \Delta CASH_{it} + \Delta STDEBT_{it}$, in which ΔCA_{it} is firm i 's change in current assets (item ACT) between year $t - 1$ and t , ΔCL_{it} is change in current liabilities (item LCT), $\Delta CASH_{it}$ is change in cash (item CHE), and $\Delta STDEBT_{it}$ is change in debt in current liabilities (item DLC, zero if missing). $CFO_{it} = NIBE_{it} - TA_{it}$, in which $NIBE_{it}$ is income before extraordinary items (item IB). $TA_{it} = \Delta CA_{it} - \Delta CL_{it} - \Delta CASH_{it} + \Delta STDEBT_{it} - DEPN_{it}$, in which $DEPN_{it}$ is depreciation and amortization expense (item DP, zero if missing).

Following Francis et al., we estimate annual cross-sectional regressions in Equation (A11) for each of Fama-French (1997) 48 industries (excluding 4 financial industries), with at least 20 firms in year t . We winsorize the regressors at the 1 and 99 percentiles of all firms each year. The annual cross-sectional regressions yield firm- and year-specific residuals, v_{it} . We measure accrual quality, $AccQ_{it} = \sigma(v_{it})$, as the standard deviation of firm i 's residuals, v_{it} , calculated over years $t - 4$ through t .

At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on AccQ for the fiscal year ending in calendar year $t - 2$. To avoid look-ahead bias, we do not sort on AccQ for the fiscal year ending in $t - 1$, because the regression in Equation (A11) requires the next year's CFO. Monthly value-weighted decile returns are calculated from July of year t to June of $t + 1$, and the deciles are rebalanced in June of $t + 1$.

A.1.5.11 Ind. At the end of June of each year t , we use the industry classifications from Kenneth French's Web site to assign stocks into 10 industry portfolios based on SIC codes (Compustat annual item SIC1) for the fiscal year ending in calendar year $t - 1$. When Compustat SIC codes are unavailable, we use CRSP SIC codes (CRSP item SICCD) for June of year t . Monthly value-weighted portfolio returns are computed from July of year t to June of year $t + 1$, and the portfolios are rebalanced in June of $t + 1$. We exclude financial firms from the last industry portfolio ("Other").

A.1.6 Trading Frictions

Table 2 lists 13 anomaly variables in this category, including ME, Ivol, Tvol, Svol, MDR, β , D- β , S-Rev, Disp, Turn, 1/P, Dvol, and Illiq.

A.1.6.1 ME. ME is price times shares outstanding from CRSP. At the end of June of each year t , we use NYSE breakpoints to sort stocks into deciles based on the June-end ME, and calculate monthly value-weighted decile returns from July of year t to June of $t + 1$.

A.1.6.2 Ivol. Following Ang et al. (2006), we measure a stock's idiosyncratic volatility (Ivol) as the standard deviation of the residuals from regressing the stock's returns on the Fama-French (1993) 3 factors. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on the Ivol estimated with daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.3 Tvol. Following Ang et al. (2006), we measure a stock's total volatility (Tvol) as the standard deviation of its daily returns. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on the Tvol estimated with the daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.4 Svol. Following Ang et al. (2006), we measure systematic volatility (Svol) as $\beta_{\Delta VXO}^i$ from the bivariate regression:

$$r_d^i = \beta_0^i + \beta_{MKT}^i MKT_d + \beta_{\Delta VXO}^i \Delta VXO_d + \epsilon_d^i, \quad (A12)$$

in which r_d^i is stock i 's return on day d , MKT_d is the market factor return, and ΔVXO_d is the aggregate volatility shock measured as the daily change in the Chicago Board Options Exchange S&P 100 volatility index (VXO). At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on $\beta_{\Delta VXO}^i$ estimated with the daily returns from month $t - 1$. We require a minimum of 15 daily returns. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$. Because the VXO data start in January 1986, the Svol decile returns start in February 1986.

A.1.6.5 MDR. Following Bali, Cakici, and Whitelaw (2011), at the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on the maximal daily return (MDR) in month $t - 1$. We require a minimum of 15 daily returns. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.6 β . Following Frazzini and Pedersen (2014), we estimate β for firm i as:

$$\hat{\beta}_i = \hat{\rho} \frac{\hat{\sigma}_i}{\hat{\sigma}_m}, \quad (\text{A13})$$

in which $\hat{\sigma}_i$ and $\hat{\sigma}_m$ are the estimated volatilities for stock i and the market, and $\hat{\rho}$ is their correlation. To estimate the volatilities, we compute the standard deviation of daily log returns over a 1-year rolling window (with at least 120 daily returns). To estimate correlations, we use overlapping 3-day log returns, $r_{it}^{3d} = \sum_{k=0}^2 \log(1 + r_{t+k}^i)$, over a 5-year rolling window (with at least 750 daily returns). At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on $\hat{\beta}_i$ estimated at the end of month $t - 1$. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.7 D- β . Following Dimson (1979), we use the lead and the lag of the market return, along with the current market return, when estimating beta (D- β):

$$r_{id} - r_{fd} = \alpha_i + \beta_{i1}(r_{md-1} - r_{fd-1}) + \beta_{i2}(r_{md} - r_{fd}) + \beta_{i3}(r_{md+1} - r_{fd+1}) + \epsilon_{id}, \quad (\text{A14})$$

in which r_{id} is the return on stock i on day d , r_{md} is the market return, and r_{fd} is the risk-free rate. We estimate the Dimson market regression for each stock using daily returns from the prior month. We require a minimum of 15 daily returns. The market beta of stock i is calculated as $\text{D-}\beta_i \equiv \hat{\beta}_{i1} + \hat{\beta}_{i2} + \hat{\beta}_{i3}$. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on $\text{D-}\beta_i$ estimated with the daily returns from month $t - 1$, and calculate monthly value-weighted decile returns for month t . The deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.8 S-Rev. To construct the Jegadeesh (1990) short-term reversal (S-Rev) deciles, at the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on the return in month $t - 1$. To be included in a decile in month t , a stock must have a valid price at the end of month $t - 2$ and a valid return for month $t - 1$. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.9 Disp. Following Diether, Malloy, and Scherbina (2002), we measure analyst earnings forecasts dispersion (Disp) as the ratio of the standard deviation of earnings forecast (IBES unadjusted file, item STDEV) to the absolute value of the consensus mean forecast (unadjusted file, item MEANEST). We use the earnings forecast for the current fiscal year (fiscal period indicator = 1). Stocks with a mean forecast of zero are assigned to the highest dispersion group. In addition, we exclude stocks with a price less than \$5. At the beginning

of each month t , we use NYSE breakpoints to sort stocks into deciles based on Disp in month $t - 1$. Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$. Because the forecast data start in January 1976, the Disp decile returns start in February 1976.

A.1.6.10 Turn. Following Datar, Naik, and Radcliffe (1998), at the beginning of each month t , we calculate the share turnover (Turn) of a stock as its average daily share turnover over the prior 6 months from $t - 6$ to $t - 1$. We require a minimum of 50 daily observations. Daily turnover is the number of shares traded on a given day divided by the number of shares outstanding on that day.¹⁵ At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on Turn and calculate value-weighted decile returns for month t .

A.1.6.11 1/P. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on the reciprocal of the share price ($1/P$) at the end of month $t - 1$. We calculate value-weighted decile returns for the current month t and rebalance the deciles at the beginning of month $t + 1$.

A.1.6.12 Dvol. At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on their average daily dollar trading volume (Dvol) over the prior 6 months from $t - 6$ to $t - 1$. We require a minimum of 50 daily observations. Dollar trading volume is share price times the number of shares traded. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010). Monthly value-weighted decile returns are calculated for the current month t , and the deciles are rebalanced at the beginning of month $t + 1$.

A.1.6.13 Illiq. We calculate the Amihud (2002) illiquidity measure (Illiq) as the ratio of absolute daily stock return to daily dollar trading volume, averaged over the prior 6 months. We require a minimum of 50 daily observations. Dollar trading volume is share price times the number of shares traded. We adjust the trading volume of NASDAQ stocks per Gao and Ritter (2010). At the beginning of each month t , we use NYSE breakpoints to sort stocks into deciles based on Illiq over the prior 6 months from $t - 6$ to $t - 1$. We calculate value-weighted decile returns for month t and rebalance the deciles at the beginning of month $t + 1$.

¹⁵Following Gao and Ritter (2010), we adjust the trading volume for NASDAQ stocks to account for the institutional features of the way that NASDAQ and NYSE-Amex volume are reported. Prior to February 1, 2001, we divide NASDAQ volume by 2.0. This adjustment accounts for the practice of counting as trades both trades with market makers and trades among market makers. On February 1, 2001, according to the director of research of NASDAQ and Frank Hathaway (the chief economist of NASDAQ), a “riskless principal” rule goes into effect and results in a reduction of approximately 10% in reported volume. As such, from February 1, 2001 to December 31, 2001, we divide NASDAQ volume by 1.8. During 2002, securities firms began to charge institutional investors commissions on NASDAQ trades, rather than the prior practice of marking up or down the net price. This practice results in a further reduction in reported volume of approximately 10%. As such, for 2002 and 2003, we divide NASDAQ volume by 1.6. For 2004 and later years, in which the volume of NASDAQ (and NYSE) stocks has mostly been occurring on crossing networks and other venues, we use a divisor of 1.0. This practice reflects the fact that there are no longer important differences in the NASDAQ and NYSE volume.

Table 1

Empirical properties of the q -factors

Size (ME) is price per share times shares outstanding. Investment-to-assets (I/A) is the annual change in total assets (Compustat annual item AT) divided by lagged total assets. ROE is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged book equity. Book equity is the shareholders' equity, plus balance-sheet deferred taxes and investment tax credit (item TXDITCQ) if available, minus the book value of preferred stock. Depending on availability, we use the stockholders' equity (item SEQQ), or common equity (item CEQQ) plus the carrying value of the preferred stock (item PSTKQ), or total assets (item ATQ) minus liabilities (item LTQ) in that order as the shareholders' equity. We use redemption value (item PSTKRQ) if available, or carrying value for the book value of the preferred stock. At the end of June of each year t , we use the median NYSE size at the end of June to split NYSE, Amex, and NASDAQ stocks into 2 groups, small and big. Independently, at the end of June of each year t , we also sort stocks into 3 I/A groups, using the NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked I/A for the fiscal year ending in calendar year $t - 1$. In addition, independently, at the beginning of each month, we sort stocks into 3 groups based on NYSE breakpoints for the low 30%, middle 40%, and high 30% of the ranked ROE. Earnings data in Compustat quarterly files are used in the sorts in the months immediately after the most recent public quarterly earnings announcement dates (item RDQ). Taking the intersections of the 2 size, 3 I/A , and 3 ROE groups, we form 18 portfolios. Monthly value-weighted returns on the 18 portfolios are calculated for the current month, and the portfolios are rebalanced monthly. The size factor, r_{ME} , is the difference (small-minus-big), each month, between the average returns on the 9 small portfolios and the average returns on the 9 big portfolios. The investment factor, $r_{I/A}$, is the difference (low-minus-high), each month, between the average returns on the 6 low I/A portfolios and the average returns on the 6 high I/A portfolios. The ROE factor, r_{ROE} , is the difference (high-minus-low), each month, between the average returns on the 6 high ROE portfolios and the average returns on the 6 low ROE portfolios. The data for SMB, HML, and UMD are from Kenneth French's Web site. MKT is the value-weighted market return minus the 1-month Treasury bill rate from CRSP. The t -statistics (in parentheses) in Panel A are adjusted for heteroscedasticity and autocorrelations, and the p -values (in parentheses) in Panel B test that a given correlation is zero.

Panel A: Descriptive statistics							Panel B: Correlation matrix (p-values)							
	Mean	α	β_{MKT}	β_{SMB}	β_{HML}	β_{UMD}	R^2		$r_{I/A}$	r_{ROE}	MKT	SMB	HML	UMD
r_{ME}	0.31 (2.12)	0.23 (1.62)	0.17 (4.33)				0.06	r_{ME}	-0.11 (0.02)	-0.31 (0.00)	0.25 (0.00)	0.95 (0.00)	-0.07 (0.13)	0.01 (0.90)
		0.04 (1.09)	0.02 (1.59)	0.99 (57.37)	0.17 (7.05)		0.93	$r_{I/A}$		0.06 (0.20)	-0.36 (0.00)	-0.22 (0.00)	0.69 (0.00)	0.05 (0.31)
		0.01 (0.15)	0.02 (2.40)	0.99 (61.51)	0.19 (7.34)	0.03 (2.16)	0.94	r_{ROE}			-0.19 (0.00)	-0.38 (0.00)	-0.09 (0.06)	0.50 (0.00)
$r_{I/A}$	0.45 (4.95)	0.52 (5.93)	-0.15 (-5.58)				0.13	MKT				0.28 (0.00)	-0.32 (0.00)	-0.15 (0.00)
		0.33 (4.85)	-0.06 (-3.66)	-0.02 (-0.81)	0.39 (11.98)		0.50	SMB					-0.23 (0.00)	-0.01 (0.79)
		0.28 (3.85)	-0.05 (-3.24)	-0.02 (-0.87)	0.41 (11.94)	0.05 (1.97)	0.52	HML						-0.15 (0.00)
r_{ROE}	0.58 (4.81)	0.63 (5.62)	-0.11 (-2.38)				0.04							
		0.77 (6.94)	-0.09 (-2.08)	-0.33 (-5.75)	-0.20 (-2.38)		0.20							
		0.50 (4.75)	-0.03 (-0.98)	-0.33 (-4.38)	-0.10 (-1.48)	0.28 (6.27)	0.40							

Table 2

List of anomalies

This table lists the anomalies that we study. The anomalies are grouped into 6 categories: (i) momentum; (ii) value-versus-growth; (iii) investment; (iv) profitability; (v) intangibles; and (vi) trading frictions. For each anomaly variable, we list its symbol, brief description, and source in the academic literature. Appendix A details variable definition and portfolio construction.

Panel A: Momentum

SUE-1	Earnings surprise (1-month holding period), Foster, Olsen, and Shevlin (1984)	SUE-6	Earnings surprise (6-month holding period), Foster, Olsen, and Shevlin (1984)
Abr-1	Cumulative abnormal stock returns around earnings announcements (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	Abr-6	Cumulative abnormal stock returns around earnings announcements (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
RE-1	Revisions in analysts' earnings forecasts (1-month holding period), Chan, Jegadeesh, and Lakonishok (1996)	RE-6	Revisions in analysts' earnings forecasts (6-month holding period), Chan, Jegadeesh, and Lakonishok (1996)
R6-1	Price momentum (6-month prior returns, 1-month holding period), Jegadeesh and Titman (1993)	R6-6	Price momentum (6-month prior returns, 6-month holding period), Jegadeesh and Titman (1993)
R11-1	Price momentum, (11-month prior returns, 1-month holding period), Fama and French (1996)	I-Mom	Industry momentum, Moskowitz and Grinblatt (1999)

Panel B: Value-versus-growth

B/M	Book-to-market equity, Rosenberg, Reid, and Lanstein (1985)	A/ME	Market leverage, Bhandari (1988)
Rev	Reversal, De Bondt and Thaler (1985)	E/P	Earnings-to-price, Basu (1983)
EF/P	Analysts' earnings forecasts-to-price, Elgers, Lo, and Pfeiffer (2001)	CF/P	Cash flow-to-price, Lakonishok, Shleifer, and Vishny (1994)
D/P	Dividend yield, Litzenberger and Ramaswamy (1979)	O/P	Payout yield, Boudoukh, Michaely, Richardson, and Roberts (2007)
NO/P	Net payout yield, Boudoukh, Michaely, Richardson, and Roberts (2007)	SG	Sales growth, Lakonishok, Shleifer, and Vishny (1994)
LTG	Long-term growth forecasts of analysts, La Porta (1996)	Dur	Equity duration, Dechow, Sloan, and Soliman (2004)

Panel C: Investment

ACI	Abnormal corporate investment, Titman, Wei, and Xie (2004)	I/A	Investment-to-assets, Cooper, Gulen, and Schill (2008)
NOA	Net operating assets, Hirshleifer et al. (2004)	Δ PI/A	Changes in property, plant, and equipment plus changes in inventory scaled by assets, Lyandres, Sun, and Zhang (2008)
IG	Investment growth, Xing (2008)	NSI	Net stock issues, Pontiff and Woodgate (2008)
CEI	Composite issuance, Daniel and Titman (2006)	NXF	Net external financing, Bradshaw, Richardson, and Sloan (2006)
IvG	Inventory growth, Belo and Lin (2011)	IvC	Inventory changes, Thomas and Zhang (2002)
OA	Operating accruals, Sloan (1996)	TA	Total accruals, Richardson et al. (2005)
POA	Percent operating accruals, Hafzalla, Lundholm, and Van Winkle (2011)	PTA	Percent total accruals, Hafzalla, Lundholm, and Van Winkle (2011)

Panel D: Profitability

ROE	Return on equity, Haugen and Baker (1996)	ROA	Return on assets, Balakrishnan, Bartov, and Faurel (2010)
RNA	Return on net operating assets, Soliman (2008)	PM	Profit margin, Soliman (2008)
ATO	Asset turnover, Soliman (2008)	CTO	Capital turnover, Haugen and Baker (1996)
GP/A	Gross profits-to-assets, Novy-Marx (2013)	<i>F</i>	<i>F</i> -score, Piotroski (2000)
TES	Tax expense surprise, Thomas and Zhang (2011)	TI/BI	Taxable income-to-book income, Green, Hand, and Zhang (2013)
RS	Revenue surprise, Jegadeesh and Livnat (2006)	NEI	Number of consecutive quarters with earnings increases, Barth, Elliott, and Finn (1999)
FP	Failure probability, Campbell, Hilscher, and Szilagyi (2008)	<i>O</i>	<i>O</i> -score, Dichev (1998)

Panel E: Intangibles

OC/A	Organizational capital-to-assets, Eisfeldt and Papanikolaou (2013)	BC/A	Brand capital-to-assets, Belo, Lin, and Vitorino (2014)
Ad/M	Advertisement expense-to-market, Chan, Lakonishok, and Sougiannis (2001)	RD/S	R&D-to-sales, Chan, Lakonishok, and Sougiannis (2001)
RD/M	R&D-to-market, Chan, Lakonishok, and Sougiannis (2001)	RC/A	R&D capital-to-assets, Li (2011)
H/N	Hiring rate, Belo, Lin, and Bazdresch (2014)	OL	Operating leverage, Novy-Marx (2011)
<i>G</i>	Corporate governance, Gompers, Ishii, and Metrick (2003)	AccQ	Accrual quality, Francis et al. (2005)
Ind	Industries, Fama and French (1997)		

Panel F: Trading frictions

ME	The market equity, Banz (1981)	Ivol	Idiosyncratic volatility, Ang et al. (2006)
Tvol	Total volatility, Ang et al. (2006)	Svol	Systematic volatility, Ang et al. (2006)
MDR	Maximum daily return, Bali, Cakici, and Whitelaw (2011)	β	Market beta, Frazzini and Pedersen (2014)
D- β	Dimson's beta, Dimson (1979)	S-Rev	Short-term reversal, Jegadeesh (1990)
Disp	Dispersion of analysts' earnings forecasts, Diether, Malloy, and Scherbina (2002)	Turn	Share turnover, Datar, Naik, and Radcliffe (1998)
1/P	1/share price, Miller and Scholes (1982)	Dvol	Dollar trading volume, Brennan, Chordia, and Subrahmanyam (1998)
Illiq	Illiquidity as absolute return-to-volume, Amihud (2002)		

Table 3**Insignificant anomalies in the broad cross section**

For each anomaly variable, we report the average return (m) of the high-minus-low decile and its t -statistic (t_m) adjusted for heteroscedasticity and autocorrelations. Table 2 provides a brief description of the symbols. We form all of the deciles using NYSE breakpoints, and calculate value-weighted portfolio returns. Appendix A details variable definition and portfolio construction.

	R6-1	A/ME	Rev	EF/P	D/P	O/P	SG	LTG	ACI	NXF
m	0.48	0.43	-0.39	0.45	0.27	0.35	-0.27	0.01	-0.27	-0.30
t_m	1.43	1.82	-1.57	1.73	0.94	1.53	-1.34	0.02	-1.70	-1.55
	TA	RNA	PM	ATO	CTO	F	TES	TI/BI	RS	O
m	-0.19	0.13	0.10	0.22	0.20	0.37	0.32	0.13	0.29	-0.08
t_m	-1.31	0.61	0.40	1.11	1.11	1.28	1.92	0.86	1.82	-0.37
	BC/A	RD/S	RC/A	H/N	G	AccQ	ME	Ivol	Tvol	MDR
m	0.18	0.01	0.32	-0.25	0.03	-0.18	-0.24	-0.54	-0.37	-0.31
t_m	0.73	0.06	1.27	-1.47	0.09	-0.79	-0.90	-1.56	-0.95	-0.94
	β	D- β	S-Rev	Disp	Turn	1/P	Dvol	Illiq		
m	-0.13	0.07	-0.31	-0.33	-0.12	-0.00	-0.26	0.27		
t_m	-0.36	0.30	-1.39	-1.24	-0.43	-0.01	-1.30	1.14		

Table 4

Significant anomalies in the broad cross section: Pricing errors and tests of overall performance

For each anomaly variable, m , α , α_{FF} , α_C , and α_q are the average return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha for the high-minus-low decile, and t_m , t , t_{FF} , t_C , and t_q are their t -statistics adjusted for heteroscedasticity and autocorrelations, respectively. $|\alpha|$, $|\alpha_{FF}|$, $|\alpha_C|$, and $|\alpha_q|$ are the average magnitude of the alphas, and p , p_{FF} , p_C , and p_q are the p -values of the GRS test on the null hypothesis that all of the alphas are jointly zero across the deciles formed on the variable. Table 2 provides a brief description of the symbols.

	SUE-1	SUE-6	Abr-1	Abr-6	RE-1	RE-6	R6-6	R11-1	I-Mom	B/M	E/P	CF/P	NO/P	Dur	I/A	NOA	Δ PI/A	IG
m	0.45	0.24	0.73	0.30	0.89	0.60	0.85	1.18	0.51	0.70	0.59	0.52	0.66	-0.54	-0.42	-0.38	-0.51	-0.41
α	0.50	0.27	0.76	0.31	1.02	0.71	0.92	1.29	0.58	0.75	0.69	0.63	0.84	-0.62	-0.50	-0.38	-0.57	-0.45
α_{FF}	0.55	0.39	0.84	0.38	1.20	0.94	1.12	1.52	0.68	0.01	0.05	0.01	0.52	-0.06	-0.15	-0.52	-0.41	-0.26
α_C	0.34	0.18	0.62	0.19	0.56	0.37	0.06	0.09	-0.18	-0.01	0.01	-0.06	0.49	-0.08	-0.09	-0.41	-0.36	-0.20
α_q	0.16	0.02	0.64	0.26	0.12	0.03	0.24	0.24	0.00	0.21	0.17	0.22	0.36	-0.27	0.14	-0.38	-0.26	0.05
t_m	3.59	2.17	5.50	3.11	3.43	2.58	3.17	3.52	2.33	2.88	2.63	2.44	3.23	-2.59	-2.45	-2.55	-3.43	-2.93
t	4.26	2.68	5.84	3.33	4.13	3.28	3.63	4.18	2.68	3.05	3.12	3.01	4.45	-2.98	-2.94	-2.52	-3.91	-3.16
t_{FF}	4.50	3.62	5.93	3.89	4.81	4.52	4.47	4.99	3.25	0.04	0.34	0.08	3.51	-0.44	-1.09	-3.30	-2.93	-1.99
t_C	2.62	1.69	4.37	2.06	2.56	2.15	0.51	0.67	-1.11	-0.06	0.03	-0.40	3.33	-0.56	-0.61	-2.69	-2.48	-1.51
t_q	1.12	0.18	4.07	2.18	0.43	0.14	0.71	0.54	0.01	1.15	0.76	1.04	2.38	-1.32	1.08	-1.90	-1.85	0.39
$ \alpha $	0.16	0.11	0.13	0.08	0.19	0.14	0.17	0.21	0.16	0.22	0.23	0.20	0.23	0.24	0.17	0.15	0.15	0.13
$ \alpha_{FF} $	0.17	0.13	0.16	0.11	0.27	0.23	0.19	0.26	0.15	0.07	0.10	0.08	0.17	0.11	0.12	0.17	0.13	0.13
$ \alpha_C $	0.11	0.09	0.12	0.08	0.11	0.09	0.10	0.13	0.06	0.06	0.09	0.07	0.15	0.08	0.10	0.14	0.12	0.11
$ \alpha_q $	0.05	0.07	0.13	0.07	0.10	0.11	0.08	0.13	0.13	0.08	0.10	0.14	0.12	0.08	0.09	0.12	0.14	0.09
p	0.00	0.00	0.00	0.01	0.04	0.21	0.00	0.00	0.09	0.04	0.01	0.05	0.00	0.00	0.00	0.00	0.00	0.00
p_{FF}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.19	0.18	0.43	0.00	0.15	0.01	0.00	0.00	0.00
p_C	0.00	0.00	0.00	0.01	0.16	0.12	0.00	0.00	0.45	0.29	0.38	0.37	0.00	0.41	0.02	0.00	0.01	0.00
p_q	0.42	0.04	0.00	0.02	0.46	0.08	0.00	0.01	0.03	0.35	0.13	0.02	0.00	0.72	0.01	0.00	0.00	0.01

	NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GP/A	NEI	FP	OC/A	Ad/M	RD/M	OL	Svol
m	-0.68	-0.57	-0.41	-0.45	-0.30	-0.46	-0.40	0.80	0.62	0.34	0.39	-0.67	0.56	0.79	0.63	0.39	-0.60
α	-0.78	-0.79	-0.47	-0.51	-0.33	-0.53	-0.50	0.96	0.78	0.32	0.40	-1.06	0.65	0.82	0.47	0.44	-0.72
α_{FF}	-0.64	-0.50	-0.29	-0.38	-0.37	-0.32	-0.29	1.17	1.00	0.50	0.63	-1.44	0.61	0.15	0.22	0.37	-0.66
α_C	-0.54	-0.40	-0.19	-0.30	-0.33	-0.25	-0.27	0.85	0.67	0.45	0.43	-0.67	0.40	0.32	0.31	0.33	-0.62
α_q	-0.26	-0.22	-0.03	-0.28	-0.56	-0.12	-0.10	0.05	0.09	0.11	0.18	-0.17	0.09	0.11	0.60	-0.05	-0.37
t_m	-4.13	-2.96	-2.77	-3.05	-2.32	-3.02	-2.57	3.11	2.70	2.18	3.31	-1.98	4.07	2.96	2.31	2.06	-2.57
t	-4.86	-4.79	-3.29	-3.35	-2.47	-3.64	-3.50	4.02	3.67	2.02	3.45	-3.80	4.69	3.08	1.81	2.22	-3.12
t_{FF}	-4.28	-3.72	-2.10	-2.61	-2.84	-2.42	-2.06	5.43	5.40	3.25	6.03	-6.44	4.52	0.79	0.93	1.91	-2.88
t_C	-3.58	-2.93	-1.34	-1.97	-2.32	-1.88	-1.82	4.03	3.59	2.85	3.73	-3.79	2.97	1.37	1.40	1.76	-2.59
t_q	-1.75	-1.50	-0.20	-1.84	-3.90	-0.87	-0.67	0.37	0.72	0.71	1.68	-0.57	0.66	0.39	2.40	-0.27	-1.42
$\overline{ \alpha }$	0.18	0.19	0.14	0.16	0.15	0.12	0.12	0.18	0.15	0.06	0.19	0.16	0.14	0.23	0.13	0.11	0.18
$\overline{ \alpha_{FF} }$	0.18	0.15	0.11	0.12	0.13	0.11	0.11	0.24	0.23	0.14	0.23	0.23	0.15	0.13	0.17	0.11	0.19
$\overline{ \alpha_C }$	0.15	0.15	0.10	0.10	0.12	0.11	0.10	0.15	0.14	0.14	0.15	0.12	0.13	0.18	0.21	0.12	0.16
$\overline{ \alpha_q }$	0.11	0.12	0.11	0.08	0.15	0.12	0.08	0.09	0.07	0.11	0.09	0.13	0.11	0.11	0.27	0.12	0.11
p	0.00	0.00	0.01	0.00	0.00	0.01	0.01	0.01	0.07	0.25	0.00	0.00	0.00	0.04	0.24	0.54	0.01
p_{FF}	0.00	0.00	0.03	0.01	0.00	0.00	0.01	0.00	0.00	0.01	0.00	0.00	0.00	0.18	0.02	0.07	0.01
p_C	0.00	0.00	0.11	0.04	0.00	0.01	0.02	0.00	0.04	0.01	0.00	0.00	0.00	0.07	0.01	0.06	0.06
p_q	0.02	0.01	0.08	0.56	0.00	0.00	0.11	0.05	0.75	0.38	0.05	0.00	0.02	0.07	0.00	0.09	0.20

Table 5

Significant anomalies in the broad cross section: An alternative q -factor model without the size factor

For each anomaly variable, α_3^q is the high-minus-low alpha from an alternative 3-factor q -model without the size factor, and t_3^q is the t -statistics adjusted for heteroscedasticity and autocorrelations. $|\alpha_3^q|$ is the average magnitude of the alphas across a given set of deciles, and p_3^q is the p -value of the GRS test on the null hypothesis that all of the alphas are jointly zero across a given set of deciles. Table 2 provides a brief description of the symbols.

	SUE-1	SUE-6	Abr-1	Abr-6	RE-1	RE-6	R6-6	R11-1	L-Mom	B/M	E/P	CF/P	NO/P	Dur	I/A	NOA	Δ PI/A	IG
α_3^q	0.21	0.07	0.67	0.30	0.05	-0.05	0.37	0.42	0.15	0.42	0.29	0.30	0.21	-0.38	0.09	-0.35	-0.28	0.00
t_3^q	1.43	0.42	3.95	2.48	0.18	-0.22	0.96	0.85	0.48	2.13	1.21	1.42	1.26	-1.72	0.68	-1.71	-2.00	0.00
$ \alpha_3^q $	0.05	0.07	0.12	0.08	0.14	0.15	0.10	0.16	0.09	0.11	0.12	0.15	0.12	0.11	0.09	0.12	0.11	0.13
p_3^q	0.62	0.13	0.00	0.00	0.20	0.02	0.00	0.00	0.10	0.15	0.04	0.01	0.01	0.31	0.01	0.01	0.03	0.00
NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GP/A	NEI	FP	OC/A	Ad/M	RD/M	OL	Svol		
α_3^q	-0.18	-0.10	0.03	-0.28	-0.44	-0.06	-0.01	-0.14	-0.09	0.12	0.14	0.03	0.21	0.37	0.89	0.07	-0.25	
t_3^q	-1.18	-0.65	0.23	-1.86	-2.83	-0.38	-0.05	-0.84	-0.60	0.79	1.26	0.09	1.31	1.24	2.86	0.37	-0.91	
$ \alpha_3^q $	0.10	0.12	0.15	0.11	0.13	0.09	0.11	0.10	0.09	0.12	0.10	0.17	0.13	0.12	0.30	0.12	0.11	
p_3^q	0.04	0.01	0.01	0.26	0.02	0.07	0.04	0.03	0.55	0.36	0.05	0.00	0.01	0.06	0.00	0.11	0.12	

Table 6

***Q*-factor loadings and underlying characteristics for significant anomalies in the broad cross section**

For the high-minus-low decile formed on each anomaly variable, β_{MKT} , β_{ME} , $\beta_{\text{I/A}}$, and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the *q*-factor model, and $t_{\beta_{\text{MKT}}}$, $t_{\beta_{\text{ME}}}$, $t_{\beta_{\text{I/A}}}$, and $t_{\beta_{\text{ROE}}}$ are their *t*-statistics, respectively. ME is the average market capitalization (in billions of dollars) across all firms in the high decile, minus that in the low decile. I/A (in percent) is the sum of changes in assets across all firms in the high decile divided by the sum of their 1-year-lagged assets, minus the sum of changes in assets across all firms in the low decile divided by the sum of their 1-year-lagged assets. ROE (in percent) is the sum of the latest announced quarterly earnings across all firms in the high decile divided by the sum of their 1-quarter-lagged book equity, minus the sum of the latest announced quarterly earnings across all firms in the low decile divided by the sum of their 1-quarter-lagged book equity. t_{ME} , $t_{\text{I/A}}$, and t_{ROE} are the corresponding *t*-statistics. All *t*-statistics are adjusted for heteroscedasticity and autocorrelations. Table 2 provides a brief description of the symbols.

	SUE-1	SUE-6	Abr-1	Abr-6	RE-1	RE-6	R6-6	R11-1	I-Mom	B/M	E/P	CF/P	NO/P	Dur	I/A	NOA	Δ PI/A	IG
β_{MKT}	-0.08	-0.06	-0.06	-0.03	-0.05	-0.07	-0.09	-0.14	-0.11	-0.03	-0.12	-0.15	-0.18	0.11	0.02	-0.02	0.05	-0.02
β_{ME}	0.10	0.09	0.07	0.09	-0.15	-0.19	0.27	0.40	0.31	0.46	0.25	0.19	-0.32	-0.23	-0.11	0.06	-0.05	-0.11
$\beta_{1/A}$	0.02	-0.11	-0.13	-0.16	0.04	-0.12	-0.07	0.04	-0.03	1.45	0.99	1.01	1.03	-0.85	-1.37	-0.01	-0.77	-0.82
β_{ROE}	0.48	0.45	0.28	0.18	1.33	1.12	1.02	1.48	0.82	-0.51	-0.09	-0.24	0.02	0.24	0.15	-0.01	0.16	-0.07
$t_{\beta_{MKT}}$	-1.82	-1.53	-1.31	-1.20	-0.76	-1.24	-1.17	-1.43	-1.72	-0.59	-2.02	-2.41	-3.86	1.67	0.62	-0.55	1.33	-0.71
$t_{\beta_{ME}}$	1.94	1.27	0.67	1.82	-1.42	-1.98	1.43	1.74	1.86	5.37	1.90	1.66	-4.40	-1.61	-1.81	0.54	-0.94	-1.95
$t_{\beta_{1/A}}$	0.18	-0.97	-1.25	-2.24	0.25	-0.82	-0.27	0.12	-0.13	12.74	5.76	6.79	10.25	-5.69	-15.50	-0.04	-6.98	-10.91
$t_{\beta_{ROE}}$	5.75	5.95	3.26	2.94	10.09	9.96	5.31	5.67	4.90	-5.98	-0.66	-1.78	0.19	1.87	2.29	-0.12	1.93	-1.06
ME	0.69	0.75	-0.01	0.03	0.77	0.87	0.40	0.52	0.62	-2.46	-0.73	-0.89	1.23	0.41	0.88	0.07	0.63	0.22
I/A	-1.46	-0.96	-1.37	-1.13	-0.80	0.72	-4.07	-3.83	-1.18	-9.70	-1.11	-5.63	-14.43	3.95	83.89	55.72	61.16	34.03
ROE	5.80	3.38	1.59	1.49	6.58	6.47	4.14	5.34	1.61	-5.68	0.21	-0.72	1.18	0.49	1.63	-1.24	0.84	0.50
t_{ME}	4.91	5.38	-0.29	1.31	8.75	9.65	4.92	4.95	3.67	-10.31	-4.09	-4.57	7.74	5.23	7.75	1.71	7.56	6.16
$t_{1/A}$	-3.30	-2.57	-2.36	-2.58	-1.22	1.13	-5.54	-4.66	-1.79	-17.06	-1.20	-5.47	-13.81	2.73	32.74	18.28	30.77	22.38
t_{ROE}	16.46	19.07	13.38	15.47	29.77	27.86	16.00	17.06	10.24	-29.57	1.30	-4.83	7.27	2.22	10.00	-7.89	5.10	3.24
NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GP/A	NEI	FP	OC/A	Ad/M	RD/M	OL	Svol		
β_{MKT}	0.04	0.24	-0.03	0.04	0.03	-0.01	0.06	-0.10	-0.14	0.05	0.02	0.44	-0.13	0.04	0.16	-0.06	0.04	
β_{ME}	0.17	0.26	0.12	0.00	0.28	0.15	0.21	-0.41	-0.38	0.03	-0.10	0.43	0.25	0.50	0.66	0.26	0.31	
$\beta_{1/A}$	-0.68	-1.06	-0.96	-0.65	-0.02	-0.90	-0.90	0.10	-0.10	-0.24	-0.30	0.17	0.35	1.42	0.21	0.16	-0.21	
β_{ROE}	-0.32	-0.12	0.05	0.18	0.29	0.05	0.04	1.50	1.31	0.52	0.63	-1.61	0.51	-0.27	-0.58	0.54	-0.43	
$t_{\beta_{MKT}}$	0.99	6.27	-0.77	1.01	0.80	-0.19	1.50	-2.57	-4.48	1.20	0.88	6.46	-3.74	0.50	2.51	-1.22	0.53	
$t_{\beta_{ME}}$	2.24	3.79	2.85	-0.07	4.41	3.20	3.28	-6.56	-6.41	0.51	-2.53	2.45	5.69	2.85	6.75	2.63	2.30	
$t_{\beta_{1/A}}$	-6.14	-13.11	-11.81	-5.49	-0.21	-9.61	-8.72	1.05	-1.23	-2.35	-3.78	0.63	3.52	6.03	1.21	1.34	-1.30	
$t_{\beta_{ROE}}$	-4.07	-1.42	0.56	1.95	4.59	1.04	0.55	20.71	16.86	7.08	10.83	-8.79	7.12	-1.37	-4.10	4.85	-3.54	
ME	-1.37	-1.79	0.26	0.19	-0.24	-0.36	-0.36	2.81	2.66	0.39	2.34	-3.09	-1.31	-1.34	-4.39	-1.31	-0.19	
I/A	27.04	14.80	37.85	44.80	10.15	11.12	16.14	3.56	5.32	-1.29	5.35	-3.91	-13.77	-10.71	-3.22	-5.71	0.66	
ROE	-1.71	-1.41	0.42	1.07	0.88	1.02	0.36	16.95	14.71	3.94	4.32	-8.74	1.52	-3.33	-2.80	1.86	-0.64	
t_{ME}	-6.44	-7.77	4.27	4.86	-5.13	-9.39	-6.76	10.56	10.50	10.84	11.58	-10.76	-9.44	-9.83	-9.47	-8.43	-3.95	
$t_{1/A}$	13.85	14.37	23.42	34.89	5.19	7.90	12.93	4.16	6.88	-2.05	11.45	-4.18	-11.38	-12.17	-2.54	-4.70	1.37	
t_{ROE}	-11.87	-9.07	3.31	8.13	5.06	7.42	2.56	29.02	27.97	23.88	27.36	-25.56	7.97	-12.60	-9.21	11.42	-4.03	

Table 7

Earnings momentum (SUE-1) and price momentum (R6-6) deciles

m , α , α_{FF} , α_C , and α_q are the average excess return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha, and t_m , t , t_{FF} , t_C , and t_q are their t -statistics, respectively. β_{MKT} , β_{ME} , $\beta_{1/A}$, and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the q -factor model, and $t_{\beta_{MKT}}$, $t_{\beta_{ME}}$, $t_{\beta_{1/A}}$, and $t_{\beta_{ROE}}$ are their t -statistics, respectively. ME is the average market capitalization (in billions of dollars) across all of the firms in a given decile. I/A (in percent) is the sum of changes in assets across all of the firms in a given decile divided by the sum of their 1-year-lagged assets. ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given decile divided by the sum of their 1-quarter-lagged book equity. All t -statistics are adjusted for heteroscedasticity and autocorrelations.

Table 8**Momentum-reversal deciles**

At the beginning of each month t , we use NYSE breakpoints to sort all stocks based on their latest earnings surprises. Monthly value-weighted decile returns are calculated, separately, for the holding periods from month t to $t + 11$ (SUE-12); from month $t + 12$ to $t + 35$ (SUE-13-36); and from month $t + 36$ to $t + 59$ (SUE-37-60). In addition, at the beginning of each month t , we use NYSE breakpoints to sort all stocks based on their prior 6-month returns from month $t - 7$ to $t - 2$. Skipping month $t - 1$, we calculate monthly value-weighted decile returns, separately, for the holding periods from month t to $t + 11$ (R6-12); from month $t + 12$ to $t + 35$ (R6-13-36); and from month $t + 36$ to $t + 59$ (R6-37-60). m , α , α_{FF} , α_C , and α_q are the average return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha for the high-minus-low decile, and t_m , t , t_{FF} , t_C , and t_q are their t -statistics adjusted for heteroscedasticity and autocorrelations, respectively. $|\alpha|$, $|\alpha_{FF}|$, $|\alpha_C|$, and $|\alpha_q|$ are the average magnitude of the alphas, and p , p_{FF} , p_C , and p_q are the p -values of the GRS test on the null that all the alphas are jointly zero across a given set of the deciles.

	Panel A: Earnings momentum			Panel B: Price momentum		
	SUE-12	SUE-13-36	SUE-37-60	R6-12	R6-13-36	R6-37-60
m	0.16	-0.11	0.13	0.57	-0.24	-0.03
α	0.19	-0.08	0.16	0.60	-0.25	-0.05
α_{FF}	0.31	-0.04	0.14	0.86	-0.04	0.07
α_C	0.10	0.01	0.14	0.09	-0.02	0.06
α_q	-0.01	0.04	0.07	0.17	-0.09	0.08
t_m	1.57	-1.58	1.69	2.67	-1.84	-0.29
t	2.09	-1.09	2.09	2.86	-1.91	-0.56
t_{FF}	3.31	-0.56	1.90	4.41	-0.37	0.85
t_C	1.17	0.11	1.83	0.83	-0.18	0.76
t_q	-0.06	0.51	0.80	0.71	-0.70	0.86
$ \alpha $	0.09	0.07	0.07	0.13	0.11	0.08
$ \alpha_{FF} $	0.11	0.09	0.09	0.17	0.09	0.08
$ \alpha_C $	0.08	0.09	0.09	0.07	0.09	0.09
$ \alpha_q $	0.07	0.07	0.07	0.06	0.08	0.08
p	0.01	0.10	0.10	0.01	0.00	0.08
p_{FF}	0.00	0.01	0.01	0.00	0.00	0.05
p_C	0.01	0.02	0.05	0.04	0.04	0.02
p_q	0.03	0.00	0.03	0.08	0.00	0.00

Table 9

Operating accrual (OA) and percent operating accrual (POA) deciles

m , α , α_{FF} , α_C , and α_q are the average excess return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha, and t_m , t , t_{FF} , t_C , and t_q are their t -statistics, respectively. β_{MKT} , β_{ME} , $\beta_{1/A}$, and β_{ROE} are the loadings on the market, size, investment, and ROE factors in the q -factor model, and $t_{\beta_{MKT}}$, $t_{\beta_{ME}}$, $t_{\beta_{1/A}}$, and $t_{\beta_{ROE}}$ are their t -statistics, respectively. ME is the average market capitalization (in billions of dollars) across all of the firms in a given decile. I/A (in percent) is the sum of changes in assets across all of the firms in a given decile divided by the sum of their 1-year-lagged assets. ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given decile divided by the sum of their 1-quarter-lagged book equity. All t -statistics are adjusted for heteroscedasticity and autocorrelations.

	Panel A: OA									Panel B: POA										
	Low	2	3	4	5	6	7	8	9	High	Low	2	3	4	5	6	7	8	9	High
m	0.50	0.56	0.66	0.60	0.57	0.60	0.52	0.40	0.40	0.20	0.64	0.56	0.60	0.61	0.46	0.52	0.55	0.43	0.39	0.18
α	-0.06	0.08	0.22	0.18	0.16	0.20	0.07	-0.03	-0.08	-0.39	0.12	0.07	0.16	0.14	0.00	0.09	0.09	-0.03	-0.10	-0.41
α_{FF}	0.10	0.14	0.22	0.13	0.11	0.16	0.10	-0.04	-0.05	-0.27	-0.01	0.03	0.13	0.12	0.01	0.11	0.22	0.08	-0.06	-0.32
α_C	0.11	0.19	0.20	0.08	0.13	0.13	0.10	0.00	-0.02	-0.22	-0.03	0.05	0.07	0.15	0.00	0.12	0.22	0.12	-0.02	-0.28
t_m	1.71	2.38	3.15	2.97	2.77	3.07	2.40	1.76	1.67	0.65	2.39	2.35	2.82	2.60	2.04	2.49	2.53	1.87	1.62	0.58
t	-0.50	0.81	2.96	2.38	2.01	2.51	1.10	-0.39	-0.96	-3.65	1.04	0.77	1.92	1.75	0.05	1.32	1.27	-0.28	-1.09	-3.26
t_{FF}	0.86	1.47	3.11	1.77	1.40	2.05	1.53	-0.47	-0.65	-3.22	-0.05	0.33	1.48	1.40	0.19	1.53	3.13	0.86	-0.73	-3.32
t_C	0.98	1.74	2.58	1.08	1.54	1.56	1.46	0.02	-0.21	-2.49	-0.30	0.50	0.75	1.68	0.06	1.85	3.20	1.15	-0.22	-2.95
The q -factor model regressions																				
α_q	0.39	0.22	0.24	-0.06	0.06	-0.02	0.00	-0.12	-0.21	-0.17	-0.04	0.07	0.12	0.26	0.06	-0.04	0.19	0.10	-0.20	-0.16
β_{MKT}	1.08	1.02	0.96	0.95	0.93	0.91	0.98	0.93	1.04	1.11	1.11	1.05	0.96	1.00	0.98	0.95	0.97	0.93	1.03	1.10
β_{ME}	0.02	-0.09	-0.12	-0.03	-0.09	-0.04	-0.09	0.05	0.03	0.30	0.23	-0.04	-0.06	-0.10	-0.07	-0.04	-0.12	-0.03	0.09	0.37
$\beta_{1/A}$	-0.56	-0.09	0.06	0.27	0.22	0.22	0.01	0.01	-0.03	-0.58	0.35	0.26	0.14	0.05	-0.05	0.06	-0.25	-0.30	-0.11	-0.55
β_{ROE}	-0.26	-0.12	-0.05	0.16	0.00	0.17	0.14	0.11	0.22	0.03	-0.12	-0.20	-0.04	-0.19	-0.02	0.17	0.09	0.07	0.22	-0.06
t_q	3.13	1.62	3.02	-0.78	0.66	-0.19	0.00	-1.23	-2.23	-1.86	-0.40	0.75	1.41	2.89	0.71	-0.55	2.54	0.95	-1.91	-1.34
$t\beta_{\text{MKT}}$	32.16	35.50	56.68	48.78	38.87	42.85	44.43	41.06	35.50	39.68	43.24	37.37	30.61	43.73	47.87	40.75	41.32	35.95	50.05	34.83
$t\beta_{\text{ME}}$	0.46	-2.09	-4.28	-1.17	-2.16	-1.19	-2.55	1.19	0.82	6.79	6.54	-0.91	-1.34	-3.12	-2.22	-1.54	-3.55	-0.85	1.94	7.96
$t\beta_{1/A}$	-6.04	-0.80	1.05	5.83	2.75	3.14	0.12	0.08	-0.42	-9.86	6.29	3.76	2.36	0.95	-0.67	1.13	-4.84	-5.17	-1.25	-6.41
$t\beta_{\text{ROE}}$	-3.55	-1.79	-1.08	4.17	0.05	4.06	3.21	2.13	3.83	0.69	-2.42	-2.70	-0.70	-4.44	-0.46	4.34	2.02	1.23	3.57	-1.24
Characteristics in the q -factor model																				
ME	0.81	1.48	1.93	2.21	2.37	2.14	2.37	1.78	1.46	0.58	0.90	1.10	1.41	1.69	1.73	1.97	2.47	2.35	1.49	0.54
I/A	13.87	9.06	7.72	7.37	7.45	7.78	8.63	7.13	11.49	24.02	5.16	6.97	6.48	8.85	8.52	8.38	10.81	11.41	14.22	16.28
ROE	1.88	2.61	2.90	2.91	2.86	2.90	3.09	2.94	2.89	2.75	1.22	1.82	2.24	2.47	2.69	3.21	3.75	3.96	3.51	2.24

Table 10

Twenty-five size and book-to-market portfolios

m , α , α_{FF} , α_C , and α_q are the average excess return, the CAPM alpha, the Fama-French alpha, the Carhart alpha, and the q -alpha, and t_m , t_{FF} , t_C , and t_q are their t -statistics, respectively. $|\alpha|$, $|\alpha_{FF}|$, $|\alpha_C|$, and $|\alpha_q|$ are the average magnitude of the alphas, and p , p_{FF} , p_C , and p_q are the p -values of the GRS test across the 25 portfolios. β_{MKT} , β_{ME} , $\beta_{I/A}$, and β_{ROE} are the market, size, investment, and ROE factor loadings in the q -factor model, and $t_{\beta_{MKT}}$, $t_{\beta_{ME}}$, $t_{\beta_{I/A}}$, and $t_{\beta_{ROE}}$ are their t -statistics, respectively. ME is the average market capitalization (in billions of dollars) across all of the firms in a given portfolio. I/A (in percent) is the sum of changes in assets across all of the firms in a given portfolio divided by the sum of their 1-year-lagged assets. ROE (in percent) is the sum of the most recently announced quarterly earnings across all of the firms in a given portfolio divided by the sum of their 1-quarter-lagged book equity. The t -statistics are adjusted for heteroscedasticity and autocorrelations.

	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L
	α ($ \alpha = 0.29$)											
	α_{FF} ($ \alpha_{FF} = 0.10$)											
Small	0.08	0.72	0.84	0.95	1.11	1.02	-0.60	0.13	0.30	0.45	0.59	1.19
2	0.32	0.69	0.86	0.87	0.99	0.67	-0.34	0.12	0.33	0.39	0.48	0.82
3	0.38	0.71	0.77	0.77	1.02	0.65	-0.25	0.16	0.28	0.32	0.56	0.81
4	0.52	0.59	0.73	0.74	0.84	0.32	-0.07	0.08	0.24	0.29	0.40	0.47
Big	0.40	0.54	0.54	0.61	0.56	0.16	-0.07	0.10	0.14	0.25	0.19	0.26
	t ($p = 0.00$)											
Small	0.20	2.02	2.57	3.07	3.30	4.59	-2.55	0.64	1.66	2.54	2.98	5.62
2	0.90	2.20	3.03	3.29	3.31	2.93	-1.94	0.77	2.34	2.77	2.60	3.70
3	1.15	2.49	3.03	3.05	3.93	2.76	-1.77	1.40	2.27	2.52	3.37	3.63
4	1.73	2.35	2.91	3.08	3.24	1.43	-0.57	0.76	2.11	2.33	2.53	2.09
Big	1.70	2.53	2.71	3.04	2.43	0.79	-0.79	1.27	1.42	2.10	1.17	1.24
	α_q ($ \alpha_q = 0.11$)											
	β_{MKT}											
Small	-0.48	0.03	0.12	0.18	0.22	0.70	-0.25	0.27	0.31	0.30	0.32	0.57
2	-0.18	0.03	0.09	0.10	0.04	0.22	-0.14	0.02	0.03	0.07	0.10	0.24
3	-0.04	0.04	0.09	0.03	0.16	0.20	-0.01	-0.03	-0.04	-0.01	0.14	0.15
4	0.15	-0.01	0.07	0.03	0.09	-0.06	0.18	-0.14	-0.01	0.02	0.06	-0.12
Big	0.17	0.07	0.07	-0.03	-0.13	-0.31	0.10	-0.04	0.06	-0.01	-0.04	-0.13
	t_q ($p_q = 0.00$)											
Small	-4.00	0.36	1.58	2.59	2.53	5.72	-1.48	2.24	3.09	3.68	2.72	2.91
2	-2.28	0.37	1.34	1.40	0.44	1.88	-1.21	0.29	0.37	0.67	0.89	1.25
3	-0.50	0.53	0.93	0.28	1.40	1.43	-0.09	-0.30	-0.37	-0.05	1.16	0.92
4	1.87	-0.16	0.74	0.24	0.75	-0.42	1.50	-1.58	-0.06	0.21	0.44	-0.61
Big	2.89	0.91	0.70	-0.36	-1.02	-2.12	1.32	-0.49	0.65	-0.06	-0.23	-0.70
	$t_{\beta_{\text{MKT}}}$											
Small	-4.00	0.36	1.58	2.59	2.53	5.72	-1.48	2.24	3.09	3.68	2.72	2.91
2	-2.28	0.37	1.34	1.40	0.44	1.88	-1.21	0.29	0.37	0.67	0.89	1.25
3	-0.50	0.53	0.93	0.28	1.40	1.43	-0.09	-0.30	-0.37	-0.05	1.16	0.92
4	1.87	-0.16	0.74	0.24	0.75	-0.42	1.50	-1.58	-0.06	0.21	0.44	-0.61
Big	2.89	0.91	0.70	-0.36	-1.02	-2.12	1.32	-0.49	0.65	-0.06	-0.23	-0.70

	Low	2	3	4	High	H-L	Low	2	3	4	High	H-L
	β_{ME}						$\beta_{I/A}$					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ME}}$						$t_{\beta_{I/A}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70	-9.48	-2.32	3.93	6.26	8.97	10.46
3	14.60	16.16	5.21	7.05	4.85	-1.80	-11.40	0.43	3.35	6.24	9.13	12.41
4	6.26	6.97	3.22	3.63	2.14	-1.54	-8.47	2.51	4.05	5.50	7.28	9.57
Big	-7.59	-2.67	-5.22	-1.63	-2.15	1.06	-9.47	2.19	4.98	4.87	6.25	8.00
	ME						I/A					
Small	0.07	0.07	0.07	0.06	0.05	-0.03	10.90	12.67	9.00	7.02	1.77	-9.13
2	0.33	0.33	0.33	0.33	0.32	-0.01	17.87	14.63	11.22	7.57	3.92	-13.94
3	0.76	0.76	0.76	0.76	0.76	0.00	18.30	13.64	9.95	8.14	4.47	-13.82
4	1.86	1.83	1.78	1.77	1.82	-0.04	15.28	11.31	8.86	7.06	5.18	-10.09
Big	15.98	13.55	11.19	9.94	8.45	-7.53	12.59	11.36	7.49	8.22	6.44	-6.14
	β_{ROE}						ROE					
Small	1.14	1.18	1.08	1.01	0.99	-0.15	-0.66	-0.32	-0.11	0.17	0.53	1.19
2	0.93	0.94	0.83	0.72	0.85	-0.07	-0.72	-0.13	0.25	0.42	0.59	1.31
3	0.72	0.63	0.49	0.43	0.47	-0.24	-0.77	0.03	0.37	0.54	0.78	1.56
4	0.40	0.33	0.24	0.20	0.18	-0.21	-0.70	0.18	0.39	0.60	0.77	1.47
Big	-0.22	-0.09	-0.21	-0.10	-0.14	0.08	-0.39	0.12	0.30	0.61	0.80	1.20
	$t_{\beta_{ROE}}$						$t_{\beta_{ROE}}$					
Small	16.49	18.22	23.29	33.68	14.79	-1.31	-5.09	-3.64	-1.46	3.06	5.90	8.59
2	16.68	32.58	19.26	12.82	13.78	-0.70</						

Maximum Sharpe ratios

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Panel A: Sharpe ratios										Panel B: Maximum Sharpe ratios										
		MKT	SMB	HML	UMD	r_{ME}	$r_{I/A}$	r_{ROE}		CAPM	FF Carhart	q								
		0.10	0.06	0.13	0.16	0.10	0.24	0.22		0.10	0.21	0.30	0.43							
Panel C: Anomaly portfolios																				
		SUE-1	SUE-6	Abr-1	Abr-6	RE-1	RE-6	R6-6	R11-1	I-Mom	B/M	E/P	CF/P	NO/P	Dur	I/A	NOA	$\Delta PI/A$	IG	
S_{H-L}		0.14	0.09	0.23	0.15	0.16	0.13	0.15	0.16	0.10	0.14	0.12	0.11	0.16	0.12	0.11	0.12	0.17	0.14	
S_m		0.27	0.26	0.28	0.23	0.25	0.21	0.31	0.28	0.20	0.21	0.24	0.21	0.30	0.24	0.24	0.29	0.25	0.26	
		NSI	CEI	IvG	IvC	OA	POA	PTA	ROE	ROA	GP/A	NEI	FP	OC/A	Ad/M	RD/M	OL	Svol	All	
S_{H-L}		0.21	0.14	0.13	0.14	0.10	0.15	0.12	0.15	0.13	0.10	0.14	0.10	0.18	0.14	0.12	0.10	0.14	0.48	
S_m		0.30	0.32	0.23	0.25	0.25	0.23	0.23	0.24	0.21	0.18	0.26	0.28	0.26	0.23	0.21	0.17	0.29	1.60	

Figure 1
The investment channel

